

COMPRESSION DYNAMICS AND RADIATION
EMISSION FROM A DEUTERIUM
PLASMA FOCUS

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To my loving father and mother,

beloved sister and brothers

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MUHAMMAD SHAHID RAFIQUE

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by
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The main aim of the project was to investigate and correlate the focus dynamics during the radial phase with the radiation (deuterons, neutrons and X-rays) emitted from our 3 kJ plasma focus device.

A magnetic spectrometer was successfully designed, constructed and implemented on the forward axis of our focus device. An automated scanning and measurement system was also successfully designed, developed and employed. This developmental work enabled us to obtain the energy spectra of the deuterons emitted in the forward direction, and to correlate the ion emission with the neutron output.

The plasma dynamics was investigated by means of a three-frame computer-controlled laser shadowgraphic system specifically designed, developed and successfully implemented in this project.

The deuteron energy spectra ranging from 80 keV to 250 keV are in strong correlation with the corresponding total neutron yield. The ion energy distribution is well fitted by an exponential function of the form $\exp(-E_d/B)$. The spectra at higher energies are in good agreement with the empirical relation $dN/dE \propto E^{-x}$.

An average neutron energy of 2.48 MeV and 3 MeV were determined in the radial and the axial directions, respectively. The neutron production was correlated with the neutron anisotropy and soft X-ray output. The discharges with high neutron yield exhibit high anisotropy and low soft X-ray production.

Higher growth rates of the instabilities favor neutron emission. The pinch lifetime is shorter for the discharges with higher growth rates. The radiation output is higher for the discharges with shorter pinch lifetime. The neutron yield is higher for smaller pinch radius, and the soft X-ray production is lower. The neutron yield is lower for the discharges with the longer pinch length, which is favorable for the soft X-ray production. The radiation output appears higher for the discharges with larger aspect ratio. The discharges with higher radial implosion speeds exhibit higher radiation output.

The experimentally measured radial and axial elongation trajectories along with the radial implosion speed are in good agreement with the results obtained from the numerical simulation of the plasma focus.

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SUMMARY

of the PhD thesis entitled

COMPRESSION DYNAMICS AND RADIATION EMISSION FROM A DEUTERIUM PLASMA FOCUS

submitted by

MUHAMMAD SHAHID RAFIQUE

SUPERVISOR: *Prof. Sing Lee*

This thesis presents the results of our investigations performed on the radiation (neutrons, deuterons, soft X-rays and hard X-rays) emitted from a 3kJ Mather-type plasma focus device.

The first objective of this project was to investigate the contribution to the neutron yield from the beam-target mechanism in the deuterium plasma focus. This aim was achieved by studying the correlation between the total neutron yield and the time-integrated energy spectra of the axially emitted deuteron beam, on a shot-to-shot basis.

To attain this goal, a magnetic spectrometer was successfully designed, constructed and implemented in our experiments. CR-39 solid state nuclear track material was used in the plane of the spectrometer to detect the energetic ions.

More importantly, an automated scanning and measurement system was also successfully developed and employed to overcome the traditional time-consuming and laborious counting and measurement of ion tracks. This automated image analysis system was controlled by a PC running a program written in *ALI* (analytical language for image processing) specifically developed for this application. This complex automated deuteron energy spectra analysis system represents the element of novelty of this work.

Deuteron spectra and total neutron yield for more than five hundred plasma focus discharges were obtained. Cumulative spectra for three sequential discharges were taken for improved statistical accuracy. A sequence of spectra was obtained for deuterium gas pressures of 3.5, 4.0, and 4.5 mbar.

Our analysis shows a clear correlation between the intensity of the axially emitted deuteron beam and the corresponding total neutron yield. The deuteron energy spectra range from 80 keV to 250 keV, with $100 \text{ keV} < E_d < 140 \text{ keV}$ as the most frequent energy range for each operating pressure.

The experimental ion energy distribution is relatively well fitted by an exponential function of the form $\exp(-E_d/B)$. It is also noted that the spectra at higher energies are in good agreement with the empirical relation $dN/dE \propto E^{-x}$, as suggested by previous workers.

Time-resolved measurements were also performed on the emission of neutrons, soft X-rays, and hard X-rays. Our objective was to correlate the electrical discharge parameters and the characteristic time periods of the focus with some of the emission parameters. The idea was to identify the focus operating regimes and the favorable conditions for enhanced emission of radiation.

Different diagnostics were employed for this purpose. The time-resolved soft X-ray measurements employed two filtered *PIN* diodes. This enabled us not only to monitor the emission characteristics of the focus in the soft X-ray range, but also to determine, in some discharges, the electron temperature of the pinch by employing the filter ratio method.

The time-resolved hard X-ray and neutron measurements were performed using three plastic scintillator-photomultiplier systems. Two of these detectors were placed radially with respect to the focus in a time-of-flight geometry. The other detector was placed in the axial direction of the focus. The idea was to estimate the neutron energies and prove the dominant effect of beam-target mechanism responsible for the neutron production.

The signals from our time-resolved diagnostics were recorded using a *1 GSa/s*, *250 MHz Hewlett Packard* digital logic analyzer [model 16500A]. Using the software library that came with the interface card, a program written in C++ was developed to transfer and store data from the scope to an *ASCII* file on a PC via a *GPIB* interface. The acquired data were analyzed using Microsoft *Excel* and *Microcal Origin 6.0*. The time resolution achieved in our work was 1 nanosecond.

An average neutron energy of 2.48 MeV in the radial direction was determined using the peak-to-peak method of analysis. This result was combined with the estimated average neutron energy of 3 MeV in axial direction. This supports the dominant effect of the beam-target mechanism responsible for neutron production.

These results correlate very well with our work on deuteron emission spectra. Deuterons of energy 100 keV were found to be responsible for the production of 3 MeV neutrons in the forward direction.

The neutron anisotropy was determined by integrating the time-resolved emission rate signals from the fast plastic scintillator-photomultiplier systems placed at the same distance from the pinch, one in the axial and the other one in the radial direction. An average value of 1.45 for the neutron anisotropy was determined in our experiments.

For each focus discharge, the neutron anisotropy was correlated with the total neutron yield and the soft X-ray production. The correlation reveals that, regardless of the operating pressure, the anisotropy is greater for discharges with higher neutron yield. The correlation between the anisotropy and the soft X-ray production shows that the anisotropy is lower for discharges with high soft X-ray production (regardless of the operating pressure).

The soft X-ray and the total neutron yield were also correlated. While performing the neutron optimization of the device, it was observed that lower operating pressures favor the soft X-ray production. As the pressure increases, the soft X-ray yield decreases and the neutron emission is enhanced. The optimum neutron production pressure is 4 mbar of deuterium. The further increase of the pressure (beyond 4 mbar) causes a reduction in both soft X-ray and neutron emission. The plasma focus device cannot be optimized, simultaneously, for both neutron and soft X-ray production.

The correlation of all our time-resolved diagnostic parameters help us to identify two regimes of the plasma focus, when operated in deuterium. One is the single-compression regime corresponding to a single-compression during the pinch phase. The

other one is the multiple-compression regime associated with two or more compressions of the focussed column. Our data show that the neutron yield is greater in discharges exhibiting the single-compression regime.

The optical investigation of the radial phase of the focus was initially performed by means of a single-frame laser shadowgraphy. The shadowgraphic images of the focus were correlated with the electrical parameters of the discharge, as well as with the X-ray and neutron emission characteristics. This enabled us to identify the important periods of the focus (i.e. the radial compression, the instant when minimum pinch radius is attained, the pinch lifetime, and the post focus phenomena), and to correlate the presence of the plasma instabilities with the radiation output.

Due to shot-to-shot variations in the focus evolution, one needs to capture multiple frames in order to get a clearer picture of the focus dynamics. For this purpose, a three-frame, computer-controlled laser shadowgraphic system was developed and implemented successfully. The implementation of this method allowed us *i)* to study the plasma dynamics during the compression phase, *ii)* to examine the temporal evolution of the plasma instabilities during the radial and pinch phases, *iii)* to monitor the radial implosion parameters (i.e. radial and axial shock positions, radial and axial shock speeds, etc.), *iv)* to determine the pinch lifetime and its geometrical characteristics, *v)* to correlate all this information with the radiation emission characteristics of the focus, and *vi)* to compare the experimental data with the results predicted by the theoretical model.

The growth rate of the instabilities was correlated with the radiation output: higher growth rates favour neutron emission. A correlation of the instability growth rate with the pinch lifetime showed that the pinch lifetime is shorter for the discharges with

higher growth rates. The radiation output is higher for the discharges with shorter pinch lifetime. The pinch dimensions (column radius, column length and aspect ratio) were also correlated with the radiation output.

The trajectories of the inward radial shock front and its axial elongation were determined from the shadowgraphic images. Radial and axial elongation velocities were also calculated. The results were then correlated with the radiation output. The discharges with higher radial implosion speeds exhibit higher radiation output.

A computational modeling of the plasma focus was also performed. The experimentally measured radial and axial elongation trajectories were compared with the simulated trajectories. The simulated results are in fairly good agreement with the experimental data.

This dissertation is structured in seven chapters. The first chapter is an introductory description of the historical elements of the plasma focus.

The focus dynamics, together with the radiation emission and related phenomena, are thoroughly discussed in the second chapter devoted to the literature review.

The theoretical descriptions of the diagnostics employed are presented in chapter three.

A computational model of the plasma focus is discussed in chapter four.

Chapter five presents the experimental set-up of the diagnostics employed.

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Chapter 1

INTRODUCTION

For many years, the greatest effort in the direction of achieving controlled thermonuclear fusion has been concentrated on the study of powerful electrical discharges. In these high power discharges, the electrodynamic forces cause both the magnetic confinement and the heating of the plasma.

The experiments were termed as linear and toroidal Z-pinches, as the current flows in the axial direction or Z-axis [1]. The investigation of powerful electrical discharges in coaxial geometries (i.e. linear Z-pinches) in atmosphere of deuterium led to the observation of neutron emission. The internal flux of neutrons was found to vary with the gas filling pressure, purity of the gas and the discharge current.

The intense burst of neutrons was found to be accompanied by X-ray emission [2]. The focused plasma column becomes unstable due to magnetohydrodynamic instabilities (sausage and kink modes) which also cause high electric fields that accelerate ions to very high energies. The highly energetic ion beams produce neutrons by interaction with the residual plasma, a mechanism known as beam-target interaction.

One of the devices developed for producing hot and dense plasmas of fusion interest is the plasma focus.

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Plasma acceleration by $\vec{j} \times \vec{B}$ forces has been experimentally and theoretically studied in many laboratories since the beginning of the controlled fusion program.

The early work in USA with coaxial gun systems was pioneered by Marshall [3]. Mather, following the work of Osher [4], investigated the fast coaxial gun mode which led to the development of the high pressure mode of operation.

The plasma focus device, in which an accelerated plasma is

magnetically compressed to a short lived (50-200 ns), rather dense ($\sim 10^{-19} \text{ cm}^{-3}$) and hot ($\sim 1 \text{ keV}$) filament (focus), was discovered independently by Mather (USA) [5] and Filippov (USSR) [6] in early 1960s.

The devices investigated by these two pioneers have significantly different geometries, though they are similar in many aspects and results. Both are composed of two coaxial electrodes (see Figure 1.1) separated by an insulator sleeve. They form an accelerator powered by a capacitor bank switched by a sparkgap. The acceleration of the current sheath, the radial compression to the axis at the end of the gun leading

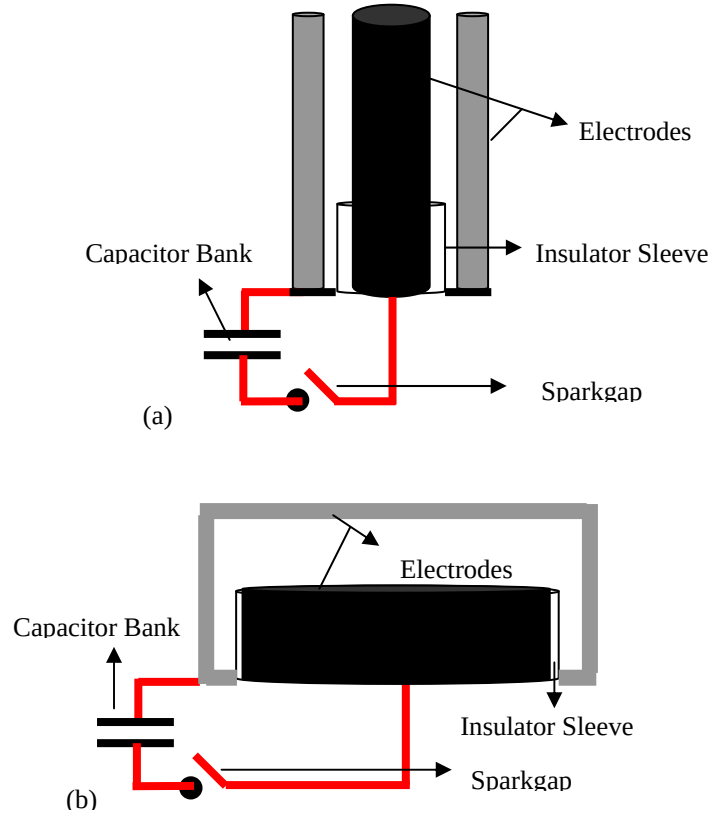


Figure 1.1: Sketch of plasma focus geometry
a) Mather-type b) Filippov-type

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to a dense Z-pinch filament, and the following turbulent plasma phase, characterize the typical plasma focus discharge.

Many plasma focus devices have since been built by other investigators, but all broadly conform to one of the two original geometries, and can be classified as either of Mather- or Filippov-type.

A large range of investigations have been performed on both types of plasma focus geometries with stored energies varying from 1kJ to 1 MJ. During 60's, the intense bursts of neutrons from the plasma focus device operated in deuterium led to the premature announcement of the achievement of the thermonuclear fusion. This, in turn, led to great progress in improving the associated pulse power technology. It also helped to understand the various physical phenomena occurring in these systems.

Over the years, scientists have developed several devices with the goal of increasing the neutron yield. For more than 20 years, the dense plasma focus (DPF) device was the strongest pulsed neutron source available. The transient nature (i.e. the neutron emission time is $\sim 0.1 \mu\text{s}$) of the plasma focus is both its beauty and a drawback.

The plasma focus device emits neutrons from fusion reaction depending on the fuel used (e. g. deuterium, mixture of deuterium and tritium, etc.). Fueling the focus with the mixture of tritium and deuterium, ~ 5 -10 times more neutrons of 10-13 MeV energy can be produced from the $D - Li$ reaction as compared to the $D - D$ reaction (2.5 MeV) [7]. Thus, plasma focus can generate neutrons above 10 MeV without the implications associated with the handling of tritium.

The half width of the neutron spectrum emitted by plasma focus is close to the one emitted by a thermonuclear plasma with $kT_i = 3 \text{ keV}$ [8]. Since, in DPF devices,

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plasma generation does not require a massive structure like the tokamak, the useful neutron spectrum is distorted little by neutron scattering in the structural material of the device.

The neutron-emitting volume is a cone, the height of which can range between 1 and 100 mm [9], depending on the experimental conditions. Within limits, plasma focus can be operated as a point source, or as a volume source.

All the plasma neutron sources developed and operated so far have achieved comparable values of the neutron flux. Plasma focus, however, offers a cost reduction of a factor of 100 for a given number of neutrons produced per square centimeter [10].

Besides being a ready source of hot dense plasma and fusion neutrons, the focus also emits copious amount of X- rays (soft and hard), especially when operated with high Z gases. This distinctive feature of the plasma focus sets it apart from other devices as a prime candidate for industrial applications.

In summary, a wide range of plasma phenomena is readily available from this easily constructed machine, and has led to its study in many laboratories around the world.

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1.1 OBJECTIVES OF THE THESIS

The main purpose of this experimental work was to study the emission of radiation (neutrons, deuterons and X-rays) from a small plasma focus device operated in deuterium and to correlate the radiation output with the focus dynamics during the radial phase.

The first objective of the project was to obtain the energy spectra of the axially emitted deuteron beams on a shot-to-shot basis and to correlate the deuteron flux with the total neutron yield.

To attain this goal, a magnetic spectrometer was designed, constructed and implemented on the forward axis of our 3 kJ plasma focus device.

In order to detect the energetic ions, CR-39 solid state nuclear track material was used in the plane of the spectrometer. To overcome the major shortcoming of very laborious and time-consuming requirement of counting/measuring tracks by visual inspection using an optical microscope at high magnification, an automated image scanning and track measurement system was developed and implemented.

The second objective of the project was to investigate the correlation between the focus dynamics and the related phenomena. This goal was achieved by studying the emission of soft X-rays, hard X-rays and neutrons and by correlating some of the emission parameters with the characteristic time periods of the focus.

Different diagnostics were employed for this purpose. The idea was to improve and extend some of the classical methods of detection by developing and implementing, in our work, new and updated high performance data acquisition and analysis systems provided by the modern technologies.

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The optical investigation of the radial phase of the focus was initially performed by means of single-shot laser shadowgraphy in a divergent beam. The new element of this work was the use of a *CCD* based image capture system.

The time-resolved soft X-ray measurements employed filtered *PIN* diodes. This enabled us not only to monitor the emission characteristics of the focus in the soft X-ray region, but also to determine, in some discharges, the electron temperature of the pinch by employing the filter ratio method.

The time-resolved hard X-rays and neutron measurements used three plastic scintillator-photomultiplier systems, of which two were placed radially at different distances from the anode (time-of-flight geometry). The other detector was placed in the axial direction of the focus. The idea was to measure the neutron energies and re-iterate the dominant effect of the beam-target mechanism responsible for the neutron production.

The total neutron yield was monitored by an In foil activation detector placed axially on top of the focus chamber.

Another new element introduced by the present work was the measurement of the anisotropy of the neutron emission. The anisotropy was determined by integrating the time-resolved emission rate signals from the scintillator-photomultiplier systems placed at the same distance from the pinch, one in the axial and the other in the radial direction.

The focus discharge and the diagnostics were triggered and controlled by a computer. Our data acquisition and analysis system enabled us to record, transfer and store data from all the diagnostics simultaneously. The time resolution achieved in our work was 1 nanosecond.

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The diagnostics employed enabled us to analyze and correlate the following radiation parameters: total neutron yield, neutron flux anisotropy and soft X-ray production.

The correlation of the shadowgraphic images of the focus with the electrical parameters of the discharge together with the X-rays and neutron emission characteristics enabled us to identify the important periods of the focus (i.e. the radial compression, the instant when minimum pinch radius is attained, the pinch lifetime, and the post focus phenomena).

With respect to this idea, we extended our work in developing a three-frame computer-controlled CCD-based laser shadowgraphy system.

Our aim was to study the correlation between the pinch characteristics (column length, column radius, aspect ratio), the radiation (neutrons and soft X-rays) emission, and the instabilities growth rates.

The third objective of this project was to perform a numerical simulation of the focus dynamics and to correlate the data with the experimental results. This aim was achieved by implementing a computational model which includes the effect of radiation.

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1.2 LAYOUT OF THE DISSERTATION

A summary of the content of each chapter is presented in the following paragraphs.

The first chapter gives some introductory and historical elements of the plasma focus. This is followed by the presentation of the objectives of the thesis.

The second chapter is devoted to the description of the plasma focus and the emission of the radiation. The first part describes the dynamics of the dense plasma focus discharge. Various phenomena like fusion reaction and emission of the fusion products, instabilities and X-ray emission are discussed in the next sections.

A detailed description of the diagnostic techniques employed in the experiments is given in the third chapter.

In chapter four, a one dimensional plasma focus model is discussed.

Chapter five provides the details of the plasma focus facility (NIE-SSC-PFF) and the experimental set-up employed in our project.

Chapter six presents the results and provides a detailed discussion and interpretation of the data.

Chapter seven is devoted to the conclusions of the present work, and proposes some directions for the future investigations of the focus phenomena, as well as some practical applications of the radiation emitted from the DPF device.

Chapter 2

PLASMA FOCUS DYNAMICS, RADIATION EMISSION AND ASSOCIATED PHENOMENA

The dense plasma focus device is a magnetohydrodynamic coaxial plasma accelerator [11, 12]. In this device, the magnetic energy is initially stored behind the moving current sheath. A fraction of this energy is converted into plasma energy during the rapid collapse (Z-pinch) of the current sheath towards the axis beyond the end of the central electrode.

The dynamics of the plasma focus discharge presented in the first section of this chapter is followed by a discussion on the radiation (neutrons and X-rays) emission and related phenomena.

2.1 DYNAMICS OF THE DPF DISCHARGE

The dense plasma focus (*DPF*) consists of two coaxial electrodes (anode and cathode) separated by an insulator sleeve. The Filippov [13] geometry has an anode aspect ratio (diameter/length) >1 , whereas the Mather-type [3] is characterised by an aspect ratio < 1 . The device under investigation in the Division of Physics, NIE/NTU is of Mather-type, as shown schematically in Figure 2.1.

The operation of the plasma focus can be divided into three stages separated both in time and location: breakdown or inverse pinch phase, axial acceleration phase, and radial collapse phase.

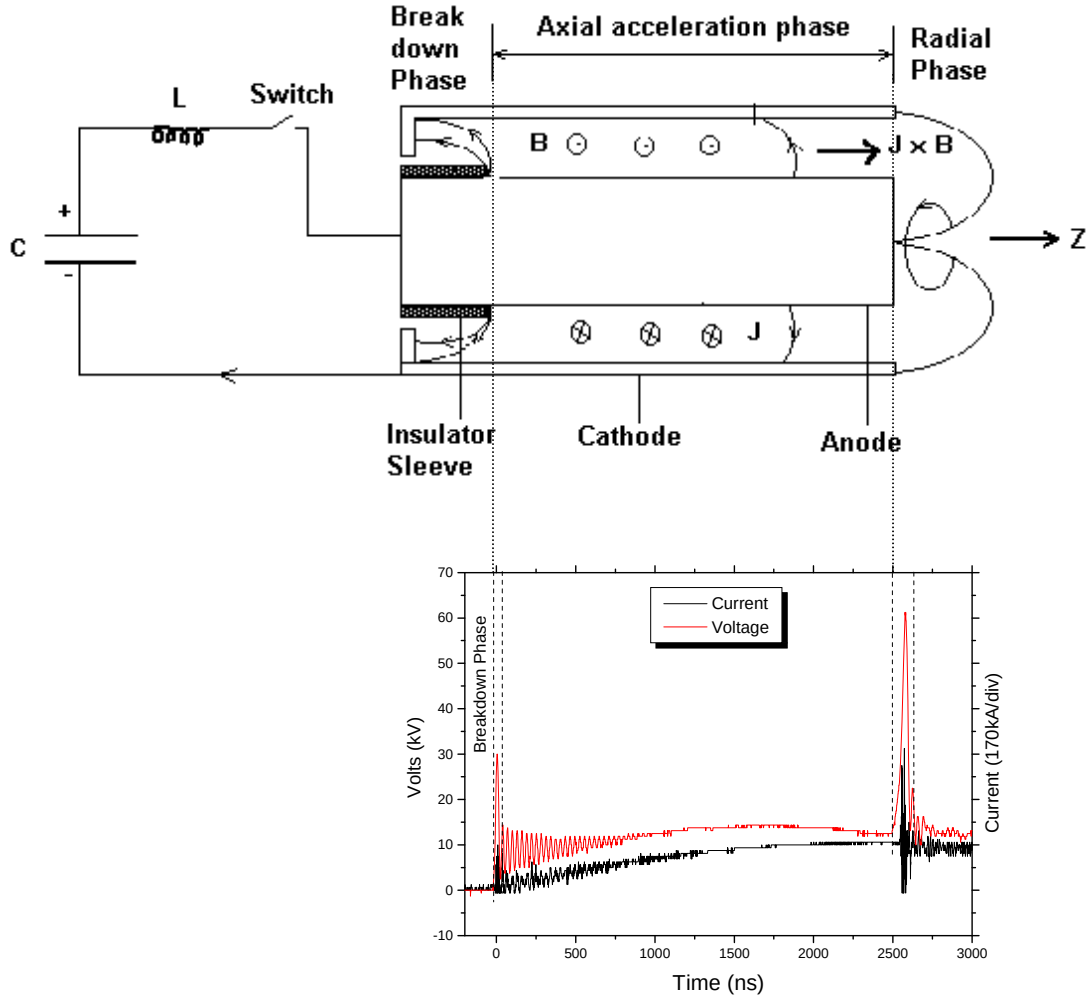


Figure 2.1: Current sheath dynamics in DPF and the corresponding current and voltage signal profiles showing the phases of the discharge.

These three phases are illustrated in Figure 2.1. The plasma focus device is usually operated with the inner electrode as anode and the outer electrode as cathode. The operation of the device with the reverse polarity decreases the neutron and X-ray output [14]. The gross dynamics of the three distinct phases is described below.

2.1.1 BREAKDOWN OR INVERSE PINCH PHASE

An azimuthally symmetric discharge can be initiated when a high voltage pulse is applied between the two coaxial electrodes of the DPF device filled with an appropriate pressure of the working gas. At the beginning of the discharge, an electrical breakdown occurs. This generates some initial plasma configuration through which the discharge current can flow. When the DPF device is operated with deuterium in the pressure range of 1-10 mbar, the value of the static breakdown voltages are lower than 1 kV (Paschen's law) [15], i.e. one order of magnitude smaller than the usual bank voltage.

The parameters of the electrodes [16, 17], the solid insulator sleeve [18 - 21] and the initial gas conditions [18, 19] are major factors on which the development of the high current discharge (within the plasma focus system) depends.

At very low pressures, the discharge can develop within the whole inter electrode volume [22]. On the other hand, at very high pressures, a radial filamentary discharge occurs between the coaxial electrodes [23, 24].

At the optimized operating pressure, a sliding discharge develops along the insulator. This sliding discharge, also called gliding discharge or flashover, can be explained by the effect of the surface capacitance. Usually, a discharge is initiated at the edge of the outer electrode due to the appearance of a high electric field.

When high voltage is applied, breakdown occurs along the insulator surface after a time delay of a few nanoseconds [18]. This time delay is almost independent of the voltage applied (within a reasonable range), but depends strongly on the initial gas pressure [24]. During this time, further electrons are created by field emission from the metallic edges and ionisation of the ambient gas. These electrons are accelerated

along the radially directed electric field lines towards the inner electrode and the insulator surface, if the central electrode is the anode. As the discharge develops, depending on the gas and surface conditions, the surface capacitance is gradually charged. The highest electric field remains at the front of the discharge. Due to this effect, an ionisation front along the insulator surface is propagated. When the sliding discharge reaches the end of the insulator, both electrodes are connected. The breakdown takes place just above the surface of the insulator between the anode and the back plate of the cathode. During this breakdown, weak current filaments are formed, which are then lifted off from the insulator surface in an inverse pinch manner propelled by the $\vec{j} \times \vec{B}$ force (see Figure 2.1).

When the current filaments reach the inner surface of the outer electrode (after 50-500 ns), they blend to form a uniform current sheath due to a high conductance of the sliding discharge. This current is a double layer structure consisting of an ionisation front and a magnetic piston [25 - 27].

The frontal section of the sheath (attached to the inner electrode) remains temporarily immobile during the inverse pinch phase. When the uniform current sheath is formed, the focus formation process enters into the second phase.

2.1.2 AXIAL ACCELERATION PHASE

The current sheath formed in the breakdown phase is accelerated by its own $\vec{j} \times \vec{B}$ force towards the open end of the electrodes. The radial component of $\vec{j} \times \vec{B}$ pushes the current sheath towards the outer electrode. The axial component of the force is stronger near the anode due to $1/r$ dependence of the force (j_r, B_θ). This leads

to a higher velocity of the current sheath near the surface of the central electrode than near the outer electrode.

The following are the two conditions that make the axial phase important for the subsequent formation of the hot and dense plasma. The first condition is that the current sheath should arrive at the axis in coincidence with the first maximum of the discharge current. This condition is a common requirement to all the pinch devices. It represents the optimisation of the energy transfer from the capacitor bank to the pinch plasma.

The second condition is that the structure and the (r, z) profile of the plasma sheath should have certain characteristics for a good focusing effect. This condition requires an axial symmetry, smooth (r, z) profiles and a thin uniform current sheath structure.

As no significant tangential stress exists within the current sheath, the forces acting on it are normal to the surface everywhere. Therefore, the evolution of the current sheath must be conceived as a sort of swelling balloon rather than a rigid piston. The mean free path, both for ion-atom and elastic collision, is very small for a normal filling pressure range. A gas dynamical shock is produced by this fast moving structure. This shock heats and compresses the neutral gas in front of it. Therefore, the sheath will have a complex structure with a compressed and hot ionising region.

The pattern of the driving current determines the complete structure of the sheath. The more uniform is the pattern, the smoother is the sheath. An azimuthal filamented pattern will produce a filamentary structured sheath [27 - 29].

For the axial rundown phase, different diagnostic techniques like: magnetic probe measurements, schlieren, shadowgraphy, interferometric investigations, image

converting cameras and other light detecting systems, etc. are usually employed. The measured speed of the sheath is 1.5-15 cm/ μ s [30, 31]. It was found that for large plasma focus devices, a large fraction (50%) of the current going into the device flows behind the current sheath. The thickness of the plasma sheath is 2-4 cm [30, 31].

At the end of this phase, one end of the current sheath sweeps around the end of the anode. The outer end of the current sheath continues along the tube sweeping along with it the greater portion of the accumulated plasma in the axial direction. Only a fraction of plasma at the end of axial acceleration phase contributes to the final focus.

2.1.3 RADIAL PHASE

At the end of the axial rundown phase, the current sheath sweeps around the end of the inner electrode (usually anode) and finally collapses due to the inward $\vec{j} \times \vec{B}$ force. This collapse occurs within 50-200 ns depending on the device characteristics. The radial collapse velocities range between 7 and 60 cm/ μ s depending on the geometry of the electrodes, initial gas pressure, structure of the current sheath and electrical characteristics of the device.

This phase is important due to its extremely high energy density, transient character, the emission of intense radiation, high energy particles and for copious nuclear fusion products as well, when operated in deuterium or *D-T* mixtures.

On the basis of the reported experimental data, the radial phase can be subdivided into four subphases, namely compression phase, quiescent phase, unstable phase and decay phase. In the following sub-sections these sub-phases are discussed in detail.

2.1.3.1 THE COMPRESSION PHASE

The radial phase starts with the rapid collapse of the azimuthally symmetrical, non-cylindrical, funnel-shaped plasma sheath towards the axis under the influence of the inward $\vec{j} \times \vec{B}$ force. The radial implosion ends when the plasma column reaches the minimum radius ($r=r_{min}$). This instant is usually taken as the time reference, i.e. when $r=r_{min}$, $t=0$.

Shock heating is the main heating mechanism inside the plasma before the front of the current sheath meets along the Z-axis. Therefore, the ~ 100 eV electrons are cold compared to ~ 300 eV ions [32]. After the plasma structure is transformed into a plasma column, joule heating becomes the main heating mechanism [27].

For the formation of the final focus, the plasma column is compressed adiabatically. The magnetic field starts diffusing into the plasma column at the end of this phase. It completely diffuses (very fast) resulting into an anomalous high plasma resistance [33]. Due to this large increase in the plasma column impedance, a sharp voltage spike and a dip in the current are observed for a typical plasma focus discharge, as it can be seen in Figure 2.1.

Spectroscopy [34], interferometry [35], laser scattering [27] and other methods are employed to determine the maximum value of the electron density and the electron and ion temperatures. The final electron temperature ranges from 1 to 2 keV [14].

Assuming a Bennett equilibrium in the pinch phase, the final temperature should only depend on the current, I , and the linear density, N , i.e. $T \sim I^2/N$; the lower the linear density, the higher the temperature. The electron temperature is independent of the minimum radius, r_{min} , of the pinched plasma column.

At the end of the compression, a plasma column is formed, then stagnates for a brief period of time. The diameter of the column is of the order of 1mm and the length is a few cm. The plasma density at this stage is $\sim 10^{19} \text{ cm}^{-3}$. The length of the pinched column as well as the minimum radius, scale to the central electrode radius [14].

Based on the snowplow equation, a computational model was developed for the radial phase. A zero radius column is obtained when this model is applied to the radial compression. To overcome this, either the introduction of a retarding kinetic energy pressure term, or the use of a criterion for the minimum radius as Larmour radius, were devised in different attempts. Lee [36], however, showed that these methods are not energy-consistent and should be replaced by an energy-balance model. This energy-balance criterion provides the correct end point for the implosion trajectory, and gives the correct quasi-equilibrium radius. A complete energy-consistent trajectory is obtained after the energy-balance criterion is combined with the snowplow model [37]. In addition, the computed pinch length and the minimum radius agree with the experimental measurements.

2.1.3.2 THE QUIESCENT PHASE

After the stagnation, this phase indicates the beginning of the expansion of the plasma column in the axial as well as in the radial direction. The rate of expansion in the radial direction is hindered by the confining magnetic pressure. But the plasma can expand unhindered in the axial direction due to the fountain-like geometry of the current sheath. Thus, an axial shock front is formed.

At the end of the compression phase, due to the sharp change in the plasma inductance, an electric field is induced in the plasma column. This electric field accelerates the ions and the electrons in the opposite directions. The relative drift velocity between the electrons and ions increases and approaches the electron thermal velocity. At this stage, microinstabilities like electron-cyclotron and various forms of beam-plasma instabilities, start to develop. The plasma column is magnetically confined, and the temperature of the plasmas further increases by ohmic heating. In addition, the magnetically confined column is hydromagnetically unstable to sausage ($m=0$) and kink ($m=1$) instability modes.

The pinch lifetime, t_p , can be defined, based on these observations, as the time between the first compression and the instant of the onset of the $m=0$ instability. This pinch life time is many times larger than the radial Alfvén transient time of the pinch column. The reasons behind this are: the strong axial flow, two dimensional curvature of the current flow, $\vec{B} \times \nabla \vec{B}$ drift and/or the influence of the self generated axial magnetic field.

2.1.3.3 THE UNSTABLE PHASE

In the plasma focus evolution, the unstable phase is the richest in associated phenomena like $D-D$ reaction products (neutrons, protons), hard and soft X-ray emission, fast deuterons and electrons. Figure 2.2 shows a schematic of the emission of the radiation from the plasma focus after the focusing event.

Due to the growth of the $m=0$ instability, the induced electric field is enhanced. This accelerates the electrons towards the inner electrode (anode) and the ions in the opposite direction. At the same

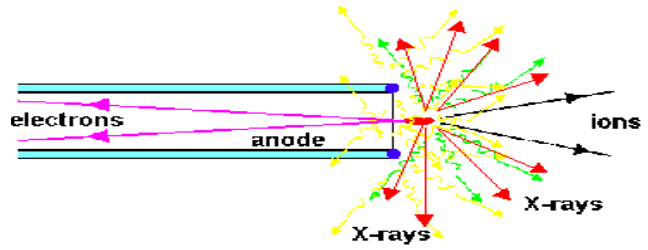


Figure 2.2: Emission of radiation from the focus.

time, an axial ionisation wave was reported [11, 27]. Using time-resolved interferometry, the measured peak velocity of this front in deuterium of about 120 cm/ μ s confirms that the ionisation is caused by the fast deuterons [27, 38]. This ionising front overtakes the axial shock front (formed due to the axial expansion of the plasma column). The beginning of this ionising front coincides with the beginning of the main hard X-ray and neutron pulses. The ionisation front develops into a bubble-like structure with several density gradients [27, 39]. The first gradient is attributed to the ionisation front that separates the ambient gas from the ionised structure.

Due to the bombardment of the accelerated runaway electrons on the central electrode (anode), a fairly large number of impurities are injected into the plasma column. This decreases the Pease-Braginski current which causes the plasma column to explode near the central electrode [40]. The explosion proceeds sequentially along the plasma column. This breaking up corresponds to the second density gradient observed in interferometric holograms [27]. This density gradient moves away from the anode at a speed of 20 cm/ μ s, much slower than the ionisation front.

The disruption of the plasma column continues until the whole plasma column is broken completely [41].

2.1.3.4 THE DECAY PHASE

The decay phase is the last phase of the radial collapse as well as the last stage of interest of the plasma focus dynamics. During the decay phase, a very large, hot and thin plasma cloud is formed due to the complete breaking of the plasma column. A large amount of Bremsstrahlung radiation is emitted during this phase. The soft X-ray emission rises abruptly during the decay. At this stage, the neutron pulse that starts from the beginning of the unstable phase reaches its peak value [42].

The high electric field induced during this period gives rise to the emission of beams of energetic ions and electrons. The implosion time is ~ 100 ns, while the lifetime of the plasma column is ~ 20 ns.

For the numerical design of the plasma focus, Lee has developed a dynamic model [43, 36-37]. In this model, the axial trajectory is computed using a snow-plow model. The radial dynamics is traced using a generalized slug model which considers a pinching plasma of increasing length with the plasma layer lying between a shock front and magnetic piston. This model can give a realistic final pinch radius. The experiments described in this dissertation were all performed on the focus device NIE-SSC-PFF designed on the basis of Lee's model.

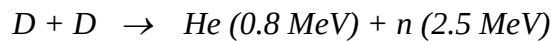
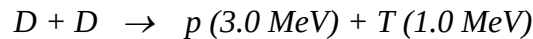
2.2 MODELS OF *D-D* FUSION

In the late 1930's, H. Bethe first elucidated the phenomenon of fusion, which is the source of energy production in the sun and stars. It is not a natural process on Earth because of substantial limitations imposed by the Coulomb barrier. In this process, two light nuclei are brought sufficiently close together to overcome their Coulomb barrier for a time comparable to the fusion reaction time.

The possible ways to obtain fusion are:

- i) An energetic light nuclei beam may be directed to the stationary light nuclei in solid or gaseous form. The beam and the target nuclei undergo fusion reactions and this process is called beam-target.
- ii) The light nuclei gas may be heated to a sufficiently high temperature and confined for a long time. The gas reaches a Maxwellian distribution. The nuclei undergo a fusion reaction and this process is known as thermonuclear fusion.

The basic deuterium-deuterium (or *D-D*) fusion reactions are:



with approximately fifty percent probability in each channel.

2.3 INSTABILITIES IN DENSE PLASMA FOCUS

The plasma instabilities are the major obstacle in the way of progress towards the controlled thermonuclear fusion. These instabilities impose several limitations on the amount of current and pressure that can be confined by a magnetic field.

In the first experimental attempts to produce the conditions needed for a controlled thermonuclear fusion during the early 1950's, a large current was driven

through a column of ionised gas in an effort to heat and confine it [44 - 47]. The current produces a magnetic field around the plasma column which thermally insulates the hot plasma from the wall and exerts an inward radial force that confines the plasma.

The limitation on the confinement time of the plasma and the disruption of the focused column are caused by the onset and occurrence of the macroinstabilities [48, 49] and the presence of plasma turbulence [48 - 52]. The possible types of instabilities in plasma focus devices are discussed in this section.

2.3.1 RAYLEIGH-TAYLOR INSTABILITY

With the help of the interferometric studies, Peacock et. al. [49] confirmed the presence of the Rayleigh-Taylor instability in the compression phase of the plasma focus. The interface between the plasma and the magnetic field is found to be unstable during the radial compression [53]. The inertial force inside the plasma column is in the direction normal to the magnetic field. This charge independent force interacts with the magnetic field causing the electron and ions to drift in opposite directions [53]. This drift results into a charge separation in the surface layer of the plasma column. The charge separation establishes an electric field, which then introduces a further plasma drift. Thus, the system becomes unstable and this instant is referred to as the onset of the Rayleigh-Taylor instability. At the end of the compression phase the boundary of plasma column is fluted due to this instability. The growth time of the Rayleigh-Taylor instability is about 30 ns [49].

2.3.2 THE $m=0$ INSTABILITY

The main problem encountered in the early fusion experiments was that the plasma column would spontaneously pinch itself off due to magnetohydrodynamic (MHD) instabilities, especially the $m=0$ (sausage instability).

These MHD instabilities are classified according to their poloidal mode number ($e^{im\theta}$).

The mechanism behind the growth of this instability is as follows. A poloidally symmetric radial perturbation constricts the plasma column slightly in one place and makes it bulge out in the other places as shown in

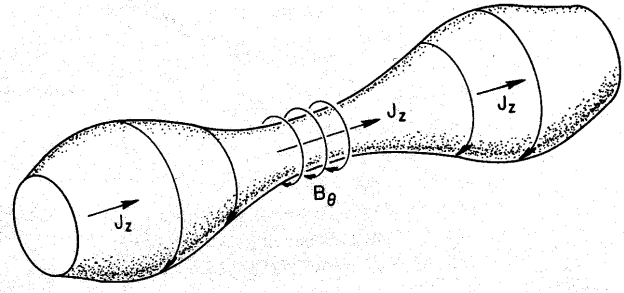


Figure 2.3: The $m=0$ instability.

Figure 2.3. The longitudinal current density (J_z) flowing through a smaller cross-section area of the constricted part produces a stronger magnetic field (B_θ) there, whereas the field along the bulged parts becomes weaker. The stronger field around the each bulged part exerts a greater inward force on the plasma column, so that the column constricts further. These subsequent rapidly changing magnetic fields at each constriction induce a large longitudinal electric field. This electric field accelerates the ions within the plasma column up to the energies of hundreds of keV and leads to the emission of bursts of neutrons.

Apparently, superthermal ions accelerated by similar instabilities led to the premature announcement of the success in the controlled thermonuclear fusion.

However, it was accepted that such violent instabilities must be suppressed before useful thermonuclear energy can be produced.

The sausage instability can be stabilised by permeating the plasma column with a longitudinal magnetic field comparable to the strength of the poloidal field around the column. One way to consider this stabilisation mechanism is by noting that any constriction of the plasma column requires energy to compress the longitudinal magnetic field within the plasma. A magnetic field in the conducting fluid exerts pressure perpendicular to the field lines and tension along the field lines. The compression of the longitudinal field within the plasma results in a restoring force from the increased magnetic pressure.

2.3.3 THE $m=1$ INSTABILITY

The addition of a small longitudinal magnetic field stabilizes the sausage instability. However, a new instability arises, known as $m=1$ (kink instability), in which the plasma column twists into a helical shape like a cork screw, as shown in Figure 2.4. The driving mechanism for this kink instability becomes clear by noting that, when any part of the plasma column is bent, the

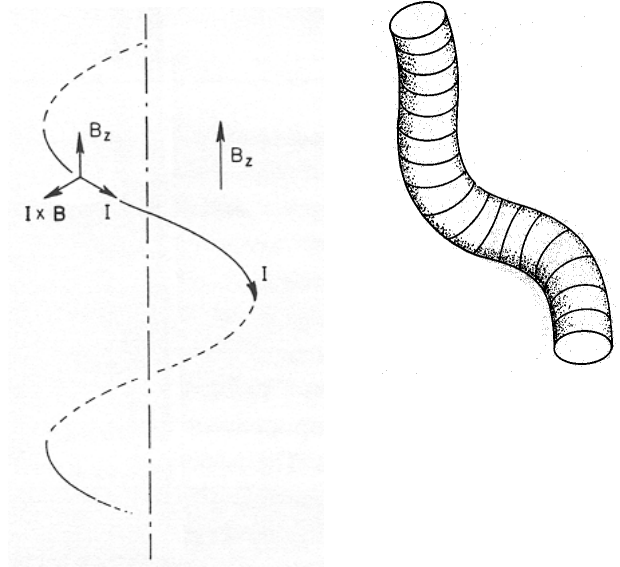


Figure 2.4: The $m=1$ instability.

poloidal field on the inner edge of the bend becomes stronger than the field at the outer edge. The resulting magnetic pressure pushes the column to bend further.

In order to stabilize this $m=1$ kink mode, longitudinal magnetic field must be made strong enough and plasma column fat enough so that no part of the magnetic field between the plasma and the wall closes upon itself once along the length of the plasma column. This is known as Kruskal–Shafranov criterion. Roughly speaking, the tension associated with the longitudinal magnetic field prevents the plasma column from bending into a kink.

2.3.4 MICROINSTABILITIES AND TURBULENCE

The existence of the microinstabilities and turbulence in the plasma focus was reported by many authors. Bernard [27] used the light scattering diagnostic techniques to study the level of the density fluctuation as a function of direction. He observed that the excitation of waves along the current direction was more intense (1.5 times) than along the direction perpendicular to it. This variation of the scattered light indicates not only the existence of the microinstabilities, but also the correlation of the microinstabilities with the drift velocity between the electrons and ions due to the current flow along the electrode axis. The observation of the anomalous resistivity [54] of the pinched plasma, non-thermal radiations in the microwave range [55] and bursts of high energy ions and electrons, also indicate the presence of microinstabilities and turbulence in the DPF.

During the radial phase, the rapid change of inductance results in an induced electric field in the plasma column. The induced field is given by

$$E = I \frac{dL}{dt}$$

where I is the discharge current and dL/dt is the rate of change of inductance.

At the end of the quiescent phase, the growth of the $m=0$ instability enhances the induced electric field locally. The emission of electrons starts as soon as the induced electric field is higher than 3×10^3 kV/m, the critical field strength. These runaway electrons move towards the anode and produce hard X-rays through electron bombardment mechanism. The electron energy estimated from hard X-ray is in the range of 50 keV to 500 keV [56]. Hence, the ordered velocity of the beams, estimated from the hard X-ray energy exceeds the thermal velocity of the plasma, determined from the electron temperature. Under this condition, it has been shown theoretically that the beam-plasma instabilities will grow rapidly.

Schonbach et. al. [62] observed reproducible microwave signals in the frequency range from 2.6×10^9 Hz to 1.8×10^{10} Hz. This microwave emission during the pinch phase is strongly correlated to the emission of the soft X-rays.

2.4 RADIATION EMISSION FROM THE DENSE PLASMA FOCUS

Considerable efforts have been concentrated from the discovery of the plasma focus on the study of the emission of radiation like neutrons, X-rays (hard and soft) and charged particles. Also, special attention has been paid to study the current sheath structure, its dynamics, pinch formation and role of MHD instabilities in the emission of radiation.

The general properties of DPF discharge such as, neutron emission times, energy spectra, and X-ray emission were first studied by Mather [57, 58] and Bernstein [59, 60].

The structure of the pinched column for small plasma focus devices (≤ 10 kJ) was investigated by several authors [11, 61 - 64]. It was found that in small scale experiments, the primary filamentation of the current sheath is destroyed during the collapse phase, when a plasma column of a few millimetres in diameter is formed [61].

Time-resolved X-ray measurements performed by means of different diagnostic techniques (scintillator-photomultiplier system and gated X-ray image intensifier) showed that X-ray hot spots have lifetime below 5-10 ns [62, 63]. Some authors have identified these plasma structures within regions of $m=0$ instabilities [11, 61, 62]. Other experimental and theoretical studies [64 - 67], however, suggested that plasma structures are of more complex nature and require further detailed investigations.

Sadowski et. al [68] investigated the fine structure of soft X-ray (0.8-3 keV) emitting zones in a neutron optimised plasma focus. They found that distinct continuous filaments are registered during the pinch phase. Their time-resolved X-ray measurements show that the filamentation exists 20-50 ns after the maximum compression.

Comisar [69] suggested that the lifetime of the resulting fountain like pinch column appear to be governed by the formation of sausage instabilities. Nevertheless, the build up time of the $m=0$ instabilities are at least 5-10 times shorter than the observed pinch lifetime (~ 50 ns).

Golberg [70] proposed that Rayleigh-Taylor instabilities developing in a layer of matter, accelerated by the pressure of the magnetic field or of a light fluid, is seen to be suppressed if the acceleration layer scoops unperturbed matter entering into the motion.

Haas et. al. [71] studied the dynamics and the structure of the plasma layer by means of the schlieren technique and found out that the growth rate of Rayleigh-Taylor instability is $\sim 10^8 \text{ s}^{-1}$, so there may be some stabilising effects due to plasma viscosity.

In the following sections, special attention is given to the plasma related phenomena like fusion products (neutrons), radiation emission (soft and hard X-ray), and particle (ions) beam emission.

2.4.1 NEUTRONS AND ENERGETIC IONS EMISSION

Theories concerning thermonuclear models (static, moving boiler models [58, 72]), the linear beam-target interaction [58] and the crossed field model were developed [73, 74]. Korzhavin [75] proposed that if the currents are raised to 10 MA, the neutron emission will be predominantly thermonuclear and the contribution of the turbulent mechanism will be very small.

In the early times of the focus research, it was considered that the neutrons are of thermonuclear origin [26]. However, this idea was dropped because it was in contradiction with the experimental data. The flux of the neutrons was found to be anisotropic [76] (the flux in the axial direction being usually higher than that in the radial direction). Therefore, it was proposed that the beam-target fusion accounts for the most of the neutron production in the plasma focus.

In the beam-target mechanism, there is a clear distinction between the distribution of the high energy particles, called beam, and that of the target which consists either of neutral gas or of low temperature plasma. In the thermonuclear model, it is proposed that the neutrons are produced by a Maxwellian population distribution of deuterium. Bernstein and Fai [77] found evidence that the neutron source is not thermonuclear, but they also doubted the credibility of beam-target model due to a much smaller neutron flux anisotropy.

In the following years, additional diagnostics like interferometry, holography, laser light scattering, temporal and spatially resolved neutron detection, ion measurements using plastic detectors and soft X-ray images, were employed.

Ion and electron temperature measurements carried out by ruby laser scattering [78, 27] led to a low value of a temperature ($T_i \leq 1$ keV) that was not suitable for a thermonuclear interpretation of the neutron production. The observation favours the beam-target model supported by observations of a long axially extended neutron source [79, 80]. However, many discrepancies arose from the too broad energy spectra and too low neutron anisotropy to be accounted for a beam-target production. Furthermore, measurements of neutron source extension [81, 82] doubted the relevance of the beam-target model.

The early statement of the essential thermonuclear production in Filippov geometry and a more distinct beam-target neutron production in Mather devices could not be confirmed uniformly by further investigations. Even a linear combination of both the models, as suggested by Maisonnier [82], could not overcome the general inconsistency between the experimental results and theory.

Again the beam target model was proposed to be responsible for the neutron production based on either intense medium energy ion beams ($E_d \sim 300$ keV) [83, 84], or low energy ($E_d \sim 100$ keV) [85] beams. Conrads et. al. [86] gave the evidences for a non-collision dominant velocity distribution (average energy of at least 20 keV) of the ions producing the neutrons in a dense plasma focus. He also indicated that for the group of ions moving in the axial direction, the average energy is at least 0.3 MeV. Tiseanu [87] et. al. concluded that in a small energy plasma focus device, the medium energy deuterons (50-100 keV) with a relatively broad angular distribution are responsible for most of the neutron yield. Yokoyama [88] et. al. proposed that the neutrons were generated by the moving plasma diode mechanism.

Kaeppler [89], analysed the phenomena in DPF leading to the neutron production and suggested that the neutron production due to beam-beam collisions play an essential role beside that due to the beam-target collisions.

Time-of-flight measurements in the axial and the radial direction indicate an anisotropy in the energies of the emitted neutrons [90, 91] resulting from the axial motion of the centre of mass of each pair of the deuterons. Decker et. al. [92] observed two distinct peaks of neutron energies in the axial direction at 2.5 and 2.7-2.8 MeV indicating that both the boiler and beam-target mechanism take part in the neutron production. Potter [93], proposed on the basis of his MHD calculations that the neutron yield should be about 10^{12} per pulse for a 32 kJ Mather-type device. However, this yield is much larger than the experimentally observed values.

Bernstein [76], proposed a crossed-field acceleration model. He stated that the high energy deuterons follow curved trajectories.

Imshenik et. al. [94] argued that the effective energy of the deuterons participating in thermonuclear reaction is comparable to the mean thermal energy of Maxwellian distribution at plasma temperature above a few keV. This shows that the integral neutron yield is slightly sensitive to the deuteron distribution function, provided the function does not have sufficiently dense forms in the high energy region. The calculated thermonuclear neutron yield is 4×10^{16} neutron per pulse for a plasma focus of 10 MJ energy.

Michel et. al. [95] supported the beam-target model as a suitable mechanism for the neutron emission from low energy focus devices. Pecorella et. al. [96] measured the time- and space-resolved neutron emission from a 120 kJ Filippov-type dense plasma focus. According to them, 70% of the neutron yield results from the thermonuclear plasma located in the region between 1 to 3 cm above the anode. About 20% is due to the beam-target interactions, and the remaining 10% is distributed uniformly along the axis of the device.

Gentilini et. al. [97], supported the beam-target model as a dominant mechanism for the neutron production. They also speculated that the major contribution to the neutron emission comes from low energy deuterons (< 50 keV).

Steinmetz [98] proposed that experimental data cannot be explained in terms of the beam-target or thermonuclear model. Instead, he was of the opinion that there was another mechanism, called the particle-trap model, in which ion streams are trapped in self-consistent magnetic fields which then follow some sort of spiralling within the dense plasma region.

Michel et. al. [99] measured the energy and angular distribution of the neutrons by means of angular neutron flux and time-of-flight measurements on a 1 kJ

device. They pointed out that the beam-target mechanism is applicable to the neutron production mechanism. Yamada et. al. [100] predicted that in the beam-target, model slow deuterons (< 74 keV) contribute more to the neutron yield.

From the anisotropy [74] of the neutron yield, it can be suggested that beam-target neutron generation is important, and a better understanding is required for the beam acceleration mechanism, the ion emitting zones in the system, the instant of ion beam emission and energy spectrum of the accelerated ions. The entire mechanism of the ion acceleration is yet to be clarified.

The conventional explanation of the ion acceleration involves electric fields which result from the destruction of the plasma column due to the development of the magnetohydrodynamic instabilities [101] especially, the $m=0$ instability.

V. Kondoh et. al. [102] suggested that the high electric field, which is causing the acceleration of the ions, is due to the degradation of the plasma conductivity caused by turbulence [103]. Ananin and Gureev [104] argued that the presence of the high energy ions in the plasma focus can be explained from the known physical principles of the system.

The question about the conditions for the production of the high energy deuteron beams in a focus is interesting both from the physics point of view and their applications. One problem is to identify the region where the acceleration of the particles takes place. This seems to be the main reason why we are not able to explain the mechanism of the acceleration of the ions to such high energies in DPF.

The solution of this problem would explain the mechanism of the deuteron acceleration in the plasma focus and will allow us to develop the plasma focus as a commercial ion beam source.

Belyaeva [105], using the principles of geometrical optics, managed to find the region of accelerated deuterons in the vicinity of the pinch. She also suggested that high energy deuteron spectra are modulated. The depth of the modulation depends on the pinch configuration, which, in turn, is controlled by the anode design. Sadowski et. al. [106] guessed that the multi-spike structure of the ion pulses in a plasma focus discharges comes from sequentially appearing $m=0$ break-offs.

A few workers, studying the ion emission from DPF, measured the angular and energy distribution [107 - 109]. The energy spectra of the ion beams were found to obey the scaling law $dN/dE \propto E^{-x}$ where $x \cong 3.5 \pm 0.5$. Ananin [104] explained that the ion energy distribution differs from the Maxwellian distribution.

A. Serban and S. Lee [110] suggested that the neutron emission begins at the end of the compression phase when the plasma reaches the minimum radius. They found the radiation output (neutron and soft X-rays) to be dependent on the sheath velocity. They proposed a scaling law $Y_{th} \sim (I_p V_{axial})^4$ for the non-beam neutron component.

Zakaullah et. al [111] investigated the emission of neutrons and ions from a low energy plasma focus device. They suggested the beam-target mechanism at low pressure, trapped gyrating particle model at optimum pressure and jet-like moving boiler model at high operating pressure.

The plasma focus was considered as a low impedance ion beam generator [83, 84] applicable for inertial confinement fusion, but other laboratories [112 - 115] could not justify the use of such strong beams in their facilities.

It was suggested by Cloth [9] that DPF, as a fusion device, can be used as a neutron source for the reactor material testing. Zucker et. al [116], proposed that a

continuously operated 1 pulse per second DPF device capable of delivering a minimum of 10^{15} neutrons per pulse is suitable for material testing purposes. Rager [117] proposed that a 30 MJ device can be used as a nuclear design test facility working much below the break-even at a level at which the alpha particles do not influence the scaling of the neutron yield and the whole dynamics of the process.

The mechanism of the emission of the bursts of neutrons has been investigated from various point of views, but without complete success. Some authors have proposed other models that describe the mechanism [58, 13, 118]. Gullickson et. al. [119] formulated their model on the basis of the experimental and theoretical points of view in order to explain the evolution from the collapse phase to the neutron generation. The central idea in almost all the models is that an intense electric field generated by an anomalous plasma resistivity accelerates the charged particles.

Yamamoto et. al. [120] studied the neutron production mechanism through the time correlation between the dynamic behaviour of the focused plasma and the burst of the deuteron beams, X-rays and neutrons. They showed that the peak of the neutron appears after the disruption of the collapsed plasma column. The disruption of the plasma is caused by a growth of a macroscopic instability in $m=0$ type. Toshikazu et. al. [121] also investigated the same sort of correlation. They suggested that the generation of the ion and the electrons beam occur in an interval of 20 ns around the peak of the voltage spike, and the plasma inductance increases monotonically, but rather slowly during the process of the $m=0$ instability growth.

The neutron emission is correlated with the emission of the ions (deuterons). The studies of high energy ions emitted from the plasma focus facilities provided some information on the dynamics of the plasma phenomena and also on the

mechanisms of the ion acceleration. On the other hand, the ion emission studies are also important for developing various technological applications of DPF facilities. Therefore, such studies were carried out in many laboratories [107, 122 - 126].

Jegar and Herold [127] studied the energy spectra of fusion reaction protons with the energy resolution of 50 keV and deuterons with the energy resolution of 70 keV, employing SSNTDs. Kellay and Marquez [128] used the CR-39 detector in combination with a Thomson spectrometer in the axial direction. They estimated the axial ion flux and energy for various filling pressures. It was concluded that the ion density at low ion energy is very high and plays an important role in the neutron production in the DPF device.

Antanasisijevic et. al. [129] studied D^+ ions emitted from 1 kJ Mather type plasma focus by using CR-39 detectors. They estimated the D^+ ions emitted from the focus region in 4π geometry to be of the order of 10^{15} per pulse and their maximum energy was found to be less than 500 keV.

Bostick et. al. [130] investigated the time-resolved energy spectrum of an axial ion beam in a 5.4 kJ DPF device. Deuterons of 0.3-9.0 MeV were detected. Styger et. al. [125] studied the ion beam using SSNTDs. They also monitored the neutron emission. They found out that the charged particle current was dependent on the discharge current of the capacitor bank.

Todorovic [131] studied the positive ions (${}^3_2\text{He}$, ${}^3_1\text{H}$, ${}^1_1\text{H}$) emitted from a low energy (3.2 kJ) plasma focus. When the detectors were exposed directly (without Al foils) to the radiation, about 10^{16} tracks /cm³ were recorded.

2.4.2 X-RAY EMISSION

For a long period after the discovery of DPF, most of the studies were done in hydrogen or its isotopes, with minor attention being paid to the X-ray emission. For the past decade, considerable potential application of DPF as an intense (soft) X-ray source [132, 133] has led to more intensive studies.

For a focussed plasma at $T_e = 1$ keV, its X-ray continuum is expected to peak at $\lambda_o = 6.2$ Å, which is in the soft X-ray region. On the other hand, for a high-Z focus plasma, the characteristic line emission also falls in the soft X-ray region. For example, for nearly fully ionized argon or neon plasma at a suitable high temperature, the K_α line radiation has wavelengths of 4.2 Å and 12.132 Å respectively.

Besides making the plasma focus an intense X-ray source which is useful in many applications such as microlithography and X-ray microscopy, the X-ray emitted also provides a convenient means for studying the plasma properties.

The radiation spectrum of the plasma focus in the X-ray region covers a wide range from below 1 keV up to over 500 keV in a time span ranging from few nanoseconds to few hundred nanoseconds. The integrated energy represents a significant fraction of the electromagnetic energy stored in the DPF driver. It can reach up to 20% in a Filippov-type geometry.

The electromagnetic radiative processes of the plasma focus are: quasi-equilibrium thermal radiation from macroscopic plasma structures and radiation due to electron beams interacting with non-plasma targets (i.e. electrodes) and/or with periodic electron density structures [134].

The following electron-ion interaction processes are considered to be responsible for the thermal X-ray component: Bremsstrahlung, recombination emission and line emission. The cause of the dominant hard X-ray emission is attributed to the non-thermal high energy electron beams striking the anode surface.

Time-resolved measurements [135] showed that the power law spectrum ($dN_{HX}/dE \sim E^{-a}$, with $a = 2 \sim 4$, E in keV and N_{HX} - the hard X-ray flux - in photons per cm^2) found for the plasma focus in the range 100 ~ 500 keV is the result of the time integration over spectra produced by quasi-monochromatic electron beams with average energies changing over a ten nanosecond time scale.

The hard X-ray emission was correlated with the neutron production when deuterium was used as the working gas: the total neutron yield is large when the hard X-ray emission is large [11].

The emission of soft X-rays from the plasma focus have long been studied and various diagnostic techniques have been employed over the years. These diagnostics include: (i) pinhole cameras (with one or multiple pinholes and appropriate filters) to record the time-integrated spatial/spectral distribution of the X-ray emitting region, (ii) microchannel plates used in conjunction with pinhole cameras for time-resolved spatial distribution of soft x-rays, (iii) semiconductor detectors with appropriate filters for space-integrated temporal/spectral distribution of the soft X-ray emission, (iv) crystal X-ray spectrographs for space-resolved spectral analysis, (v) X-ray streak photography for temporal distribution of the X-rays in one space dimension (radial or axial), and (vi) X-ray films, X-ray resists and X-ray pulse calorimeters for absolute soft X-ray yield measurements.

The soft X-ray (i.e. with energy below 10 keV) emitting source is situated on the electrode axis at the open end of the electrode system in the vicinity of the inner electrode face. The time-integrated images show that it has a roughly cylindrical shape. Its dimensions vary from below 1mm to over 10 mm in diameter. These gross dimensions have been shown to be related to the anode radius [42], with an axial length from few mm to few cm. Spectral filtering of the X-ray image reveals a fine structure presenting intensely radiating small macroscopic entities (called by various authors "hot spots" [136]), "plasma beads" [61], "bright spots" or "micropinches" [134]) as well as filaments inside and along the axis of the pinch [137].

Choi et. al. [138, 139] reported X-ray "hot spots" with typical dimension of about 100 μm which occur within 5 ns of the first compression (the focus formation) during the period of the stable phase of the pinch. They concluded that the "hot spots" are not the manifestation of the $m=0$ instabilities which disrupt the pinch column.

A detailed grey level analyses of X-ray images [140] indicated the presence of helical structures of high emissivity at the center of these images. These help in explaining the production of the axial magnetic field and the relaxation of the pinch in the plasma focus.

For a low energy deuterium-filled plasma focus device the soft X-ray emission extends over a time interval of about 60-70 ns and features three peaks: the first is emitted by the dense pinched plasma column at the end of the compression phase; the second by the unstable plasma column; and the third (corresponding to the disruption phase) is due to emission from the face of the inner electrode [141].

Favre et. al. [142] investigated the temporal and spatial characteristics of the X-ray emission in a 3 kJ DPF operated in hydrogen with an argon admixture. They reported two main periods in the soft X-ray emission corresponding to two successive compressions in the focus.

Zoita et. al. [143] studied the influence of the radiation processes on the plasma evolution during the pinch and post-pinch phases in experiments with neon-seeded deuterium discharges carried out on a 28 kJ / 60 kV device. They reported dramatic changes in the radiation characteristics of the discharge as well as in the pinch configuration. High aspect ratio (length/diameter > 10) pinches emitting large amounts of soft X-rays accompanied by efficient emission of neutrons (and hard X-rays) in the absence of macroscopic instabilities were produced. This confirmed the physical picture obtained from experiments performed with a lower energy plasma focus device [144]. They also concluded that the line radiation as well as the main continuum radiation in the spectral range of 3-4 keV are emitted mainly from the intense bright spots having dimensions of about 50 μm . Similar ideas were reported by Presura et. al. [145] from their investigation of the spectral characteristics of the soft X-ray emission of a plasma focus device operating in deuterium with various additions of argon and argon-krypton.

The X-ray emission characteristics depend strongly on the plasma focus device operating régimes and parameters (e.g. gas filling composition and pressure, stored energy, peak discharge current, driver impedance, electrode material, shape, profile and configuration, polarity of the inner electrode, etc.). Among them, the composition of the gas and the filling pressure have the strongest influence. When

deuterium is used, the working pressure for the highest X-ray yield does not coincide with the neutron-optimized output pressure [146, 147].

For an X-ray optimized plasma focus device, the scaling law of the X-ray output Y_X as a function of the peak discharge current I_{max} and the pinch radius, r , can be written empirically [148] as:

$$Y_X = \frac{I_{max}^4}{r^2}$$

This dependence, similar to the scaling law found for Z-pinches, indicates a better operation at higher voltages for a given stored energy.

However, the exact mechanisms by which high intensity X-rays are emitted in the plasma focus are still controversial and more experimental work is required in order to optimize the X-ray production for practical applications.

The scope of the plasma focus regarding fusion and the pulse power technology is still under active consideration. The reason behind is that the technological development has enabled the successful operation of large DPF devices leading to better understanding and formulations of scaling laws of DPF performance.

Plasma focus is not only a fusion device (among low- β tokamak and inertial fusion machines), but also facilitates the understanding of the problems of plasma physics, pulse power technology and related phenomena. Irrespective of the size, all plasma focus devices, small (1-10 kJ), medium (10-100 kJ) and large (above 100 kJ), produce similar results within a small range of densities [149, 150]. For the

educational purposes, the small devices are given preference due to cost-effectiveness and simplicity [151]. A large variety of plasma phenomena can be studied by employing simple diagnostic techniques. The other features that make this machine a better choice compared to the large machines are easy: maintenance, compact design, flexibility and short turn-around time-characteristics.

Chapter 3

PLASMA DIAGNOSTIC TECHNIQUES

The diagnosis of hot, dense and short-lived plasmas requires special and carefully designed experimental techniques and instrumentation. This is mainly because the plasma conditions vary over a large range. Temporal and spatial resolution of better than a nanosecond and a micrometer, respectively, are required. The continuous development of novel instrumentation and techniques is crucial for the diagnosis of plasmas produced by devices like the plasma focus.

For the investigation of the radiation and charged particles emitted from our plasma focus device, different diagnostics were employed. In this chapter, the theoretical description of these diagnostics is presented.

3.1 ELECTRICAL DIAGNOSTICS

Two basic electrical diagnostics were used in our experiments. One is for the measurement of the discharge current, and the other one for the voltage measurement. These two diagnostics are discussed in the following sections.

3.1.1 ROGOWSKI COIL

The Rogowski coil [152] is a multi turn solenoid bent into the shape of a torus. It is used to measure the discharge current from the capacitor bank. Figure 3.1 shows the schematic of the Rogowski coil.

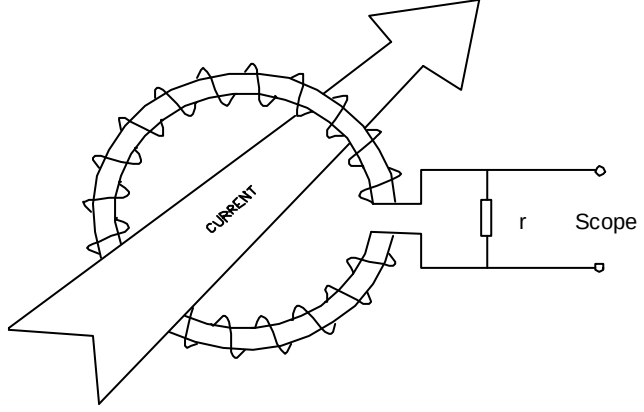


Figure 3.1: Schematic diagram of a Rogowski coil.

The coil is terminated with a low inductance resistor, r . The equivalent circuit is given in Figure 3.2, where L_c and r_c are the inductance and the resistance of the coil, respectively with I the discharge current and i the circuit current through the coil, the circuit equation can be written as:

$$L_c \frac{di}{dt} + (r + r_c)i = k \frac{dI}{dt} \quad (3.1)$$

a) If $L_c \frac{di}{dt} \gg (r + r_c)i$, then from (3.1) one can obtain $i = (k/L_c) \times I$, so the output voltage is proportional to I :

$$V_0 = ri = \frac{rk}{L_c} I \quad (3.2)$$

In this case the Rogowski coil acts as a current transformer.

b) If $(r + r_c)i \gg L_c \frac{di}{dt}$, then from 3.1 the output is

$$V_0 = ri = \left(\frac{rk}{r_c + r} \right) dI/dt \quad (3.3)$$

Under this condition, the coil can be used to measure the derivative of the current (dI/dt).

When the Rogowski coil is used as a current transformer, the following formula [153] may be used for the calibration.

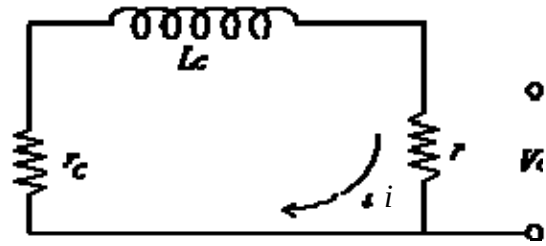


Figure 3.2: Equivalent circuit of the Rogowski coil.

$$I_1 = \frac{\pi C_0 V_0 (1 + f)}{T} \quad (3.4)$$

Where

$$C_0 = 30\mu\text{F}$$

V_0 = charging voltage

f = reversal ratio $f = 1/4 (V_5/V_4 + V_4/V_3 + V_3/V_2 + V_2/V_1)$, where V_i ($i = 1, 2, 3, 4, 5$)

are the absolute values of the decreasing amplitudes of the consecutive half

cycles of the Rogowski coil signals, and I_1 is the first peak value of the discharge current.

T = periodic time (average of several cycles taken)

Thus, by measuring f and T from the oscillogram, I_1 may be computed. Then the calibration factor of the coil may be written as

$$k_1 = I_1/V_1 \text{ (A/Volt)} \quad (3.5)$$

The calibration factor of the coil used in our experiments is $k_1 = 15.6 \times 10^3$ A/Volt.

3.1.2 VOLTAGE PROBE

The voltage probe is simply a resistive voltage divider, which measures the voltage between the high voltage plate and the ground plate

of the DPF device. Figure 3.3 shows the structure of the divider, consisting of a series of ten resistors of 510 ohms each with a shunt resistor of 51 ohm. It, therefore, attenuates the voltage by a factor of 100. The divider is enclosed in a copper tube of 20 mm diameter.

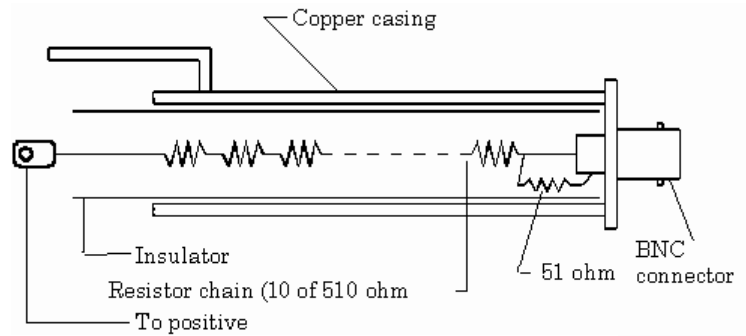


Figure 3.3: Schematic of voltage probe.

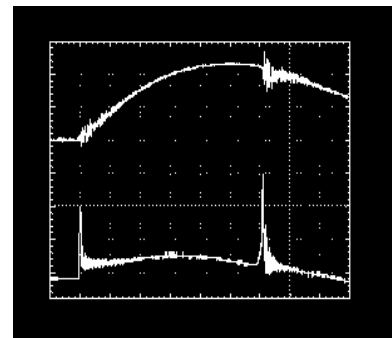


Figure 3.4: Typical waveforms of the current (upper trace) and the voltage (lower trace).

Typical signals of the Rogowski coil and the high voltage probe are shown in Figure 3.4.

3.2 TOTAL NEUTRON YIELD MEASUREMENTS

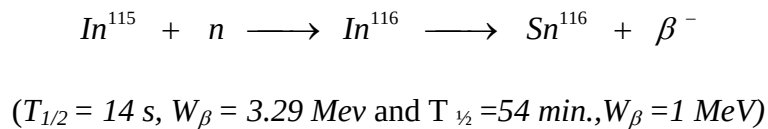
The presence of the neutrons in the reaction products is fortunate from the diagnostic standpoint, since they can be transmitted relatively freely through the machine environment to the detector located outside the machine.

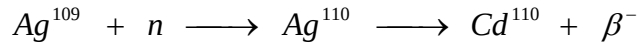
Various techniques can be used to determine the average total yield of neutrons per discharge. A commonly used diagnostic is the neutron activation method.

This type of detection depends upon the neutron activation of an appropriate material by a thermalized neutron flux and the subsequent counting of the level of the induced β activity. Any detector (photomultiplier, GM tube, etc.) with the output properly amplified and scaled could be used for the detection of energetic β radiation. One technique is to incorporate the moderator, the foil for activation and the Geiger-Muller tube.

For the plasma focus experiments, it is convenient to use either indium or silver as an activation material.

The two reactions of interest are [152]:





$$(T_{1/2} = 24.5 \text{ s}, W_{\beta} = 2.8 \text{ MeV and } T_{1/2} = 2.3 \text{ min.}, W_{\beta} = 1.62 \text{ MeV})$$

The β counting is done for an appropriate time interval. The detecting system is usually calibrated by means of a known radioactive neutron source.

A detailed description of the experimental set-up for the total neutron yield measurement is given in chapter 5.

3.3 CHARGED PARTICLE DETECTION

For the detection and analysis of various radiation, different types of detectors have been developed. They may be grouped as i) dynamic, or active detectors and ii) static, or passive detectors.

The dynamic detectors do not provide a permanent record of radiation. These are used to study the temporal behavior of the radiation emission and their energy analysis. Geiger-Muller tube, scintillation detectors coupled with photomultiplier tubes etc. are all members of this class.

The static detectors give a permanent record of the incident radiation but the information about temporal behavior of the radiation flux is normally not available with these detectors. This family includes` photographic films and solid state nuclear track detectors (SSNTDs).

There are three principal types of processes by which charged particles (i.e. protons, deuterons, alpha particles, etc.) transfer their energy to the stopping medium.

- 1 Coulomb electrostatic force between the particle and the electrons of the target atom may cause excitation or ionization.
- 2 When particles are decelerated, they lose their energy by emitting electromagnetic radiation (bremsstrahlung).
- 3 A direct electrostatic force between the moving charged particle and the target nuclei can result in the ejection of target atoms from the lattice sites, or out of the molecular chain.

3.3.1 FORMATION OF TRACKS IN SSNTDs

Heavy ions traversing an insulating medium produce narrow regions of atomic defects (lattice vacancies, interstitial, etc.) or broken molecular chains known as latent damage trails. To amplify these latent damage trails and produce observable tracks, a selective etching process is used. A three basic steps approach to the track formation is represented in Figure 3.5.

The starting point of the track formation process is the transfer of energy from the projectile ion to

the target electrons
and nuclei in binary

encounters as
described in the

previous section.

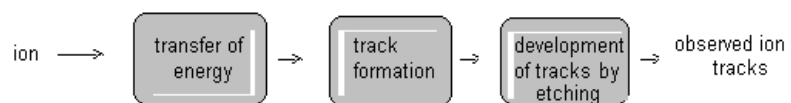


Figure 3.5: Three steps approach for track creation and observation.

Various authors have proposed different mechanisms for track formation.

F. Seitz and J. S. Kohler [154] introduced the concept of “thermal spike” as a mechanism by which energetic particles could produce a considerable disruption of a

crystal lattice. In this model, the passage of an energetic particle is assumed to produce intense heating of a localized region of the lattice. Therefore, the temperature of the region becomes very high from which it cools rapidly due to heat conduction. As a result of this heating episode, various atomic processes are activated, and damage to the lattice is produced. Bonfiglioli et. al. [155] and Chadderton et. al. [156, 157, 158] studied fission-fragment tracks in crystals. They presented more detailed considerations of the manner in which the electronic excitation produced by fission fragments is transferred to the lattice atoms.

In insulators, electrons can interact readily with the polar and acoustic modes of lattice vibration. Once the delta ray energy falls below the band-gap width, the release of further electrons into the crystals conduction band becomes impossible. Therefore, the electron-phonon collisions are expected to be the predominant energy loss process in an insulator. Excitation is communicated to the lattice more efficiently in such materials. In insulators, a narrow and intense spike is produced. This spike leads to a sufficiently severe localized radiation damage. Due to the radiation damage, etchable latent damage trails are formed.

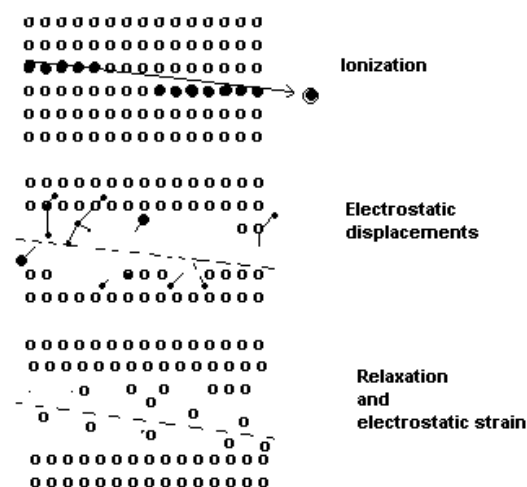


Figure 3.6: Ion explosion mechanism for track formation.

Fleischer, Price and Walker [159] proposed in 1965, a semi quantitative track formation mechanism. According to their investigations, a heavily ionizing particle

leaves a narrow region containing a high concentration of positive ions in its wake. Mutual repulsion can drive these ions into interstitial positions provided the electron-positive-ion recombination time is long in comparison to the lattice vibration time. Subsequently, there is a neutralization of the positive ions and relaxation of the surrounding lattice into the disrupted region. The lattice strains are set-up along the track core as shown in Figure 3.6.

3.3.2 TRACK ETCHING TECHNIQUES

The damage trails produced by irradiation in SSNTDs are very small (of the order of a few tens of angstroms). Therefore, only a transmission electron microscope or an atomic force microscope can “see” these tracks. For the enlargement of these tracks so that an optical microscope may “observe” them, a technique called etching

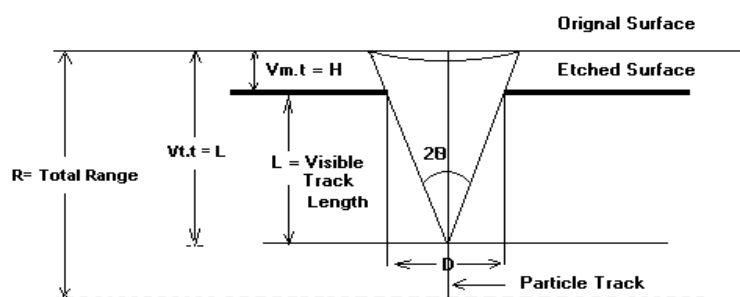


Figure 3.7: Track etching geometry for normally incident ions.

is used. When SSNTDs are treated with some suitable chemical etching solution, which is the most widely used method, the material around the latent track is attacked preferentially and a cylindrical or a cone-shaped hole of the order of 1-30 μm length and 0.1-10 μm diameter is formed.

The track geometry for normally incident ions is shown in Figure 3.7. When a detector is placed in an etching solution and is heated to a suitable fixed temperature,

the undamaged material is etched with a rate v_m ($\mu\text{m/h}$). The damaged material around the track is etched at a much faster rate v_t ($\mu\text{m/h}$). The material etching rate v_m depends on the plastic type etchant normality and the temperature of the solution. The track etching rate v_t depends, along with these factors, on the energy loss of the ion as well.

During etching, the undamaged material is etched isotropically at the rate v_m , whereas the track etching is in action with the same rate v_t and the material is continuously etched also with the rate v_m . The simultaneous action of two processes develop cones at the track sites. Their shape depends on the angle of the track with the surface of the detector. For obliquely incident ion tracks (shown in Figure 3.8), if the angle of the track is less than a critical angle, θ_c [160], the track would not be visible after etching

treatment. This is due to the fact that the component of the track etching rate perpendicular to the detector surface

$v_{\perp t}$ becomes smaller than the bulk etch

rate v_b if the angle of incidence is less than

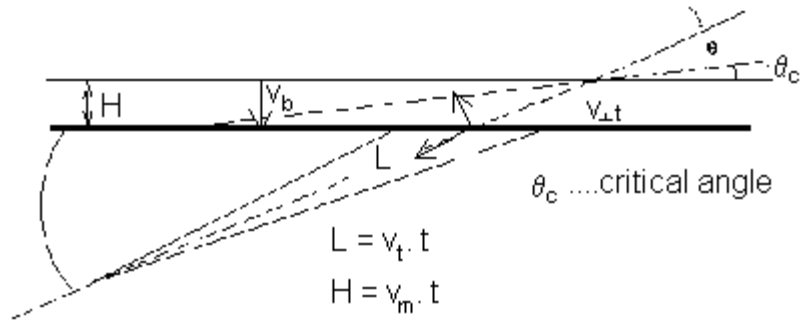


Figure 3.8: Side view of an etched track to explain the phenomenon of critical angle.

$$\theta_c = \sin^{-1} (v_b/v_t)$$

where v_b is bulk etch rate and v_t is track etch rate.

Different etchants can be used for the enlargement of the tracks for different detectors, but for plastics, the aqueous solution of *NaOH* with concentration of 6 normality is the most frequently used etchant [161]. The temperature range employed for it is 40 °C to 80 °C. It is necessary to control the temperature of the etchant ($\pm 1^\circ\text{C}$). The time for etching varies from a few hours to many hours, but for high efficiency and accuracy long etching time is preferable.

For *CR-39* detectors 6.25-normal (6.25 N) solution of *NaOH* as an etchant has been found to be very effective. A suitable etching time for *CR-39* is 8 hours. However, this period may be prolonged to increase the track size even further. The corresponding temperature of the solution should be maintained at 70 °C ($\pm 1^\circ\text{C}$) [161].

For good results, it is important to control the quality of the etchant. High concentration of the etch product is built up after prolonged use of the etchant and it is therefore, necessary that fresh etchant should be used frequently. Some workers, such as Enge et. al. [162], use constant stirring during the etching process to prevent etch product build up. After completion of the etching, the detectors are required to be washed in an ultrasonic bath (10-15 minutes) for the removal of the etchant molecules from inside the etch pits.

3.4 TIME-RESOLVED HARD X-RAYS AND NEUTRON MEASUREMENTS

The time-resolved X-ray and neutron measurements correlated with other diagnostics improve the overall picture of the temporal evolution of the transient

plasmas. These measurements are helpful in finding out the mechanisms of the emission of the X-rays and the neutrons.

The principle of the technique of the time-resolved neutron flux measurement from the plasma focus is described in the following paragraphs.

Neutrons and high energy photons falling upon a suitable scintillation material produce light pulses which can be detected with a photomultiplier.

When good time resolution is required, plastic scintillators are employed. A plastic scintillator is commonly bonded by an optical joint (with a silicon vacuum grease or a high viscosity silicon grease such as Dow Corning DC-200) to the photomultiplier photocathode. For high light collection efficiency, the scintillator is generally enclosed in an aluminum foil or surrounded by a material such as MgO , TiO_2 , or coated with white tygon paint. As far as the linearity of plastic like the NE102 is concerned, Bailey [163] reported that light output versus energy deposition is linear for photon energies as low as 3 keV.

The choice of the photomultiplier is dependent on the experimental requirements. The photo-tubes vary widely as to the spectral response, quantum efficiency, gain, noise level, maximum current output, physical size and overall structure. When a suitable tube has been chosen, there remains a selection of a tube base circuit appropriate to the signal duty cycle of the experiment. For example, the provision of linear anode output to high current pulses over a long duty cycle requires capacitors on each stage large enough to ensure that $\int idt/C \ll V$ for each stage. The capacitors of several μF , or a separate power supply may be necessary for the last output stage to maintain proper dynode voltage for the constant gain. The output

voltage pulse for a typical scintillator-photomultiplier combination can be calculated from the following equation:

$$V = \frac{dN}{dt} g \varepsilon e G R$$

where

N - number of incident photons,

g - light-collection efficiency,

ε - quantum efficiency of photocathode,

G - gain of photomultiplier tube,

R - terminating resistance,

It is possible to have a few nanoseconds resolution with the combination of a fast multiplier and a selected scintillation material.

The detection efficiency for fast neutrons in a plastic scintillator depends upon the neutron energy, the geometry and the scintillation material.

The detector signal must be checked for spurious signals such as Compton recoils from the hard X-rays. The contribution due to the hard X-rays can be measured through the determination of the signal attenuation with lead absorbers. When necessary, the hard X-ray signal contribution can be cut off with sufficient lead shielding.

3.4.1 NEUTRON TIME-OF-FLIGHT METHOD

Particle velocities can be measured by their time of flight over a known distance through the use of fast time-resolved detectors. The use of the time- resolved secondary emission particle detectors allows a comparison of the transit time spread over the observed signal structure from the source at various distances. A measurement of the time interval for similar signal structures then determines the velocity of the particle. The expression for determining the energy of the particle of mass m having velocity v with total kinetic energy W , is $W=1/2 mv^2$.

In principle, it is possible to obtain the entire energy spectrum if the particles are emitted in a burst by recording the time dependent rate of particles that are incident on the detector placed at distance x from the source.

The energy resolution is given by $\Delta E/E = 2\Delta T/\tau$, where ΔT is the emission duration, $\tau = x/v$ is the average time of flight of the particle from the source to the detector, and v is the average velocity.

The best accuracy of such a measurement is limited by the value of the emission duration or by the intensity. This theoretical examination of the energy resolution shows that the time-of-flight method can be used for a source emitting short particle pulses with a broad energy distribution and not too large average energy. This energy resolution can be improved by increasing the flight path x , but the maximum value of the flight path is limited by the intensity of the emission of the source. Thus, for a particle source characterized by long emission duration, or high average velocities and narrow spectra, this method is quite difficult to use. However, the average energy of the bulk of neutrons can be determined using a peak-to-peak method of analysis.

3.5 TIME-RESOLVED SOFT X-RAY MEASUREMENTS

For the detection of the soft X-rays there is an increasing number of semiconductor detectors e.g., surface barrier, ion-implanted and avalanche junction diodes as well as charge coupled devices.

When a semiconductor detector absorbs an X-ray photon of energy $h\nu$, a number of $h\nu/\omega$ electron-hole pairs are produced. Here ω is the average energy needed to create an electron-hole pair. The collected charge is proportional to $h\nu$. For silicon at room temperature $\omega = 3.62$ eV [164]. The quantum detection efficiency is given by [164]

$$Q = \frac{h\nu}{\omega} \exp(-\mu_d t_d) \cdot [1 - \exp(-\mu_s t_s)] \quad (3.6)$$

where the first exponential term describes the absorption in the dead layer (on the front surface of the detection area) of thickness t_d , and the second one gives the efficiency for stopping X-rays in the depletion (sensitive) region of thickness t_s .

It can be seen that there exists an energy range over which semiconductor detectors can be used. The lower limit is determined by absorption in the dead layer and the upper limit by incomplete absorption. Lower energy X-rays may be detected if the dead layer is thinner.

Since both the electrons and the holes have large mobilities, and the collection distances are short, it is possible to achieve relatively short collection times. This is an important advantage for the diagnosis of pulsed X-ray sources.

The pair creation energy ω is, in general, a function of the photon energy [165]. It is usually assumed that ω is constant for photon energies well above the band gap of Si ($\sim 10 \times E_g$) [166]. Recently, however, Monte Carlo simulations of the pair creation energy [167] show significant photon energy dependence up to a few hundred eV. It was assumed a photon energy-independent value of 3.70 ± 0.07 eV.

BPX65 PIN photodiode (a detector normally used in the optical range) was chosen for our system. The glass window was removed for application to X-ray detection. The typical parameters of this diode are listed below:

radiant sensitive area	1 mm ²
intrinsic Si wafer thickness	10 μ m
dead layer thickness	0.5 μ m
risetime (typical) (@ 900 nm)	0.5 ns
dark current	< 5 nA (nom. 1 nA)

A five-channel X-ray spectrometer with filtered PIN diodes was used in our experiment. The details about this spectrometer are provided in chapter 5.

The measurements of the soft X-rays emitted from the plasma focus device can be correlated with the current derivative, hard X-rays and neutron emission for the better understanding of the plasma focus process.

3.6 LASER SHADOWGRAPHY METHOD

The development of the shadow method is attributed to Dvorak, who used it to take photographs of the air disturbances in 1880. Shadowgraphy, among all other optical diagnostics for the plasma, is the simplest in principle. A uniform parallel light beam passes through the test object and falls onto a screen or a detector. For a test object of uniform refractive index the screen is uniformly illuminated. If there is a constant refractive index gradient transverse to the beam, all light rays suffer equal deflection with no change in the screen illumination. However, if the first order derivative of the refractive index, varies transverse to the beam, the light rays are not equally displayed at the screen, resulting in variation in illumination.

Consider a parallel monochromatic light beam propagating parallel to Z-axis of a Cartesian co-ordinate system, encountering a system in which the refractive index $n(X, Y, Z)$ is a function of position (see Figure 3.9). If there is a constant refractive index gradient $\partial n / \partial X$, in the X direction, all light rays suffer equal deflection with no change in the screen illumination. However, if the first derivative of the refractive index varies transverse to the beam, all light rays are not equally displaced at the screen, resulting in variation in illumination. The relative variation in the light intensity is given by [168] (all symbols are shown in figure). The shadowgraphy records the line integral of the second derivative of the refractive index. The derivation of the above equation has assumed the linearity in the change of the refractive index gradients in the traverse direction.

In a real system, the light beam used to illuminate the region containing the refractive disturbance, diverges from true parallelism by the angle θ .

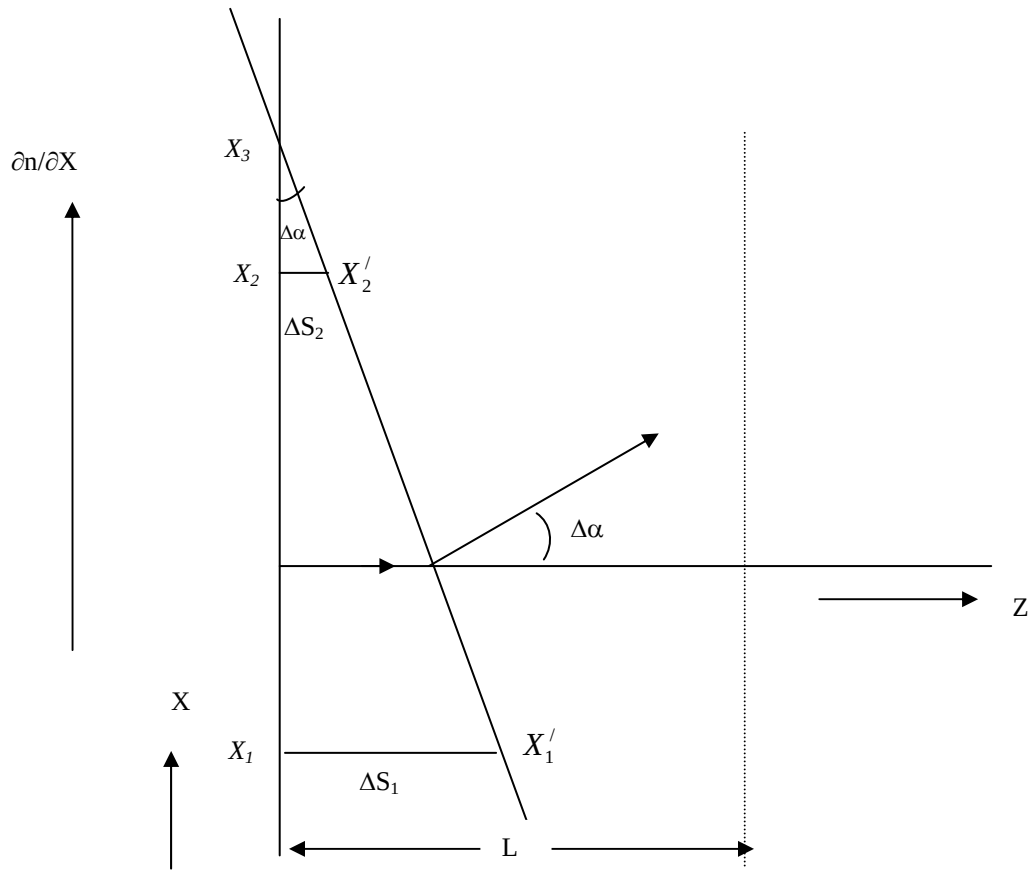


Figure 3.9: Angular deviation of a wave front due to refractive index gradient.

$$\frac{\Delta I}{I} = l \int_0^L \frac{1}{n(X, Y, Z)} \left(\frac{\partial^2}{\partial X^2} n(X, Y, Z) + \frac{\partial}{\partial Y^2} n(X, Y, Z) \right) dZ \quad (3.7)$$

The beam is usually obtained using a condenser to focus a light source on to a pinhole of aperture d , which itself is at the focus of the lens of focal length f , thus $\theta = d/f$.

At the screen, the light rays having passed through the same point in the test section are spread up to a distance $S = l\theta$ apart, due to the beam divergence. Hence, S is the minimum object dimension resolvable at the screen. The resolution may be

improved by decreasing either length l , $\mathcal{G}$, or both. But the sensitivity of the system is proportional to l (equation 3.7). If l' is the minimum separation at which proper sensitivity is achieved to resolve inhomogeneities of dimension δ , then

$$\mathcal{G} < \delta / l' \quad (3.8)$$

It is impossible to resolve infinitesimally small objects by reducing $\mathcal{G}$ due to two reasons: firstly, reducing $\mathcal{G}$ implies reducing the source aperture d . Since the light flux at the screen is directly proportional to d^2 , therefore, the sensitivity of the screen sets a lower limit to d and hence $\mathcal{G}$. Secondly, the diffraction at the object limits the maximum resolution achievable at the screen. An inhomogeneity of dimension δ causes a deflection of the order of λ/δ to light of wavelength λ . To produce a shadowgraph effect larger than the diffraction effect, a refractive deflection ε is needed, such that $\varepsilon > \lambda/\delta$, or

$$\varepsilon \delta > \lambda \quad (3.9)$$

The inhomogeneity produces a detectable shadow effect if the above criterion (3.21) is satisfied. However, the dimension of the inhomogeneity as recorded on the screen could be misleading. To resolve an inhomogeneity in the test section, at the screen, the diffractive displacement of light rays at the screen must not exceed δ , i.e. $\delta > l' \lambda / \delta$ or

$$\delta > \sqrt{l'\lambda} \quad (3.10)$$

Therefore, if criterion (3.9) is satisfied, inhomogeneity having smaller dimension than δ , in which there are strong second derivatives of n , will be detected as a broadened shadow of dimension δ , on the screen.

For the refractive disturbances to be studied in plasma in which n is not a function of Z , the equation (3.7) reduces to the following relation:

$$\frac{\Delta I}{I} \propto Ll \left(\frac{\partial^2}{\partial X^2} n_e(X, Y) + \frac{\partial^2}{\partial Y^2} n_e(X, Y) \right)$$

Since for plasma $n \approx 1$ and $n-1 \propto n_e$, the local change in the screen illumination is proportional to the second derivative of the electron density.

In principle, it is possible to obtain values of n_e from plasma in which $\nabla^2 n_e \neq 0$ by suitably integrating the observed intensity changes, according to equation (3.7). However, for the extraction of quantitative values of density from shadowgraphs depends on having a uniform or very simple Z dependence for the density, so that the necessary double integration may be performed. Also, the two constants of integration must be obtained from other measurements, and this is a serious disadvantage. Furthermore, quantitative results are only obtainable if the displacement of the ray from its undisturbed position on the screen is sufficiently small to avoid confusion by the overlapping of the rays from various source locations. Since the sensitivity of any shadowgraph system is proportional to l , the object to screen distance, therefore, by

suitably choosing l , large variations in screen intensity can be displayed without overlapping. However, in case of plasma in which large variations in $\underline{\nabla}^2 n_e$ occur, it is not usually possible to display the weaker phenomena and to avoid this overlapping.

Generally, it is quite difficult to use shadowgraphy for the quantitative purpose. Usually it is used to locate the position of the rapidly varying density gradients. This technique is well suited for the study of shocks and turbulence in a plasma.

The minimum detectable second derivative of the electron density using shadowgraphy can be estimated. If n_e has no Z dependence, then from equation (3.7)

$$\left(\underline{\nabla}^2 n_e(X, Y)\right)_{\min} = -\left(\frac{\Delta I}{I}\right)_{\min} \frac{1}{4.48 \times 10^{-14} \lambda^2 L l'} \quad (3.11)$$

Using criterion (3.10), the optimum value of $l' = \delta^2 / \lambda$, then (3.10) becomes

$$\left(\underline{\nabla}^2 n_e(X, Y)\right)_{\min} = -\left(\frac{\Delta I}{I}\right)_{\min} \frac{1}{4.48 \times 10^{-16} \lambda \delta^2 L}$$

Consider the plasma focus dense pinch of diameter $L=0.2$ cm and with fine scale structure $\delta = 0.01$ cm, illuminated with light at 500 nm. Assuming that the minimum detectable change in the light intensity is 5%, the equation (3.11) becomes

$$\left(\nabla^2 n_e(X,Y)\right)_{\min} \cong 1.12 \times 10^{31} \text{ cm}^{-5}$$

The main attraction of studying the plasma by the shadowgraphy technique is its relative simplicity compared with the schlieren photography and interferometry. Furthermore, strong variations of the electron density gradients occur in plasma. The other two techniques cannot always be used because various plasma regions act as lenses and cause distortion to the produced image.

An arrangement for the laser shadowgraphy, to study pre- and post-focus phenomena in the plasma focus, using a nitrogen dye laser, was developed. A detailed description of the shadowgraphic system is presented in chapter 5.

3.7 ELECTRON TEMPERATURE MEASUREMENTS

The measurement of the plasma electron temperature is one of the fundamental diagnostic requirements. There are different methods and techniques for the determination of the electron temperature in a plasma. For example, measurement of absolute or relative X-ray continuum emission from the plasma, measurement of relative spectral line intensities, absorption of continuum by a set of different metal foils etc. A brief description of the measurement of the plasma electron temperature by two foil absorption technique, or filter ratio method [169, 170], is presented in the following paragraphs.

The electron temperature from the plasma can be derived theoretically by the filter ratio method using the ratio of the X-ray intensities transmitted through two foil absorbers having different absorption coefficients.

The signal ratio or X-ray transmission ratio of the X-ray intensities $E(\lambda)$ integrated over a transmission spectra of the filters from the two detectors is given by

$$R_T = \frac{\int E(\lambda) e^{-\sum_a \mu_a x_a - \sum_b \mu_b x_b} d\lambda}{\int E(\lambda) e^{-\sum_a \mu_a x_a} d\lambda} \quad (3.12)$$

where, for free-free radiation

$$E(\lambda) = 1.88 \times 10^{-29} \frac{n_e n_i Z_i^2}{\lambda^2 T_e^{1/2}} g_{ff} e^{\frac{-12398.4}{\lambda T_e}} \quad (3.13)$$

and for free-bound radiations

$$E(\lambda) = 1.38 \times 10^{-30} \frac{n_e n_i Z_{i+1}^4}{\lambda^2 T_e^{3/2}} g_{bf} \sum_n \frac{\chi_{i,n}^2 \xi_n}{n} e^{\frac{\chi_{i,n} - \frac{12398.4}{\lambda}}{T_e}} \quad (3.14)$$

where T_e - electron temperature (eV),

λ - wavelength ($\text{\AA}$),

n_e - electron density (cm^{-3}),

n_i - ion density (cm^{-3}),

χ_{in} - ionization potential of the n-th state of the i-th species,

ξ_n - number of vacancies in n-th state,

$Z_{i, i+1}$ - effective charge of the ion,

PLASMA DIAGNOSTIC TECHNIQUES

$x_{a,b}$ - thickness of a and b foils, respectively (cm)

$\mu_{a,b}$ - mass absorption coefficients,

g_{bf} - Gaunt factor for free-bound transition,

g_{ff} - Gaunt factor for free-free transition, and

n - electronic state.

When the plasma focus is operated in deuterium, the X-rays are mostly emitted by free-free transitions. Therefore, the equation (3.13) has to be taken into consideration for $E(\lambda)$ in equation (3.12).

Chapter 4

THEORETICAL MODEL OF PLASMA FOCUS

This chapter deals with the physical model of the plasma focus taking into account the effect of the radiation emission. Experiments were carried out along with the development and implementation of the physical model. Some of the experimental results were used as input parameters of the code. The results from numerical simulation, in turn, can be used to help improve the experiments.

A comparison of the numerical simulation with the experimental results is presented in chapter 6.

4.1 INTRODUCTION

The dynamics process in the plasma focus is usually divided into breakdown phase, axial rundown phase and radial phase, where the radial phase contains compression, quiescent, unstable and decay sub-phases.

For the purpose of developing and implementing the computational model, the related evolution processes (axial rundown phase to radial compression phase) were simplified as shown in Figure 4.1. This approach is based on and summarizes the works by Lee [171-174], Lee and Liu [175] and Liu [176]. These processes are treated as four phases in modeling: axial phase, radial inward shock phase (radial phase), radial reflected shock phase (reflection phase), and radial slow compression phase (slow compression phase).

The features of these phases are:

i) The axial phase begins after the breakdown phase, which is

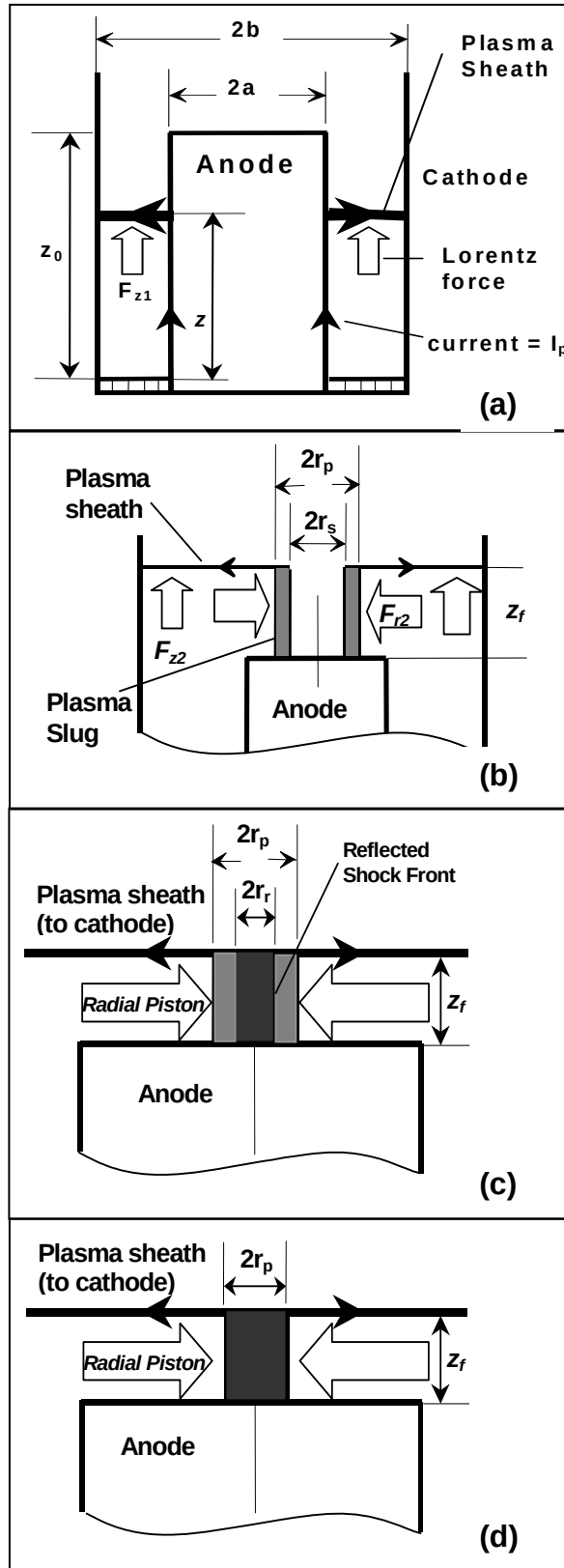


Figure. 4.1: Simplified processes in plasma.

not taken into account in this model. During this phase, the up-moving (Figure 4.1(a)) plasma layer is simplified as a flat annular conductive sheath connecting the anode to the cathode. It sweeps from the anode bottom to the top, and accumulates a fraction f_{m1} of all the encountered mass of the working gas. The axial position of the sheath is z , where the z -axis is pointing upwards in the direction towards the open end of the electrode system.

The axial phase begins at $z=0$, taken as a back wall insulator and ends at $z = z_0$, where z_0 is the anode length. Therefore, the mass of the plasma sheath at z is $f_{m1} n_0 m z \pi (b^2 - a^2)$, where, n_0 is the atomic number density of the filling gas, m is the atomic mass of the filling gas, and a and b are the anode and cathode radii, respectively.

ii) The radial phase is divided into three sub-phases: radial inward shock phase, radial reflected shock phase and slow compression phase.

In the radial inward shock phase, the plasma is treated as a slug with outer radius r_p (piston position), inner radius r_s (shock front) and height z_f measured from the open end of the anode. The slug is compressed radially inward by the radial magnetic piston. A fraction f_{m2} of the gas in the path is collected into the slug. The inward shock phase begins at $r_p = r_s = a$, and ends when the shock front meets at the axis center at $r_s = 0$.

iii) When the shock front arrives the axis, the particles in the slug transfer most of their kinetic energy E_k into plasma internal energy by collisions. This would significantly increase the temperature and density of the plasma at the center. This process is called the reflected shock phase.

In this phase, the plasma is considered as two parts: the central part is the relatively stationary hot and dense plasma column; the outer part is the fast compressing plasma slug with lower temperature and density. The boundary that separates the hot and dense center column with the outer plasma slug is defined as the reflected shock front. The reflected phase ends when all particles join the center plasma column, i.e. reflected shock front and magnetic piston meet at a certain radial position, hence $r_r = r_p$.

iv) After the reflected shock phase, there is the slow compression phase. In this phase, the plasma column continues to compress until it is disassembled by the instabilities.

4.2 ELECTRICAL PROPERTIES AND CIRCUIT EQUATIONS

4.2.1 EQUIVALENT CIRCUIT EQUATIONS

The equivalent electrical circuit of the plasma focus during the discharge is drawn schematically in Figure 4.2 [174, 176]. In this figure, C_0 , is the capacitance of the energy

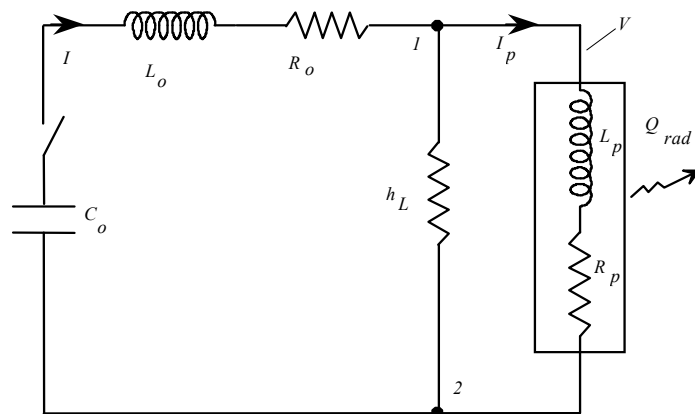


Figure 4.2: Equivalent circuit structure of the plasma focus.

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bank. R_0 and R_p are the circuit and plasma resistance, respectively. The circuit inductance comprises of the fixed circuit inductance, L_0 , and the changing plasma tube inductance, L_p . h_L is the leakage resistance in the plasma tube, and is directly related to the current leakage along the insulator surface.

The energy bank is initially charged to a high voltage V_0 . According to Kirchhoff's law, the equation of this circuit can be written as

$$\frac{1}{C_0} \int_0^t I dt - V_0 + R_0 I + \frac{d}{dt}(L_0 I) + R_p I_p + \frac{d}{dt}(L_p I_p) = 0 \quad (4.1)$$

and the tube voltage (voltage across the points 1 and 2) is

$$V = V_0 - \frac{1}{C_0} \int_0^t I dt - R_0 I - \frac{d}{dt}(L_0 I) = R_p I_p + \frac{d}{dt}(L_p I_p) \quad (4.2)$$

For the simplified plasma geometry described in section 3.1, the electrical parameters of the plasma tube and the circuit equations in each phase can be derived.

4.2.1.1 AXIAL PHASE

In axial phase, the configurations of the plasma tube and plasma sheath are simplified as shown in Figure. 4.1(a). The conductor of the current is composed of the conductive cylinder (electrodes) and the plasma sheath at position z (measured from

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the bottom of the anode). This electrical system is considered as a standard cylindrical coaxial conducting tube with height z , outer radius b and inner radius a .

The induced magnetic field between the electrodes is

$$B_{\theta} = \frac{\mu}{2\pi} \times \frac{I_p}{r} \quad (4.3)$$

where $I_p = f_c I$ is the current flowing through the plasma, and f_c is a current shedding factor.

The resultant $\vec{j} \times \vec{B}$ force on the plasma sheath is directed axially upward to the open end of the anode.

The force exerted by the magnetic piston to drive the plasma sheath is:

$$F_{zI} = \int_a^b I_p \times \frac{\mu I_p}{2\pi r} dr = \frac{\mu I_p^2}{2\pi} \ln\left(\frac{b}{a}\right) \quad (4.4)$$

The tube inductance is:

$$L_{pI} = \frac{\mu}{2\pi} z \ln\left(\frac{b}{a}\right) \quad (4.5)$$

By substituting this expression into the general circuit equation (4.1), one gets the circuit equation for the axial phase:

$$\frac{dI}{dt} = \frac{V_0 - \frac{1}{C_0} \int_0^t I dt - (R_0 + f_c R_p) I - \frac{\mu}{2\pi} f_c I \ln\left(\frac{b}{a}\right) \frac{dz}{dt}}{L_0 + \frac{\mu}{2\pi} f_c z \ln\left(\frac{b}{a}\right)} \quad (4.6)$$

4.2.1.2 RADIAL PHASE

In the radial phase (shown in Figure. 4.1(b)), the current flows through anode (radius a , height z_0), plasma slug (radius r_p , height z_f), plasma sheath ($r_p \rightarrow b$ at position z_f) and back to cathode (radius b). The induced magnetic field can also be expressed by equation 4.3.

The $\vec{j} \times \vec{B}$ force on the plasma slug directed radially inwards is given by:

$$F_{r2} = \int_0^{z_f} I_p \times \frac{\mu I_p}{2\pi r_p} dz = \frac{\mu I_p^2}{2\pi} \frac{z_f}{r_p} \quad (4.7)$$

This is the force exerted by the magnetic piston compressing the plasma slug. The force on the plasma sheath which connects the slug to the cathode pointing axially upwards is written as:

$$F_{z2} = \int_{r_p}^b I_p \times \frac{\mu I_p}{2\pi r} dr = \frac{\mu I_p^2}{2\pi} \ln\left(\frac{b}{r_p}\right) \quad (4.8)$$

The total plasma inductance is composed of two parts: the conductive cylinder of the electrodes of height, z_0 , and the plasma slug of height, z_f , and outer radius r_p :

$$L_{p2} = \frac{\mu}{2\pi} \left(z_0 \ln\left(\frac{b}{a}\right) + z_f \ln\left(\frac{b}{r_p}\right) \right) \quad (4.9)$$

From equations (4.9) and (4.1), one gets the circuit equation for the radial phase as:

$$\frac{dI}{dt} = \frac{V_0 - \frac{1}{C_0} \int_0^t I dt - (R_0 + f_c R_p)I + \frac{\mu}{2\pi} f_c I \left(\frac{z_f}{r_p} \frac{dr_p}{dt} - \ln\left(\frac{b}{r_p}\right) \frac{dz_f}{dt} \right)}{L_0 + \frac{\mu}{2\pi} f_c \left(z_0 \ln\left(\frac{b}{a}\right) + z_f \ln\left(\frac{b}{r_p}\right) \right)} \quad (4.10)$$

4.2.2 PLASMA RESISTANCE

The plasma resistance can be expressed using Spitzer's resistivity:

$$\eta = 1.033 \times 10^{-4} \times \ln G \times \frac{Z_{eff}}{(T_{eV})^{3/2}} \quad (4.11)$$

where $\ln G = 29.27 + 1.5 \times \ln T_{eV} - 0.5 \ln[(1 + Z_{eff}) \times N_i]$, T_{eV} is the plasma temperature expressed in eV , N_i is the plasma ion density and Z_{eff} is the effective ionic charge.

Therefore, the resistance of the plasma slug during the radial phase is

$$R_{slug} = \frac{1.033 \times 10^{-4} \times \ln G \times Z_{eff}}{(T_{eV})^{3/2}} \times \frac{z_f}{\pi(r_p^2 - r_s^2)} \quad (4.12)$$

where r_s is the inner radius of the plasma slug. Here, it is assumed that the current is uniformly flowing in the slug, although it should have concentration on slug surface because of skin effect (the thickness of the surface effect is estimated to be

$$\delta \sim \sqrt{\frac{2\eta}{\mu\omega}}).$$

4.3 SHOCK WAVE AND SHOCK WAVE EQUATIONS

The motion of the plasma in focus device is driven by the strong magnetic piston at highly supersonic speed. The strong shock wave scheme, therefore, is employed to describe the plasma dynamics.

4.3.1 EQUATIONS FOR 1D SHOCK WAVE

It is assumed that the piston is moving at a very fast speed v_p ($v_p \gg c_s$ – speed of sound) in a long tube. A shock wave will be formed (shown in Figure 4.3). The shocked gas and the ambient gas are separated by a fast moving shock front. The speed of the shock front, v_s , is higher than v_p , because the shock front is continuously collecting new ambient gas into the shocked gas region. The shocked gas and the piston are moving with a steady speed, v_p . The pressure at the piston is P_p .

For the 1D configuration shown in Figure 4.3, the shock-jump equations across the shock front can be setup in the shock-fixed coordinates (q_0 and q are the shocked and ambient gas speeds in the shock-fixed coordinates).

$$\text{Mass:} \quad \rho_0 q_0 = \rho q \quad (4.13)$$

$$\text{Momentum:} \quad \rho_0 q_0^2 + P_0 = \rho q^2 + P \quad (4.14)$$

$$\text{Energy in unit mass:} \quad \rho_0 q_0 \left(\frac{1}{2} q_0^2 + h_0 \right) = \rho q \left(\frac{1}{2} q^2 + h \right) \quad (4.15)$$

where ρ_0 and P_0 , respectively, are the gas density and pressure of the ambient gas, $q_0 = v_s - v_0$ is the relative speed of the ambient gas to the shock front, and h_0 is the enthalpy per unit mass.

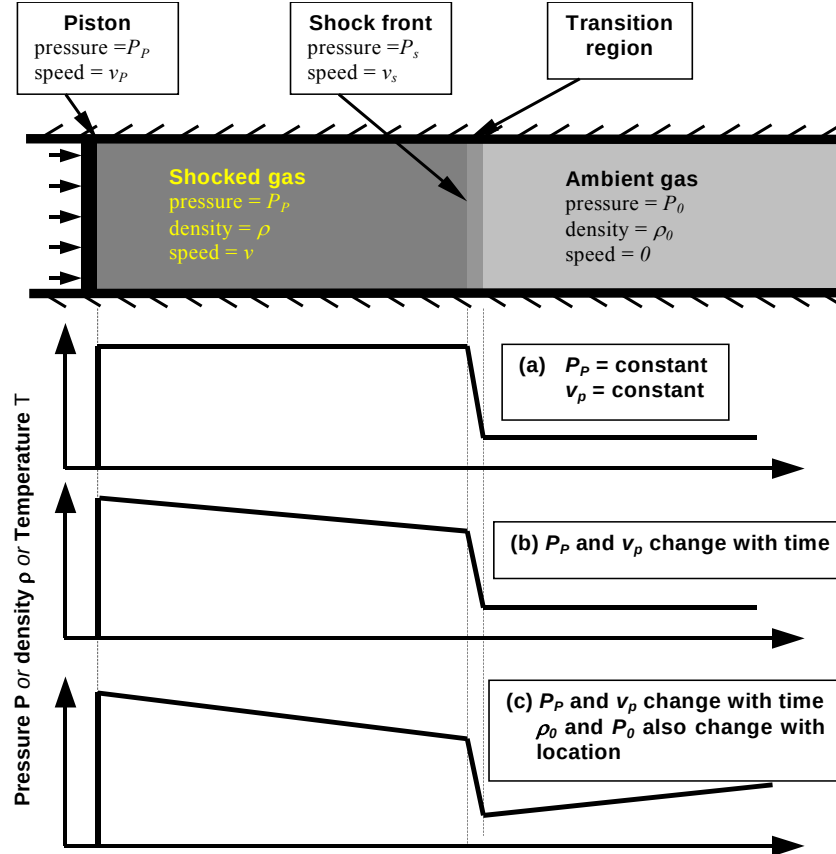


Figure 4.3: Shock wave induced by a fast piston with (a); constant pressure and ambient parameters; (b); changing pressure and constant ambient parameters and (c); changing pressure and changing ambient parameters.

The other two equations that describe the shocked gas are:

$$\text{Thermal state: } P = nkT = \rho \frac{R_0}{M} TD \quad (4.16)$$

$$\text{Caloric state: } h = \frac{\gamma}{\gamma - 1} \frac{P}{\rho} \quad (4.17)$$

where M is the molecular weight, $R_0 = 8.31 \text{ J/mol}\cdot\text{K}$ is the universal gas constant and D is the departure coefficient (particles come from one neutral atom, $D = Z_{eff} + 1$).

4.3.2 PARAMETERS OF 1D SHOCK WAVE

If the piston pressure, P_p , is constant, a steady state motion is established as shown in Figure 4.3(a).

$$\begin{cases} P = P_p = \text{const} \\ q = v_s - v_p = \text{const} \\ q_0 = v_s = \text{const} \end{cases} \quad (4.18)$$

For strong shocks, $P \gg P_0$ and $h \gg h_0$. Hence P_0 and h_0 can be omitted from equations (4.14) and (4.15), respectively. Thus, from equations (4.13) and (4.18) one gets the parameters of the 1D strong shock wave gas as

$$\text{Shock density ratio: } \Gamma = \frac{\rho}{\rho_0} = \frac{\gamma + 1}{\gamma - 1} \quad (4.19)$$

$$\text{Shock pressure: } P_s = P_p = \rho_0 \left(1 - \frac{1}{\Gamma} \right) v_s^2 = \frac{2\rho_0}{\gamma + 1} v_s^2 \quad (4.20)$$

$$\text{Shock temperature: } T = \frac{M}{R_0} \frac{(\Gamma - 1)}{\Gamma^2} \frac{q_0^2}{D} = \frac{2m_i}{k_B(1 + Z_{eff})} \frac{(\gamma - 1)}{(\gamma + 1)^2} v_s^2 \quad (4.21)$$

The equations (4.20) and (4.21) are very important in describing the plasma behavior. The former revealed the fact that the plasma dynamics is determined by the piston pressure, mass density, as well as the thermodynamic state of the plasma itself. The latter describes the

plasma temperature, which forms basis for calculation of other parameters. Figure 4.4 shows the relationship between shock speed and shock temperature for several gases.

In this steady 1D shock wave, the piston is moving with v_p , the shock front is moving with v_s and continuously collecting ambient gas into the shocked region. In a short time dt , the recruited gas is

$$dm = \rho_0 S_p v_s dt \quad (4.22)$$

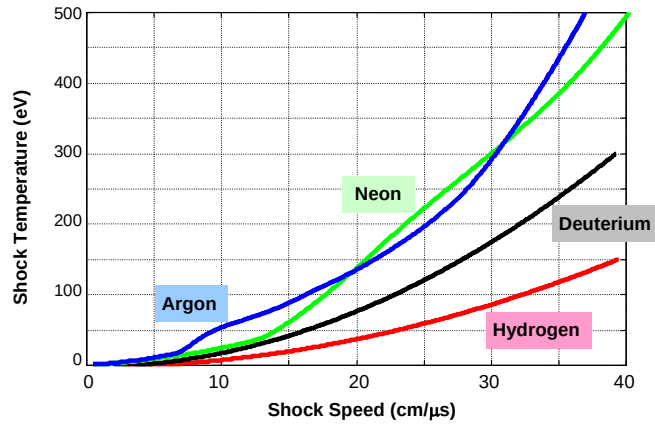


Figure. 4.4: Relationship between shock speed and shock temperature for hydrogen, deuterium, neon and argon plasma [179].

where S_p is the piston area. Then, the work done by the piston per unit mass (that is also the energy level difference between shocked and unshocked gas) is given by:

$$E = \frac{P_p \times S_p \times v_p dt}{dm} = \left(1 - \frac{1}{\Gamma}\right) v_s v_p = v_p^2 \quad (4.23)$$

On the other hand, the kinetic energy of the shocked gas equals $\frac{1}{2} v_p^2$.

Therefore, one half of the work done by the piston is converted into the kinetic energy, and the other half is transferred into the internal energy to heat the gas.

4.3.3 SHOCK WAVE WITH CHANGING PISTON PRESSURE OR CHANGING AMBIENT GAS PRESSURE

If the piston pressure is changing, $P_p = P_p(t)$, then the parameters in the shocked gas region also change with time and location as shown in Figure 4.3(b). The pressure, density and temperature of the shocked gas, as a function of location and time, should be written as $P(x,t)$, $\rho(x,t)$, and $T(x,t)$, respectively.

Assuming that the signal of the changing P_p propagates in the shocked gas region with a speed c_s (sound speed in the gas), then the transmission time from piston to shock front is

$$t_{p-s} = (x_p - x_s) / c_s \quad (4.24)$$

where the sound speed c_s in the gas can be calculated as [176],

$$c_s = \sqrt{\frac{\gamma (1 + Z_{eff}) kT}{m_i}} \quad (4.25)$$

If the time scale of t_{p-s} is comparable with the changing speed of piston, the effect of changing piston pressure must be taken into account for the calculation. The pressure distribution $P(x,t)$ in the shocked gas can be simplified as:

$$P(x,t) = P_p(t - x/c_s) \quad (4.26)$$

The other parameters can be calculated by assuming local thermal equilibrium.

If the ambient gas density and pressure have a distribution, $P_0 = P_0(x)$ and $\rho_0 = \rho_0(x)$. The 1D equations remain valid within the thin gas layers close to the shock front (for both sides).

Generally speaking, it takes time for the shocked gas to reach new equilibrium with the new ambient parameters (for example, $\Gamma = \rho/\rho_0$). However, if the time for new equilibrium is much smaller than the time scale of change in ambient parameters, one can still use the constant ambient environment equations for calculations.

4.4 PLASMA AS THERMODYNAMIC GAS

The plasma slug is finally developed into a hot and dense plasma column. In this section, the thermodynamic properties of the plasmas confined in the slug, or a final column, are discussed.

4.4.1 THERMODYNAMIC PROPERTIES OF PLASMA GAS

The plasma in the column can be considered as a thermodynamic gas. The force that drives the compression is the magnetic piston, whose pressure is:

$$P_B = \frac{1}{2\mu} \left(\frac{\mu I_p}{2\pi r_p} \right)^2 \quad (4.27)$$

With the supersonic compression, the shock front and the inner and upper sides of the slug are continuously collecting new gas. The total number of the collected atoms (ions) is:

$$N_{ion0} = \pi n_0 \int_t \left((r_p^2 - r_s^2) \frac{dz_f}{dt} - 2z_f r_s \frac{dr_s}{dt} \right) dt \quad (4.28)$$

At the same time, the number of charged particles in this plasma gas also increases by the ionization

$$N_e = N_{ion} \times Z_{eff} \quad (4.29)$$

A fraction, $(1-f_{m2})$, of the collected gases is leaked during the compressing process. The number of leaked particles is:

$$N_{leak} = N_{ion0} \times (1 - f_{m2}) \quad (4.30)$$

There are energy sources and sinks in the plasma gas. The total energy that is transferred into the plasma is named as EINP. This energy source, basically, is a combined contribution of the work done by the magnetic piston and the joule heating. This energy is used to heat, ionize and push the plasma. The relevant energy sinks are the thermal energy, ionization energy, E_i , radiation energy loss, Q_{rad} , plasma kinetic energy, E_k , and energy loss along with the particle leakage, E_S . The characteristics of the plasma gas in the slug are summarized in Table 4.1.

In this table, m_i is the mass of atom or ion. The fraction of the plasma ionized to the Z -th ionized state is represented by α_z . $Z_{eff} = \sum (z\alpha_z)$ is the effective (average) charge number of one ion. I_z is the total energy required to raise one ion from its unionized state to its z -th ionized state, and E_z is the average excitation energy per Z -th ionized state ion. These energies can be computed from the tabulated values of atomic and ionic energy levels and the statistical weights of these levels, once a suitably converging summation scheme is adopted.

4.4.2 EFFECTIVE SPECIFIC HEAT RATIO OF PLASMA GAS

The ionization and excitation processes play an important role in the energy balance in the plasma gas. This effect can be described by the effective specific heat ratio, γ . In ideal gas, the difference between C_v and C_p is caused by the extra work done by gas expansion. In plasma gases, this difference becomes smaller because the ionization/excitation energies are quite large compared to the extra work

By taking the ionization effect into consideration, the effective specific heat ratio, γ , can account for the energy and particle changes by ionization, and can be related directly to plasma temperature. Lee has proposed a simplified method to compute the values of γ as function of temperature [177, 178]. According to a theory based on energy balance [173], the radius ratio r_p/r_o of a constant-current plasma pinch depends on the value of the effective specific ratio, γ , of the pinched plasma.

The calculation is improved by taking into account the ionization effect on the basis of the corona plasma equilibrium [179].

The enthalpy of the plasma can be expressed as:

$$h = N_{ion} \left[\frac{5}{2} (1 + Z_{eff}) kT + \sum_{z=1}^{Z_n} (\alpha_z I_z) + \sum_{z=1}^{Z_n} (\alpha_z E_z) \right] \quad (4.31)$$

On the other hand, the value of γ is defined in terms of the enthalpy by:

$$h = \frac{\gamma}{\gamma - 1} PV = \frac{\gamma}{\gamma - 1} (1 + Z_{eff}) N_{ion} kT \quad (4.32)$$

Combining these two equations, the value of γ can be computed as:

$$\frac{\gamma}{\gamma - 1} = \frac{5}{2} + \frac{1}{kT(1 + Z_{eff})} \left(\sum_{z=1}^{Z_n} (\alpha_z I_z) + \sum_{z=1}^{Z_n} (\alpha_z E_z) \right) \quad (4.33)$$

By omitting the excitation energies, E_z , (which are much smaller than ionization energies), this equation can be written as:

$$\frac{\gamma}{\gamma - 1} \approx \frac{5}{2} + \frac{\sum_{z=1}^{Z_n} (\alpha_z I_z)}{kT(1 + Z_{eff})} = \frac{5}{2} + \frac{E_i}{kT(1 + Z_{eff})} \quad (4.34)$$

Table 4.1: Thermodynamic quantities in plasma column.

Properties	Expressions	
Gas Pressure	Pressure of the magnetic piston: $P_B = \frac{1}{2\mu} \left(\frac{\mu I_p}{2\pi r_p} \right)^2$	
Particle Source/Sink (N _{ion} : ions in the slug)	Gas collected by shock front: $N_{ion0} = \pi n_0 \int_t \left((r_p^2 - r_s^2) \frac{dz_f}{dt} - 2z_f r_s \frac{dr_s}{dt} \right) dt$	
	Ionization and recombination: $N_e = N_{ion} \times Z_{eff}$	
	Particle leakage during compression: $N_{leak} = N_{ion0} \times (1 - f_{m2})$	
Energy Source/Sink	Piston work:	$EINP = \int_t I_p^2 \left(R_p - \frac{\mu}{4\pi} \frac{z_f}{r_p} \frac{dr_p}{dt} \right) dt$
	Joule heating:	
	Thermal energy: $E_{thermal} = \frac{3}{2} N kT$	
	Ionization and excitation: $E_i = N_{ion} \times \left[\sum_{z=1}^{Z_n} (\alpha_z I_z) + \sum_{z=1}^{Z_n} (\alpha_z E_z) \right]$	
	Radiation: $Q_{rad} = \int_t (P_b + P_l + P_r) dt$	
	Kinetic energy: $E_k = \frac{1}{2} m_i N_{ion} \left(\frac{dr_p}{dt} \right)^2$	
	Energy loss along with the particle leakage: $E_{loss} = \frac{3}{2} N_{loss} kT$	

4.4.3 THERMODYNAMIC EQUATION FOR PLASMA GAS

The average thermal states of the plasma can be calculated by applying the first thermodynamics law to the plasma gas in the slug

$$dh = dQ + VdP \quad (4.35)$$

where h is the enthalpy of the plasma gas, $h = \frac{\gamma}{\gamma - 1} PV$; $dQ = d(Q_{joule} - Q_{rad} - E_I - E_s)$

is the energy increase by energy source/sink, P is the pressure of magnetic piston (equation. (4.27)). Substituting these expressions into equation (4.35), one gets the thermodynamic equation for the plasma gas in the plasma slug as:

$$\gamma PdV + VdP = (\gamma - 1)dQ \quad (4.36)$$

4.5 ENERGY AND TEMPERATURE OF THE PLASMA IN PLASMA FOCUS

4.5.1 MECHANISMS OF ENERGY TRANSFER INTO PLASMA AND PLASMA TUBE

In the plasma focus system, the whole energy comes from the electrical energy stored initially in the energy bank, which is written as:

$$E_{total} = \frac{1}{2} C_0 V_0^2 \quad (4.37)$$

The power input into the whole plasma system is (tube voltage V is expressed by equation. (4.2)),

$$P_{total} = I_p V = I_p L_p \frac{dI_p}{dt} + I_p^2 \left(R_p + \frac{dL_p}{dt} \right) \quad (4.38)$$

As the energy stored in the tube inductance is $\frac{1}{2} L_p I_p^2$, the power transferred into plasma is:

$$P_{plasma} = P_{total} - \frac{d}{dt} \left(\frac{1}{2} L_p I_p^2 \right) = I_p^2 \left(R_p + \frac{1}{2} \frac{dL_p}{dt} \right) \quad (4.39)$$

The first term in this equation is induced apparently by the joule heating which heats the plasma directly. The second term is the power contributing to the dynamics of the system (kinetic energy, and also the thermal energy, ionization energy, etc.).

The term $\frac{1}{2} \frac{dL_p}{dt}$ can be seen as a "dynamic resistance" of the plasma [173].

Therefore, the total energy transferred into plasma is:

$$EINP = \int_0^t I_p^2 \left(R_p + \frac{1}{2} \frac{dL_p}{dt} \right) dt \quad (4.40)$$

From equations (4.5), (4.9) and equation (4.40), $EINP$ is written as:

$$\textbf{Axial Phase: } EINP_1 = \int_0^t I_p^2 \left(R_p + \frac{\mu}{4\pi} \ln \left(\frac{b}{a} \right) \frac{dz}{dt} \right) dt \quad (4.41)$$

$$\textbf{Radial Phase: } EINP_2 = \int_{t_{20}}^t I_p^2 \left(R_p + \frac{\mu}{4\pi} \ln \left(\frac{b}{r_p} \right) \frac{dz_f}{dt} - \frac{\mu}{4\pi} \frac{z_f}{r_p} \frac{dr_p}{dt} \right) dt \quad (4.42)$$

where t_{20} is the initial time of the radial phase.

On the other hand, the work done is calculated by the magnetic piston. In axial phase, the force on plasma sheath is expressed by equation. (4.4), which is in Z direction. Then the work done by the axial magnetic piston F_{z1} is

$$W_1 = \int_0^z F_{z1} dz = \int_0^z \frac{\mu I_p^2}{2\pi} \ln \left(\frac{b}{a} \right) dz = \int_0^t I_p^2 \frac{dL_{p1}}{dt} dt \quad (4.43)$$

Similarly, from equations (4.7) and (4.8), the work done in the radial phase is given by F_{z2} and F_{r2}

$$W_2 = \int_0^{z_f} F_{z2} dz_f + \int_0^{r_p} F_{r2} dr_p = \int_{t20}^t \frac{\mu I_p^2}{2\pi} \left[\ln\left(\frac{b}{r_p}\right) \frac{dz_f}{dt} + \frac{z_f}{r_p} \frac{dr_p}{dt} \right] dt = \int_{t20}^t I_p^2 \frac{dL_{p2}}{dt} dt \quad (4.44)$$

Comparing equation (4.40) with equations (4.43) and (4.44), one finds that the second term of $EINP$ in equation (4.40) is just the half of the work done by the magnetic piston, while the other half is stored in the tube inductance. Therefore, there are two mechanisms for energy coupling into plasma: joule heating and magnetic piston.

In the radial phase, the magnetic piston pushes the plasma in both the axial and the radial directions. Only the radial direction is of interest because the radial compression finally develops the plasma slug into a hot and dense plasma column.

Comparing equation (4.42) with equation (4.44), the radial work (work done by F_{r2}) is considered to be the energy into the plasma slug

$$EINP_{slug} = \int_{t20}^t I_p^2 \left(R_{slug} - \frac{\mu}{4\pi} \frac{z_f}{r_p} \frac{dr_p}{dt} \right) dt \quad (4.46)$$

This expression is identical with that derived by Lee [173].

4.5.2 DRIVING PARAMETER

The concept of the driving parameter was proposed by Lee [180, 42] to describe the intensity of the driving force in plasma focus. It was found that the driving parameter, S , is nearly constant at $90 \text{ kA}/(\text{cm}\cdot\text{torr})^{1/2}$ for optimized operation

of deuterium over a wide range of machines, from small one (3kJ) to big machine (300 kJ) [42]. The driving parameter is defined as:

$$S = \frac{I/a}{\sqrt{\rho}} \quad (4.47)$$

This parameter has been observed experimentally to be a measure of speed, which has an optimum value for each working gas. It can also be considered as a measure of the drive magnetic pressure (see equation (4.27)) per unit density, or the drive magnetic energy per unit mass. Thus, any model, based on either the snowplow or slug model, will have the driven speeds primarily dependent on the factor S. It is a valuable parameter to quickly estimate the plasma state, or the required working condition in plasma focus, as well as any other electromagnetic driven plasma system.

In order to deduce here the drive parameter from the viewpoint of energy input, equation (4.39) is considered. It is assumed that the discharge current is constant in the radial phase, and the energy input by joule heating is omitted. The energy in the radial phase can be estimated as:

$$EINP_2 \sim \frac{1}{2} I_p^2 \int_{t_{20}}^t dL_p = \frac{1}{2} L_{p2b} I_p^2 = \frac{\mu}{4\pi} I_p^2 z_f \ln\left(\frac{b}{r_p}\right) \quad (4.47)$$

where L_{p2b} is the plasma slug inductance, which is the second term in equation (4.9).

The energy in the plasma slug can be considered as a certain portion of $EINP_2$ (the other part is stored in the plasma sheath). On the other hand, the mass in the column is measured by $\rho \pi a^2 z_f$. Thus, the energy per unit mass in the final plasma column can be expressed as

$$\frac{EINP_{slug}}{mass} \sim const \times \frac{\frac{\mu}{4\pi} I_p^2 z_f \ln\left(\frac{b}{r_{min}}\right)}{\rho \pi a^2 z_f} \sim const 2 \times \frac{I^2}{\rho a^2} = const 2 \times S^2 \quad (4.48)$$

where r_{min} is the final column radius. Here $\ln(b/r_{min})$ varies at a very slow rate compared to a^2 , and can be considered constant.

From equation (4.48), it is clear that the drive parameter can be directly correlated with the energy density in the final plasma column. Therefore, to increase the energy density (and consequently the plasma temperature), one can increase the current, or decrease the anode radius and gas pressure.

4.5.3 ENERGY TRANSFER PROCESS

The energy balance in the plasma focus is very important in the analysis of its behavior. Based on the discussions in previous sections, the energy transfer process is summarized in Figure 4.5.

The energy goes into plasma focus tube by three ways: inductive impedance, piston work and joule heating. The inductive impedance represents the change of the

THEORETICAL MODEL OF PLASMA FOCUS

energy stored in L_p by the change of the discharge current. The piston work is the main path to transfer energy into the plasma focus tube. According to the discussion in last section, half of the piston work is used to increase L_p by pushing the conductive plasma sheath, consequently it increases the energy stored in L_p . The other half converts into plasma energy. The joule heating is induced by the plasma resistance R_p and the flow of current through it. It directly heats the plasma.

The energy stored in the tube inductance, E_L , is determined by both the inductance and the current. It must be noticed that this energy is not dissipative; if the current is dropping, it will act as an energy source to output energy.

$EINP$, the energy into plasma, is composed of two parts: the joule heating and half of the piston work. The plasma energy exists in 4 forms: kinetic energy, E_k , internal energy, U , ionization and excitation energies, E_i and E_x , respectively, and radiation energy, Q_{rad} .

Using all the values of these energies in the plasma, the plasma temperature can be calculated from its internal energy, U , by using:

$$T = \frac{U}{\frac{5}{2} Nk} = \frac{EINP - E_k - E_i - Q_{rad} - E_s}{\frac{5}{2} (1 + Z_{eff}) N_{ion} k} \quad (4.49)$$

Using equations (4.49) and (4.25), the plasma temperature is evaluated.

4.6 EQUATIONS AND EXPRESSIONS

A group of consistent equations describes the state and behavior of the plasma in focus machine. These equations and parameter expressions can be derived on the basis of the scheme and processes described in previous sections.

4.6.1 AXIAL PHASE

During the axial phase, not all the gas is driven by the shock front (the mass factor is f_m). The swept gas is assumed to be concentrated in the plasma sheath at the axial position z .

The number of ions in the plasma sheath is given by:

$$N_{ion1} = \pi f_{m1} n_0 z (b^2 - a^2) \quad (4.50)$$

Then, the mass of the plasma sheath is

$$mass_1 = m_i \times N_{ion1} = \pi (b^2 - a^2) f_m n_0 z m_i \quad (4.51)$$

THEORETICAL MODEL OF PLASMA FOCUS

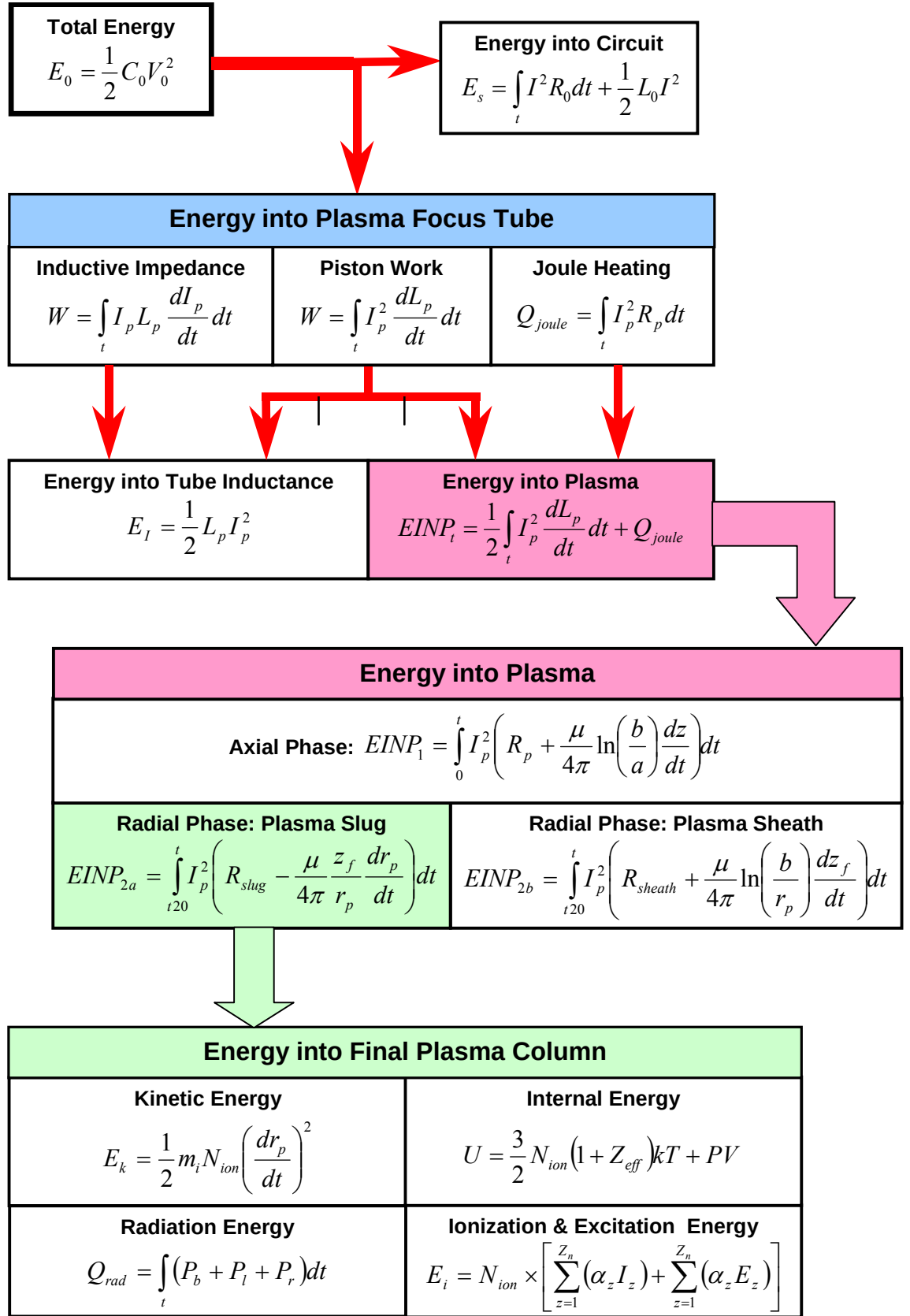


Figure 4.5: Energy transfer process in plasma focus

where n_0 is the atom density of the filled gas in the plasma focus and m_i is the mass of an ion or atom.

The momentum change of the plasma sheath in the coordinate z is:

$$\frac{d}{dt} \left(mass_1 \frac{dz}{dt} \right) = F_{z1} \quad (4.52)$$

From equations (4.4), (4.51) and (4.52), one gets

$$\frac{d}{dt} \left[\pi (b^2 - a^2) f_m n_0 z m_i \frac{dz}{dt} \right] = \frac{\mu f_c^2 I^2}{4\pi} \ln c \quad (4.53)$$

or

$$\frac{d^2 z}{dt^2} = \frac{\frac{\mu f_c^2 I^2}{4\pi} \ln \left(\frac{b}{a} \right) - \frac{mass_1}{z} \left(\frac{dz}{dt} \right)^2}{mass_1} \quad (4.54)$$

From equation (4.54) and the axial phase circuit equation (equation. (4.6)), the basic parameters $I(t)$ and $z(t)$ can be calculated.

4.6.2 RADIAL INWARD SHOCK PHASE

4.6.2.1 SHOCK WAVE EQUATIONS

In the radial inward shock phase, the plasma slug is driven by the radial magnetic piston. The shock-jump equations are still valid for the gas near the shock front, although the shock wave is in a 2D cylindrical geometry. The shock front speed can be related to the pressure according to equation (4.20). Considering the piston-shock front transmission time (t_{p-s}) effect (equation (4.26) and equation. (4.27)), the pressure at shock front is

$$P_s(t) = P_B(t - t_{p-s}) = \frac{1}{2\mu} \left[\left(\frac{I_p}{2\pi r_p} \right)^2 \right]_{t-t_{p-s}} \quad (4.55)$$

Combined with equation (4.20), one gets the plasma dynamics equation as (the driving parameter is actually the driving force account for plasma dynamics):

$$\frac{dr_s}{dt} = -\frac{1}{4\pi} \sqrt{\frac{\mu(\gamma+1)}{f_{m2} n_0 m_i}} \left[\frac{I_p}{r_p} \right]_{t-t_{p-s}} \quad (4.56)$$

Since, the compression is open at the upper end, an axial shock propagating upward in the Z-direction is expected. Furthermore, it is assumed that the pressure

driving the radial shock is the same as that driving the axial shock. Thus, the length of the plasma slug, z_f , increases during the compression. One may write:

$$\frac{dz_f}{dt} = -\frac{2}{\gamma+1} \frac{dr_s}{dt} \quad (4.57)$$

4.6.2.2 THERMODYNAMIC EQUATIONS

Based on equation (4.37), one can derive the thermodynamic equations of the plasma gas in the plasma slug.

The volume of the the plasma slug is $V = \pi(r_p^2 - r_s^2)z_f$. Then, the change in volume of a fixed mass in the plasma slug can be written as:

$$dV = 2\pi(r_p dr_p - r_s dr_s)z_f + \pi(r_p^2 - r_s^2)dz_f \quad (4.58)$$

But, the increments dz_f and dr_s also introduce more gas into the plasma slug. This new gas will take up some volume in the plasma slug (here $dr_s < 0$ and $dz_f > 0$)

$$dV' = \frac{[-2\pi r_s z_f dr_s + \pi(r_p^2 - r_s^2)dz_f]}{\Gamma} = \frac{\gamma-1}{\gamma+1} [-2\pi r_s z_f dr_s + \pi(r_p^2 - r_s^2)dz_f] \quad (4.59)$$

where $\Gamma = (\gamma + 1)/(\gamma - 1)$ is the compression ratio of the shock wave. Therefore, the actual volume change of the original gas is

$$\frac{dV - dV'}{V} = \frac{2\pi z_f \left(r_p dr_p - \frac{2}{\gamma + 1} r_s dr_s \right) + \frac{2\pi}{\gamma + 1} (r_p^2 - r_s^2) dz_f}{z_f (r_p^2 - r_s^2)} \quad (4.60)$$

From equations (4.60), (4.27) and (4.36), one gets

$$\frac{dr_p}{dt} = \frac{\frac{2}{(\gamma + 1)} \left(\frac{r_s}{r_p} \right) \frac{dr_s}{dt} - \left(1 - \frac{r_s^2}{r_p^2} \right) \left[\frac{r_p}{\gamma I} \frac{dI}{dt} + \frac{1}{\gamma + 1} \frac{r_p}{z_f} \frac{dz_f}{dt} - \frac{4\pi(\gamma - 1)}{\mu \gamma I_p^2} \frac{r_p}{z_f} \frac{dQ}{dt} \right]}{1 - \frac{1}{\gamma} \left(1 - \frac{r_s^2}{r_p^2} \right)} \quad (4.61)$$

where, $dQ = d(Q_{joule} - Q_{rad} - E_i - E_s)$ is the energy increased other than piston work.

Similarly, by taking into account the transmission time of piston to shock front, the thermodynamic equation becomes:

$$\frac{dr_p}{dt} = \frac{\frac{2 \left(P_B r_s \frac{dr_s}{dt} \right)_{t-t_{p-s}}}{P_B r_p (\gamma + 1)} - \left(1 - \frac{r_s^2}{r_p^2} \right) \left[\frac{r_p}{\gamma I} \frac{dI}{dt} + \frac{1}{\gamma + 1} \frac{r_p}{z_f} \frac{dz_f}{dt} - \frac{4\pi(\gamma - 1)}{\mu \gamma I_p^2} \frac{r_p}{z_f} \frac{dQ}{dt} \right]}{1 - \frac{1}{\gamma} \left(1 - \frac{r_s^2}{r_p^2} \right)} \quad (4.62)$$

This equation is suitable for the plasma gas in the slug in the radial inward shock and reflected shock phase. It can also be applied to describe the dense plasma column in the reflected shock and slow compression phases.

4.6.2.3 EXPRESSIONS OF PLASMA PARAMETERS

Based on the discussion in previous sections, the expressions of the plasma parameters can be summarized as following:

$$\text{Ions in plasma slug:} \quad N_{ion2} = \pi \int_{m2}^t n_0 \left((r_p^2 - r_s^2) \frac{dz_f}{dt} - 2z_f r_s \frac{dr_s}{dt} \right) dt \quad (4.63)$$

$$\text{Volume of plasma slug:} \quad V_{slug} = \pi (r_p^2 - r_s^2) z_f \quad (4.64)$$

$$\text{Energy into plasma:} \quad EINP_2 = \int_{t20}^t I_p^2 \left(R_{slug} - \frac{\mu}{4\pi} \frac{z_f}{r_p} \frac{dr_p}{dt} \right) dt \quad (4.65)$$

$$\text{Ion density:} \quad n_i = N_{ion} / V_{slug} \quad (4.66)$$

$$\text{Electron density:} \quad n_{electron} = Z_{eff} \times n_i \quad (4.67)$$

$$\text{Tube voltage:} \quad V = V_0 - \frac{CH}{C_0} - R_0 I - L_0 \frac{dI}{dt} \quad (4.68)$$

$$\text{Internal energy temperature:} \quad T_2 = \frac{2}{5N_{ion}(1 + Z_{eff})k} \times U \quad (4.69)$$

$$\text{Shock Speed Temperature:} \quad T_1 = \frac{2m_i}{k_B(1 + Z_{eff})} \frac{\gamma - 1}{(\gamma + 1)^2} v_s^2 \quad (4.70)$$

4.6.3 REFLECTED SHOCK EQUATIONS

In the reflected shock phase, the reflected shock front (the boundary of the collision) travels outwards with a fraction of the on-axis incident shock speed [171]. Here, this fraction is set as $f_{RS} \sim 0.3$ empirically and the reflected shock speed is written as:

$$\frac{dr_r}{dt} = -f_{RS} \left(\frac{dr_s}{dt} \right)_{on-axis} \quad (4.71)$$

The corresponding axial elongation equation becomes:

$$\frac{dz_f}{dt} = -\frac{2}{\gamma + 1} \left(\frac{dr_s}{dt} \right)_{on-axis} \quad (4.72)$$

At the same time, the piston is allowed to move towards the axis by assuming a "virtual" forward shock front moving with the on-axis speed of the incident shock. The thermodynamic equation (equation (4.62)) is still suitable to describe the plasma gas in the plasma slug (the gases between piston and reflect front), except that r_s is replaced by r_r in the equation.

If it is assumed that there is no mass leakage at piston and the axial shock front is collecting gas, then the number of ions in the plasma slug is:

$$N_{ion3a} = \frac{(r_p^2 - r_r^2)}{r_{p2E}^2} N_{ion2E} + \int_{t30}^t f_{m2} n_0 \times \pi (r_p^2 - r_r^2) \left(\frac{dz_f}{dt} \right) dt \quad (4.73)$$

where, N_{ion2E} is the ion number in the plasma slug at the end of radial inward shock phase and r_{p2E} is the piston radius at that time. Then, the number of ions in the center region is

$$N_{ion3b} = \left(1 - \frac{r_p^2 - r_r^2}{r_{p2E}^2} \right) N_{ion2E} + \int_{t30}^t f_{m2} n_0 \times \pi r_r^2 \left(\frac{dz_f}{dt} \right) dt \quad (4.74)$$

The other plasma parameters can be calculated by using quations (4.64) – (4.70).

At the end of reflected phase, $r_p = r_r$, all particles are compressed into the center of the plasma column. The number of ions in the column is:

$$N_{ion4} = N_{ion3E} + \int_{t40}^t f_{m2} n_0 \times \pi r_p^2 \left(\frac{dz_f}{dt} \right) dt \quad (4.75)$$

4.6.4 SLOW COMPRESSION EQUATIONS

Assuming that the diffusion of the magnetic field into the plasma is not significant in this radiative phase (~ 20 ns) then, the plasma is still confined by a thin magnetic piston. The plasma gas confined in this piston can still be considered as a thermodynamic gas. Using equation (4.62) for this phase by setting $r_s = 0$, $dr_s/dt = 0$, one gets:

$$\frac{dr_p}{dt} = \frac{-\frac{r_p}{\gamma I} \frac{dI}{dt} - \frac{1}{\gamma + 1} \frac{r_p}{z_f} \frac{dz_f}{dt} + \frac{4\pi(\gamma - 1) r_p}{\mu \gamma I_p^2} \frac{dQ}{z_f dt}}{\frac{\gamma - 1}{\gamma}} \quad (4.76)$$

Equation (4.72) can also be used to describe the axial elongation in this phase. The ion number in the plasma column is expressed by equation (4.75).

The radial phase circuit equation (equation (4.10)) is valid for the whole radial phase. Combined with all the radial phase equations and expressions, the plasma parameters can be calculated.

4.6.5 DEPENDENCE OF THE PLASMA DYNAMICS ON GAS PROPERTIES

In the radial phase, the dynamics of the plasma is mainly determined by equations (4.56) and (4.62). For different working gases, the plasma dynamics is different because of their different mass density (at same pressure). Another important factor which affects the dynamics is the energy sink in the plasma. This effect

includes mainly the ionization and emission processes. This mechanism is temperature dependent and implies that the dynamics is different for different gases even if the mass density are identical. In the shock front dynamics equation (equation (4.56)), the gas dependent parameter is the specific heat ratio γ .

Piston-shock front transmission time t_{p-s} (equation (4.24)) is another effect, which is inversely proportional to the square root of temperature. When the plasma slug approaches the axis, the piston pressure increases rapidly. Thus, the pressure at the shock front is always smaller than at the piston (by equation (4.26)). Therefore, the shock front moves more slowly for the low temperature plasma.

In equation (4.62), the piston speed, dr_p/dt , is directly related to the shock front speed, dr_s/dt . The two factors discussed above have same effects on the piston speed. However, in this equation, the energy loss, dQ , by emission and ionization accelerates the piston (the inward dr_p/dt has a negative sign). The reason for this is that the colder thermodynamic gas between the piston and shock front is easier to be compressed. This mechanism begins to affect the dynamics when the shock front is close to the axis ($r_p \gg r_s$) and dominates the plasma column dynamics in the slow compression phase.

In conclusion, the ionization and radiation processes slow down the compression speed during the radial inward shock phase. This factor speeds up the compression in the final slow compression phase.

4.7 SIMULATION USING *NIE-SSC-PFF* PARAMETERS

4.7.1 NUMERICAL CALCULATION METHOD

Based on the equations presented in the previous sections, the behavior of the plasma focus is numerically computed.

A simple linear approximation integration method [172, 43] is employed to calculate all variables by small time steps. The parameter values at a series time points $t1, t2, t3 \dots$ with an increment Δt are calculated according to the equations.

For the variables in the differential equations (i.e. I, z, z_f, r_s, r_p, r_r), their values at time t are related to the previous corresponding values at time $t-\Delta t$ as (for example, for the current, I):

$$I(t) = I(t - \Delta t) + \frac{dI}{dt}(t - \Delta t) \times \Delta t \quad (4.77)$$

where Δt is the time increment for calculation. The value $dI/dt (t-\Delta t)$ is calculated by the circuit equation. In this way, one can get the values of the discharge current and the plasma dimension parameters.

The other plasma parameters (such as $N_{ion}, T_{eV}, EINP$, etc.) are, therefore, calculated by these solved values accordingly.

In the calculation, the time increments, Δt , for each phase are set as:

$$\Delta t = \begin{cases} 10^{-9} & \text{sec} & (\text{Axial Phase}) \\ 2 \times 10^{-10} & \text{sec} & (\text{Radial Phase}) \end{cases} \quad (4.77)$$

For the axial phase, the initial conditions are:

$$\begin{cases} t = 0; \\ z = 0; & \frac{dz}{dt} = 0; & \frac{d^2 z}{dt^2} = \frac{f_c V_0}{2\pi L_0} \sqrt{\frac{2\mu \ln(b/a)}{3f_{m1} n_0 m_i (b^2 - a^2)}} \\ CH = 0; & I = 0; & dI/dt = V_0 / L_0 \end{cases} \quad (4.78)$$

where $CH = \int Idt$ is electrical charge flowing into tube.

The axial phase ends at $z=z_0$, i.e. the plasma sheath reaches the anode top.

Then, the radial phase starts at

$$r_s = r_p = a \text{ and } z_f = 0 \quad (4.79)$$

The initial values of the other parameters such as I , dI/dt etc. are taken as the final values at the end of the axial phase. The reflected shock begins when $r_s=0$ and inherits all the parameters from the radial phase. Similarly, the slow compression phase starts when $r_r=r_p$ and inherits the parameters from the reflected shock phase.

4.7.2 RESULTS FOR *NIE-SSC-PFF*

The model is applied to simulate the performance of the *NIE-SSC-PFF* plasma focus machine.

The simulation parameters for our machine are set to be:

$$f_{m1} = 0.095; \quad f_{m2} = 0.2; \quad f_c = 0.78; \quad f_{RS} = 0.3; \quad (4.80)$$

where, f_{m1} and f_{m2} are the mass factors, for the axial and the radial phases, respectively, f_c is the current shedding factor, and f_{RS} is reflected shock factor.

In modeling the slow compression phase, a radiation power loss of 10^9 W is assumed based on the experimental measurements [179]. The calculation is carried on until the plasma column is 1 mm in diameter. At this instant, the plasma column final length is approximately 20 mm.

The model presented in this chapter gives numerical values of the same order of magnitude as the measured values, thus, giving a basic understanding of the plasma focus.

Chapter 5

EXPERIMENTAL SET-UP

In order to study the emission of radiation from the plasma focus, various experiments were performed using the 3 kJ/15kV *NIE-SSC-PFF*. This device is a replica of the *UNU/ICTP PFF* machine [181].

Different diagnostics were employed for this purpose. This chapter deals with the experimental set-up of the diagnostics employed.

5.1 DETAILED DESCRIPTION OF *NIE-SSC-PFF*

Figure 5.1 shows a block diagram of the experimental set-up.

Basically, the *NIE-SSC-PFF* consists of the following subsystems:

- Driver (high voltage charger, capacitor and high voltage cables);
- Focus tube (discharge chamber, input flanges and electrode system);
- Switch (sparkgap);
- Vacuum system (vacuum pump, manometers, valves, working gas container, etc.);
- Triggering electronics (high voltage transformer);
- Control electronics (master pulse generator, high voltage SCR unit);

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- Diagnostics (Rogowski coil, high voltage probe, neutron detector, scintillator-photomultiplier system, X-ray spectrometer, and a laser shadowgraphic system);
- Data acquisition system (Digital Storage Oscilloscope – DSO, computer and software packages).

5.1.1 SWITCHING SYSTEM

A simple parallel plate sparkgap in a swinging cascade configuration [45] was used, providing a

low inductance switch at a high current. The ratio of the gap is 3:2 (4.5:3 mm). The gap is triggered through an isolating capacitor from an 800 V SCR unit (of 50 ns rise time) via a TV transformer with a step up ratio of

17 times. The jitter was found to be less than 50 ns. The circuit diagram is shown in Figure 5.2.

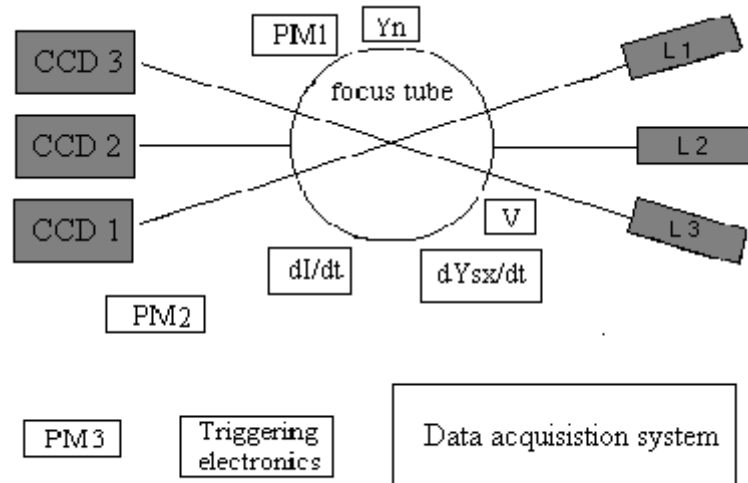


Figure 5.1: Block diagram of *NIE-SSC-PFF* and the diagnostics employed. L1, L2 and L3 are three lasers used for the shadowgraphy. PM1, PM2 and PM3 are the scintillator-photomultiplier systems for time-resolved neutron and hard X-ray detection. Yn is the total neutron yield detector. dY_{sx}/dt is the time-resolved soft X-ray detection system.

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The arrangement of the capacitor, connecting plates, spark gap and the output coaxial cables is shown in Figure 5.3(A). To keep inductance low, the

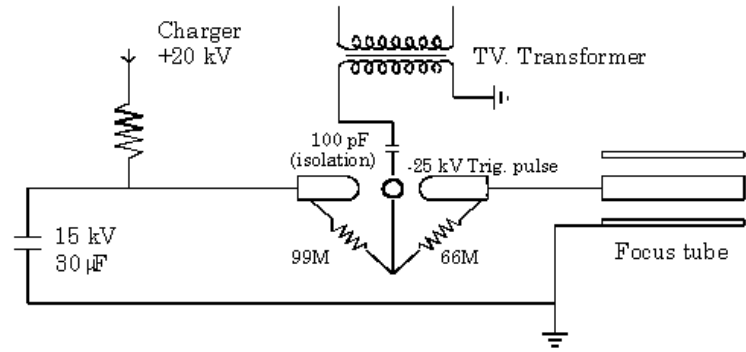


Figure 5.2: Triggering system.

earth plate of the capacitor (5) is extended up to the anode. The insulation is provided by a nylon cap (4) around the anode stud. The cap is dipped into a pool of oil (7), which is prevented from splashing out by means of an O-ring (8). Mylar sheets (6) of 2 inches wider all around than the conducting plates sandwiched by the polyethylene sheets (10) complete the insulation between the high voltage plate (12) and the ground plate (5) as shown in figure 5.3(A). The earth plate runs unbroken to the output position where the earth of the coaxial cables are connected. On the top of the insulating sheet, the high voltage (HV) plate is connected to the spark gap. Between the spark gap electrodes, a copper tubing (15) of 0.5 inches diameter is used as the trigger electrode. The output plate of the spark gap is connected to the focus tube by means of 16 coaxial cables (16) connected in parallel. The coaxial cables are connected to the anode collector plate (18) and the cathode collector plate (22).

An essential feature of the focus tube is the insulator. This glass insulator (26) between the cathode and the anode plays an important role in the symmetrical formation of the current sheath. In the present design, this glass insulator is mounted in a rubber holder compressed tightly and symmetrically by a brass flange (25). The

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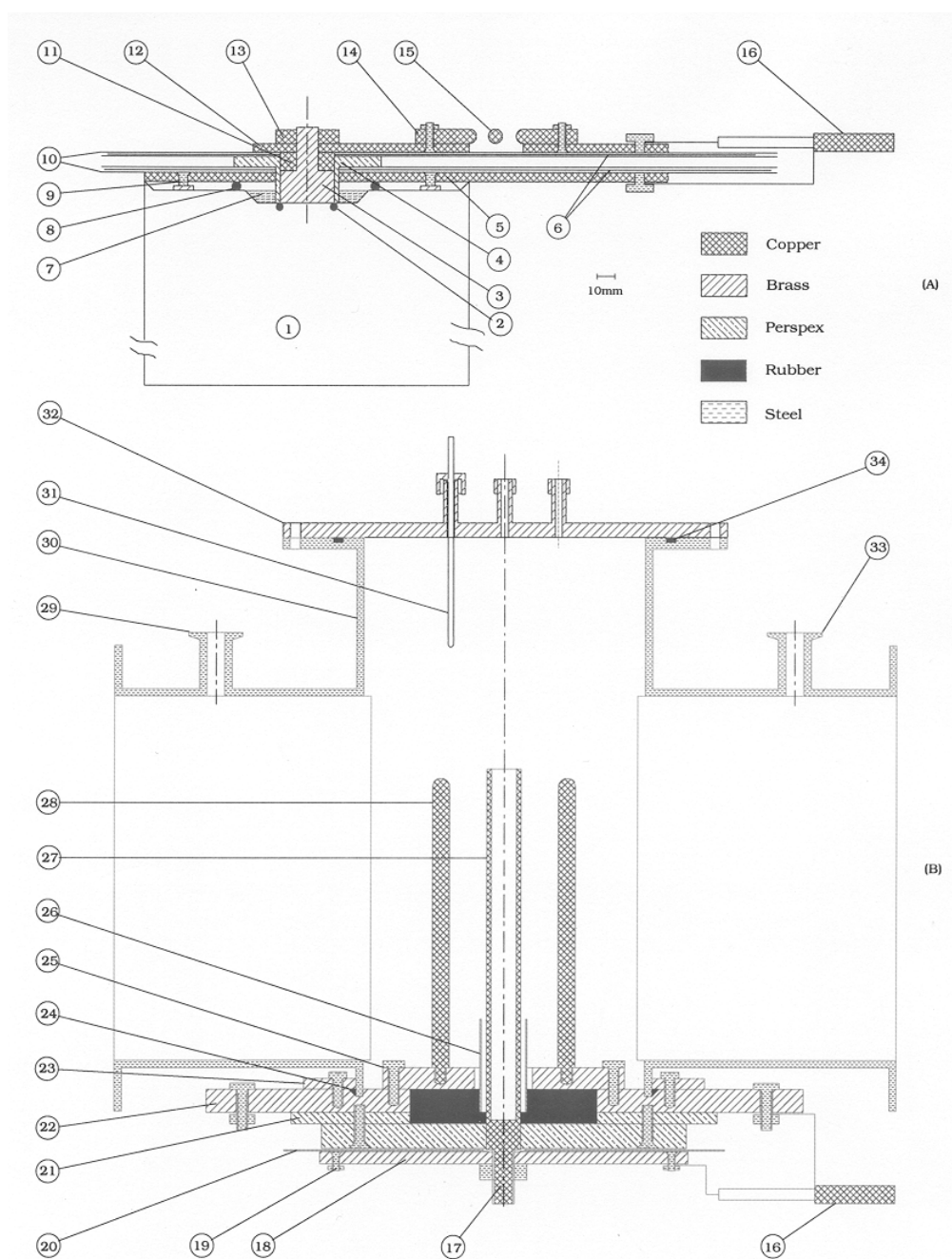


Figure 5.3: Diagram of NIE-SSC-PFF: (A) The capacitor connecting plates, the spark gap, and output coaxial cables. 1=capacitor; 2=capacitor O-ring seal; 3=nut; 4=nylon cap; 5=earth plate; 6=Mylar film; 7=oil; 8=O-ring seal; 9=earth stud; 10=polyethelene film; 11=copper ring; 12=high-voltage output plates; 13=lock nut; 14=HV electrode for swinging cascade spark gap; 15=trigger electrode; 16=output cables; (B) The plasma focus tube. 17=stud of anode; 18=anode collector plate; 19=connecting points for coaxial cables; 20=5-mil Mylar film; 21=perspex spacer; 22=cathode collector plate; 23=flange; 24=O-ring seal; 25=cathode support plates; 26=glass insulator; 27=focus anode; 28=focus cathode(6 rods); 29=outlet to diaphragm gauge and vacuum pump; 30=chamber; 31=glass jacket for magnetic probe; 32=top flange; 33=inlet for test gas; 34=O-ring.

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rubber holder also acts as a vacuum and high voltage seal. The whole arrangement is shown in Figure 5.3(B).

The discharge chamber (30) consists of a 30 cm long and of 6.5 inches diameter chromed mild steel tubing. Vacuum is provided by a single-stage rotary pump reaching an ultimate base pressure of $\sim 10^{-2}$ mbar. This level of vacuum has proven to be sufficient for obtaining a good focus and a good neutron yield when operated in deuterium.

5.2 PARAMETERS OF *NIE-SSC-PFF*

The design parameters of NIE-SSC-PFF are summarized in Table 5.1. The values of the electrical parameters of the device are listed in Table 5.2.

Table 5.1: Design parameters of *NIE-SSC-PFF*.

Parameters	Values
Capacitance	$C_0 = 30 \mu\text{F}$
Inductance	$L_0 = 110 \text{ nH}$
Impedance	$Z_0 = \sqrt{L_0 / C_0} = 60.5 \text{ m}\Omega$
Charging voltage	$V_0 = 14.5 \text{ kV}$
Anode radius	$a = 0.95 \text{ cm}$
Cathode radius	$b = 3.2 \text{ cm}$
Anode length	$z_0 = 16 \text{ cm}$

Table 5.2: Experimentally measured parameters of *NIE-SSC-PFF*.

Measured Parameters	Measured Values
Resistance	$R_0 = 11.9 \text{ m}\Omega$
Peak discharge current in short-circuit	$I_{\max} = 200 \text{ kA}$
Typical rise time	$t_{a \text{ exp}} = 2.9 \text{ }\mu\text{s}$
Typical peak axial velocity	$V_{\text{ax}} \sim 6\text{-}10 \text{ cm}/\mu\text{s}$
Typical peak radial velocity	$V_{\text{rad}} \sim 15\text{-}35 \text{ cm}/\mu\text{s}$

5.3 TOTAL NEUTRON YIELD MEASUREMENTS

The measurements of the total neutron production were performed with an indium foil activation detector. The schematic diagram of this detector is shown in Figure 5.4. The detector was placed at a distance of 20 cm from the pinch in the axial direction on the top of the focus chamber. The whole arrangement of the set-up is shown in Figure 5.5.

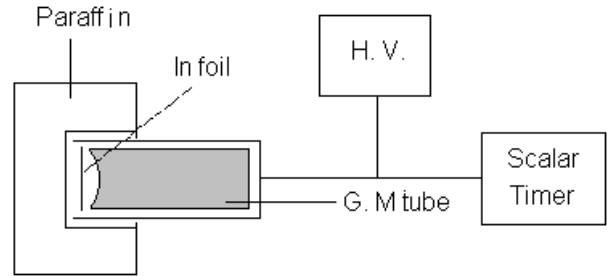


Figure 5.4: A schematic of Indium foil activation detector.

The neutron activation cross-section for In is higher for slow neutrons. Therefore, the neutrons of energy above 2.45 MeV emitted from the plasma are firstly thermalised by a 5 cm thick layer of paraffin.

^{116}In has a half-life of 14 seconds and emits β particles with a maximum energy of 3.3 MeV. A thin window Geiger-Muller (GM) tube is used to detect the

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β radiation. The voltage applied on the GM tube is 1400 V. The tube is connected to a counter and a timer. The discrimination level is adjusted to get 70 to 80 background counts in 60 s. The counter is started 1 ms after the focus discharge, so that

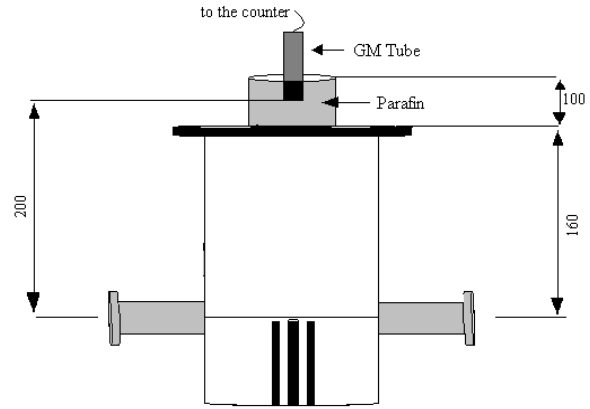


Figure 5.5: A schematic of the set-up for the total neutron yield measurement.

interference from any electrical noise or hard X-ray from the focus is avoided.

The average value of the total yield for our 3 kJ device at 4 mbar/14.5 kV is 1.0×10^8 neutrons/discharge [41]. Based on this, a calibration factor of 1.5×10^5 neutrons/count was determined.

5.4 DEUTERON MEASUREMENTS

Our experimental set-up for time-integrated deuteron energy spectra measurements consists of two main systems: one for detection and one for analysis.

The essential element of the detection system is the magnetic spectrometer.

The analysis system, representing one of the elements of novelty of this project, is the automated track counting system, also referred to as the automated image analysis system. A detailed description of the whole experimental set-up is presented in the following sections.

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5.4.1 MAGNETIC SPECTROMETER

The magnetic spectrometer implemented on our plasma focus device is illustrated schematically in Figure 5.6.

A small vacuum chamber, referred to as the upper chamber, was placed above

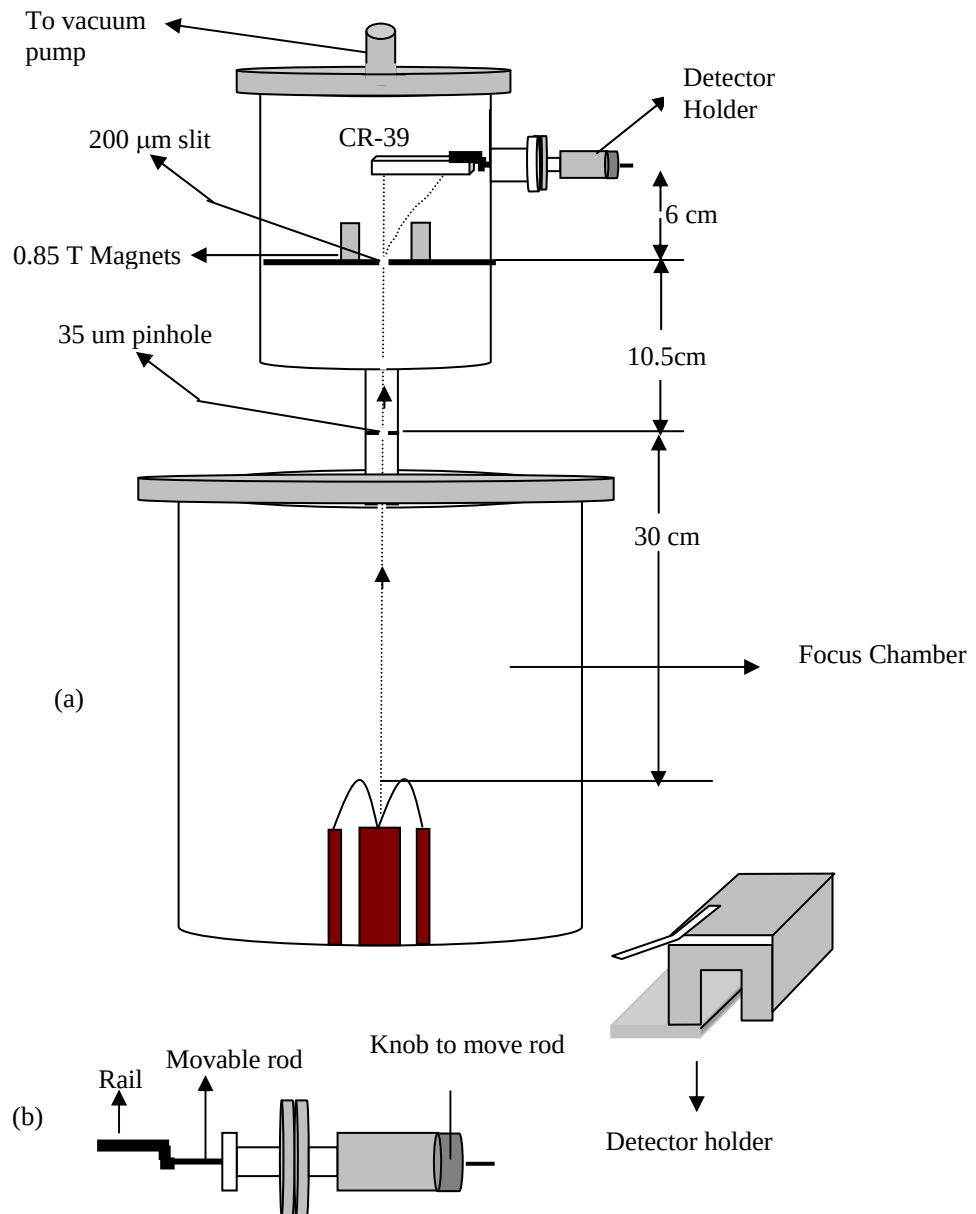


Figure 5.6: (a) The magnetic spectrometer fixed above the focus chamber; (b) the detector holder and the translator.

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the focus chamber to host the spectrometer. The upper chamber was connected to the plasma focus chamber by a tube containing a 35 microns pinhole. The pinhole was aligned with the deuteron beam from the pinch region. A rectangular slit of 200 microns in width was placed in the upper chamber 10.5 cm above the plane of the pinhole. This slit was also aligned along the direction of the deuteron beam.

A pair of permanent magnets providing a 0.85 T magnetic field perpendicular to the direction of the deuteron beam was placed directly above the plane of the slit. A detector holder, fitted with a translator, was placed 3 cm above the magnets. This provided a distance of 4.75 cm between the rectangular slit and the surface of the CR-39 detector. The actual movement of the translator was 2.5 cm. With the help of a 7 cm long aluminum rail (over which the detector holder was able to slide) attached to the end of the movable rod of the translator, it was possible to extend the range of motion of the detector from 2.5 to 4.5 cm.

A schematic of the whole arrangement is shown in Figure 5.6.

In order to minimize the ion energy loss by collision with the neutral gas molecules, differential vacuum (10^{-3} mbar) was maintained in the upper chamber.

5.4.2 AUTOMATED ANALYSIS SYSTEM

The power and elegance of the present computer imaging systems are rudimentary at best. Yet, the computer imaging system employed in our work is able to extract better quantitative information as compared to our eyes. Our computer imaging system has also automated and economized many tedious tasks that would otherwise not be performed by visual observation, or would be fraught with error.

EXPERIMENTAL SET-UP

Our automated track counting system schematically shown in Figure 5.7, comprises a Leica *DMR* microscope and a motorized stage controlled by a software known as OPTIMAS. The microscope is fitted with a *CCD* camera, which grabs the image from the microscope and displays it on the computer screen. The *CCD* camera (*XC-75 CE*) uses a 1/2 inch high-resolution imager.

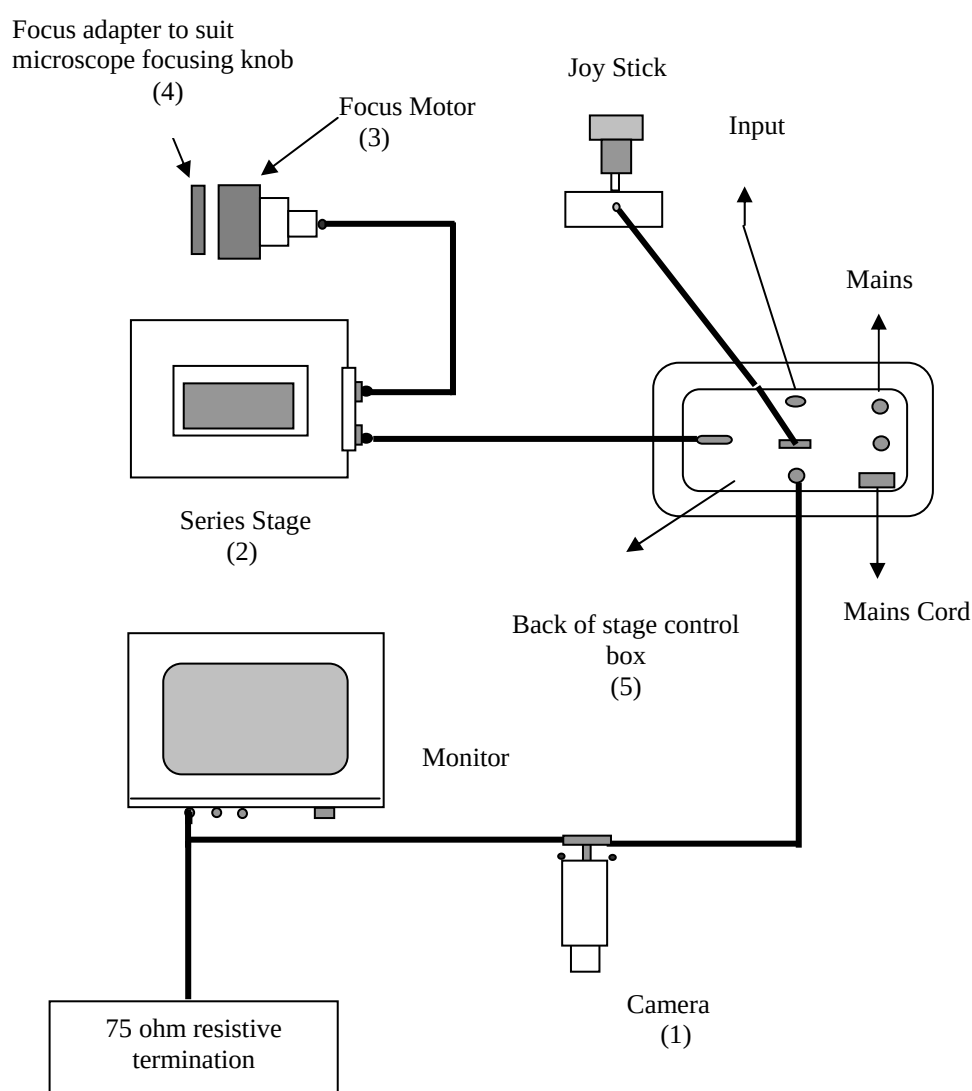


Figure 5.7: Block diagram of the automated image analysis system.

EXPERIMENTAL SET-UP

The *Leica DMR* microscope has an RS 232-C interface for the connection to a computer.

OPTIMAS provides full facility to identify and acquire, the region of interest (ROI), as well as to measure different features. *OPTIMAS* supports multiple documents interface (MDI). *OPTIMAS* macros consist of a high level programming language called *ALI* (the AnalYTical Language for Images). The auto-stage used in our system, is a software product from *Data Cell*, which facilitates computer control of the *PRIOR* motorized microscope stage (2).

With the combination of the *OPTIMAS*, auto-stage and our developed program (macros written in ALI), the automatic scanning begins at the first programmed location; the first image will be acquired, programmed, processed and the results will be saved to a file. The stage platform will then traverse to the next programmed location and re-performs the functions as at the first location. This operation will continue until the final programmed location is reached. In case of raster processing, if more than one slide is enabled, all slides will be processed in the same way. Auto focusing is done by a focus motor (3), which in-turn is attached to a focus adapter ring (4) that fits well to the microscope focus knob (see Figure 5.7). The auto-stage implements the full commands set supported by the *PRIOR* motor controller (5) in the form of macros. This stage provides a unique facility to develop such an image analysis system that can be operated unattended for very long periods of time.

The program was developed in order to scan the detector for 44×14 fields of view. Each field of view was $241 \times 181 \mu\text{m}$. Our software automatically starts scanning the image from the centroid of the neutral band. The program selects the

EXPERIMENTAL SET-UP

field, focuses of the image, applies a filter (*Sobel filter*), which enhances the edges of the tracks, binaries the tracks, and then counts them.

Another interesting feature of this program is that it provides a histogram of the number of tracks for each position. Each position corresponds to a bin of 100 micron wide. The histograms obtained from this image scanning system were used to extract information on the deuteron energy.

Different energy deuterons have different angles of deflection while passing through a magnetic field. Therefore, different bands (corresponding to respective

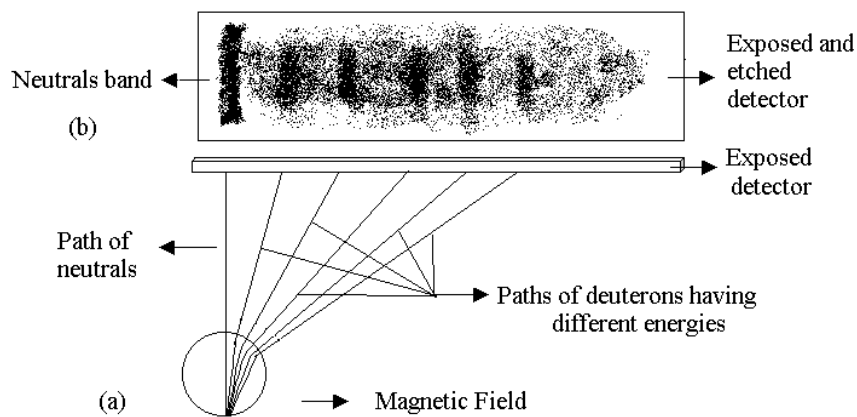


Figure 5.8: a) Paths of neutrals and different energy deuterons, forming tracks on CR-39; b) front view of CR-39 exposed at one position after etching. Neutral band, shown in figure, consists of particles which become neutral due to charge exchange with the surrounding gas and are not deflected as they pass through the magnetic field thereby forming a reference from which the displacement of the other charged particles can be measured.

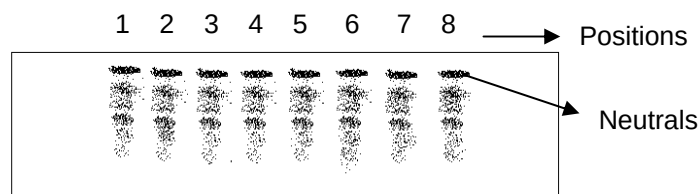


Figure 5.9: A schematic of front view of a detector exposed for 8 positions.

EXPERIMENTAL SET-UP

energies) are formed on the detector. The diagram in Figure 5.8 shows the detector as well as the mechanism for the formation of different energy tracks on the CR-39 detector. Figure 5.9 shows a detector after the etching and

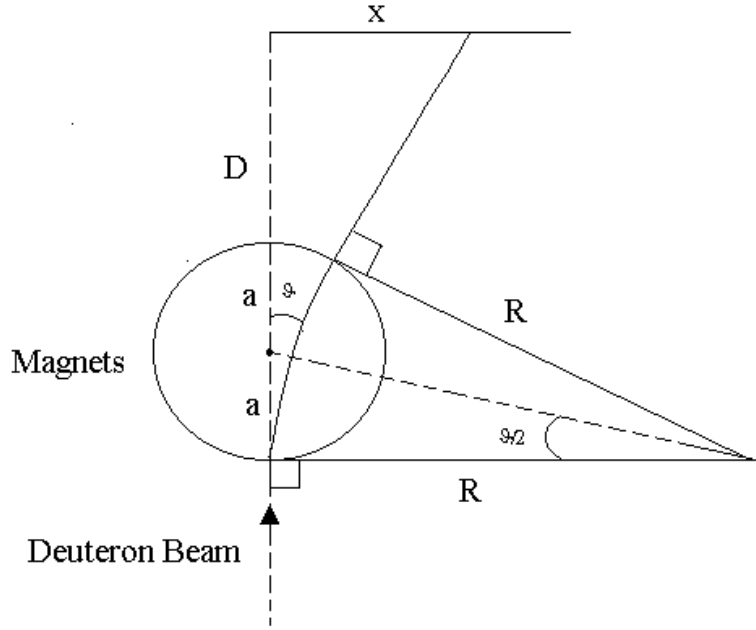


Figure 5.10: A geometrical diagram representing different parameters to calculate energy of the deuterons passing through a magnetic field.

washing processes exposed at 8 positions.

A formula, relating distances of the deflected deuterons from the neutral band with respect to their energies, was derived using a simple geometry.

Consider a deuteron beam passing through the center of a magnetic field of radius a . The vertical distance between the CR-39 detector and the center of the magnetic field is D . The radius of curvature of the deuteron beam in the magnetic field is R . The deflection angle of a particle from the center of the field is θ , as shown in Figure 5.10. From the geometry R is given by:

$$R = a \left[\frac{D}{X} + \sqrt{\frac{D^2}{X^2} + 1} \right] \quad (5.1)$$

EXPERIMENTAL SET-UP

The kinetic energy of a particle of mass, M , and charge, e , passing through a magnetic field of strength, B , is:

$$T = \frac{e^2 B^2 R^2}{2M} \quad (5.2)$$

With $B \cong 0.85\text{T}$, one obtains for deuterons:

$T = (17,305\text{keV m}^{-2}) R^2$, with R is expressed in meters.

Using the above formula, the deuteron energies for each group of the three overlapped shots were calculated. Three sets of spectra corresponding to 3.5, 4 and 4.5 mbar were obtained. Each set was sorted in five groups based on the total neutron yield in the following manner: for each set of data, 20% of spectra with the highest neutron yield were classified as percentile division I, 20% with the next highest yield as percentile division II, and so on.

5.5 TIME-RESOLVED HARD X-RAY AND NEUTRON MEASUREMENTS

Three detectors were used for the time-resolved neutron and hard X-ray flux measurements. Each detector consists of a plastic scintillator (NE 102A of 50 mm diameter and 40 mm thickness) and a photomultiplier (EMI 9813B) tube. The plastic scintillator was wrapped in an aluminum foil, with the exception of the side facing towards the window of the photomultiplier tube. For proper optical coupling, Dow Corning 200 silicon grease was used between the plastic scintillator and the tube. Each detector was enclosed in an aluminum casing. Before placing the detectors at

EXPERIMENTAL SET-UP

their respective positions, these detectors were “normalized” to give almost identical output when placed at the same distance from the focus.

The detector, referred to as PM1, was placed at a distance of 1.94 m in the axial direction (at 0° with respect to the pinch). The

other two (PM2 and PM3) were placed radially (at 90° with respect to the pinch) at the distances of 1.94 m and 3.82 m, respectively.

If it is assumed that our plasma focus discharge produces an average yield of 10^8 neutrons per shot isotropically [41], the estimated number of neutrons reaching the detectors is approximately 1300 for PM1 and PM2 and 550 for PM3, respectively. Therefore, for the output signals (to the scope) to be of the same amplitude for all the three detectors, neutral filters were placed between the plastic scintillator and the windows of the photomultipliers. To cut off most of the hard X-rays, lead sheets (3 mm thick) were placed in front of PM1 and PM2.

Figure 5.11 shows the assembly of the detector (a) and the schematic of the set-up for the time-resolved neutron and hard X-ray measurements (b).

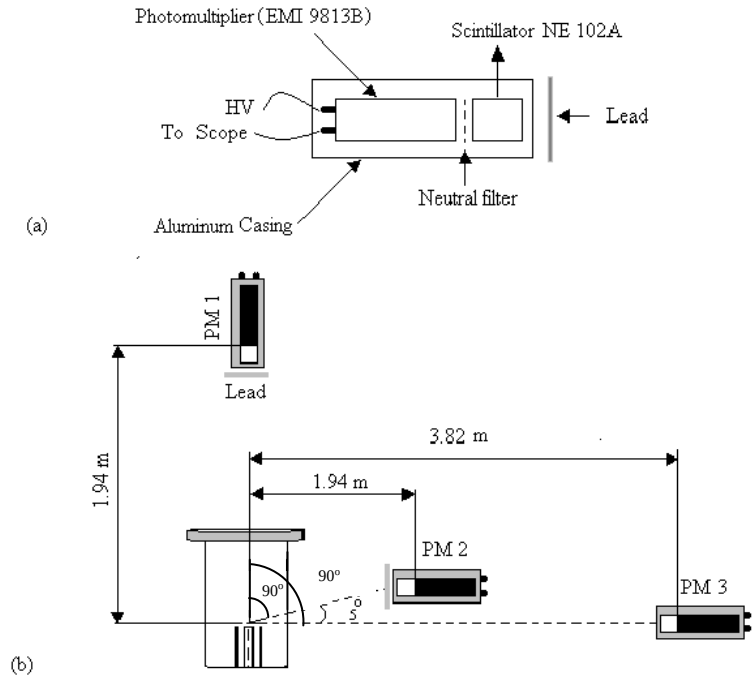


Figure 5.11: a) An assembly of the scintillator-photomultiplier system; b) Arrangement for the time-resolved neutron and hard X-rays measurements.

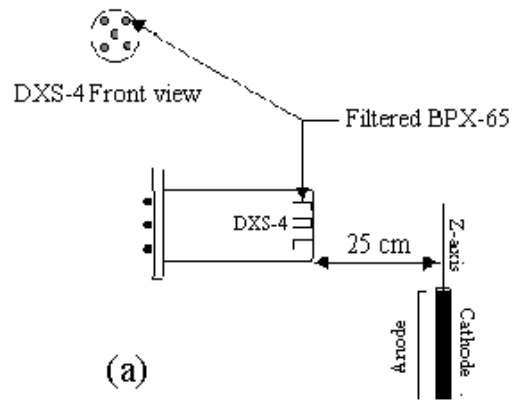
EXPERIMENTAL SET-UP

After normalization, measurements of the internal transit time (52 ns) of the photomultipliers were performed. Subsequently, appropriate delays were introduced for all the other diagnostics (i. e. dI/dt , V , dY_{sx}/dt , etc.). A 1 ns accuracy in synchronizing of all diagnostic signals was obtained.

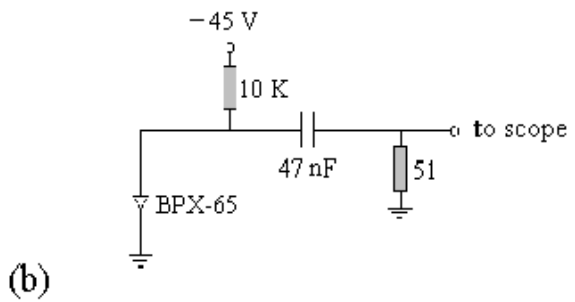
5.6 TIME-RESOLVED SOFT X-RAY MEASUREMENTS

A five-channel diode soft X-ray spectrometer (DXS-4) was employed for the time-resolved soft X-ray measurements. The technical parameters of the BPX-65 diode used in the spectrometer are presented in chapter 3.

The glass windows of the TO-18 casings of the diodes were removed, so that the diodes could be used for the X-ray detection. The spectrometer was fixed radially at a distance of 25 cm from the



(a)



(b)

Figure 5.12: a) The arrangement for the soft X-ray measurements; b) biasing circuit for the diode.

pinch zone. A schematic diagram of the arrangement for the time-resolved soft X-ray measurement is shown in Figure 5.12(a). The biasing circuit for a diode is shown in

EXPERIMENTAL SET-UP

Figure 5.12(b) [182]. The absorption foil combinations of the five channels are tabulated in Table 5.3.

Only channels 3 and 4 were used in our experiment. These channels were chosen specifically because, when the plasma focus is operated in deuterium, the X-rays are emitted predominantly by free-free transition (Bremsstrahlung). The plasma spectrum is most likely to be contaminated by the copper impurities from the electrode.

Table 5.3: The absorption foil combinations of the five channels in DSX4 spectrometer.

Channel	Filter Combination
X1 (channel 1) :	49 μm aluminized mylar
X2 (channel 2) :	24 μm aluminized mylar + 10.5 μm Ti
X3 (channel 3) :	24 μm aluminized mylar
X4 (channel 4) :	24 μm aluminized mylar + 10 μm Cu
X5 (channel 5) :	974 μm aluminized mylar

Channel 4 equipped with a 10 μm copper foil allows to check the extent of this contamination. The sensitivity of the BPX-65 diode (without filters) is shown in Figure 5.13.

The calculated [183] overall sensitivity of the filtered photodiode combination as a function of wavelength is shown in Figure 5.14. With the absolute transmission curves, the soft X-ray signals may be: (i) correlated with the discharge current, current

EXPERIMENTAL SET-UP

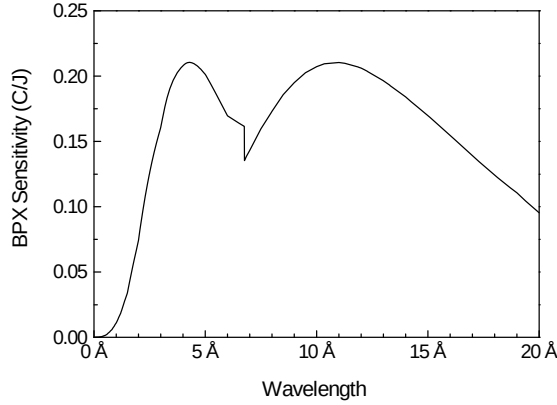


Figure 5.13: BPX-65 sensitivity in wavelength range $< 20 \text{ Å}$.

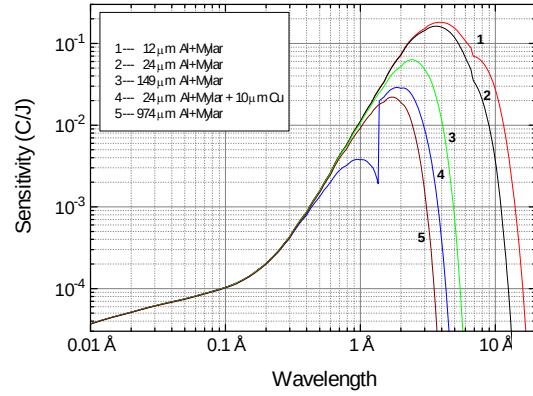


Figure 5.14: Sensitivity curve of BPX-65 with different filters.

derivative, voltage and other time-resolved signals for a better understanding of the plasma focus processes, and (ii) used to estimate total SXR yield.

5.7 OPTICAL MEASUREMENTS

Optical measurements (using shadowgraphy method) were carried out with two different set-ups. The preliminary measurements were performed using single-frame shadowgraphy. Due to shot-to-shot variation in the plasma focus dynamics, one needs to capture multiple frames to get a clearer picture of the focus dynamics. For this purpose, a three-frame computer-controlled laser shadowgraphy system was successfully designed, developed and incorporated in our experiment.

It is worth mentioning the conceptualization and the implementation of the three-frame computer controlled shadowgraphy system, which represents another significant effort put in the developmental work of this project. The discharge chamber was redesigned and modified; the lasers and optical benches were constructed; the triggering and synchronization circuits together with the computer

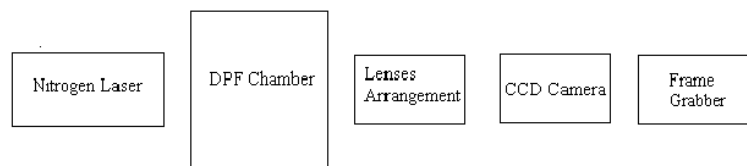
EXPERIMENTAL SET-UP

controlled image acquisition and storage systems were designed, built and accurately assembled. The two experimental set-ups are discussed in the following sections.

5.7.1 SINGLE FRAME LASER SHADOWGRAPHY

The experimental set-up for the single frame laser shadowgraphy (see block diagram in Figure 5.15),

consists of the following elements:



- Laser

An LN 203C N₂-dye (LD2S) laser with the

Figure 5.15: Block diagram of the laser shadowgraphy system.

spectral output ranging from 357 nm to 710 nm was operated as a green light source of 500 nm wavelength. In order to explore the region of interest, the natural divergence of the laser beam was used.

- Lenses arrangement.
- Filters

To cut off most of the unwanted plasma light, interference filters of 33 nm bandwidth and 68% transmission peak at 505 nm were employed in the experiment.

- CCD-based image capture system (CCD and Grabber).

To record the shadowgraphs, a PULNIX TM 6 EX CCD camera was used in the experiment. The high resolution (754×574 pixels, pixel size $8.3 \mu\text{m} \times 8.39 \mu\text{m}$) silicon based 0.8 cm format CCD camera has a peak response at 500 nm and a minimum sensitivity of 0.5 lux.

EXPERIMENTAL SET-UP

A beam of about 50 mm diameter was obtained at the region above the anode. A lens L_o of 600 mm focal length and 95 mm diameter aperture was placed at the

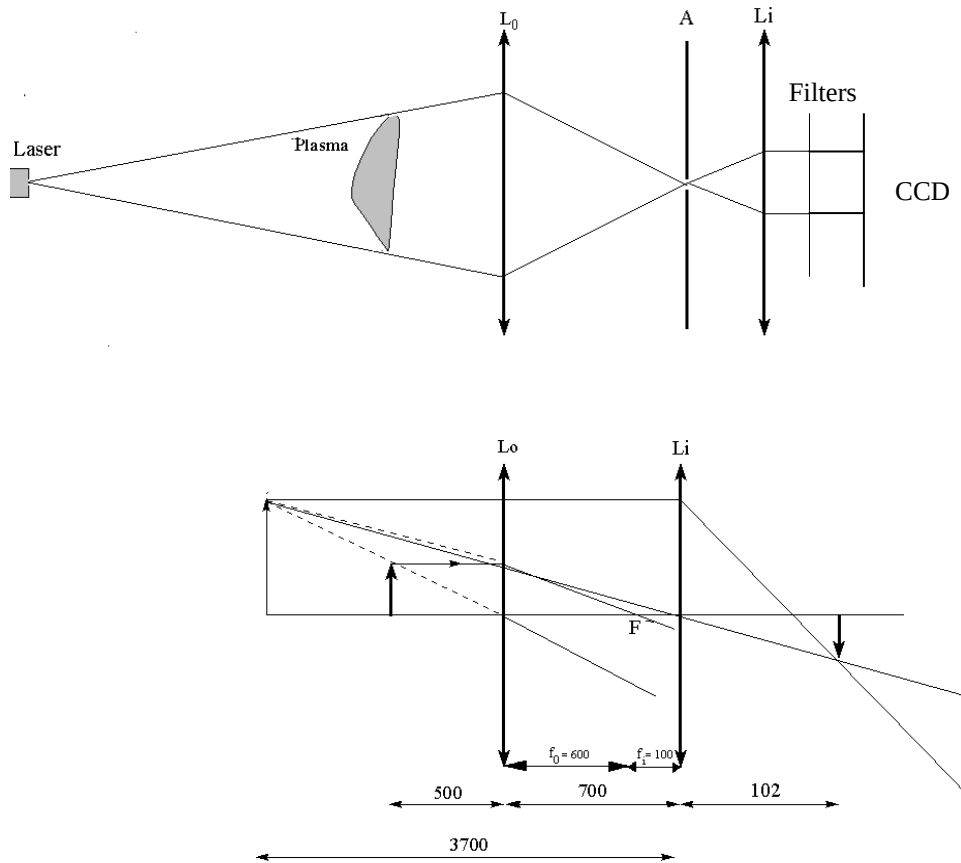


Figure 5.16: a) Optical arrangement for the laser shadowgraphy; b) the corresponding ray diagram of the image formation. All distances are expressed in mm.

distance of 600 mm from the region of interest. Another lens L_i of 100 mm focal length and 50 mm diameter was placed at a distance of 700 mm from L_o . The distance between these two lenses was adjusted to get the magnification of the image suitable for the field of view of the CCD camera.

EXPERIMENTAL SET-UP

In order to cut off most of the plasma light, an aperture stop of approximately 1 mm diameter was placed in the focal plane of the lens L_o . The CCD camera and the frame grabber attached to the computer were used to record the shadowgraphic images. The elaborate arrangement of the system and the timing diagrams are shown in Figures 5.16 and 5.17, respectively.

For the adjustment of the timing between the CCD camera signal, the plasma focus and the frame grabber, a special synchronization circuit [184] was employed.

The signal from the master trigger was carried to the synchronization circuit. This circuit divides the trigger signal into three parts. The first part goes

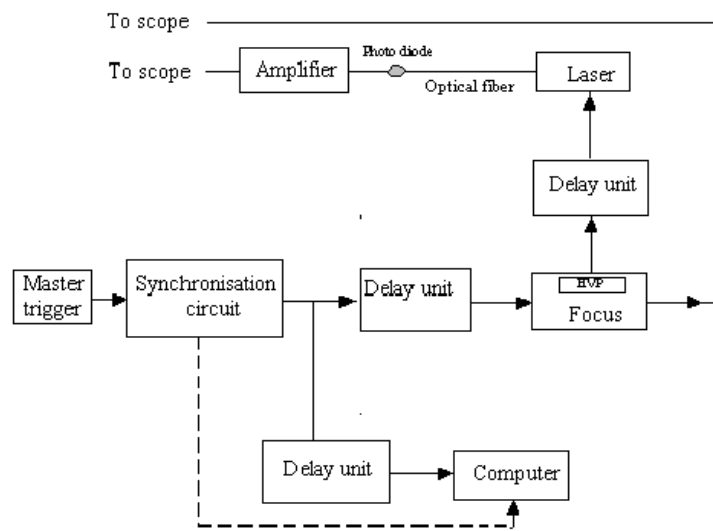


Figure 5.17: Block diagram of the electronic time sequence for the shadowgraphy system.

to the frame grabber and opens it for 20 ms. Within these 20 ms, it is necessary to trigger the focus and the laser. The second part of the signal triggers the focus after a 3 ms delay from the delay unit. The third part of the signal is used to activate the software to capture the image. It is also necessary to synchronize the laser flash time and the DPF discharge.

EXPERIMENTAL SET-UP

The high voltage probe signal was used to trigger the laser. The laser flash time instant, measured with the help of a photodiode without the delay unit, was found to be $1.5\ \mu\text{s}$ with a jitter of 50 ns averaged over 70 shots. Therefore, in order to capture the images, the required delay was introduced in the laser line.

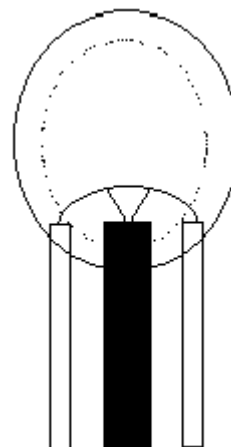


Figure 5.18: A schematic of the field of view of the laser shadowgraphy.

Figure 5.18 shows a schematic of the field of view of the laser shadowgraphy. The solid circle represents the field of view for the shadowgraphs taken for the pre-focus phase, and the dotted circle represents the field of view for the shadowgraphs taken during the focus and post-focus phases.

5.7.2 THREE-FRAME SHADOWGRAPHY

This set-up consists of four parts:

- a) Construction of Nitrogen lasers;
- b) Synchronization and triggering system;
- c) Optical arrangements;
- d) Computer-controlled three-frame capture system.

5.7.2.1 CONSTRUCTION OF THE NITROGEN LASERS

Two TEA (Transversely Excited Atmospheric) nitrogen dye lasers were constructed and another one [41] was modified.

EXPERIMENTAL SET-UP

The design of the laser is of parallel transmission line type [185]. The only difference from the previous design [41] is that a low-pressure switching device called Pseudo SparkGap (PSG) with a cold cathode was used as a switch. The PSG works with the hydrogen gas filling under the pressure of 0.2 to 0.6 torr. The basic parameters of PSG are listed below:

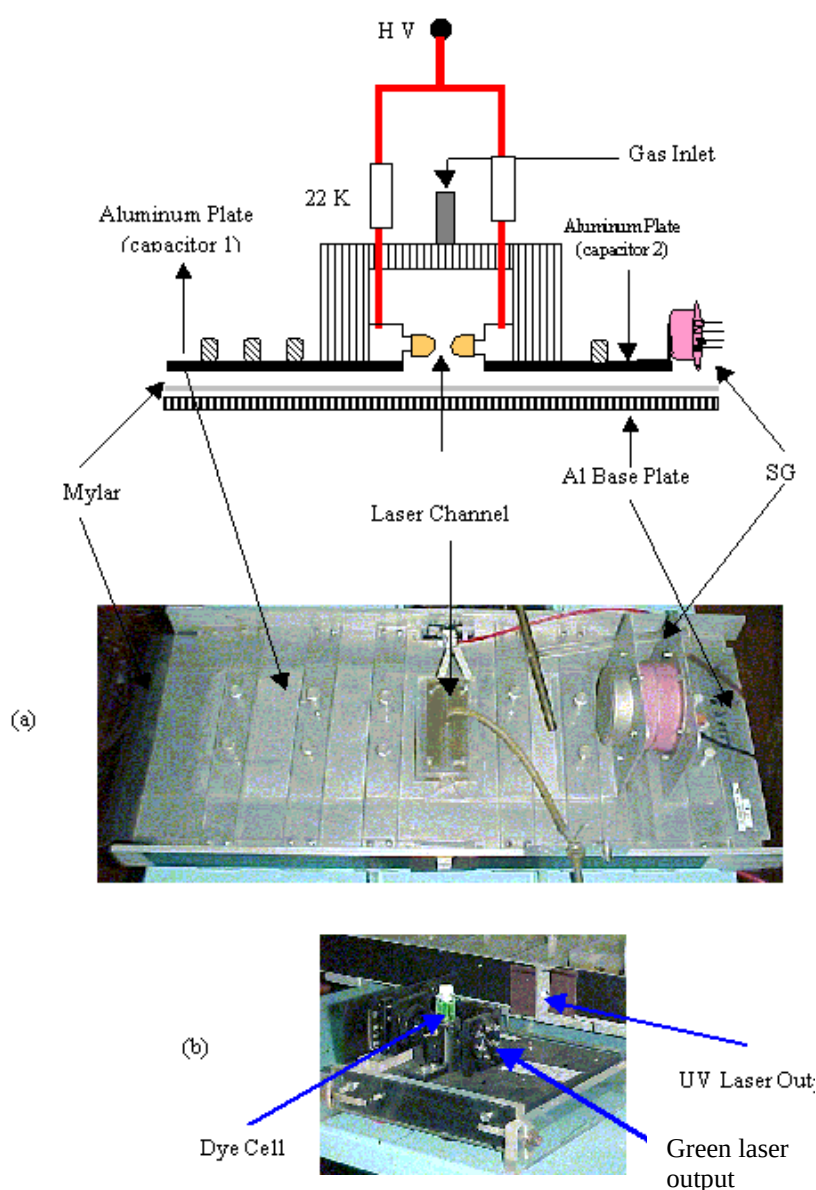


Figure 5.19: a) A view of the nitrogen laser; b) dye module.

EXPERIMENTAL SET-UP

- Hydrogen generator heating current (under $V = 6.3\text{V}$) $< 1.6\text{ A}$;
- Heating voltage of hydrogen generator 4.75 V ;
- Anode voltage (max) 20 kV ;
- Anode current amplitude (max) 150 kA ;
- Pre-heating time $3\text{--}4\text{ min}$;
- Anode current delay time $< 0.2\text{ }\mu\text{s}$;
- Repetition rate $> 10\text{ Hz}$.

The advantages of the PSG over an ordinary sparkgap are: i) extremely low jitter of few nanoseconds and ii) no maintenance is required.

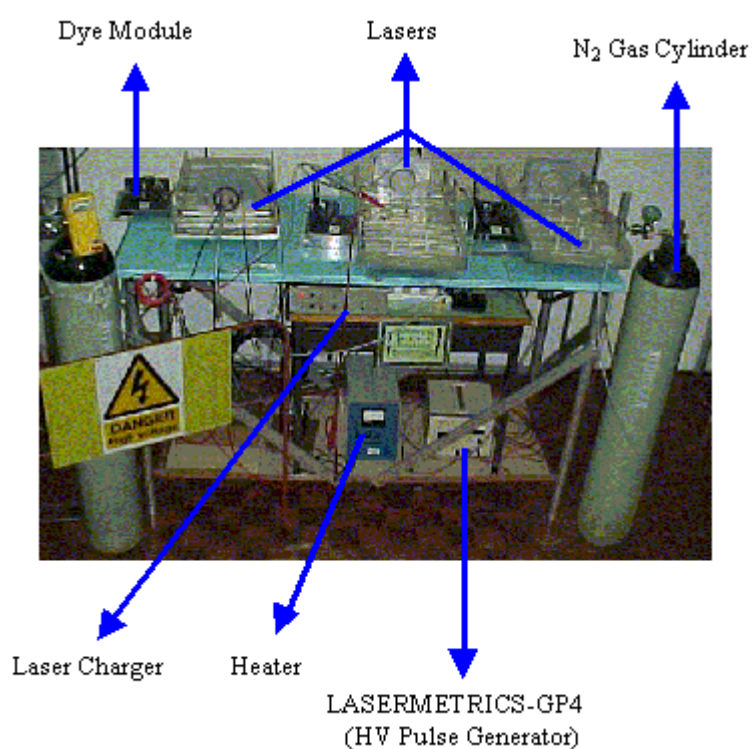


Figure 5.20: A view of the three lasers used for the shadowgraphy set-up.

EXPERIMENTAL SET-UP

The schematic view and a photograph of the nitrogen laser are shown in Figure 5.19(a). Each nitrogen laser was fitted with a dye module.

The whole set-up of the three lasers is illustrated in Figure 5.20.

5.7.2.2 SYNCHRONIZATION AND TRIGGERING SYSTEM:

A schematic diagram of the synchronization of the three lasers is shown in Figure 5.21. Another PSG (PSG 4) was used as the main trigger of the laser lines.

A 1.8 nF capacitor was charged to 17 kV using a NITROGEN LASER UMPPT NL-100 charger (not shown). The main switch PSG 4 was triggered by a

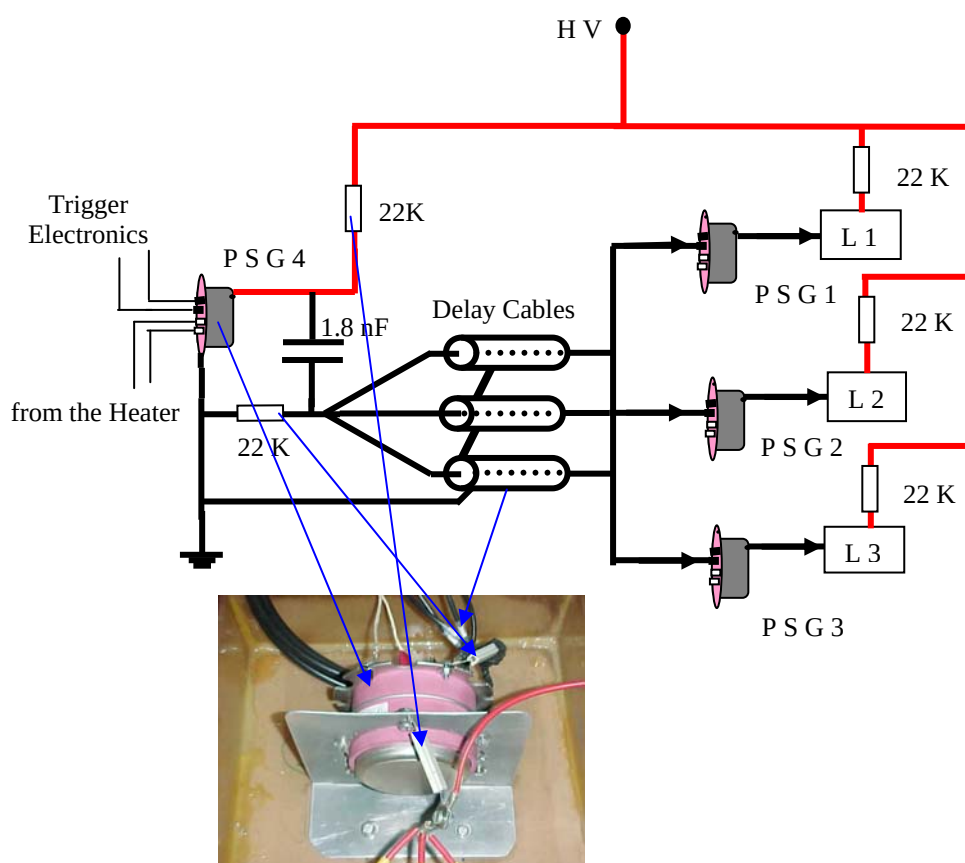


Figure 5.21: Synchronization circuit for the three lasers.

EXPERIMENTAL SET-UP

negative 3.5 kV (50 ns rise time) pulse generated by a high voltage pulse generator *LASERMATRICES GP-4* (not shown), which, in turn, was triggered by the high voltage probe signal (breakdown voltage ~ 3 kV) from the focus discharge. The capacitor was discharged through a 22 k Ω resistor. Thus, a negative 30 kV (10 ns rise time) pulse was produced and used to trigger the three lasers (L_1 , L_2 and L_3) via the corresponding sparkgaps *PSG 1*, *PSG 2* and *PSG 3*, respectively. The lasers L_2 and L_3 were delayed with respect to L_1 using delay cables of 7 ns and 14 ns, respectively. Figure 5.22 shows a schematic diagram of the timing of the triggering system of the three lasers.

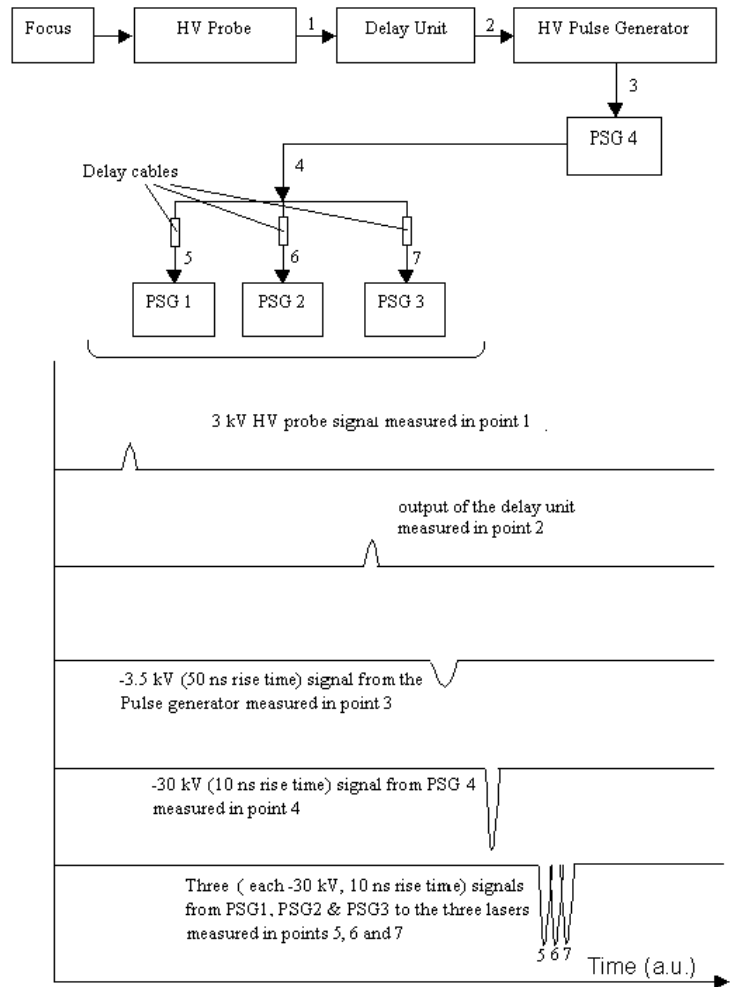


Figure 5.22: Triggering system of the three lasers.

EXPERIMENTAL SET-UP

5.7.2.3 OPTICAL ARRANGEMENT

The optical arrangement of each arm of the set-up of the three-frame shadowgraphy is the same as the one described in section 5.7.1. A view of the set-up of three-frame shadowgraphy is shown in figure 5.23. The schematic of the top view of three-frame shadowgraphy is shown in figure 5.23. The schematic of the top view

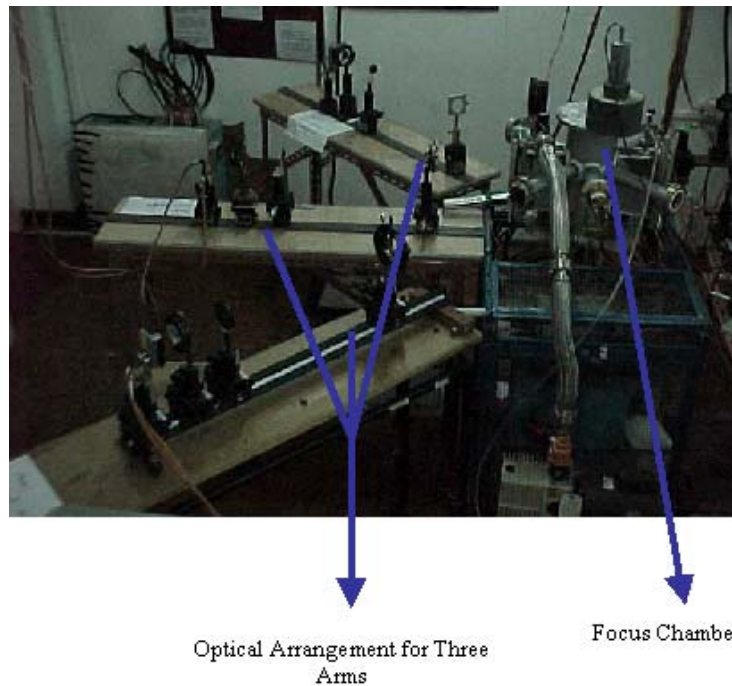


Figure 5.23: A view of the three-frame laser shadowgraphy set-up.

of the multiple frame shadowgraphy used in the experiment is shown in Figure 5.24.

5.7.2.4 THREE-FRAME CAPTURE SYSTEM

The main parts of the three-frame CCD based computer-controlled video capture system are: CCD camera, a consumer video capture card called “fly video” EZ II series, and a Pentium PC. The block diagram of the experimental set-up for the image capture system is shown in Figure 5.25. The card is based on the Brooktree BT 848 single video capture chip [186].

EXPERIMENTAL SET-UP

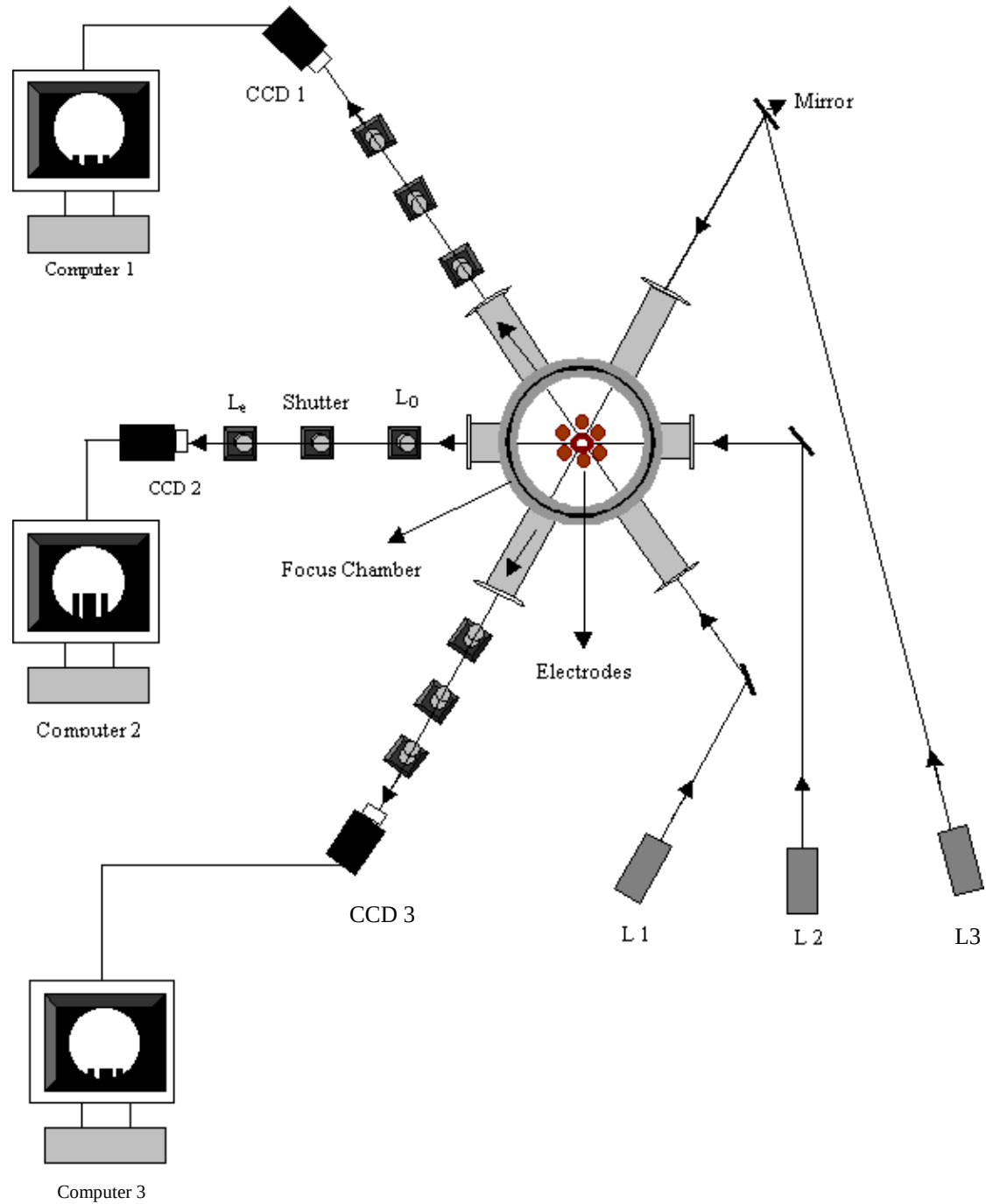


Figure 5.24: Schematic diagram of the top view of the three-frame laser shadowgraphy set-up.

EXPERIMENTAL SET-UP

This card enabled us to capture the image up to the size of 768×586 pixels to the computer memory at a full frame rate of 33 Megabytes per second through a PCI bus that supports a transfer rate of 133 Megabytes per second.

Any CCD camera with a composite output signal can be used with this video card. The composite signal consists of two parts: the positive and negative part. The positive part of the signal represents the image, and the negative part is the synchronization signal that provides the information about the starting of the each frame field and the video line.

The CCD camera used in our experiment is sensitive to visible light, X-rays and charged particles. The sensitivity of the silicon based CCD cameras to light is

limited by the band gap of the silicon below infrared [186]. Due to the protective glass cover on the CCD, the sensitivity to the ultraviolet drops. To overcome this problem, the glass cover was removed from the CCD.

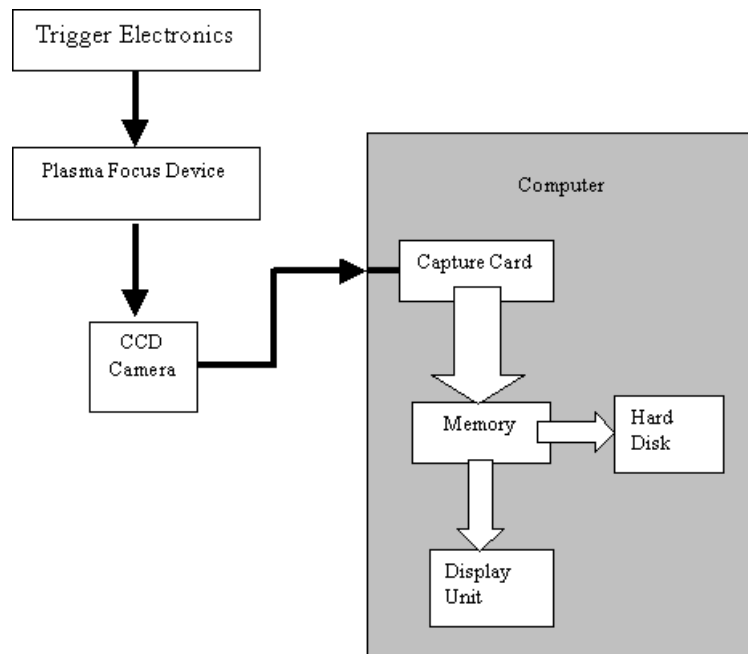


Figure 5.25: A block diagram of a set-up for the image capture system.

The CCD data acquisition and transfer were performed using software written in Basic [187]. This program made use of the Microsoft video for the Windows

EXPERIMENTAL SET-UP

standard supported by Dynamic Link Library (DLL) supplied by the video capture card manufacturer.

5.8 DATA ACQUISITION AND ANALYSIS SYSTEM

A 1 GSa/s, 250 MHz Hewlett Packard (HP) digital logic analyzer (DLA) model 16500A was used to record 8 channels of data per discharge. The time-resolution achieved with our system is better than 1 ns.

The data recorded by the DSO were then downloaded to a PC via a GPIB interface, National Instruments NI 488.2 card. Using the software library that came with the interface card, a program written in C⁺⁺ was developed to transfer and store data from the scope to an ASCII file. The acquired data were analyzed using Microsoft Excel and Microcal Origin 6.0.

Chapter 6

RESULTS AND DISCUSSION

6.1 INTRODUCTION

The results of the investigations performed on the focus dynamics and radiation (neutrons, deuterons and X-rays) emitted from *NIE-SSC-PFF*, as well as their analysis and discussion are presented in this chapter. The machine was operated at 14.5 kV. The experiments were carried out at five different pressures using deuterium as a working gas.

The first part of the chapter presents the results of the time-integrated measurements of the total neutron yield and the deuteron energy spectra. In the second part, the time-resolved neutron, hard and soft X-ray measurement are revealed and discussed. The next part presents the correlation of different radiation parameters followed by the correlation of the focus dynamics with the radiation emission. This section includes the results obtained from optical measurements for the three plasma focus phases. Pinch characteristics and instability growth rate are also discussed in this section.

A comparison of the experimental results with the numerical simulations based on the theoretical model (described in chapter 4) is presented in the last part of the chapter.

6.2 TIME INTEGRATED MEASUREMENTS

6.2.1 TOTAL NEUTRON YIELD

The neutron emission from the plasma focus device depends strongly on the operating pressure. Figure 6.1 presents the histograms of the total neutron yield

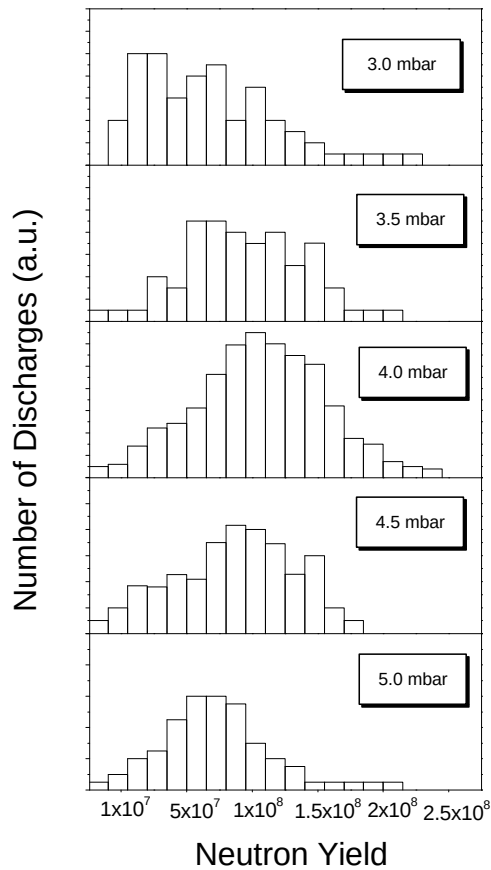


Figure 6.1: Histograms of the neutron yield for different pressures. Plasma focus operated at 14.5 kV in deuterium.

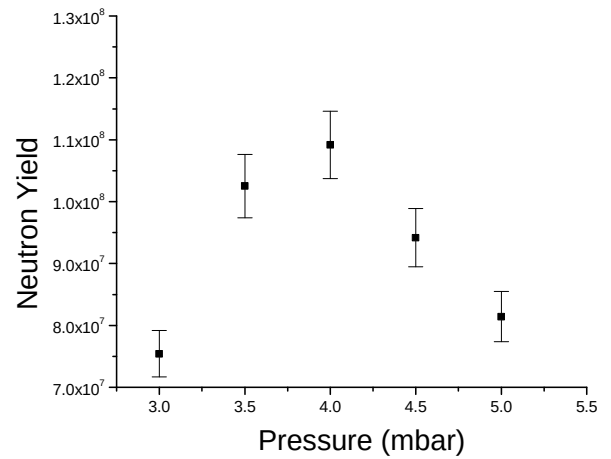


Figure 6.2: Total neutron yield as a function of pressure. The error bars represent the standard deviation.

obtained for five pressures of deuterium. Figure 6.2 illustrates the total neutron yield

RESULTS AND DISCUSSION

as a function of pressure. It is clearly seen that 4.0 mbar is the neutron-optimised pressure.

It was observed that after a shot with very high neutron production, the following one or two shots gave poor yield. The reasons could not be found, but this phenomenon agrees with the observations reported by other workers [41, 188].

6.2.2 DEUTERON ENERGY SPECTRA

The magnetic spectrometer was used to determine the energy spectra of deuterons ejected from our 3 kJ *Mather-type* plasma focus operated at 14.5 kV in deuterium. The *CR-39/PM355* solid state nuclear track detector and the activation detector were employed to detect the deuterons and the neutron flux, respectively. The *CR-39* detector was placed at 0° with respect to the anode axis. The discharges were fired at three different pressures 3.5, 4.0 and 4.5 mbar. For each pressure, 8 detectors were exposed. Each detector was exposed for 8 different positions. Three shots with three minutes delay between them were fired for each position. The three minutes delay between the shots allowed the activity of the foil activation detector to decay to near background level. To minimise the effect of impurities, the gas was changed after 6 discharges.

The tracks formed on these detectors were not visible even with a microscope. In order to enlarge these tracks, the detectors were etched for 4 to 8 hours in 6.25N *NaOH* solution maintained at $(70 \pm 1)^\circ\text{C}$. Then each detector was washed (for 10 minutes) in an ultrasonic bath to remove the *NaOH* contents from tracks.

Each set was sorted based on the total neutron yield. For each set of spectra, five averaged spectra representing 20% of spectra with the highest neutron yield

RESULTS AND DISCUSSION

(Percentile division I), 20% with the next highest yield (Percentile division II) and so on, were obtained.

Figures 6.3-6.5(a) presents the set of averaged spectra of the deuteron energy for 3.5, 4.0 and 4.5 mbar, respectively. The neutron yield corresponding to each spectrum is represented in Figure 6.3-6.5(b).

The pinch region was considered as a uniformly bright region of emission. This was taken into consideration due to the angular deflection of the deuteron beam by the ambient gas. Because of this deflection, the source appeared to be spread out. The pinhole image of the source was larger than the true image of the pinch region. Consequently, the energy spectrum became $dN/dE \cdot dw \cdot dA$ and expressed in units of $(\text{keV})^{-1}(\text{steradians})^{-1}(\text{m})^{-2}$. The width (ΔX) of the projection of the collimator aperture on the surface of the detector was 300 microns. Therefore, the energy resolution of our analyser was estimated as $\Delta E (\Delta X)/E \cong 3 - 6 \%$.

After sorting the spectra with respect to neutron yield, the deuteron energy spectrum was found to be $80 \text{ keV} < E_d < 250 \text{ keV}$ with $100 \text{ keV} < E_d < 140 \text{ keV}$ as the most frequent energy range for each pressure.

The higher energy values (above 250 keV) were not considered because the preliminary experiments indicated that the number of deuterons with energies above 250 keV is negligible. During the preliminary experiments, the CR-39 detector was covered with a layer of 3.5 micron Mylar. After exposure for three shots, the detector was etched. No tracks were found on the detector. This showed that the Mylar sheet absorbed all the deuterons. Thus, it was inferred that the maximum energy of the deuterons is less than 300 keV [189].

RESULTS AND DISCUSSION

Our results show that, regardless of the pressure and neutron yield, the ion distribution exhibits a maximum value in the energy range of $100 \text{ keV} < E_d < 140 \text{ keV}$ in more than 70% of the discharges.

The data plotted in Figures 6.3 - 6.5 clearly show, at all operating pressures, a direct correlation of the highest neutron yield with the highest deuteron flux. Furthermore, the comparison of the percentile decrease in the deuteron flux (for two consecutive spectra) with the corresponding percentile decrease in the neutron yield, reveals interesting features.

Figure 6.6(a) represents the spectra corresponding to the percentile division I, II, III, IV and V, respectively, as a function of pressure. The corresponding neutron yield is shown in Figure 6.6(b).

Table 6.1 gives the comparison of the percentile decrease in the deuteron fluxes (DF) and the neutron yield (Y_n) for each pressure.

RESULTS AND DISCUSSION

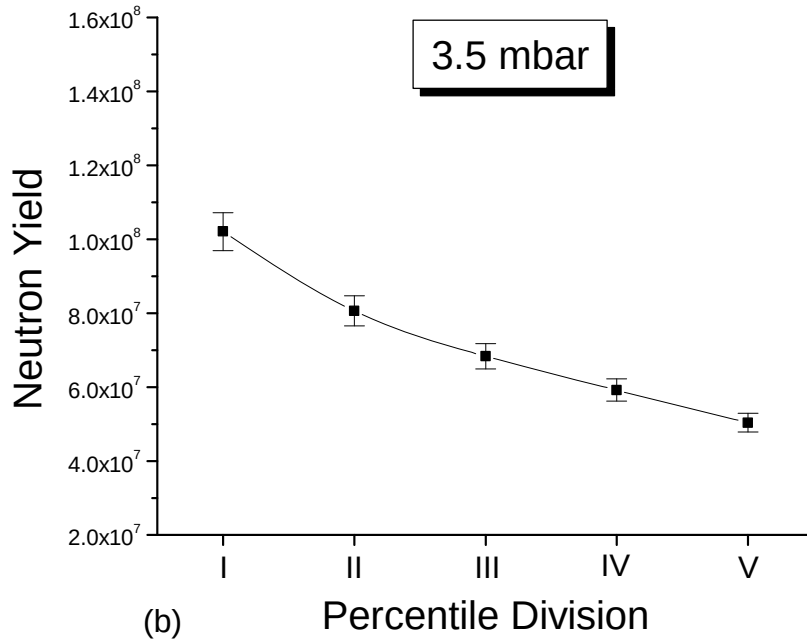
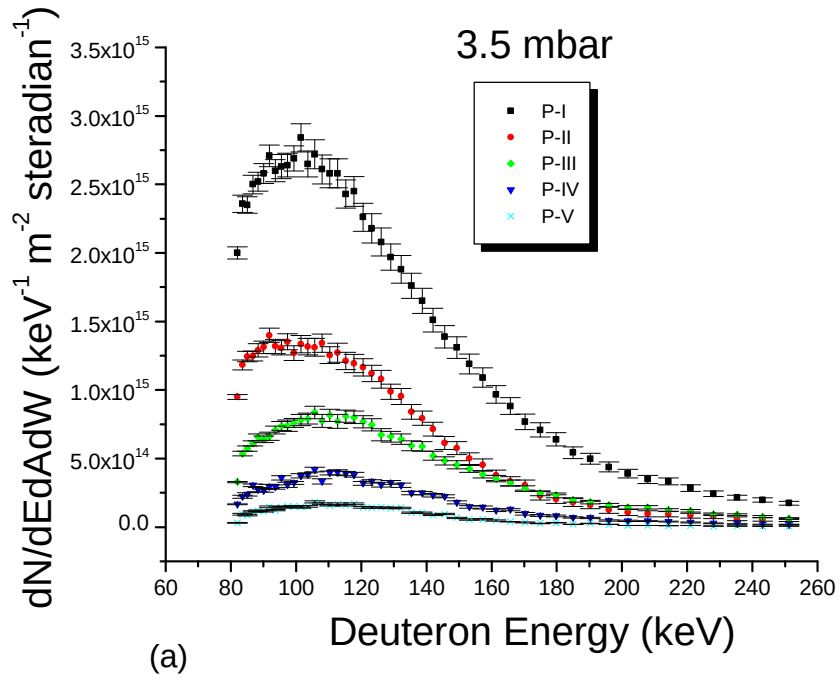


Figure 6.3: (a) Five averaged deuteron energy spectra; (b) Neutron yield corresponding to each spectrum. Plasma focus operated at 14.5 kV and 3.5 mbar of deuterium.

RESULTS AND DISCUSSION

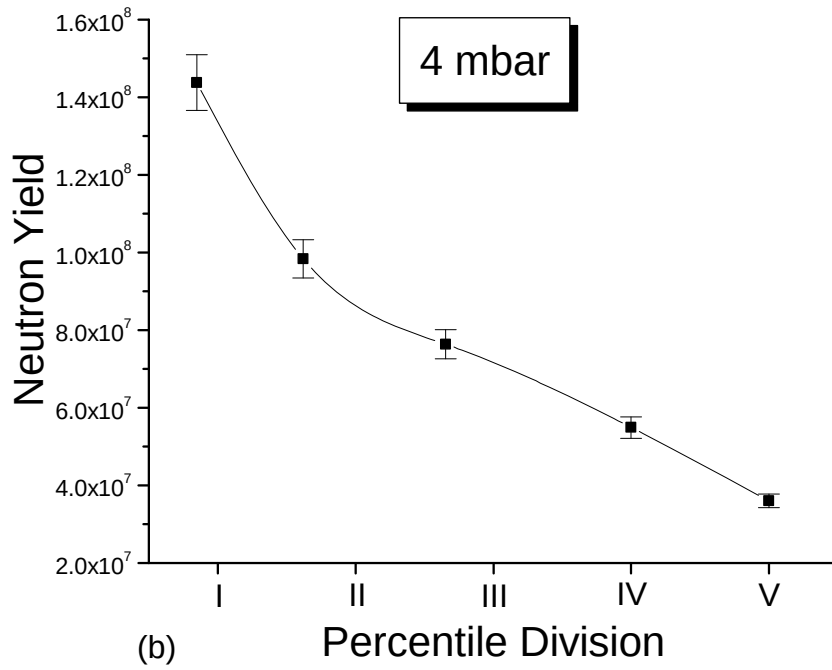
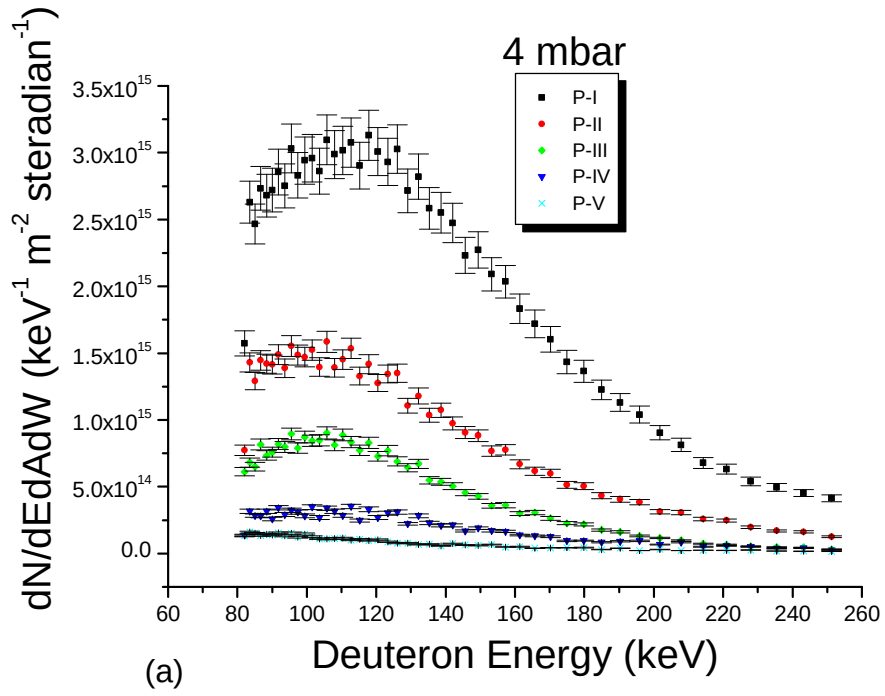


Figure 6.4: (a) Five averaged deuteron energy spectra; (b) Neutron yield corresponding to each spectrum. Plasma focus operated at 14.5 kV and 4 mbar of deuterium.

RESULTS AND DISCUSSION

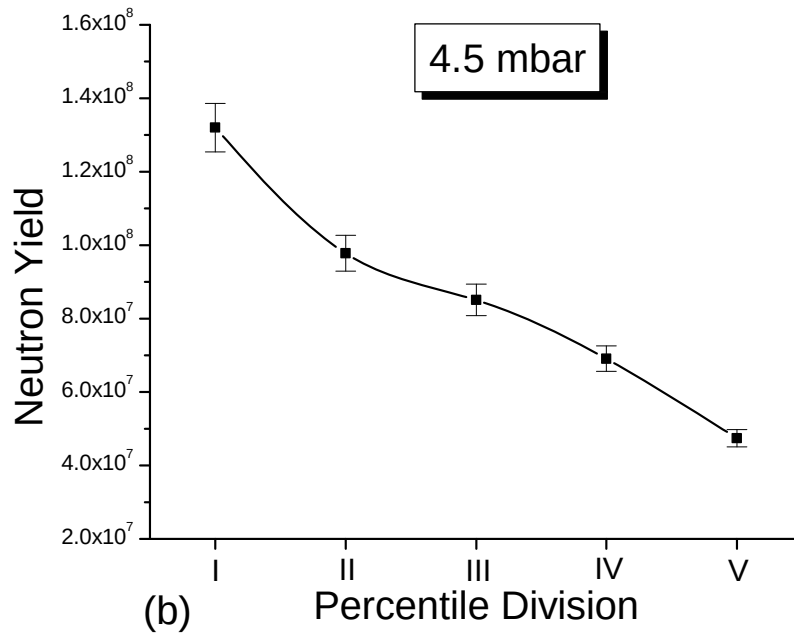
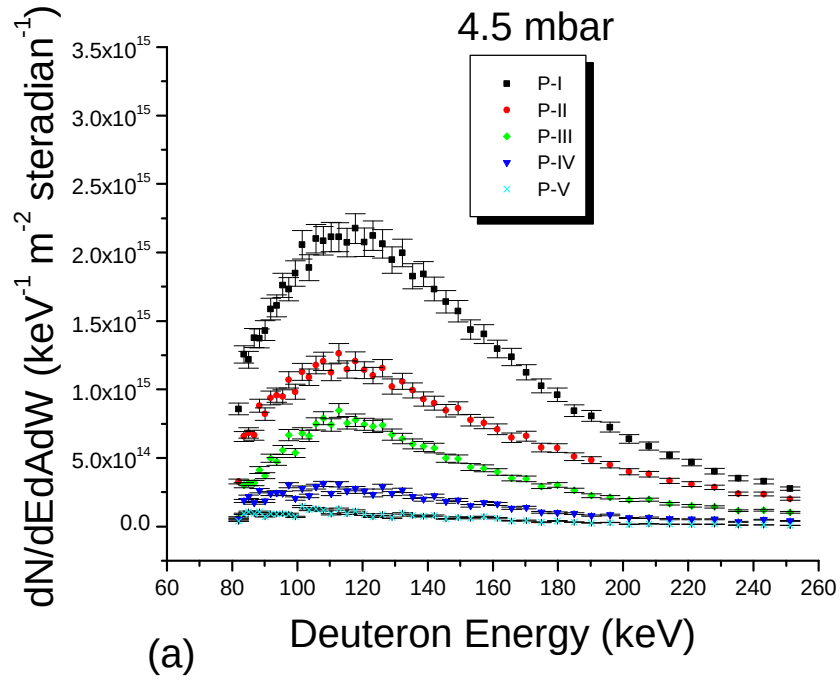
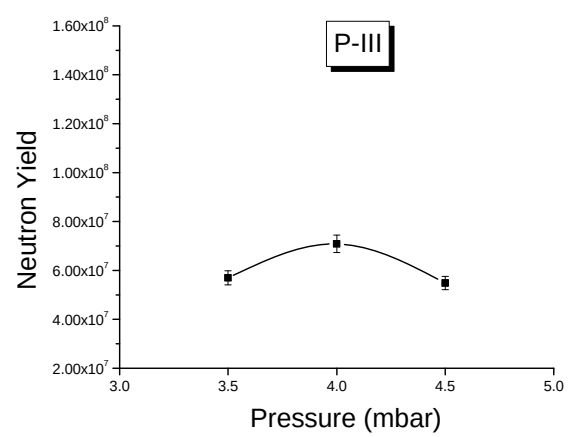
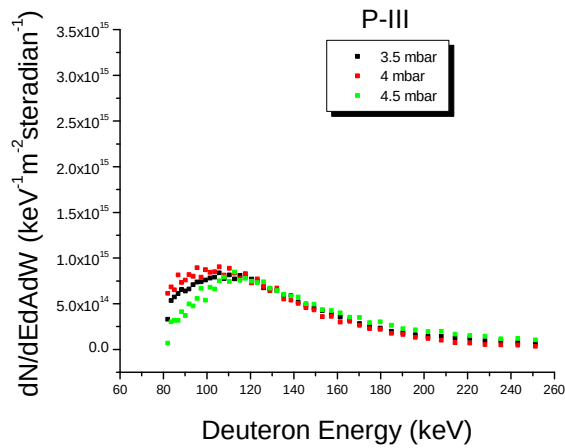
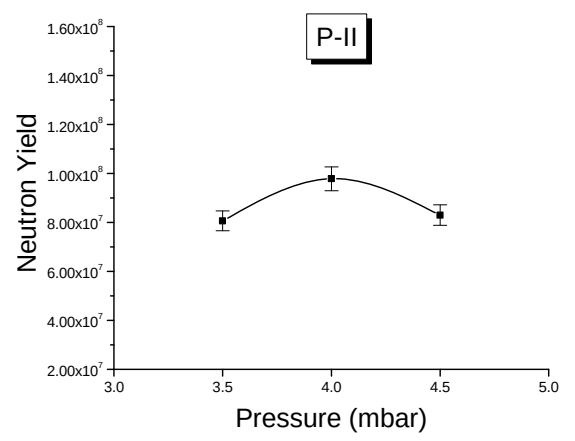
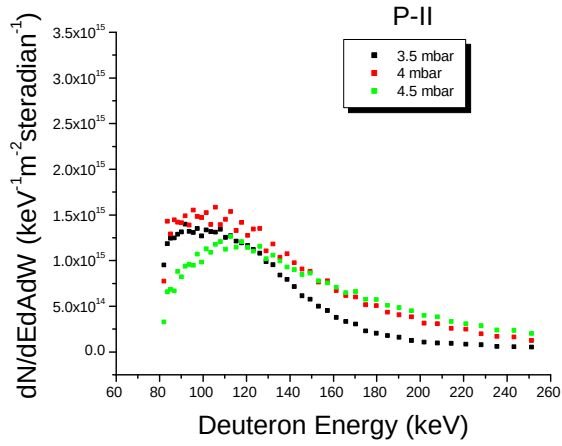
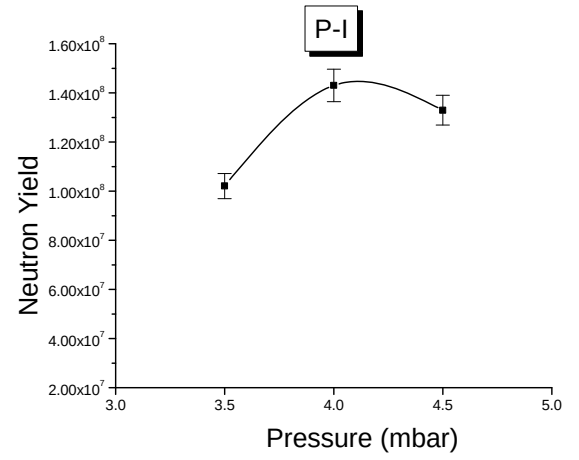
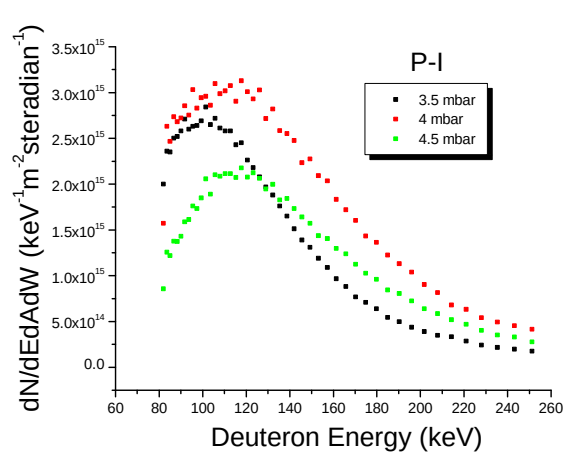


Figure 6.5: (a) Five averaged deuteron energy spectra; (b) Neutron yield corresponding to each spectrum. Plasma focus operated at 14.5 kV and 4.5 mbar of deuterium.

RESULTS AND DISCUSSION



RESULTS AND DISCUSSION

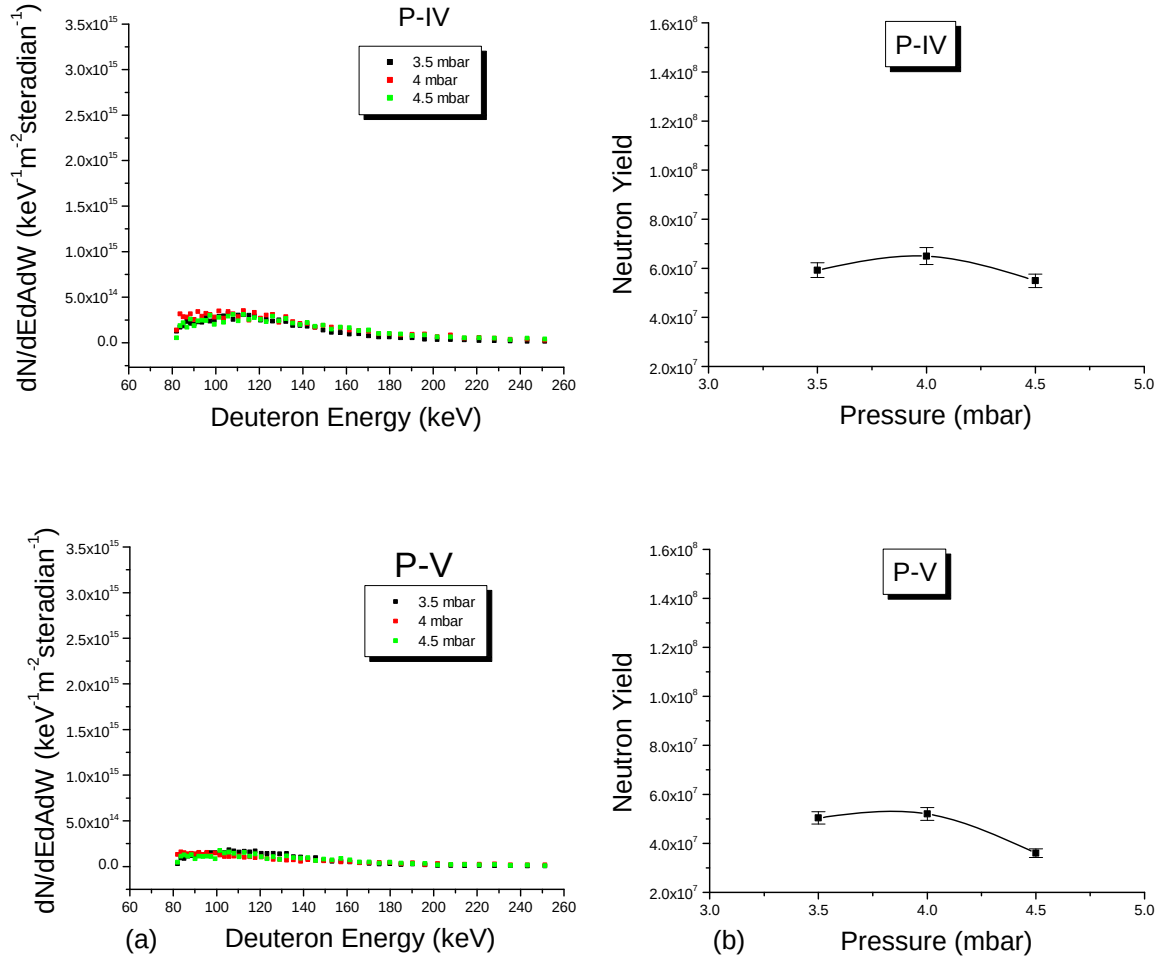


Figure 6.6: (a) Percentile divisions I, II, III, IV and V, respectively, as a function of pressure; (b) Corresponding neutron yield.

RESULTS AND DISCUSSION

Table 6.1: Comparison of the percentile decrease in the deuteron flux and the corresponding neutron yield for the two consecutive spectra.

P (mbar)	Percentile decrease (I-II)		Percentile decrease (II-III)		Percentile decrease (III-IV)		Percentile decrease (IV-V)	
	DF	Y _n	DF	Y _n	DF	Y _n	DF	Y _n
3.5	47	21	27	13	29	15	60	25
4.0	43	31	45	21	37	17	54	34
4.5	55	25	47	13	59	19	70	31

The comparison shows immediately that there is a correlation between the decrease in the neutron yield and the decrease in the corresponding deuteron flux. But, if the ratios of the corresponding percentile decrease are considered, it is clear that the decrease in the deuteron flux is much higher (for all the spectra) as compared to the decrease in the neutron yield. It can be explained by considering that the neutron production is not solely due to these deuterons detected in the forward direction. They might be produced by the scattered deuterons. In other words, one can say that the beam-target production mechanism does not play a 100% role in the production of the neutrons. There are other mechanisms involved in neutron production, but those mechanisms cannot be explained on the basis of our data.

Based on the documented experimental results [9, 190 - 193], it is reasonable to interpret our data from the perspective of a deuteron beam-target mechanism, while not excluding the possibility of other non-beam mechanisms, such as thermonuclear fusion. Our results of neutron yield correlated with the deuteron beams are consistent with the fact that the beam-target mechanism plays a major role in the neutron production.

RESULTS AND DISCUSSION

Previous studies [194, 195] on the high energy deuterons in the plasma focus devices showed that the deuteron energy distribution is rather complicated and they were unable to give proper theoretical interpretation.

The dimensions of the source of the high energy deuteron beams emitted from the plasma focus are interesting both from the point of view of fundamental physics as well as some applications. The solution to this problem would explain the mechanism of the acceleration of the deuterons in the plasma focus, and allow us to find out the possibilities of applying the plasma focus as a commercial ion beam source. It has not yet been clarified whether the accelerated deuterons are localised inside the pinch volume or in the surface layer across its boundary with the anode.

If the deuteron acceleration stems from the thin layer near the anode, then the deuteron energy will be sensitive to shape and size of the pinch, because the deuterons will pass through the pinch where they are scattered by the electromagnetic fields on their way from the anode to the recording device. If one establishes how the deuteron energy depends on the pinch configuration and size, it might help to identify the zone where the particles acceleration takes place.

The deuteron acceleration mechanisms are still under consideration. The first acceleration mechanism is attributed to the ion reflections inside the current sheath due to the electric fields generated inductively. The acceleration mechanism for the generation of the subsequent deuteron pulses are more complicated, e. g. electric field connected with the anomalous resistivity, local electric fields induced by the fast decay of the plasma structures, etc. In general, various mechanisms are probable during different phases of the DPF discharge. Unfortunately, they cannot be identified on the basis of the present experimental data.

RESULTS AND DISCUSSION

A beam of collimated ions is emitted during the focus phase [196]. This beam carries an electric charge up to about 10^{-4} C and a total energy up to 1 J in the case of an efficient discharge. Compared to the total energy in the focus region, which is in the range of 10 J, a considerable energy is converted into unidirectional beams with particle energy above 50 keV. The total number of particles in the focus volume is in the order of 10^{17} .

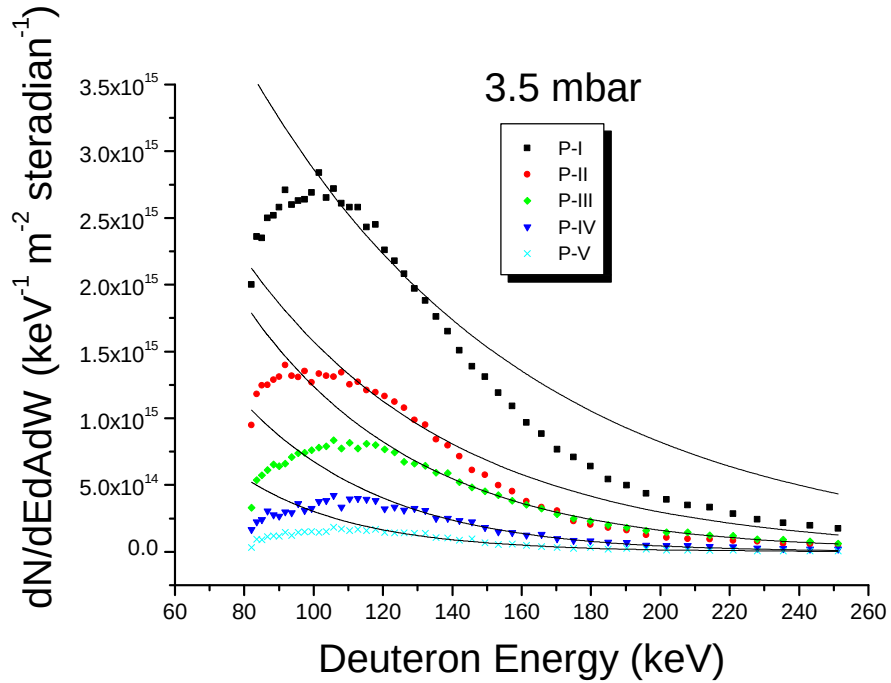


Figure 6.7: Five averaged deuteron energy spectra (dotted curves) and $\exp(-E_d/B)$ fitting curves (line). Plasma focus operated at 14.5 kV and 3.5 mbar deuterium.

RESULTS AND DISCUSSION

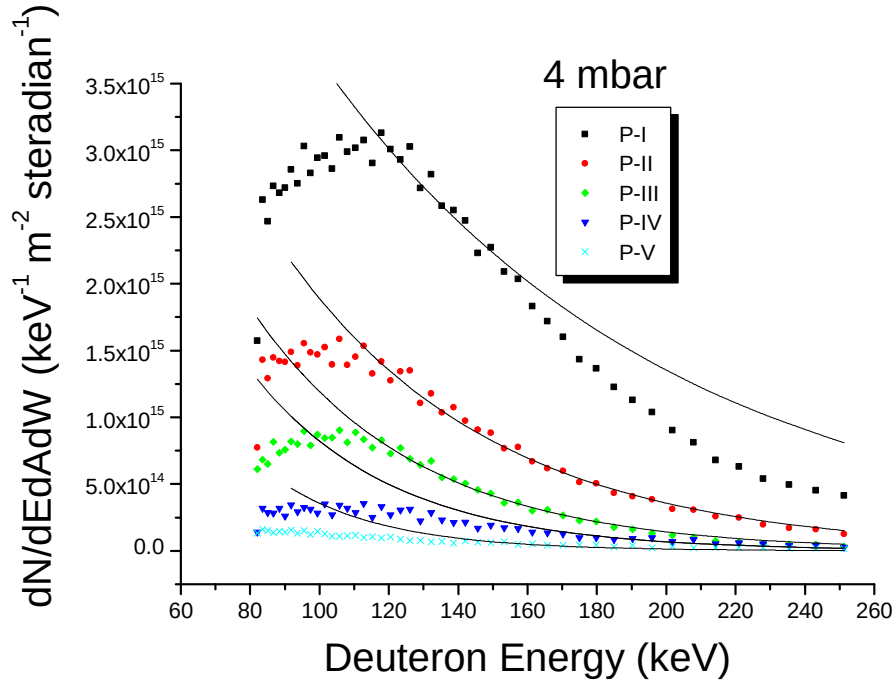


Figure 6.8: Five averaged deuteron energy spectra (dotted curves) and $\exp(-E_d/B)$ fitting curves (line). Plasma focus operated at 14.5 kV and 4 mbar deuterium.

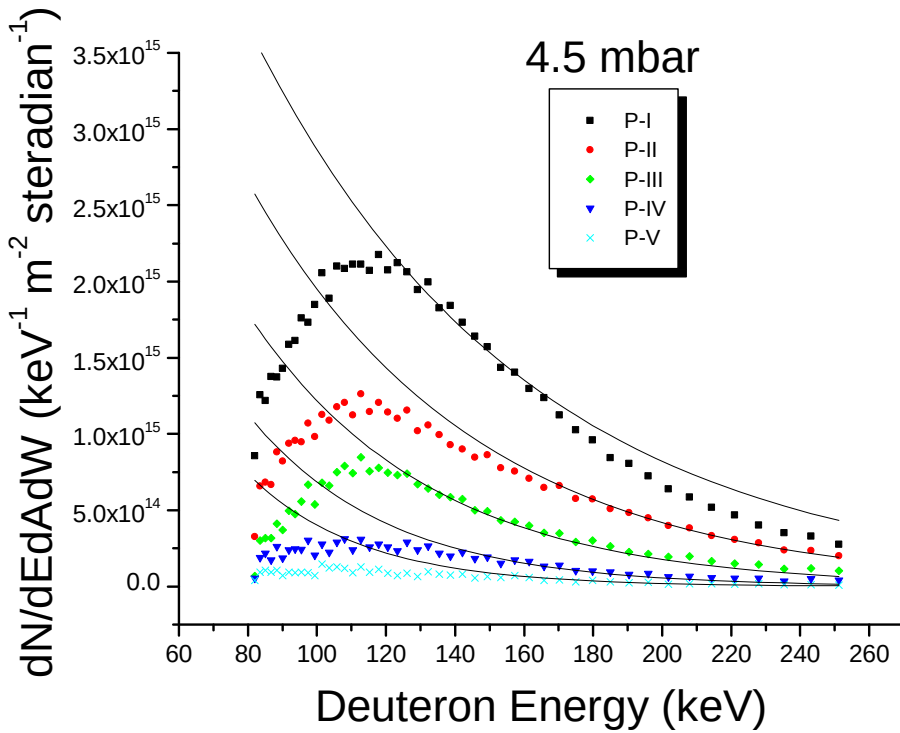


Figure 6.9: Five averaged deuteron energy spectra (dotted curves) and $\exp(-E_d/B)$ fitting curves (line). Plasma focus operated at 14.5 kV and 4.5 mbar deuterium.

RESULTS AND DISCUSSION

The ions travelling in the beam with energy greater than 50 keV are well above the threshold for the neutron production by collision with ions and atoms at rest [197]. The experimental ion distribution is relatively well fitted by an exponential function of the form $\exp(-E_d/B)$ [87] with the values of B ranging from 30 to 100 keV. The fitting curve along with the experimental data are shown in Figures 6.7-6.9.

The calculated values of B are tabulated in Table 6.2.

Table 6.2: Values of B (expressed in keV) for the function $\exp(-E_d/B)$.

Spectrum	3.5 mbar	4.0 mbar	4.5 mbar
I	80	100	80
II	66	60	65
III	49	48	53
IV	53	38	44
V	33	40	34

From the graphs it is clear that for the pressures of 3.5 and 4.0 mbar, the curves fit only for the energy range $100 < E_d < 170$ keV for the percentile division I and II. For the percentile divisions III, IV and V the curves fit very well in the energy range $125 < E_d < 250$ keV.

For the 4.5 mbar, the curve fits in the energy range $125 < E_d < 160$ keV for the percentile division I, but for the divisions II, III, IV and V, the curves fit well in the energy range $125 < E_d < 250$ keV.

The deuterons' stream in the axial direction follows an empirical law:

$$dN/dE \propto E^{-x} \text{ [163], where } x = 3.5 \pm 0.5.$$

RESULTS AND DISCUSSION

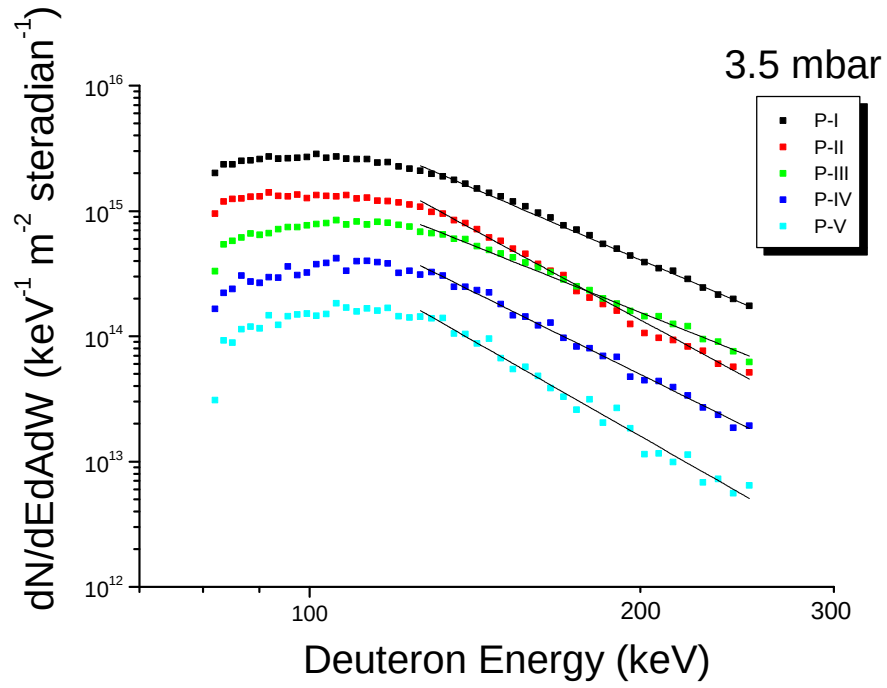


Figure 6.10: Deuteron spectra (dots), and the E^{-x} fitting curve (solid line). Plasma focus operated at 14.5 kV and 3.5 mbar deuterium.

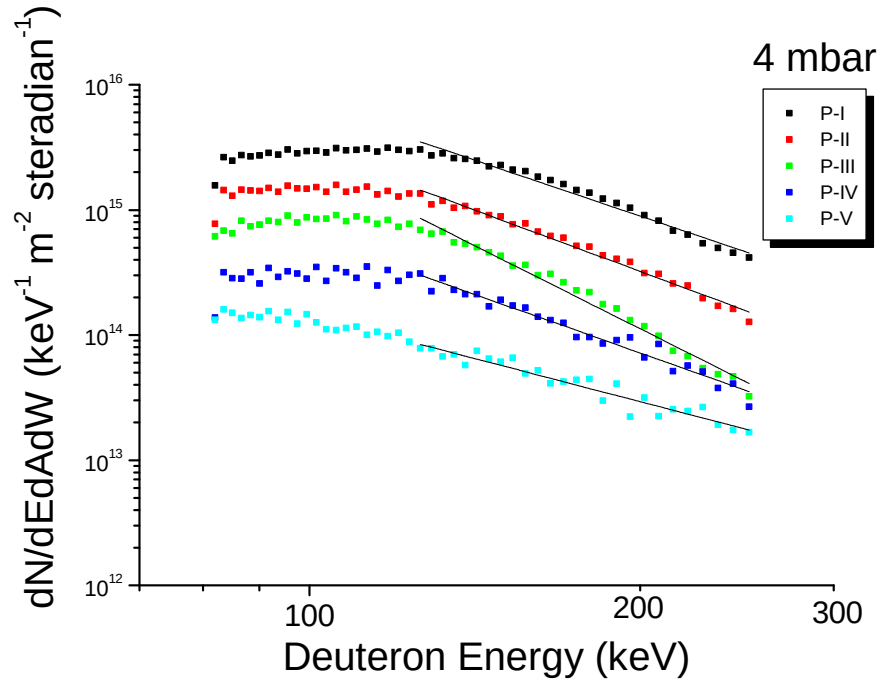


Figure 6.11: Deuteron spectra (dots), and the E^{-x} fitting curve (solid line). Plasma focus operated at 14.5 kV and 4.0 mbar deuterium.

RESULTS AND DISCUSSION

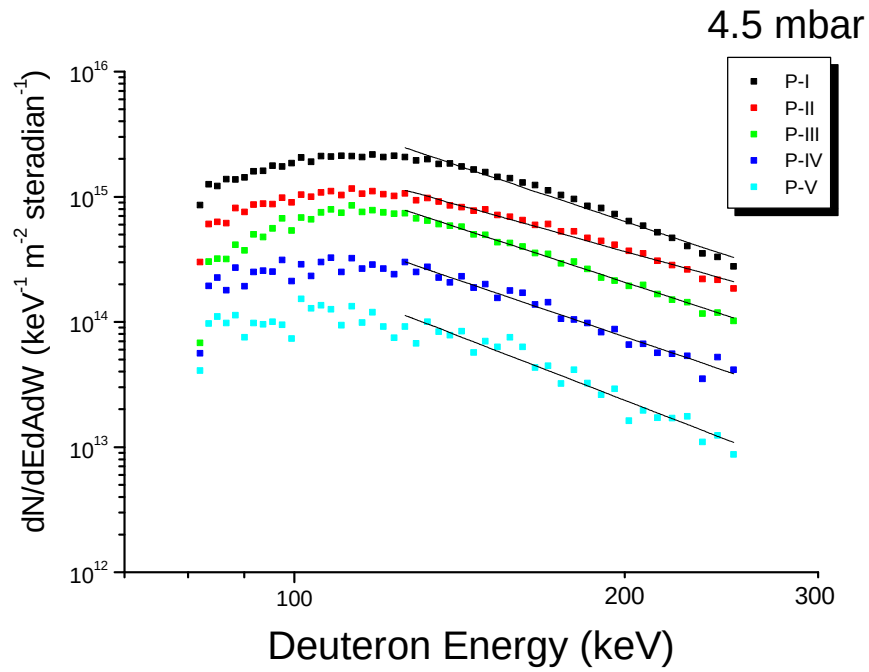


Figure 6.12: Deuteron spectra (dots), and the E^{-x} fitting curve (solid line). Plasma focus operated at 14.5 kV and 4.5 mbar deuterium.

Considering this exponential law, solid lines are drawn for each spectrum (see Figures 6.10-6.12). These lines fit well in the energy range of $125 < E_d < 250$ keV for all the spectra, at all three pressures. Table 6.3 gives the values of x for each curve fit.

Table 6.3: The values of x for the solid lines drawn in Figures 6.10-6.12.

Spectrum	3 mbar	3.5 mbar	4 mbar
I	3.7	2.9	2.9
II	4.7	3.2	2.4
III	3.5	4.4	2.8
IV	4.3	3.1	2.9
V	5.0	2.2	3.4

RESULTS AND DISCUSSION

A key objective of this project has been successfully attained in the development and implementation of an automated measurement system for etched ion tracks in *CR-39*.

The results show that there is a strong correlation between the neutron yield and the forwardly directed high energy deuteron beam emitted from the pinch region of the plasma focus. The averaged deuteron energy spectra and associated neutron yields show that this correlation is present for all three pressures (3.5, 4.0 and 4.5 mbar) investigated in this work. The higher energy region of the spectra (above 125 keV) can be well fitted by the empirical relation $dN/dE \propto E^{-x}$ with an x value broadly within the range of 3 to 4, which is in agreement with results obtained by other workers.

A more puzzling aspect of the present measurements is that for energies below 125 keV, the spectra level off and then decrease towards lower energies. This is in contradiction with the most of theoretical calculations, which predict that the energy spectra should increase monotonically as the deuteron energy decreases.

Another difference occurs from the comparison of our findings with the results of previous workers who claim that deuterons in the approximate energy range of 50 to 75 keV are responsible for the bulk of neutron production via the beam-target interaction. This discrepancy can be worked out by the pre-acceleration of all ions by applying a high voltage (~30 kV) to a special transparent gap positioned within the ion drift space. Thus, the ion spectrum will be shifted towards higher energies, which will result in its reliable registration beginning from 50 keV. It will also prevent the loss of the low energy particles from the beam during their flight to the detector.

RESULTS AND DISCUSSION

One possible explanation of this difference is related to the decreasing efficiency of the automated image processing system in recognising and counting the tracks on the surface of the *CR-39* detectors, because the range and specific energy loss of the particles decrease at lower energies. It should be remembered that the range of an 180 keV deuteron is less than 2 microns in *CR-39*, and the resulting etch pit will be a shallow spherical surface of approximately 7 microns in diameter at the surface, and of approximately 1 to 2 microns in depth. It is therefore, not surprising that the tracks of lower energy deuterons will become increasingly difficult to be distinguished from the spurious features on the surface of the detector.

The detection efficiency depends on many factors, some of which are difficult to quantify. It is clearly dependent on the diameter and depth of the track, which, in turn, is related to the bulk and track etch rates of *CR-39*, in addition to deuteron energy. Unfortunately, the bulk and track etch rates for this material appear to vary even within a single sheet (20 cm by 30 cm) of the material from which the detectors used in this experiment were cut. This may in part be due to small variations in the etching procedure, most importantly, the temperature and *pH* of the NaOH etchant. Typically, two detectors were etched each day (for 8 hours overnight) during the course of these experiments, requiring the mixing of new etching solution from NaOH granules every few days. Further work is required to ensure the consistency of etching conditions throughout a long sequence of experiments.

The detection efficiency also clearly depends on the “surface quality” of the *CR-39* detectors which were found to vary considerably even within the same *CR-39* sheet. Some of the detectors displayed defects after etching. These defects appear under the optical microscope as lines of bubbles/scratches. These lines were

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contiguous across the detectors cut from the same area of CR-39 sheet, and appeared even on unexposed detectors. Such variations in surface quality in the supplied material was clearly beyond the control of the experimenters. The solution adopted was to discard the detectors of insufficient surface quality. However, even on the remaining detectors, the density of surface features liable to be counted as spurious tracks is variable. Furthermore, even with very careful handling, it was unavoidable that throughout the cutting, exposing, etching, and washing processes, further small scratches accumulate on the detector surface.

The image processing system was designed to mimic the distinction between real and spurious tracks as judged by the human observer, by making repeated comparisons of exposed and unexposed areas of the detector surface, and over as wide selection of the detectors as endurance permitted. In this manner, the image processing algorithm and associated parameters were fine tuned.

The selected algorithm comprises sequentially: a Sobel filter, binary threshold, binary fill, binary erode and dilate (opening operation). However, even for the human observer, the distinction between real and spurious tracks becomes increasingly difficult for lower deuteron energies, and clearly, some trade-off between detection efficiency and rejection of spurious tracks must be made.

The final algorithm employed for these measurements gives, by comparison with human observation, a high ($> 90\%$) counting efficiency for $E_d > 150$ keV tracks, with minimal counting of spurious tracks.

6.3 TIME-RESOLVED MEASUREMENTS

In the previous section, extensive results on the ion emission from the plasma focus were discussed.

In order to widen the view on the focus dynamics and radiation emission, our investigations were extended. The information from multiple-frame shadowgraphy of the radial compression, pinch and post focus phases corroborated with the results of a systematic time-resolved and time integrated analysis of neutron and soft X-ray emission helped achieve our goal. Different diagnostics were employed for this purpose.

The following parameters were recorded simultaneously; current derivative (dI/dt), voltage (V), time-resolved neutron and hard X-ray emission rate (dY_n/dt , dY_{HX}/dt) from three plastic scintillator-photomultiplier systems (PM1, PM2 and PM3) and time-resolved soft X-ray emission rate (dY_{SX}/dt) using 2 filtered PIN diodes. The total neutron production was monitored using an In foil activation detector. For each focus discharge, the image of the radial phase was initially recorded using the single-shot shadowgraphy method. Later, a three-frame shadowgraphy system was developed and successfully implemented in our experimental set-up.

The focus was operated at 14.5 kV using deuterium as a working gas for five different pressures: 3.0, 3.5, 4.0, 4.5 and 5.0 mbar. In order to allow the activity of the neutron detector to decay to the background level the device was operated with a 3 minute interval between the discharges. The background readings were recorded daily. The discharge chamber was refilled after every ten discharges. Data from a total number of more than 400 discharges were recorded and analysed.

RESULTS AND DISCUSSION

The results obtained from the diagnostics employed are discussed in the following sections.

For the investigation of the emission of hard X-rays and neutrons from the plasma focus, time-resolved hard X-ray and neutron measurements were carried out using three detectors. A description of the set-up for these measurements has

been given in chapter 5. Figure 6.13 shows the synchronised signals from the three scintillator-photomultiplier detectors obtained during the preliminary experiments.

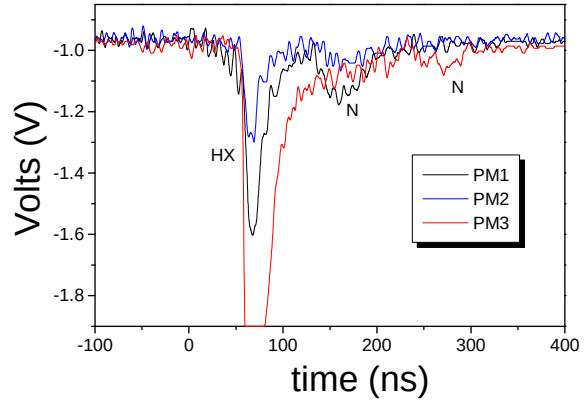


Figure 6.13: Synchronised signals from PM1, PM2 and PM3.

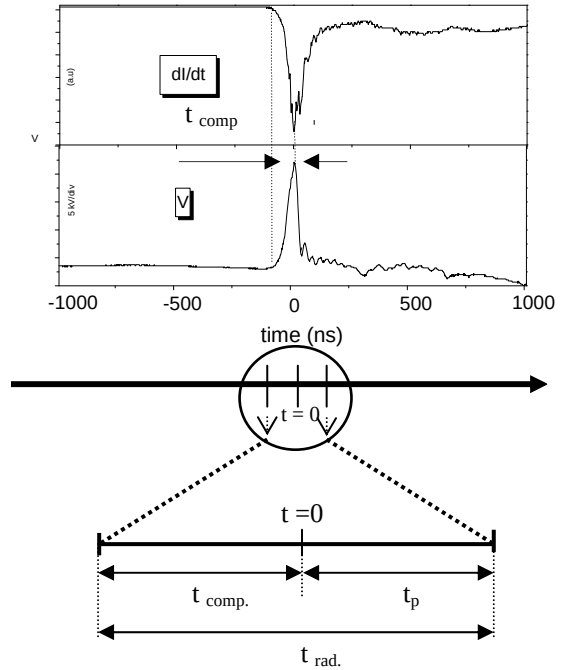


Figure 6.14: Current derivative and voltage signal profiles (top) and the time periods of the focus (bottom).

The first pulse is the hard X-ray pulse (HX), and the second is the neutron (N) pulse. The hard X-ray pulse from PM3 (the farthest detector placed in the radial plane) is more intense than the other two (from PM1 and PM2), because lead sheets were placed in front of PM1 and PM2 to cut off most of the hard X-rays in order to avoid the saturation of the photomultiplier.

RESULTS AND DISCUSSION

Based on the current derivative and the voltage signal profiles, two distinct regimes, or patterns, were identified for the plasma focus discharges at 14.5 kV / 4.0 mbar. These regimes are discussed in the following paragraphs.

6.3.1 SINGLE-COMPRESSION REGIME

In this regime, the voltage probe signal exhibits a single, sharp and fast risetime spike. The corresponding current derivative (dI/dt) signal has a distinct and sharp negative spike. An example of such a shot is shown in Figure 6.14. Almost all the shots in this regime exhibit a very high neutron yield and a low emission in the soft X-ray range.

Figure 6.14 also shows the important time periods of the focus during the radial and the pinch phase. The time reference is taken at the instant when the plasma column reaches the minimum radius, i.e. $t=0$, $r=r_{min}$. Here t_{comp} represents the duration of the radial compression phase. The instant corresponding to the beginning of this phase is determined from the current derivative signal correlated with the beginning of the rise time of the voltage spike and the shadowgraphs. The pinch lifetime, t_p , is defined as the duration from the instant $t=0$ to the instant corresponding to the development of the $m=0$ instability.

6.3.1.1 TIME-RESOLVED HARD X-RAYS AND NEUTRON MEASUREMENTS

An example of a discharge at 14.5 kV/ 4.0 mbar with a single-compression is shown in Figure 6.15. This figure represents a 500 ns time window of waveforms of the current derivative, hard X-rays (HX), neutrons (N) and voltage probe signals.

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Usually, for such kind of discharges, the neutron yield is higher (2.3×10^8 for the discharge shown) than the average.

The first hard X-ray burst shows two peaks and has a sharp risetime of 6 ns (usually 5-10 ns) and a FWHM of 40 ns (usually 30-50 ns). The first peak is related to the first compression to the minimum radius, and the second is associated with the onset of the $m=0$ instability. In the case of a double hard X-ray pulse, two neutron pulses (almost merged together) are observed. The neutron signals (from three detectors) have a small peak on the leading edge which is associated with the first hard X-ray pulse that appears at the instant of maximum compression. The second peak is related to the onset of the $m=0$ instability.

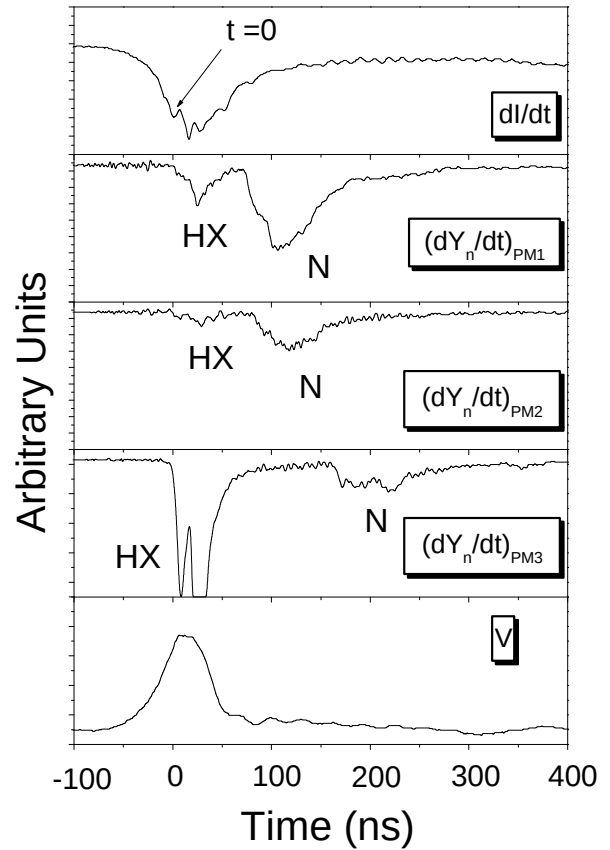


Figure 6.15: Signal profiles of dl/dt , V and dY_n/dt . $Y_n = 2.3 \times 10^8$.

The two neutron pulses (for each detector) are merged together due to a small pinch lifetime. The first neutron pulse (corresponding to the first compression) is smaller than the second one (associated with the $m=0$ instability) indicating that the majority of the neutrons are produced at and after the instant corresponding to the development

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of the $m=0$ instability. It can be explained by saying that the ion beams are generated in the axial direction due to the presence of strong electric fields in the $m=0$ instability zone. Therefore, most of the neutron production is generally assigned to the axial beam-target mechanism.

The total neutron yield Y_n is large when the hard X-ray emission is large. Optimum dynamic conditions for having high Y_n are met when only a few hard X-ray bursts with high intensity occur [199]. Discharges with many X-ray bursts (over a large period of time) have usually a smaller yield of neutrons. Similar results were previously reported by Bostick et. al. [200]. This indicates that the rate of

decay of the magnetic structure and the inductive fields, which accelerate the particles, can reach the highest value when the multiplicity of the regions of the decay does not exceed an optimum number. By an estimation of the correlation coefficient for the X-ray burst and the neutron yield, one knows that the typical electron beam parameters (intensity multiplicity, time distribution etc.) have a bearing on the total neutron yield [200].

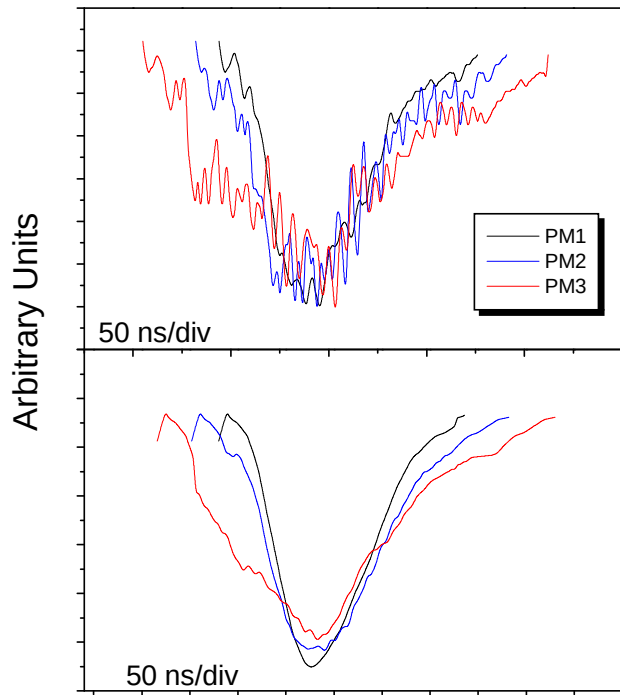


Figure 6.16: Three normalised (a) and smooth (b) neutron signals from PM1, PM2 and PM3.

RESULTS AND DISCUSSION

Figure 6.16 presents the three neutron signals from PM1, PM2 and PM3, after normalisation (a) and smoothing (b). The smoothing procedure was performed by interpolating the experimental data with higher order polynomial functions.

The fast rise time and smaller width (FWHM) of the PM1 signal (compared to the PM2 and the PM3 signals) is attributed to the emission of a bulk of neutrons in the axial direction with less dispersion in the energy.

The slow rising and wider signals from the other detectors (PM2 and PM3) show large dispersion of the neutron energy in the radial direction. A significant difference of ~ 40 ns in the width (FWHM) of the signals from PM2 and PM3 is observed. The difference is seen mainly at the beginning of the leading edge and at the end of the trailing edge. An additional cause of the broadening of the neutron pulse is the scattering of the neutron by the surroundings due their larger flight path to the detector.

The average amplitude of the neutron pulse from PM1 (axial direction) is found to be 0.725 V with the standard deviation of 0.18 V. The average amplitude of the signals from PM2 and PM3 is 0.321V with the standard deviation of 0.14 V. The difference in amplitudes is attributable to the observed anisotropy of the neutron flux. The results on the neutron anisotropy will be presented later.

The neutron pulse from PM1 has a rise time of 20 ns (typically 20-25 ns) and a width (FWHM) of 60 ns (typically 50-65 ns). The signals from PM2 and PM3 have rise times of 28 ns (usually 26-30 ns) and 36 ns (usually 35-40 ns) and their widths are 80 ns (usually 76-84 ns) and 120 ns (usually 115-125 ns), respectively.

For the shots at pressures other than the optimum pressure, it was observed that the neutron yield was low, but the emission time was longer. At higher (higher

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than the optimum) pressures, this fact was observed for more than 60% discharges. This might be explained by saying that with the increase in pressure, the role of the beam mechanism in the neutron production decreases.

In general, depending on the type of the device and gas filling pressure, it is considered that the neutrons emitted by the plasma focus are partly generated by an isotropic ion velocity distribution and partly by a directed beam of ions. The plasma parameters, neutron production, number density, average particle energy, characteristic dimensions and time, are such in the device as to rule out the existence of any thermal equilibrium.

The thermal mechanism gives a neutron spectrum corresponding to that of a wide ion energy distribution, direction and number of particles. The beam-target is an opposite limiting mechanism in which the neutron spectrum is as if it was produced by an almost monochromatic beam ($E/\Delta E \gg 1$) concentrated in space around a given angular direction and distributed over a small number of particles. There is further evidence that shorter the neutron production time, the stronger are the beam aspects. The longer the neutron production time, the more are the quasi-thermonuclear aspects [201].

6.3.1.2 TIME-RESOLVED SOFT X-RAY MEASUREMENTS

The soft X-ray measurements were performed using two (X3 and X4) of five channel PIN diode X-ray spectrometer.

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In Figure 6.17, a 500 ns time window of dI/dt , V and dY_{SX}/dt from two filtered PIN diodes signal profiles is shown.

Usually, three peaks of dY_{SX}/dt signal associated with the first burst are observed [41]. The first peak (1) observed corresponds to the plasma focus compression to the minimum radius. It has a very sharp rise time of 1 ns (usually

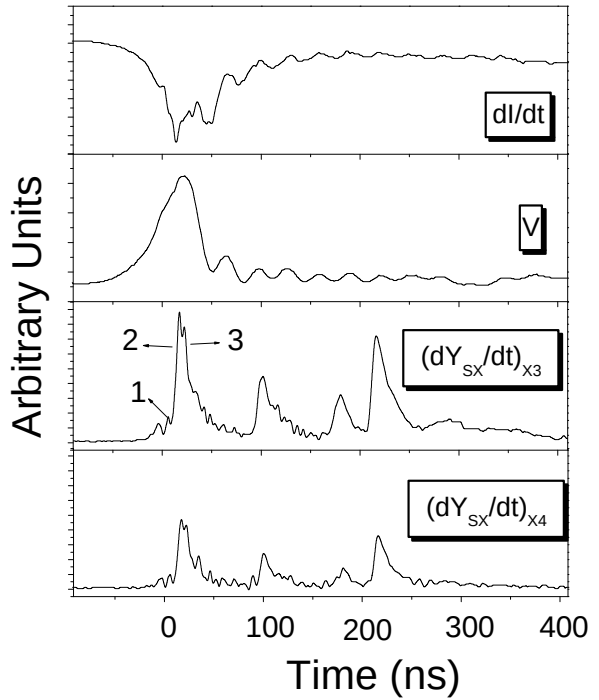


Figure 6.17: Signal profiles of dI/dt , V and dY_{SX}/dt . $Y_n = 1.3 \times 10^8$.

1-2 ns) and approximately 3 ns (usually 3-5 ns) FWHM. The plasma emission in the X-ray range may also be caused by the fast electrons which appear at the instant of the development of instabilities and interact with the surrounding plasma. As a result of the collective interaction, 90% of the beam energy may be lost; the plasma is heated; and the X-ray emission is observed over the interaction range. The turbulent structure in the plasma stream was observed in shadowgraphs and also reported by other authors [202]. The turbulence excited in the accelerator channel causes a further acceleration of the plasma as a whole, which in turn promotes a more effective contraction of the plasma at the focus. The soft X-rays and the neutrons come from the turbulently heated plasma.

The second soft X-ray peak (2) is associated with the quiescent phase. It has a rise time of 3 ns (usually 3-5 ns) with approximately 10 ns (9-13 ns) FWHM. Its peak

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is related to the onset of the $m=0$ instability. The pinch lifetime, t_p , can be estimated by measuring the instants at which the two peaks appear.

The third soft X-ray peak (3) is detected during the unstable or decay phase. The total duration of the first X-ray burst is 55 ns (typically 50-75 ns).

The second soft X-ray burst is observed 80 ns (typically 78-130 ns) after the first burst and lasts for 45 ns (typically 30-60 ns). This burst is usually lower in amplitude and smaller in width as compared to the first burst. Since there are no indications of the second pinch (see voltage signal), this pulse is related to the breaking up of the remnant plasma filaments into small structures.

The third X-ray emission period starts 160 ns (typically 100-300 ns) after the first burst and lasts for 100 ns (typically 90-200 ns). This pulse is associated with the vaporised copper jet emitted from inside or the rim of the anode.

6.3.2 MULTIPLE-COMPRESSION REGIME

In this plasma focus regime, two or more voltage spikes are observed corresponding to two, or more compressions of the plasma. For such kind of discharges, the dY_n/dt signal shows two or even three neutron pulses extended over a period of more than 250 ns.

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6.3.2.1 TIME-RESOLVED HARD X-RAYS AND NEUTRON MEASUREMENTS

An example of a discharge exhibiting multiple compression is presented in Figure 6.18. For the first hard X-ray burst, two peaks (almost merged together) are observed. The first peak appears at the instant of maximum plasma compression. The second peak is related to the onset of the $m=0$ instability. Usually, the first peak is relatively less intense than the second. The second hard X-ray burst is detected 120 ns (usually 90-150 ns) after the first pinch and is associated with a second pinch. The hard X-ray emission during the second pinch is usually less intense as compared to the first one. The second focus is ascribed as the breaking up of the filaments into hot spots.

Two distinct neutron pulses are also observed. The first neutron pulse shows the same characteristics as the pulse corresponding to the first plasma focus regime (single-compression). It is always the largest in terms of the amplitude and width. The second neutron pulse carries less neutrons and is associated with the second hard X-ray burst which appears due to the second plasma focus compression.

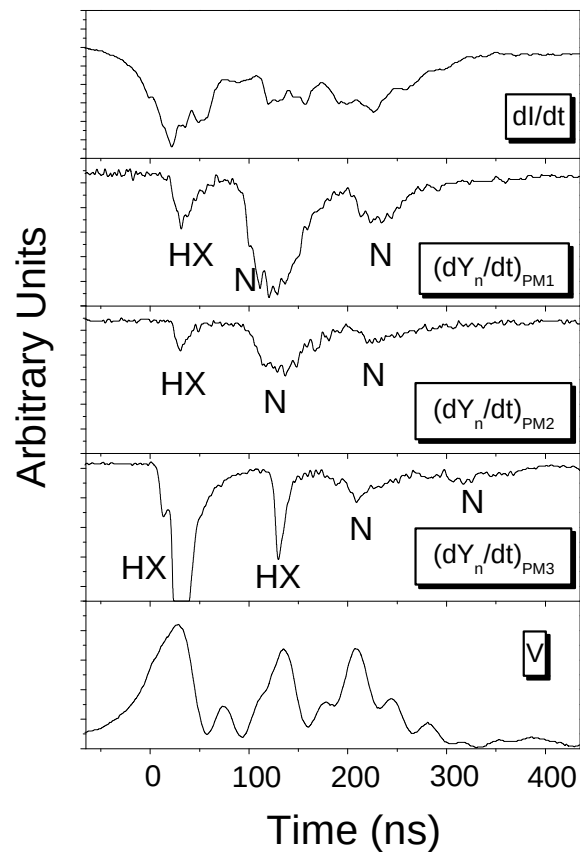


Figure 6.18: Signal profiles of dl/dt , dY_n/dt , dY_{HX}/dt and V . $Y_n = 1.8 \times 10^8$.

RESULTS AND DISCUSSION

Usually, the total neutron production for the discharges exhibiting multiple-compression is not as high as the yield generated in the discharges with a single-compression. Most of the discharges at 14.5 kV/4.0 mbar of deuterium exhibit a single-compression regime with a high neutron and hard X-ray yield. For other pressures (3.0, 3.5, 4.5 and 5.0 mbar), the second kind of pattern was found to be dominant.

6.3.2.2 TIME-RESOLVED SOFT X-RAY MEASUREMENTS

The current derivative, voltage and two (X3, X4) of the five channel PIN diode soft X-ray spectrometer signal profiles are shown in Figure 6.19.

The interpretation of the first soft X-ray burst corresponding to the first plasma compression is the same as described for the regime exhibiting single-compression. At the instant $t = 100$ ns (usually 90-250 ns) after the first pinch, another soft X-ray burst is detected. The second pulse has 15 ns (usually 10-20 ns) rise time and 20 ns (usually 15-30 ns) FWHM. This pulse is related to the second compression of the plasma. The second compression is related to the breaking up of the plasma filaments into hot spots. The third soft X-ray pulse appears 200 ns (usually 150-250 ns) after the first compression. This pulse indicates a larger emission lasting for more than 150 ns (usually 100-200 ns). This soft X-ray burst is ascribed to the copper jet emitted from the anode.

RESULTS AND DISCUSSION

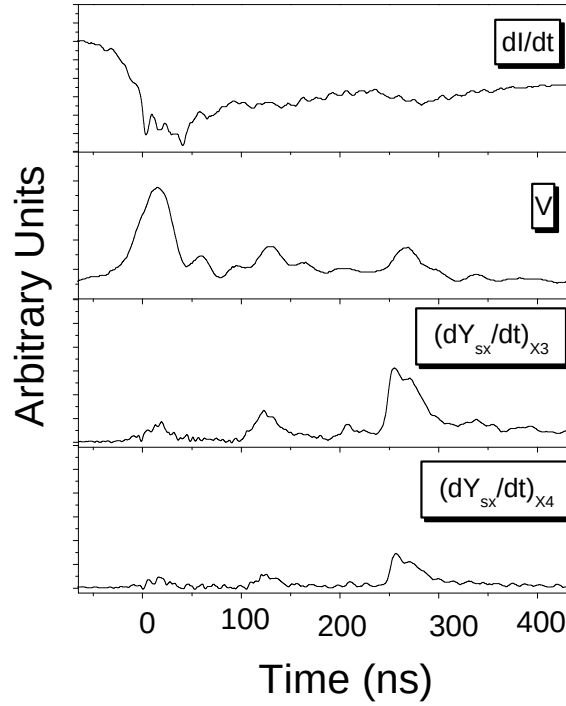


Figure 6.19: Signal profiles of dl/dt , V and dY_{sx}/dt . $Y_n = 1.5 \times 10^8$.

6.3.3 ELECTRON TEMPERATURE MEASUREMENTS

Attempts were made to measure the plasma electron temperature using the filter ratio method. Two soft X-ray signals (from X3 and X4) were integrated, and the ratio $R = X3/X4$ was calculated. For most of the discharges the ratio was less than 6, which indicates dominant $Cu-K_\alpha$ contribution from the anode [183]. However, for a few discharges the ratio was found to be greater than 6, and the electron temperature was determined to be ranging from 1-2 keV.

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Figure 6.20 shows a 100 ns window of the signal profiles of $(dY_{SX}/dt)_3$ and of the $(dY_{SX}/dt)_4$ along with the electron temperature trace determined using filter ratio method. The total pinch lifetime for this discharge is 21 ns. The instant $t = 0$ corresponds to $r = r_{min}$. It is clear that at the instant of maximum compression, the temperature increases abruptly to 1 keV. During the expansion,

the temperature decreases to 0.8 keV. The temperature range determined agrees with the results reported earlier [203].

In Figure 6.21, a comparison of the experimentally determined and simulated electron temperature is shown. The simulation is performed using a plasma focus model discussed in chapter 4. It is clear that at the instant $t=0$ and the instant of the

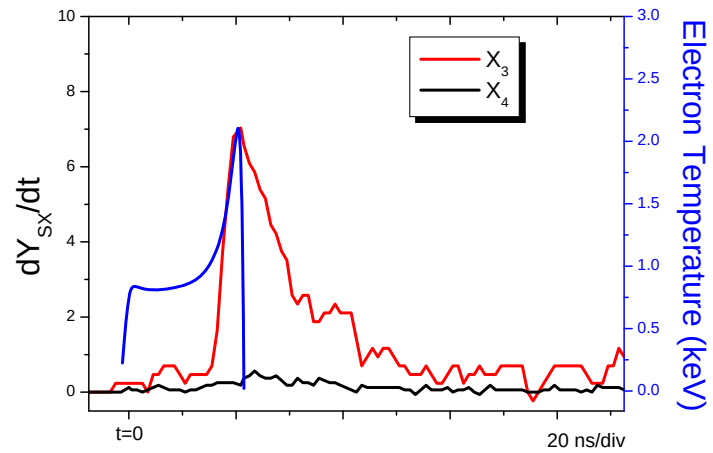


Figure 6.20: A 100 ns window of soft X-ray signal profiles (red and black) and the smoothened electron temperature curve determined using filter ratio method (blue trace).

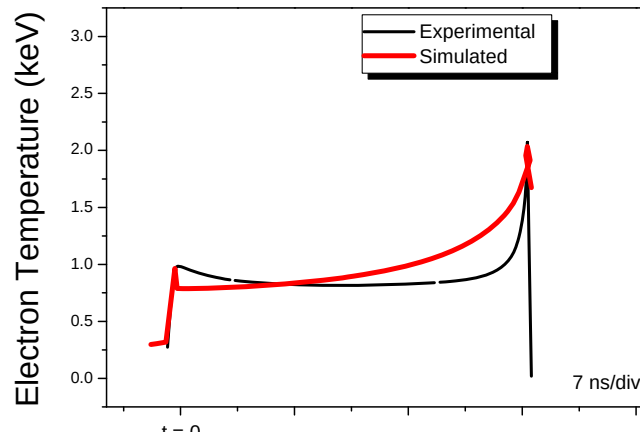


Figure 6.21: Experimentally determined (black line) and simulated (red line) electron temperature curves.

RESULTS AND DISCUSSION

development of the $m=0$ instability, the experimental and simulated curves are in good agreement.

The reason for the high electron temperature can only be seen in the turbulent heating, since, such a high temperature values cannot be explained by classical ohmic heating or by compression effect.

The measured value of the temperature is in contradiction with the absolute X-ray yield of the device and with most of the previous results of the experiments performed on this type of device, but it could be important for certain application of DPF. This experimental result gives us a certain information of the X-ray emission in the spectral range of 1-10 keV. It seems that the copper filter (one of the filters used in the experiment) could not eliminate the influence of the copper line emission because of the DPF chamber wall-plasma interaction.

6.4 ANALYSIS AND CORRELATION OF RADIATION PARAMETERS

6.4.1 SOFT X-RAY AND NEUTRON PRODUCTION

In this section, a correlation between simultaneously measured soft X-ray production and the total neutron yield is presented. The soft X-ray production is obtained by integrating the dY_{sx}/dt signal from one of the filtered PIN diode (X4). In order to obtain the total soft X-ray yield, integration was performed for the whole emission period of the first soft X-ray burst. Figure 6.22 shows a plot representing the variation of the soft X-ray production with the total neutron yield as a function of pressure. The plot shows that at 3.0 and 3.5 mbar, the soft X-ray production is 2.5 and 1.6 times, respectively, higher than the value at 4 mbar. For the higher pressures of

RESULTS AND DISCUSSION

4.5 and 5.0 mbar, the soft X-ray production is 2.6 and 3.2 times, respectively, higher than the value at 4.0 mbar. A simple result can be drawn: the optimization conditions for the emission of soft X-ray from the plasma focus are different from those for the neutron-optimised mode. Higher operating pressures favour the soft X-ray production.

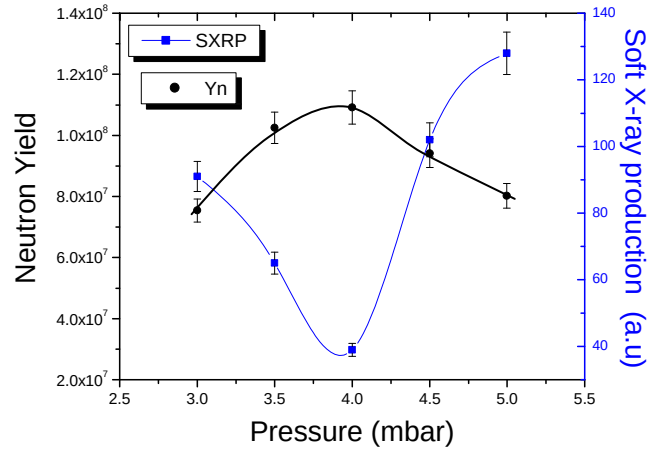


Figure 6.22: Variation of the soft X-ray production (SXRP) and total neutron yield as a function of pressure.

The variation of the soft X-ray production with the total neutron yield at different pressures is shown in Figure 6.23. This figure shows that for 3, 4.5 and 5 mbar, the soft X-ray production is higher for the lower values of the neutron yield. As the neutron yield increases, the soft X-ray production continues to decrease. But, a different trend is seen for 3.5 mbar. For the discharges with the neutron yield lower than the average value, the decrease in the soft X-ray production with the increasing neutron yield is more abrupt as compared to the discharges with higher (higher than the average value) neutron yield.

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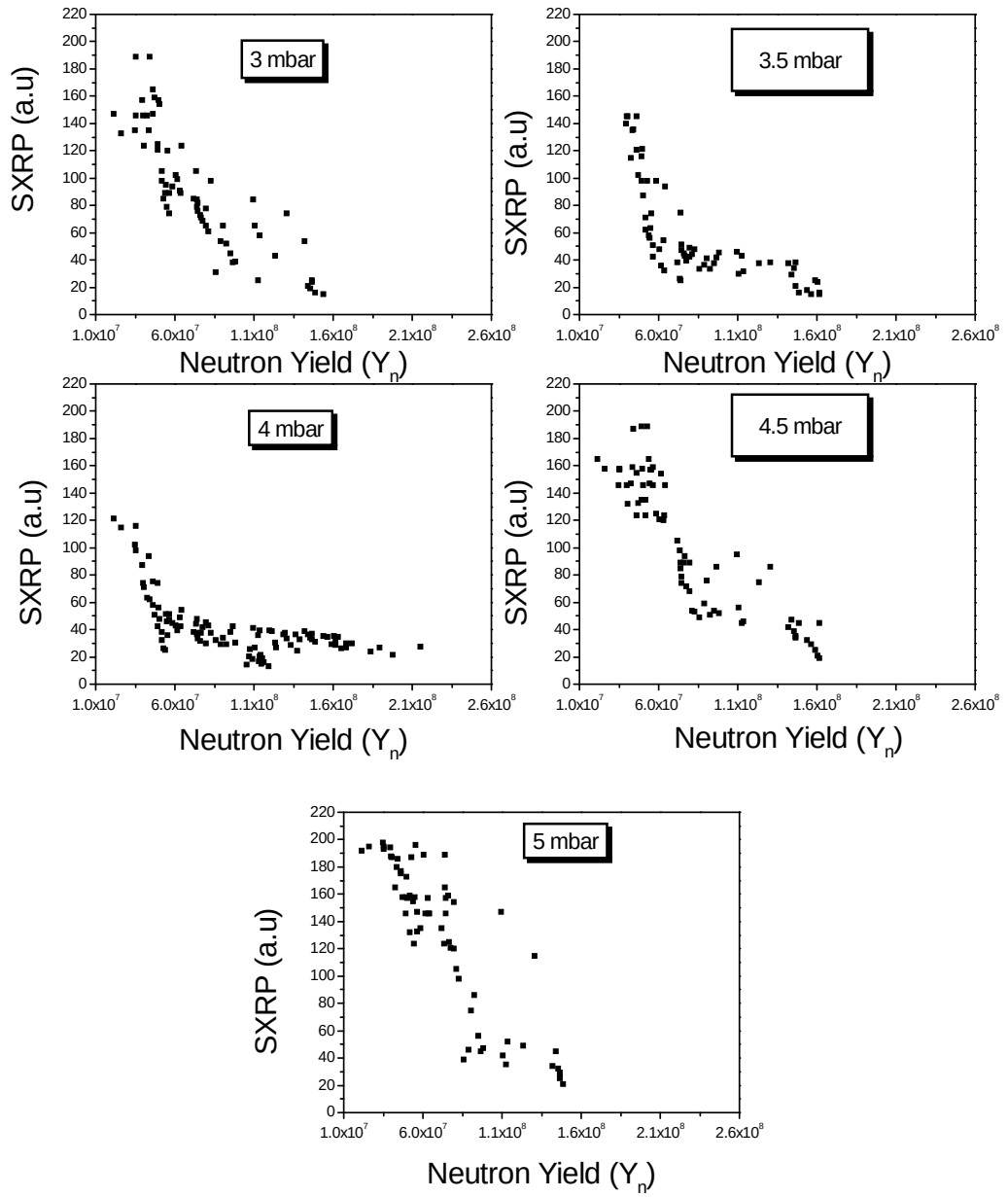


Figure 6.23: Soft X-ray production (SXP) as a function of the total neutron yield at different pressures.

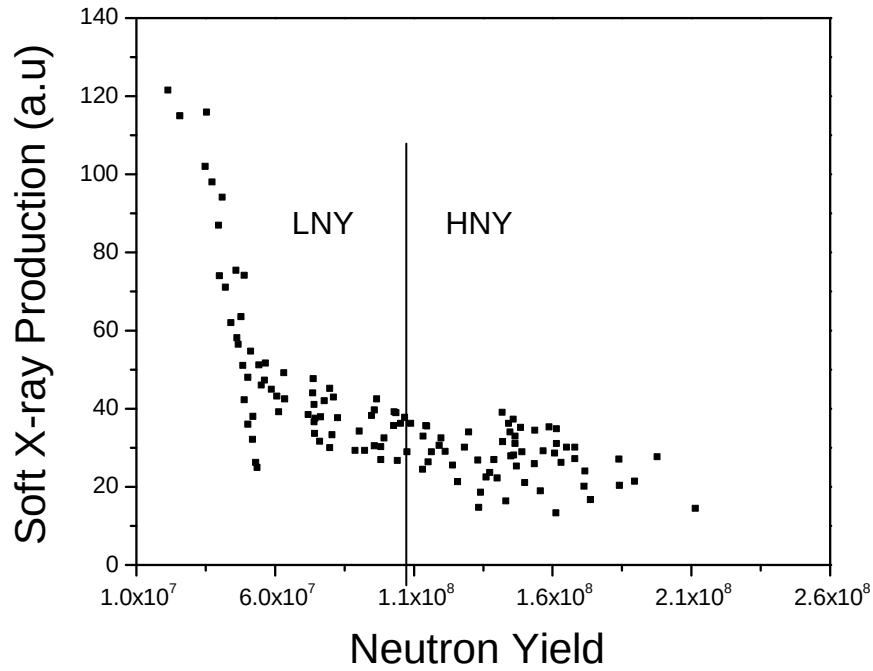


Figure 6.24: Soft X-ray production as a function of total neutron yield.

The average neutron yield for the optimum pressure was 10^8 neutrons per shot with the standard deviation of 5×10^7 neutrons per shot. The category LNY represents the discharges with the neutron yield below 10^8 neutrons per shot and the HNY represents the discharges with the neutron yield higher than 10^8 neutrons per shot.

The graphs in Figure 6.24 show the variation of the soft X-ray production with the neutron yield at optimum pressure. It is seen that for both categories LNY and HNY, the soft X-ray production decreases with the increasing neutron yield. But, the decreases in the soft X-ray production is sharper for LNY as compared to the decrease for HNY category.

6.4.2 NEUTRON ENERGIES

The average energy of the neutrons was calculated using the neutron time-of-flight peak-to-peak method of analysis (signals from PM2 and PM3). The histograms in Figure 6.25 show the measured values of the energy in the axial and radial direction. For more than 60% of the recorded discharges, the average energy of the neutron (in the radial direction) was estimated to be 2.48 MeV with the standard deviation of 0.0483 MeV.

The peak of the signal from PM1 appears 10 ns earlier than the signal from PM2 indicating that the average energy of the neutrons emitted in the axial direction is higher than the neutrons emitted in the radial direction. The peak of PM1, for more than 60% recorded

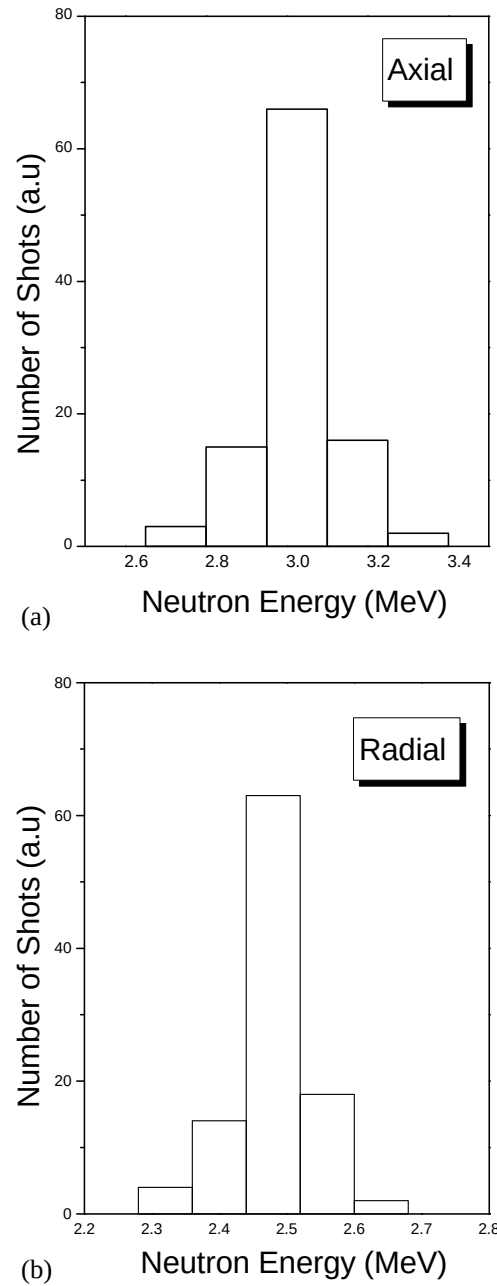


Figure 6.25: Histograms of the experimentally determined neutron energies in (a) axial and (b) radial direction.

discharges, corresponds to the neutrons of average energy 3 MeV with the standard deviation of 0.090 MeV. The result agrees with the earlier workers [204-206]. For

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$D(D,n)He^3$ reaction, the energies of the neutron moving in the forward direction for a single interaction were calculated [206]. Figure 6.26 shows a graph of the neutron energy moving in the forward direction against the energy of the deuteron interacting with the stationary deuteron target (beam-target).

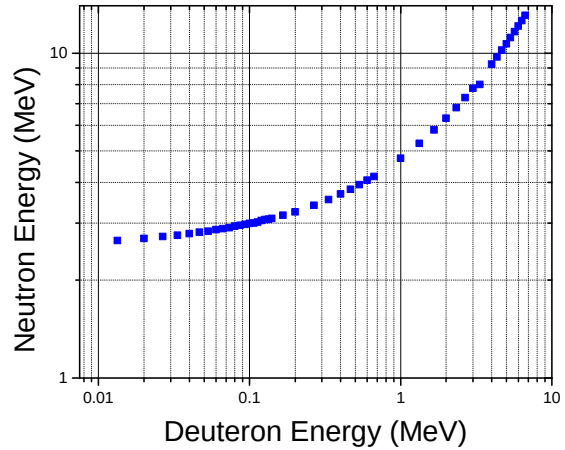


Figure: 6.26: Energies of the forwardly moving neutrons from $D-D$ beam-target reaction as a function of the deuteron beam energies.

The graph shows that the energy of the neutrons increases with the energy of the reacting deuteron

beam, and the 100 keV deuterons are responsible for the production of 3 MeV neutrons. This value of deuterons' energy is within the spectrum of the beam energies (80-250 keV), with 90-120 keV the most frequent values, measured directly from our plasma focus (discussed in section 6.2.2).

The presence of the high energy neutrons in the axial direction is thus attributable to the high energy reacting deuterons moving in the axial direction.

6.4.3 NEUTRON ANISOTROPY

Three time-resolved neutron detectors were employed to investigate the emission of the neutron from the plasma focus. The main aim of the investigation was to measure the anisotropy of the neutron emission. Neutron flux anisotropy is of paramount importance in the determination of the roles played by different neutron production mechanisms in the plasma focus. The physical mechanisms of the $D-D$

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reaction are directly correlated to the ion distribution function responsible for the neutron production.

In order to measure the neutron anisotropy, the two detectors, PM1 in the axial direction (at 0° with respect to pinch zone) and the nearest (PM2) in the radial direction (at 90° with respect to pinch zone), were placed at the same distance from the pinch. The signals (dY_n/dt) from these detectors were integrated over a fixed time interval Δt (duration of the neutron pulse). The anisotropy α was calculated by:

$$\alpha = \frac{(Y_n)_{PM_1}}{(Y_n)_{PM_2}} = \frac{Y_n \begin{pmatrix} 0 \\ 0 \end{pmatrix}}{Y_n \begin{pmatrix} 0 \\ 90 \end{pmatrix}}.$$

Generally, the amplitude of the PMT signal is dependent on the energy of the neutron. The plastic scintillator (in contrary to the NaI) have very weak dependence of amplitude on energy [207]. Plasma focus is almost monochromatic source of neutrons. Therefore, for our measurements the effect of this fact is negligible.

However, if the energy dependence is taken into account, there are two opposite phenomena affecting the scintillator output signal.

If we suppose that the dependence is linear, the 3.0 MeV neutrons are overestimated because less neutrons will produce more scintillation as compared to 2.48 MeV neutrons. Therefore, the yield is overestimated by a factor of 3.0/2.48. But the interaction of the low energy neutrons with the scintillator material is weaker. Since the scintillator thickness is much smaller than the neutron mean free path, if we assume again a linear dependence of this parameter for neutron energy range 2.48-3.0

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MeV, the 2.48 MeV neutrons are under estimated by a factor of 2.48/3.0. Therefore, the two factors cancel each other out and the equation used for the calculation of anisotropy is accurate and does not require any correction factor.

The neutrons emitted in the axial direction (0° with respect to the anode axis) have to penetrate through a 2 cm thick brass plate on the top of the focus chamber before entering the detector. Also, the neutrons emitted in the radial direction have to penetrate through the iron walls of the chamber. The scattering of the neutrons in these media results in the attenuation of the beam. The attenuation was calculated using the average total neutron reaction cross-sections (for 2.45-5.0 MeV neutrons) for both copper and iron. A correction factor was calculated for the exact measurement of the anisotropy. The measured average neutron anisotropy was $1.48 (\pm 3 \%)$. This result is in fairly good agreement with the results reported by Tiseanu et. al. [87].

As already mentioned, different models have been suggested for the production of the neutrons in the plasma focus. Some models (e.g. boiler model) are based on the assumption that ions and electrons are thermalized in the plasma focus at energies of a few keV, and the anisotropy of the neutron yield results from plasma CM motion. On the other hand, the beam-target model considers a high current of charged particles accelerated in an electric field giving a different production mechanism.

Accurate spectral distribution can distinguish between these models. However, the average value of the neutron anisotropy determined in our experiment is much higher than the expected value 1.12 from the moving boiler model of Bottoms [208].

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The neutron flux anisotropy was calculated for five different pressures 3.0, 3.5, 4.0, 4.5 and 5.0 mbar. Figure 6.27 presents the variation of the average anisotropy with the total neutron yield as a function of pressure. The plots show a similar trend for the variation in anisotropy and the total neutron yield as a function of pressure. For the neutron-optimised pressure (4 mbar), both the neutron yield and the anisotropy show their maximum values. At higher (higher than the optimum pressure) or lower (lower than the optimum pressure) pressures, there is a corresponding decrease both in the total neutron yield and the anisotropy. The reason may be related to the low neutron production in these regimes.

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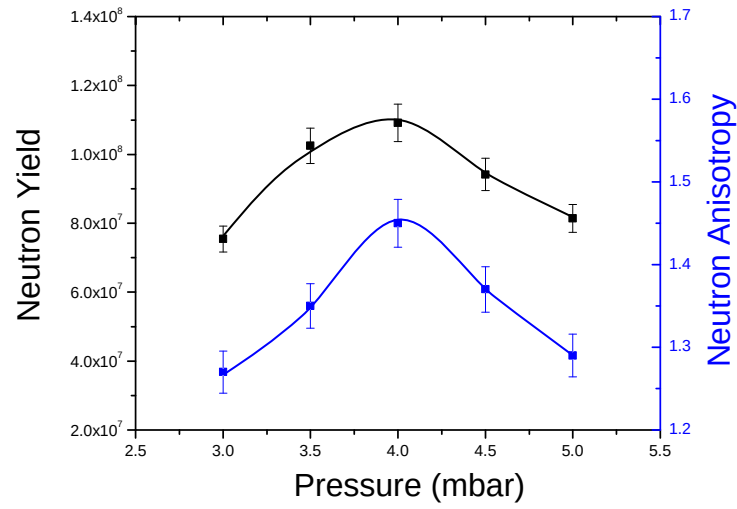


Figure 6.27: Variation of the anisotropy and the total neutron yield as a function of pressure.

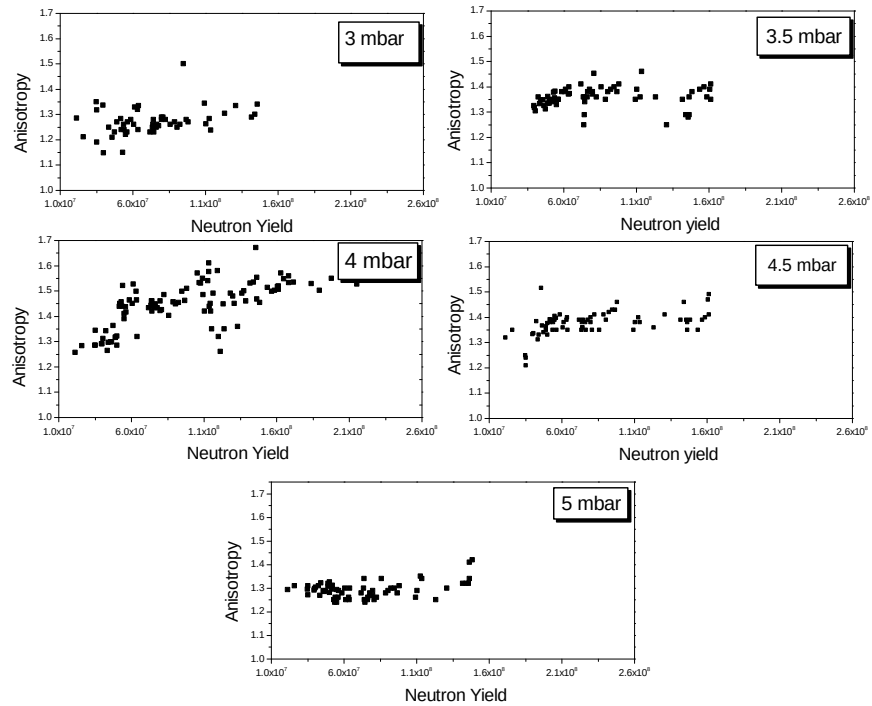


Figure 6.28: Anisotropy as a function of the total neutron yield at different pressures.

RESULTS AND DISCUSSION

Figure 6.28 illustrates the variation of the anisotropy with the neutron yield for five different pressures.

For all pressures, a common trend is observed: the value of the anisotropy is higher for the discharges with higher neutron yield. Since, the neutron yield increases with the increase in the energy of the reacting deuterons [87], therefore, the increase in the anisotropy with the increase in the total neutron yield may be related to the increase in the energy of the axially moving reacting deuterons for the discharges with the high neutron yield.

In the following paragraphs, the results obtained from the discharges at the neutron-optimised pressure are presented. For the discussion of the results, the data were classified again into two categories, namely: low neutron yield (LNY), and high neutron yield (HNY).

The average value of anisotropy for the category LNY is 1.43 with the standard deviation of 0.09. For HNY, the average value is 1.53 with a standard deviation of 0.075. Figure 6.29 presents the dependence of the neutron anisotropy on the neutron yield.

The other workers [28, 57 - 62] have also reported measurements of the neutron anisotropy from *D-D* nuclear fusion in DPF. From the anisotropy of the neutron yield, it is suggested that the beam-target neutron generation accounts for the dominant part of the neutron production in the plasma focus.

In our earlier reported results (discussed in section 6.2.2), deuteron energy spectra were correlated with the neutron production. Our results showed that the deuterons of energy 80-250 keV, moving in the axial direction, are responsible for the production of the majority of the neutrons. A strong correlation between the neutron

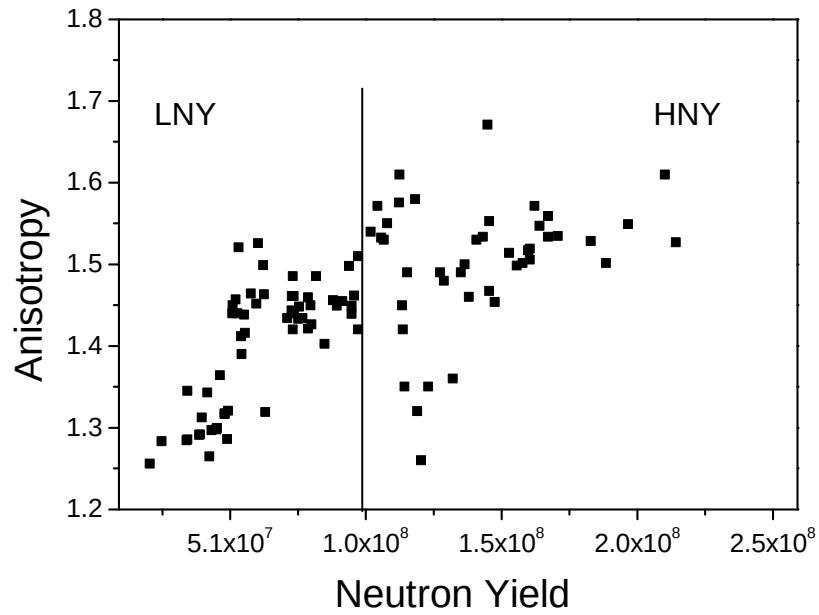


Figure 6.29: Anisotropy as a function of the total neutron yield.

yield and the forwardly directed high energy deuteron beam emitted from the pinch region of the plasma focus was found. The experimentally observed anisotropy supports the existence of strong axial ion beams, which are responsible for the production of the neutrons. Thus, the results obtained from this experiment strongly support our previous results. The charged particle flux at optimum pressure of high neutron emission, exhibits a fluence anisotropy of 6 [209], whereas the neutron anisotropy is 1.5. One may explain the observed data by considering the existence of some anomalous magnetic field in the pinch region. This magnetic field is introduced in the pinch region from the very beginning of the discharge [210], and that its flux density rises to a very high value due to the trapping and the compression of the field lines by the plasma sheath. One may speculate that the pressure of the working gas generates favourable conditions for the plasma (density and pinch filament diameter)

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to trap the energetic deuterons. These gyrating deuterons then react with the plasma column generating high neutron emission with low fluence anisotropy in accordance with the generalised beam-target mechanism [127].

The low anisotropy values for the low neutron yield can be explained on the basis of the dependence of the neutron emission on the deuteron energy. The neutron emission in the forward direction is greater for the high energy deuterons. The low value for the anisotropy in LNY regime might be explained by the lack of the accelerated particle within the hole of an ion Larmor diameter. But, this simple hypothesis cannot however be proven, because the plasma parameters in the ion micro sources are not known exactly. Considering the experimental values for our plasma focus, (i.e. the minimum radius $r_{min} = 1-2$ mm and the discharge current of 150 kA) one can estimate that the magnetic field at the pinch surface reaches $B_{max} \sim 150$ kG. Under this condition, for deuterons of energy 50-250 keV, one gets the Larmor radius of 0.6-1.5 mm. If this is the situation, then most of the deuterons of energy less than 80 keV will be confined (trapped) by the current sheath. The probability for the high energy (> 80 keV) deuterons to be trapped is much less, depending on the pinch radius. Hence, one can suspect the neutron anisotropy to be low for the discharges with the low neutron yield. But, in order to get a real picture, more data are required.

6.4.4 NEUTRON ANISOTROPY AND SOFT X-RAY PRODUCTION

The graph in Figure 6.30 shows the variation of the anisotropy and the soft X-ray production with pressure. It is seen that for lower (lower than 4.0 mbar) and higher (higher than 4.0 mbar) pressures, the soft X-ray production is higher as compared to the value at

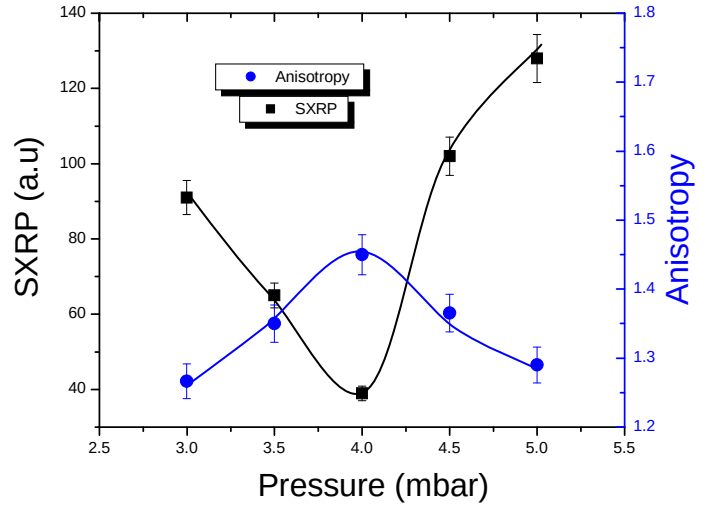


Figure 6.30: Anisotropy and the soft X-ray production as a function of pressure.

4.0 mbar. The anisotropy has a lower value at lower and higher pressures as compared to the value at the neutron-optimised pressure. The low value of the anisotropy for higher and lower pressures is related to the low neutron yield in these regimes.

The variation of the anisotropy with the soft X-ray production at different pressures is shown in Figure 6.31. For the pressures other than 4 mbar (the neutron-optimised pressure), the variation of the anisotropy with the soft X-ray production is not significant. But, for the neutron-optimised pressure, it is clear from the graph shown in Figure 6.32 that the soft X-ray production decreases with the increasing value of the anisotropy. Thus, the increase in the neutron yield is associated with an increase in the anisotropy and decrease in the soft X-ray production.

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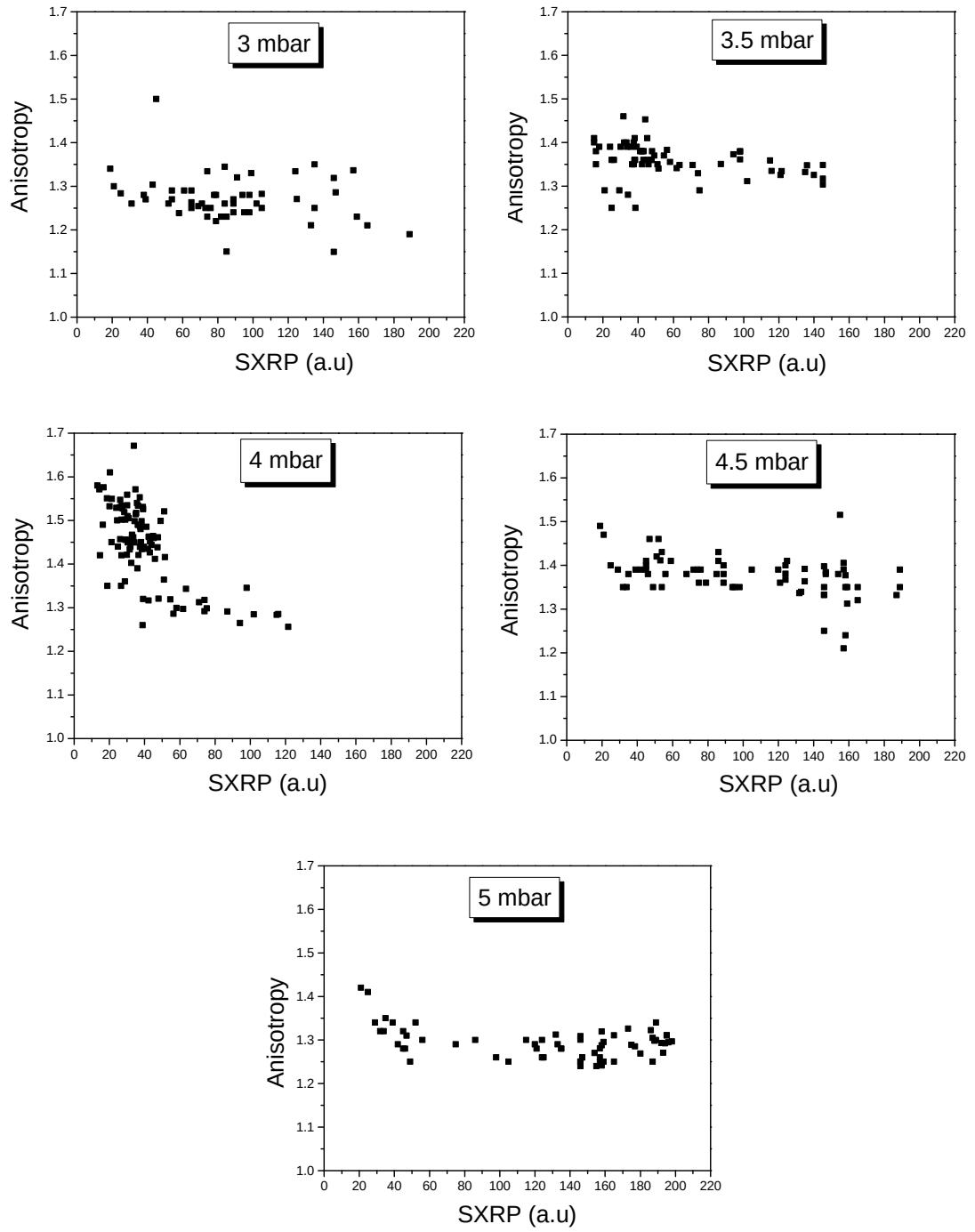


Figure 6.31: Anisotropy as a function of the soft X-ray production at different pressures.

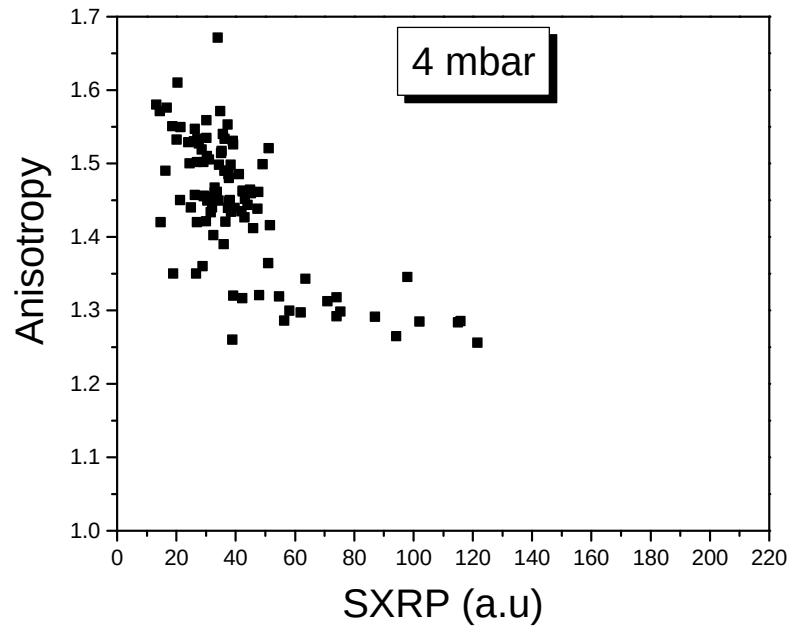


Figure 6.32: Variation of the anisotropy with the soft X-ray production.

6.5 FOCUS DYNAMICS AND RADIATION EMISSION

Optical investigations were performed for the understanding of the plasma focus dynamics. The laser shadowgraphy method was employed to record the shadowgraphs of a specific region at the open end of the electrode system. A detailed description of the experimental set-up has been presented in chapter 5.

6.5.1 PRELIMINARY INVESTIGATIONS OF GROSS DYNAMICS

Based on the plasma focus dynamics, the discussion of the shadowgraphic observations are divided into three main phases:

- i) Pre focus (radial) phase,
- ii) Focus (pinch) phase,
- iii) Post focus (decay) phase.

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This classification evaluates the plasma focus dynamics from the beginning of the radial phase until the formation of the bubble-like structures.

All shadowgraphs were taken at 14.5 kV/4.0 mbar (neutron-optimised pressure) using deuterium as a working gas.

6.5.1.1 PRE-FOCUS (RADIAL) PHASE

When the current sheath reaches the open end of the central electrode, it moves both in the radial and the axial direction, developing a symmetrical funnel shaped profile driven by $\vec{j} \times \vec{B}$ force.

During this phase, the discharges with the high neutron yield show an axis-symmetrical, large radius of curvature and smooth profile. Examples of such shadowgraphs are shown in Figure 6.33.

In some discharges, the development of the Rayleigh-Taylor

instabilities were observed. These instabilities develop due to the fast moving current sheath and density gradients. Such kind of instabilities occur in two manners: symmetrically or asymmetrically. Examples of the shadowgraphs exhibiting symmetrical (bottom frame) and asymmetrical (top frame) instabilities are shown in Figure 6.34. This figure (bottom) illustrates a symmetrical instability distorting the sheath, whereas, top frame shadowgraph is an example of asymmetrical instability.

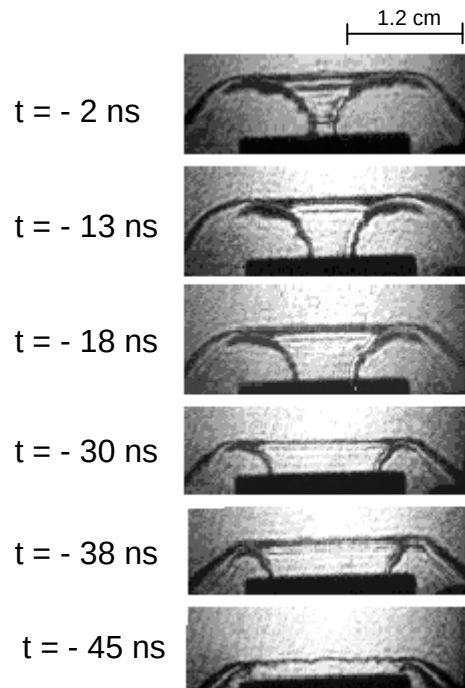


Figure 6.33: Shadowgraphs during the radial phase exhibiting high neutron yield. $Y_n = 1.6 \times 10^8$.

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Due to these instabilities, only one sector of the plasma sheath is disturbed. During the normal evolution of the plasma focus discharges producing average or higher than the average neutrons yield, the sheath is fluted by the interchange instabilities inducing distortions in the radius of curvature of the plasma sheath. This distortion results in the necking effect close to the anode surface, or even inside the inner electrode. In such a situation, the hard X-ray yield is enhanced, and also the soft X-ray production increases due to the large emission from the anode material.

The third kind of discharges observed, exhibit low neutron yield. For such discharges, the shadowgraphs show an asymmetrical, diffused current sheath, which may cause the plasma column to break long before the end of the compression phase with no proper pinch formation. Examples of such shadowgraphs are shown in Figure 6.35. This asymmetry might rise due to non-symmetrical breakdown in the beginning of the axial phase.

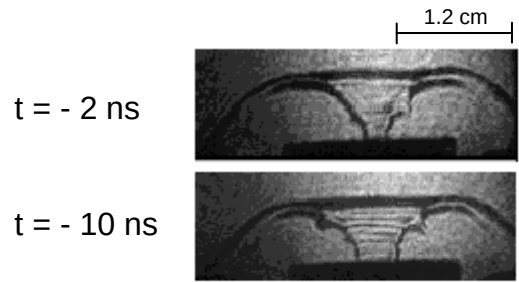


Figure 6.34: Shadowgraphs illustrating Rayleigh-Taylor instabilities. $Y_n = 2 \times 10^8$.

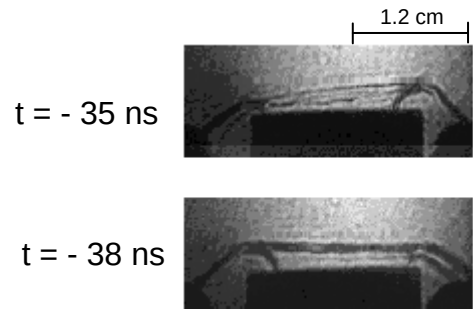


Figure 6.35: Shadowgraphs during the radial phase with asymmetrical current sheath. $Y_n = 0.2 \times 10^8$.

6.5.1.2 FOCUS (PINCH) PHASE

Figure 6.36 shows a symmetrical necking effect at the instant $t = +2$ ns about 2 mm away from the anode. The plasma column is 6 mm in length and 2 mm in diameter. A high neutron yield accompanied by an intense hard X-ray burst is observed for such kind of discharges.

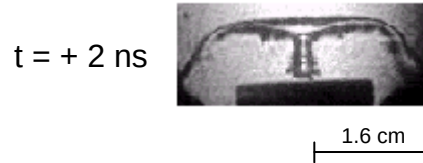


Figure 6.36: Shadowgraph exhibiting a necking effect. $Y_n = 1.8 \times 10^8$.

A series of the discharges with very low neutron yield is shown in

Figure 6.37. This figure illustrates the plasma focus dynamics at the end of the radial phase and during the pinch stage.

The axial length of the plasma column is 10-11 mm (longer than the length observed for the discharges with higher neutron yield). This is correlated to a longer time for the radial compression phase. In these cases, the plasma column is smooth at the end of the radial phase (a), or it breaks during the compression phase far away from the anode surface (b). Figure 6.37 (c and d) show that the column does not break and goes smoothly up till the maximum compression and then expands (e). There is no indication of the growth of extremely short wavelength instabilities. The neutron yield for these discharges is very low. The reason may be related to the absence of instabilities, which cause the acceleration of the charged particles responsible for the neutron and hard X-ray emission. After the maximum compression, the plasma column starts to expand until the growth of the $m=0$ instability.

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Figure 6.38 shows the expansion of the plasma column (a) and the growth of the $m=0$ instability (b and c) 8 and 12 ns, respectively, after the first compression to the minimum radius. A Rayleigh-Taylor instability distorting the right wing of the plasma sheath is also seen (c). For such kind of discharges, the neutron yield is higher, because the localised instabilities especially, the $m=0$ compress the plasma column immediately after the pinch. These instabilities cause high electric fields that accelerate the ions to very high velocities. These highly energetic ion beams produce neutrons by interaction with the residual plasma (beam-target interaction).

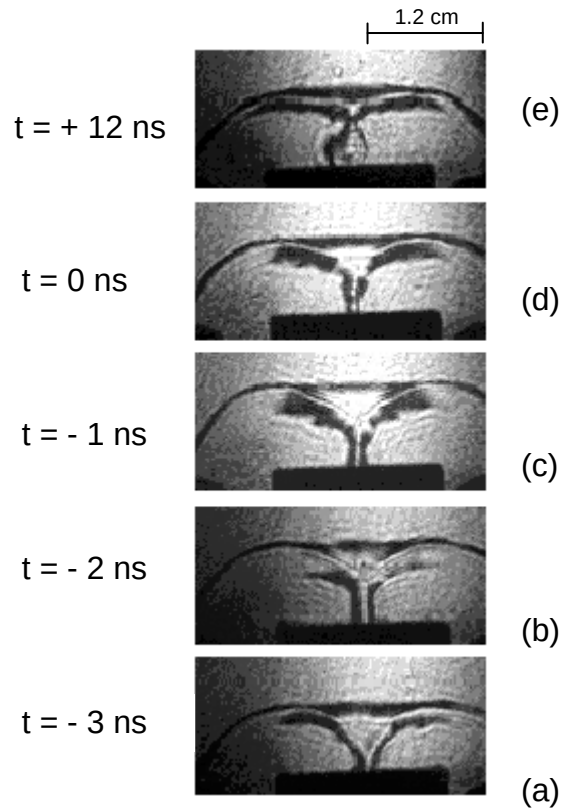


Figure 6.37: Shadowgraphs during the pinch phase exhibiting low neutron yield. Average $Y_n = 0.3 \times 10^8$.

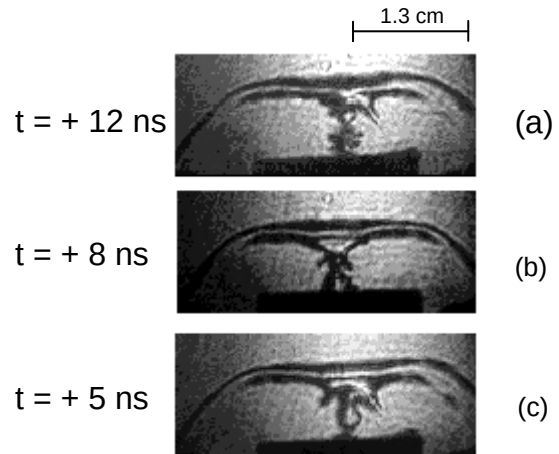


Figure 6.38: Shadowgraphs during the pinch phase for discharges exhibiting high neutron yield. $Y_n = 2 \times 10^8$.

6.5.1.3 POST-FOCUS (DECAY) PHASE

After the plasma column breaks, the remaining plasma of the disrupted pinch moves axially forming a bubble-like structure. Figure 6.39 shows the movement of the filaments up till the formation of the bubble-like structures.

Figure 6.40 shows the shadowgraphs of the bubble-like structure, which appear 40–60 ns after the compression and last for 100-150 ns. Usually, this formation occurs after a strong focusing event [212].

Figure 6.41 shows the shadowgraphs of the emission of the copper vapours from the inner electrode (150-250 ns after the pinch formation).

At the instant at which, due to the electron bombardment, copper vapours are emitted from the anode, copious amounts of the soft X-rays are produced due to the electron beam bombardment as indicated by the soft X-ray signals from the spectrometer (see Figure 6.19).

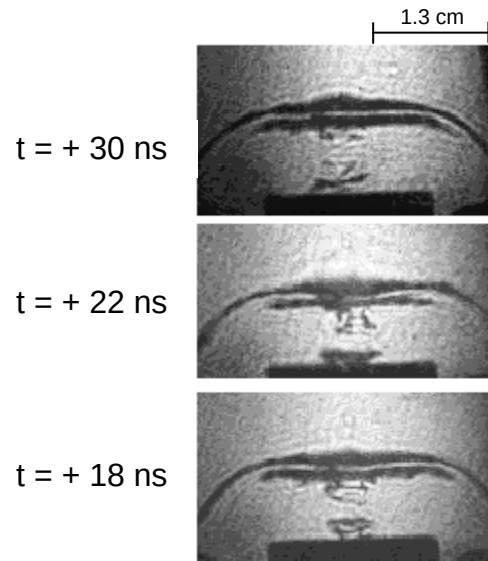


Figure 6.39: Shadowgraphs during the decay phase. Average $Y_n = 1 \times 10^8$.

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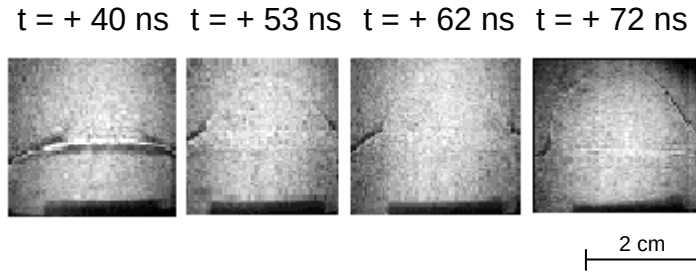


Figure 6.40: Shadowgraphs during the post focus phase forming bubble-like structures. Average $Y_n = 1.6 \times 10^8$.

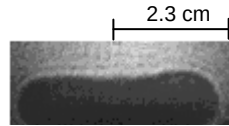


Figure 6.41: Shadowgraphs showing the emission of the copper vapours from the inner electrode. $Y_n = 1 \times 10^8$.

6.5.2 FOCUS EVOLUTION AND RADIATION OUTPUT

Single frame shadowgraphy system enabled to study the dynamics of the focus on shot-to-shot basis. But, due to shot-to-shot variation, conclusive elements on the real dynamics could not be stated. Therefore, it was necessary to take more than one shadowgraph of the same discharge at different instants. For this purpose, a three-frame shadowgraphy system was designed, developed and implemented.

The observations were recorded with a temporal accuracy of 1 ns.

In the following sections, shadowgraphic results obtained at neutron-optimised pressure are presented.

In Figure 6.42, a sequence of a shadowgraphs during the radial phase is compiled. At the beginning of the compression phase, the plasma is radially accelerated. During the initial stage of the radial phase from -30 ns to -12 ns,

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velocities of 10-23 cm/ μ s, respectively, were determined in our experiment. Usually, during this part of the radial inward motion, Rayleigh-Taylor instabilities occur due to the variation in the values of the radial density gradients.

At the later stage of the radial phase (starting from -5 ns up till the compression), velocities up to 32-42 cm/ μ s were observed in our experiment.

The instant at which the maximum compression occurs, i.e., at $t = 0$, a minimum radius $r_{min} \approx 0.4$ -1.5 mm and a maximum length $l \approx 6$ -10 mm, were observed.

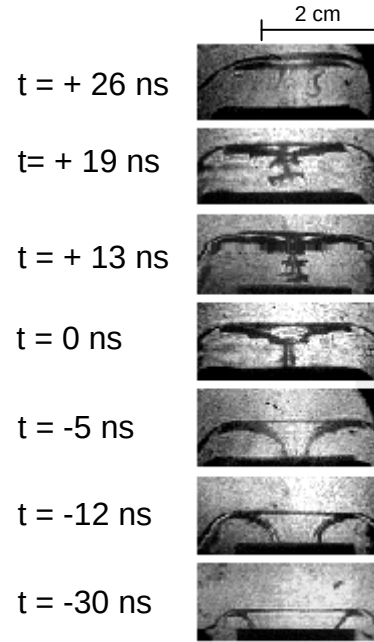


Figure 6.42: Sequence of compression in plasma focus. $Y_n = 1.2 \times 10^8$.

At the instant of the maximum compression, an intense hard X-ray burst is observed. After the compression, an $m=0$ instability develops rapidly and causes another hard X-ray burst. At this stage, the plasma is heated to 1-2 keV ranges, high electric fields are generated, and particles are accelerated to high energies. Fast electrons give rise to the hard X-rays bursts. The fast ions propagating in the axial direction with the velocities up to 2×10^8 cm/s ionize the gas on their path [213]. These fast moving ions (deuterons) are responsible for the emission of neutrons. The emission duration lasts until the column decays completely.

In order to study the different phenomena involved in the focus dynamics and to identify their role in the radiation emission, the results are again classified as: I) pre-focus phase, II) focus phase and III) post-focus phase. Each of these phases is

subdivided (based on the neutron yield) into two categories as: low neutron yield (LNY) and high neutron yield (HNY).

6.5.2.1 PRE-FOCUS (RADIAL) PHASE

This is the phase in which the current sheath starts from the open end of the anode and accelerates towards the center. This phase ends when the current sheath meets the axis ($t=0$).

A) LOW NEUTRON YIELD (LNY)

Figure 6.43 shows a three-frame sequence of a discharge during pre-focus phase. The neutron yield for the discharge is 50% of the average value, and the soft X-ray production is 2.3 times the average value. The value of the anisotropy is 1.23. The voltage signal profile shows multiple spikes indicating that the discharge belongs to the multiple-compression regime. The experimentally measured radial velocity at instant $t_{av} = -7.5$ ns is 26 cm/ μ s ($\pm 15\%$). At the end of the phase ($t = -5$ ns), a Raleigh-Taylor instability (encircled) developed earlier seems to be stabilised, as the plasma sheath looks smoother.

Figure 6.44 shows another discharge with LNY during the pre-focus phase. In this discharge an interesting feature is observed. The current sheath is much thicker than “normal”. At the instant of maximum compression (i.e. at $t=0$), the column seems to be compressed at one point relatively far from the face of the anode, while, near the tip of the anode the sheath is yet to reach the axis. It is worth noting the axial elongation of the plasma as well as the short radius of curvature of the sheath. Also, no short wavelength interchange instabilities are observed. The low neutron yield

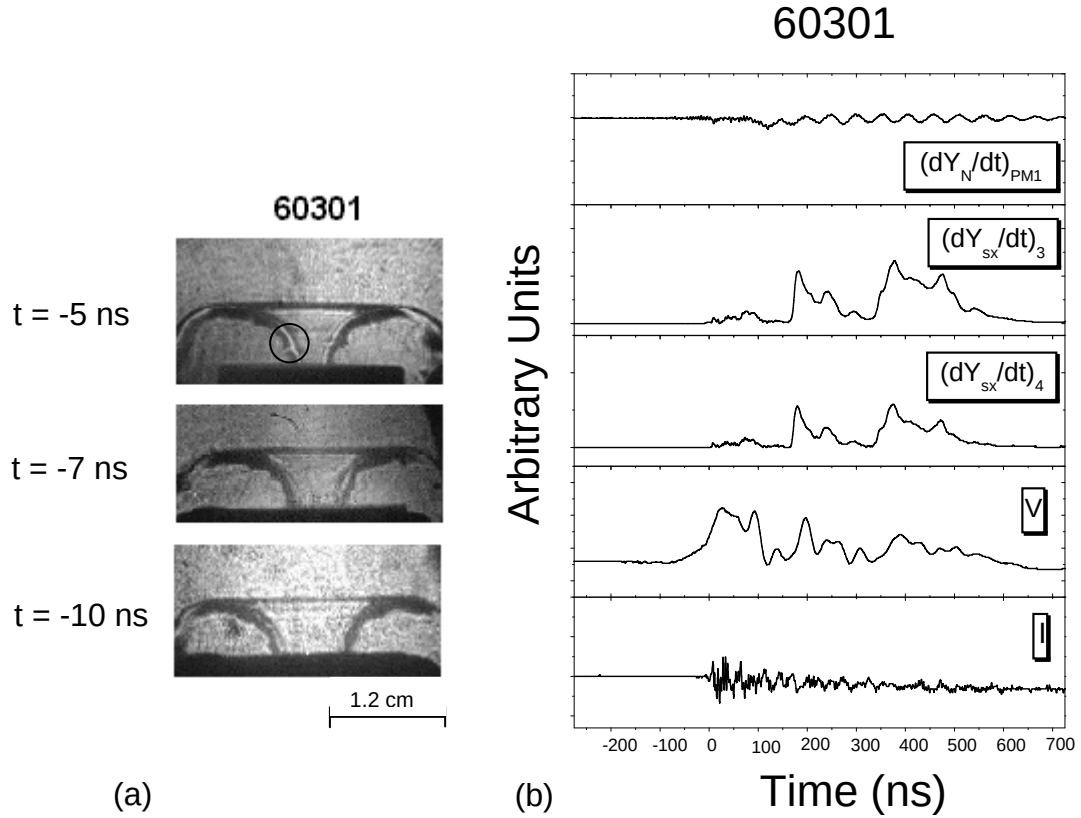


Figure 6.43: A three-frame sequence of a discharge during pre-focus phase; b) Corresponding signal profiles of dY_N/dt , dY_{HX}/dt , dY_{SX}/dt , V and I . $Y_n = 0.5 \times 10^8$.

could then be explained by a significant loss of the plasma in the axial direction (towards and away from the anode), which reduces the number of particles to be confined, thus, the final density and temperature have low values. For this discharge, the neutron yield is 50% of the average value. The soft X-ray production is only 1.2 times the average. The value of the anisotropy is 1.25. Experimentally measured average radial velocity for this discharge at $t_{av} = -2$ ns is $30 \text{ cm}/\mu\text{s}$ ($\pm 15\%$).

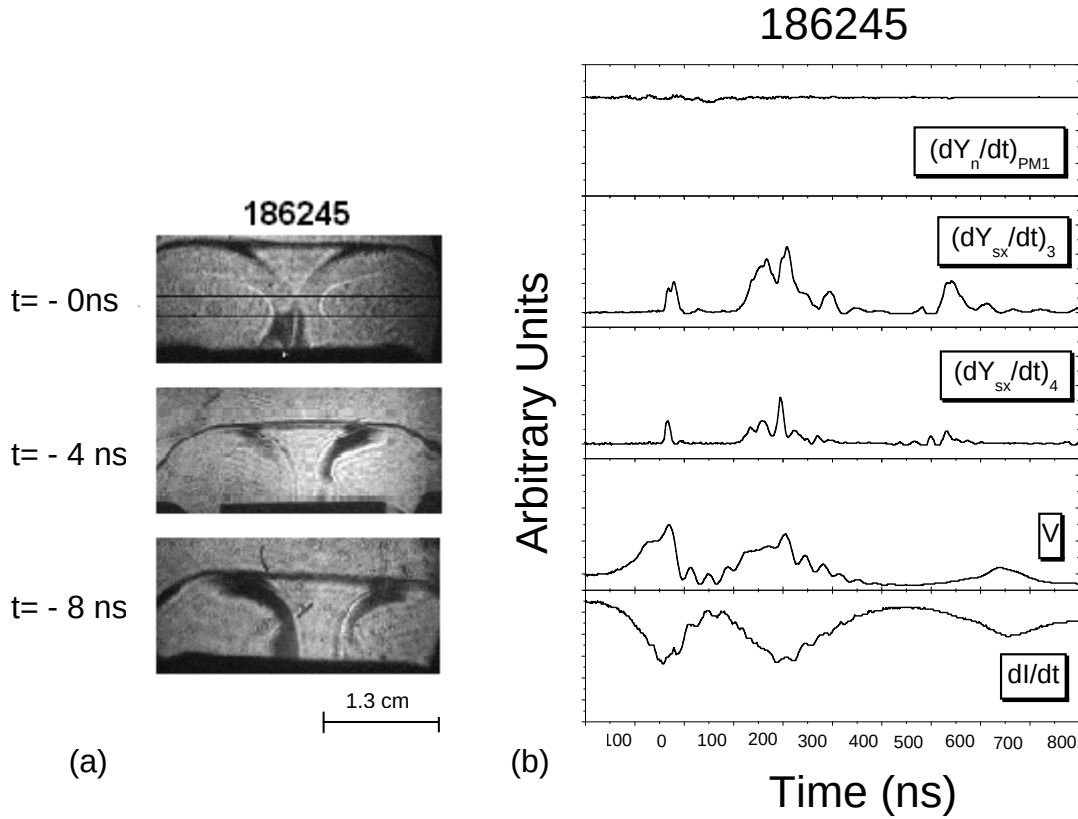


Figure 6.44: A three-frame sequence of a discharge during pre-focus phase; b) Corresponding signal profiles of dY_n/dt , dY_{HX}/dt , dY_{SX}/dt , V and dI/dt . $Y_n = 0.4 \times 10^8$.

The comparison of the two low neutron yield discharges shows that the implosion speed alone is not a critical parameter for neutron production. The compression may be fast, but if the geometrical profile of the sheath favours significant losses, the confinement efficiency decreases. In both the discharges, no very high local electric fields are produced, so not many electrons can be accelerated at high energies to produce a significantly high hard and soft X-ray output.

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Therefore, one can say that in these type of discharges, the energy transfer from the driver to the plasma is extremely inefficient.

Figure 6.45 shows a three-frame sequence of a discharge which exhibits average neutron yield and soft X-ray production. The value of the anisotropy is 1.45. The sheath profile is smooth, without the presence of Raleigh-Taylor (RT) instability. Experimentally measured average

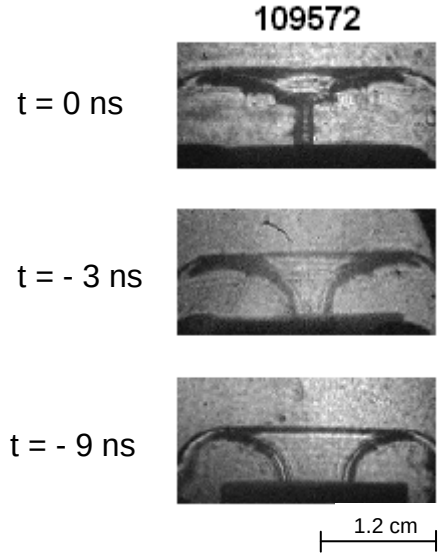


Figure 6.45: A three-frame sequence during pre-focus phase. $Y_n = 1 \times 10^8$.

radial velocity at the instant $t_{ave} = -6$ ns is -28 cm/ μ s ($\pm 15\%$). The diameter of the plasma column is 1.6 mm ($\pm 10\%$), and its effective length is 5.3 mm ($\pm 10\%$).

B) HIGH NEUTRON YIELD (HNY)

Figure 6.46 shows a three-frame sequence of a discharge exhibiting high neutron yield. In this particular discharge, the development of RT instability (encircled) is seen on both the wings of the sheath. The wavelength of the instability at $t = -6$ ns is 2.58 mm ($\pm 10\%$). Normally, the growth of instability is associated with a high neutron yield and intense hard X-ray bursts. The role of these instabilities on the neutron yield and X-rays (soft and hard) will be discussed in detail later. The neutron yield for this discharge is 1.6 times the average value, and the soft X-ray production is 60% of the average value. The anisotropy value for this discharge is 1.54.

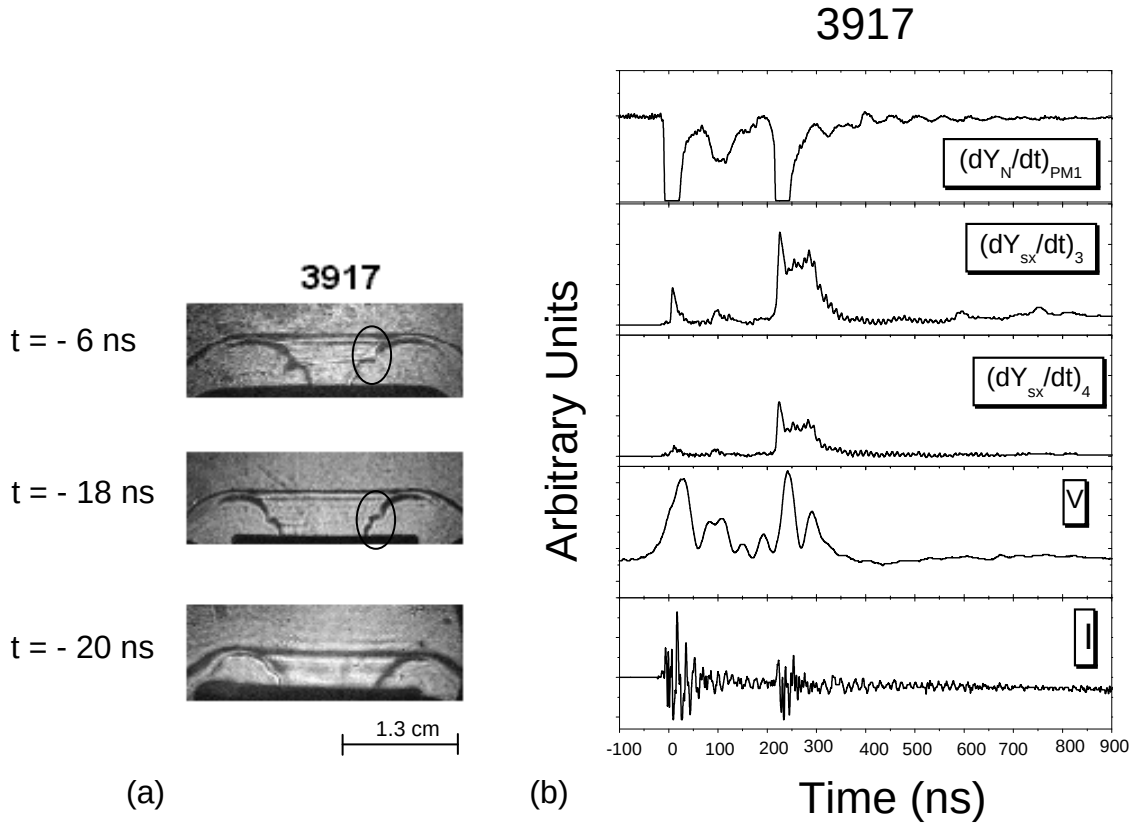


Figure 6.46: A three-frame sequence of a discharge during pre-focus phase; b) Corresponding signal profiles of dY_n/dt , dY_{HX}/dt , dY_{SX}/dt , V and I . $Y_n = 1.6 \times 10^8$.

The average radial velocity at the instant $t_{ave} = -19$ ns is $21\text{cm}/\mu\text{s}$ ($\pm 15\%$) is 19% higher than the value of the average radial velocity at the same instant in this category. This value of the average radial velocity is 36% higher than the measured velocity at the same instants for the LNY category.

6.5.2.2 FOCUS (PINCH) PHASE

The results presented in this section describe the shadowgraphs taken during the pinch phase of the radial motion of the current sheath. Again, the results are divided into two categories namely: LNY and HNY.

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A) LOW NEUTRON YIELD (LNY)

Figure 6.47 shows a sequence of a three-frame shadowgraph, along with the corresponding time-resolved neutron, X-ray (soft and hard), current and voltage signal profiles, taken during the focus phase. The discharge belongs to multiple-compression regime, as indicated by the voltage signal profile. The neutron yield for this discharge is 45% of the average value, but the soft X-ray production is 3 times the average value. The neutron anisotropy is 1.23.

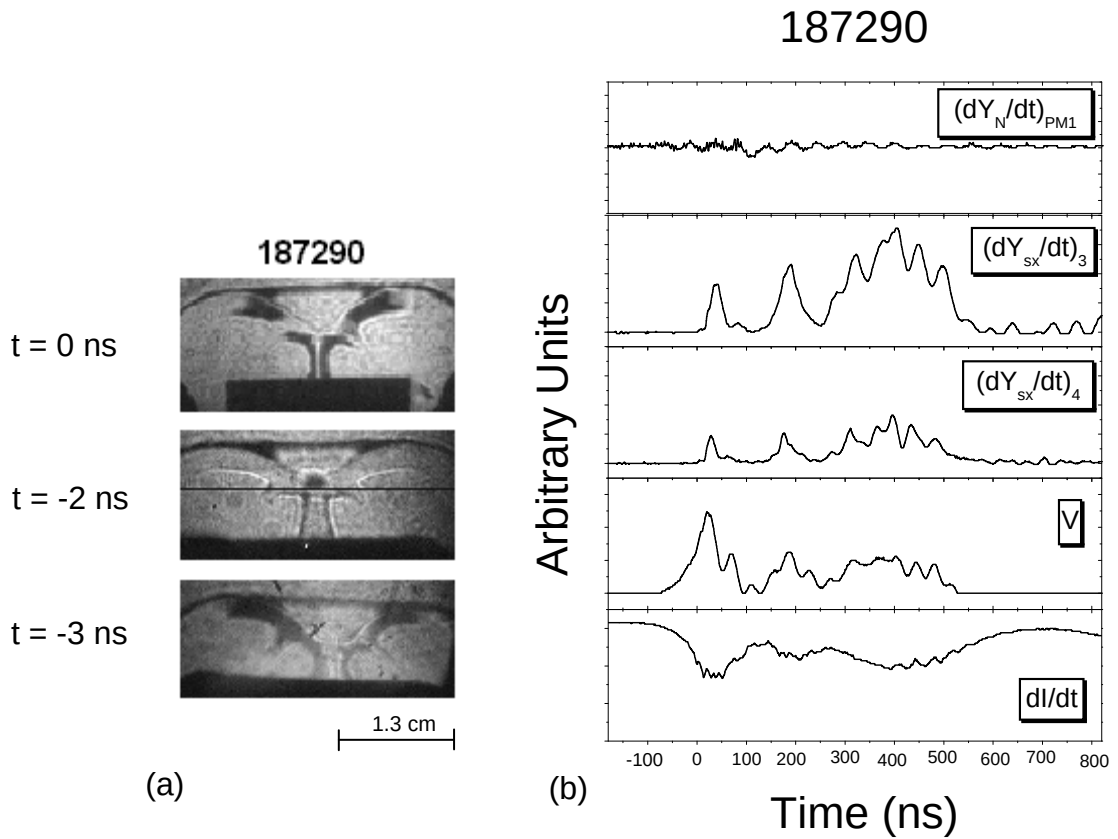


Figure 6.47: A three-frame sequence of a discharge during focus phase; b) Corresponding signal profiles of dY_N/dt , dY_{HX}/dt , dY_{SX}/dt , V and I . $Y_N = 0.45 \times 10^8$.

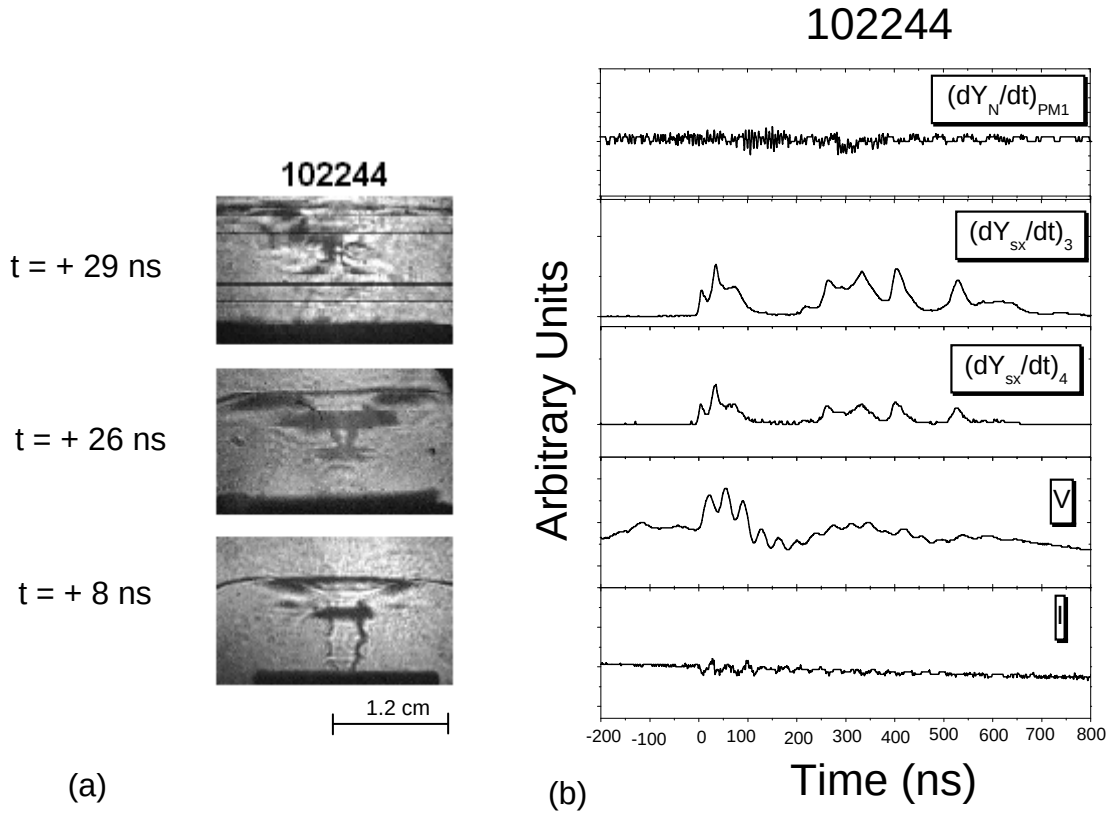


Figure 6.48: A three-frame sequence of a discharge during focus phase; b) Corresponding signal profiles of dY_N/dt , dY_{HX}/dt , dY_{SX}/dt , V and I . $Y_n = 0.45 \times 10^8$.

The development of RT instabilities is seen on both wings of the section of the current sheath. Usually, the high neutron yield is attributed to these instabilities, but for this discharge, even though there are instabilities of large amplitude, the neutron yield is very low.

The radius of the focused column is $0.9 (\pm 10\%)$ mm and the length is $5 (\pm 10\%)$ mm. The pinch radius is $1.36 (\pm 10\%)$ times smaller than the average radius, and the length is 46% shorter as compared to the average value in this category. These kind of discharges emit intense burst of soft X-rays. The soft X-ray

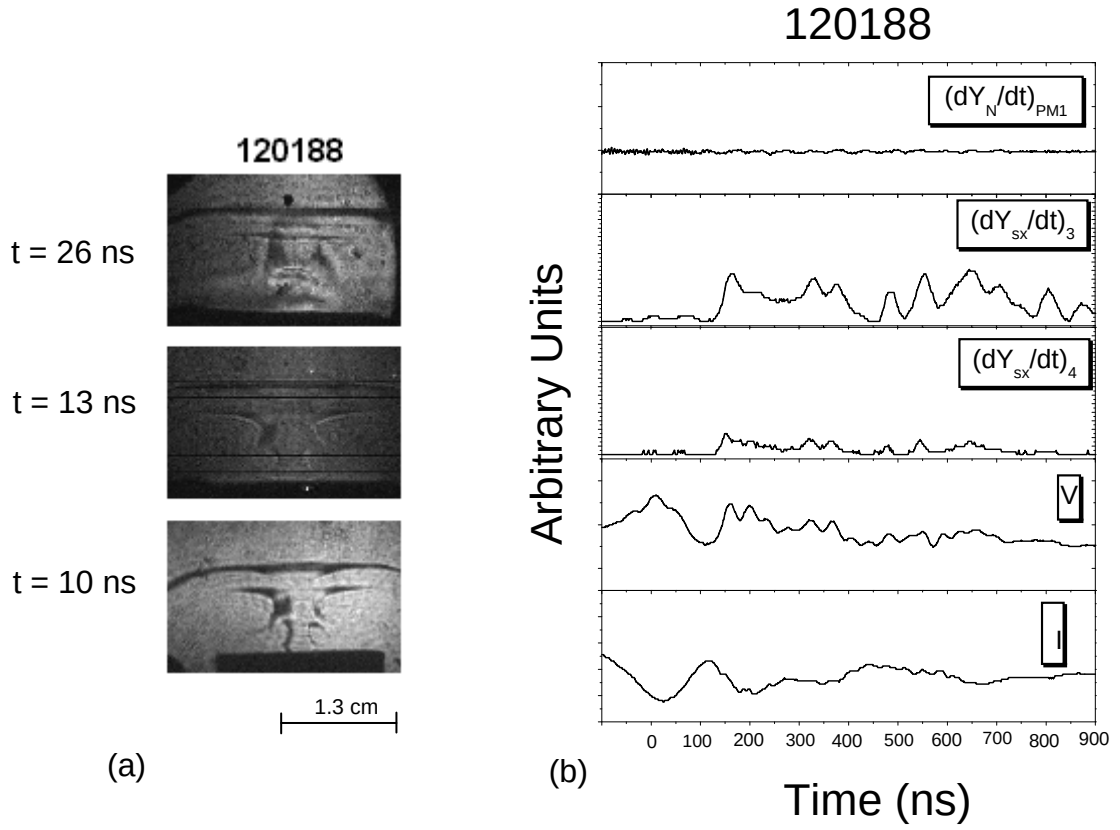


Figure 6.49: A three-frame sequence of a discharge during focus phase; b) Corresponding signal profiles of dY_n/dt , dY_{HX}/dt , dY_{SX}/dt , V and dl/dt . $Y_n = 0.45 \times 10^8$.

production is 3 times the average value. The strong X-ray emission is related to the emission of strong relativistic electron beams immediately after the pinch formation.

Figures 6.48 and 6.49 show another two discharges for the LNY category. In Figure 6.48, there are distinct ripples on the surface of the plasma column. These kind of ripples are well known and were reported by H. Herold et. al. from schlieren pictures [214]. These ripples are caused by flute instabilities during the compression or at the later stage. The disturbances are seen on both sides of the surface of the column. The measured wavelength of both disturbances is 2.33 mm ($\pm 10\%$). This kind of ripples could also be caused by the $m=0$ instability, because in the middle and

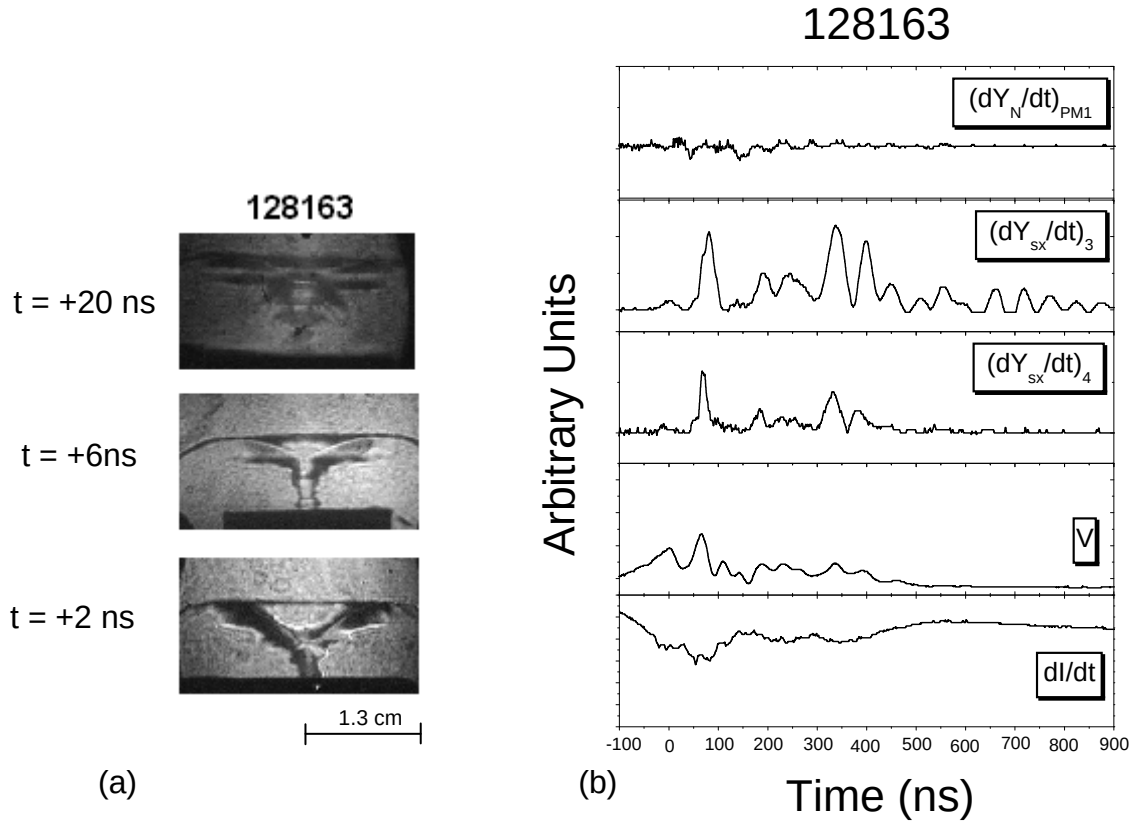


Figure 6.50: A three-frame sequence of a discharge during focus phase; b) Corresponding signal profiles of dY_N/dt , dY_{HX}/dt , dY_{SX}/dt , V and dI/dt . $Y_n = 0.45 \times 10^8$.

the top frame it seems that the column breaks up in the same manner as it does for the $m=0$ instability.

The neutron yield for this discharge is 30% of the average value, while the soft X-ray production is 3.5 times the average value. The value of the anisotropy is 1.20. It appears that the ripples are associated with rich X-rays output rather than neutrons. At this stage this assumption could not be confirmed from our data.

Since the X-ray emission depends on both the electron concentration and electron temperature, the plasma condition may correspond to the increased values of

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both the parameters. These ripples might have been developed from primary current sheath ripples maintained during the compression.

In Figure 6.49, the same kind of situation is seen. The difference is that the column has a broken surface on both the sides (bottom frame). It seems that this kind of ripple is a continuity of a RT instability developed during the initial stage of the radial phase. The signs of the development of the $m=0$ instability is also seen in the same frame. At the later stage of the pinch phase (middle and top frame), the column is broken due to the $m=0$ instability. For this discharge the neutron yield is 30% of the average value and the soft X-ray production is 1.4 times the average value. The anisotropy is 1.25. It appears that this kind of discharges are neither suitable for the neutron yield, nor for the X-ray production. This is in contradiction with the discharge illustrated in Figure 6.46. However, the percentage of this kind of discharges is very low.

Figure 6.50 shows another sequence in LNY category in which the development of $m=0$ instability is clearly seen at the later stage of the pinch (top frame). The neutron yield for the shot shown is 20% of the average value, and the soft X-ray production is 3 times the average value. The anisotropy is 1.30.

The first compression takes place very close to the anode. The “visible” part of the pinch is only a few mm above the anode rim. This geometry of the plasma is obviously in favour of a large particle loss mechanism in the forward direction. During the quiescent phase, the axial part of the current “stays” at the same place. It appears that the upper part of the pinch is compressed later than the lower part (close to the anode), but the diameter of the plasma column does not change. The compression of the upper part of the pinch accelerates the electrons towards the

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anode, and this explains the high X-ray output. However, when the “necking” moves away from the anode rim, it pushes the deuterons in the forward direction. If no significant compression along the column is observed, it means there is no increase in the local electric field.

In the following paragraphs, the results obtained from the measurements of the pinched column dimensions i.e., length (L), radius (R) and aspect ratio (length/radius) (A), for the LNY category are discussed in detail.

Figure 6.51 presents a graph, which shows the variation of the pinched column length with the average radius.

Figures 6.52 and 6.53 shows the dependence of the neutron yield and soft X-ray production, respectively, on the column radius. These plots suggest that the neutron yield is higher for smaller radius, and on the other hand, the soft X-ray production is lower. The compression of the column is one of the important parameter to set the favourable conditions for the plasma focus to work in the neutron-optimised mode. The smaller the radius, the greater is the possibility for the development of the instabilities which create a strong electric field which, in turn, are responsible for the emission of the particle beams (deuterons and electrons).

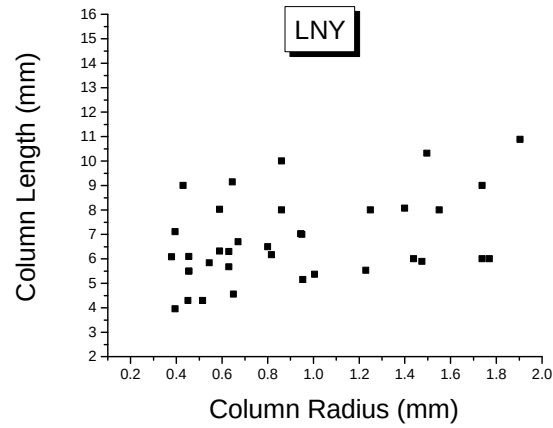


Figure 6.51: Variation of column length with the column radius.

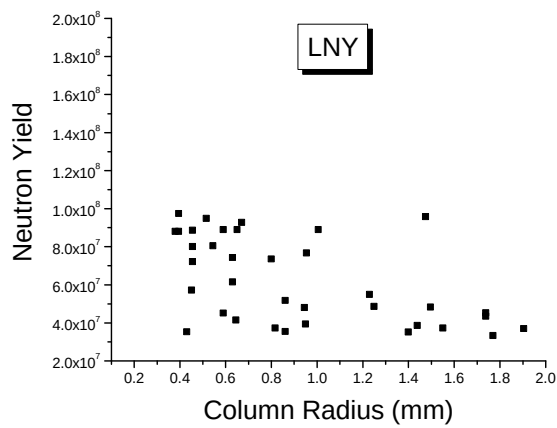


Figure 6.52: Neutron yield as a function of column radius.

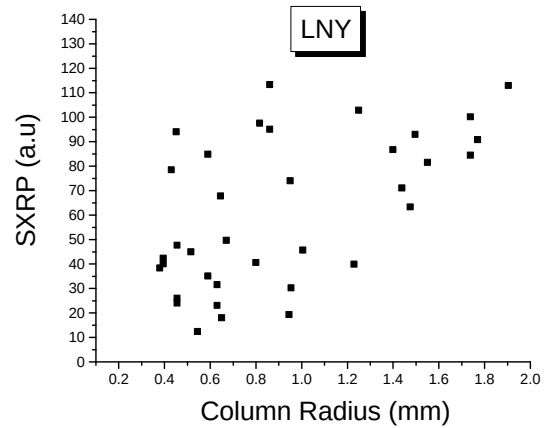


Figure 6.53: Soft X-ray production as a function of column radius.

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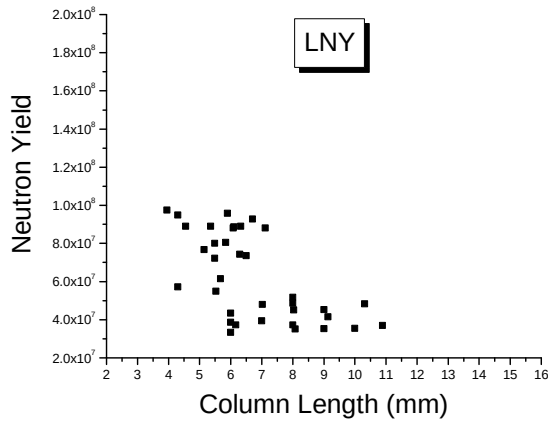


Figure 6.54: Neutron yield as a function of column length.

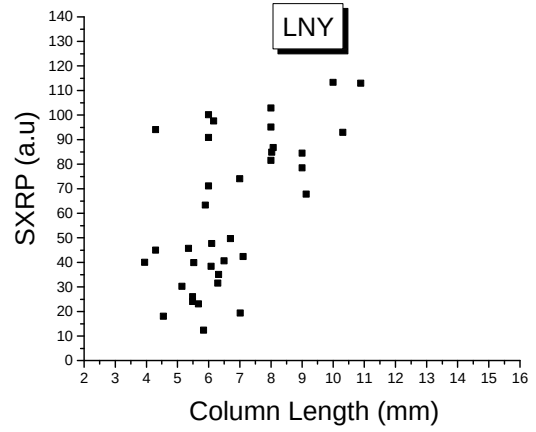


Figure 6.55: Soft X-ray production as a function of column length.

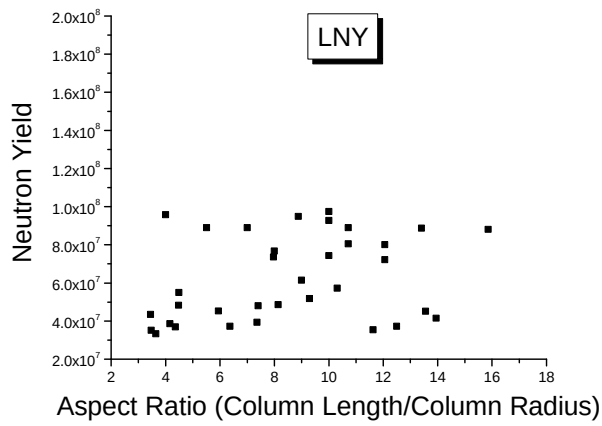


Figure 6.56: Neutron yield as a function of aspect ratio.

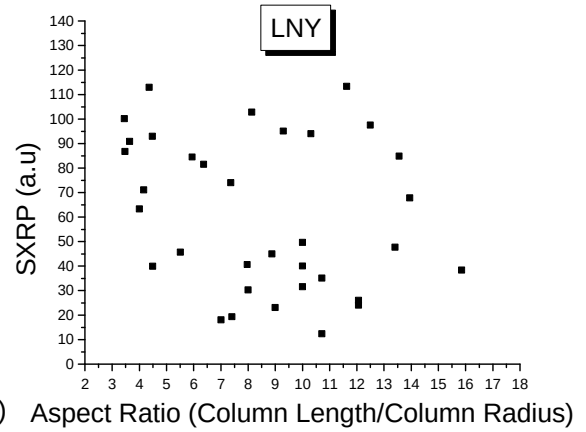


Figure 6.57: Soft X-ray production as a function of aspect ratio.

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The length is another parameter, which could be related to the emission of radiation. Figures 6.54 and 6.55 show the variation of the neutron yield and the soft X-ray production, respectively, with the length of the column. The neutron yield seems to be lower for the discharges with the longer pinch length, which appears to be more favourable for the soft X-ray production. However, the length of the column seems more closely related to the neutron production rather than the soft X-ray emission. Our data also indicate that the column radius has more effect on the radiation emission (both soft X-rays and neutrons) compared to the pinch length.

Figure 6.56 and 6.57 show the neutron yield and the soft X-ray production, respectively, as a function of the aspect ratio (length/radius). The radiation output (both the neutron yield and the soft X-ray)

appears higher for the discharges with larger aspect ratio. The scattered aspect of the plotted data show that there is no clear and conclusive correlation between the two geometrical parameters of the pinch and the radiation output. This, in turn, shows that there are many different ways in which the final pinch is formed and the radiation mechanisms are not simple. Other parameters, like the compression dynamics, implosion velocity, sheath symmetry,

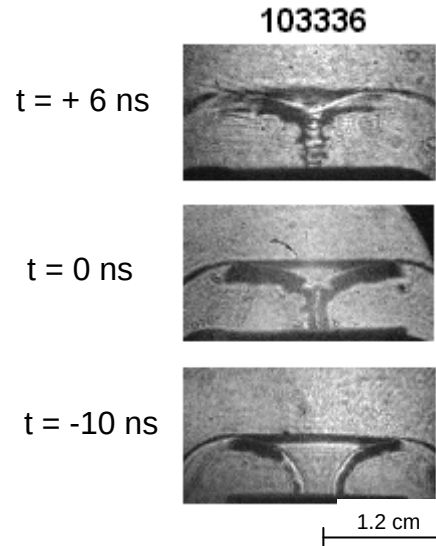
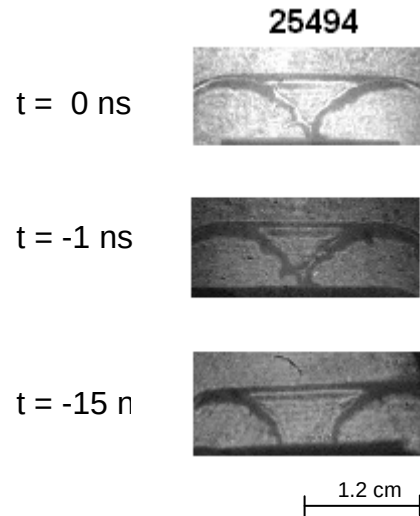


Figure 6.58: A three-frame sequence of a discharge during focus phase. $Y_n = 0.6 \times 10^8$.

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development and evolution of instabilities, pinch lifetime (confinement time), number of particles in the pinch volume, plasma density and temperature are of greater importance in the neutron production mechanism/s and radiation emission.

Figure 6.58 shows a three-frame sequence during the pinch phase for LNY category. The neutron yield for this discharge is 40% lower than the average value. The soft X-ray production is 1.5 times higher than the average value. The value of the anisotropy is 1.40. There are no signs of the development of the RT instability during the radial phase. The sheath profile is smooth. At the instant $t=0$ the column is a bit bent. It seems as if $m=1$ instability is developing but, in the later stage (top frame) the column is broken due to a very weak $m=0$ instability. During the pinch phase (top frame), the column is fluted by instabilities and turbulence developed after the first compression. The pattern was observed in many discharges exhibiting almost the same neutron yield.



In Figure 6.59, an unusual discharge is shown. This kind of discharges is seen very rarely (only 2% of the recorded discharges). The neutron yield and the soft X-ray production

have the average values. The sheath dynamics seems to be normal during the initial stage, as can be seen in the bottom frame. But, 1 ns before the maximum compression the right wing of the sheath moves faster than the left one and the compression occurs in a very peculiar manner. The column is formed somewhere inside the hollow anode.

Figure 6.59: A three-frame sequence of a discharge during focus phase. $Y_n = 1 \times 10^8$.

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A RT instability disturbing the left sector can be seen in the middle and the top frames.

In Figure 6.60 (bottom and middle frames), the column seems to behave like a “tornado”. The column looks twisted and highly compressed at the tip of the anode (bottom frame). This kind of discharges are not commonly observed (only 5% of the recorded discharges). Both, the neutron yield and the soft X-ray production have average values. The anisotropy for this discharge is 1.49.

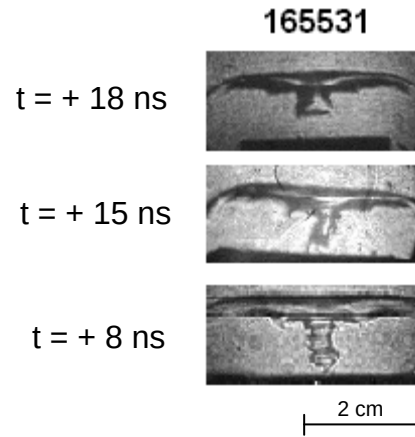


Figure 6.60: A three-frame sequence of a discharge during the focus phase. $Y_n = 1 \times 10^8$.

B) HIGH NEUTRON YIELD (HNY)

Figure 6.61 shows a three-frame sequence for the HNY category. The neutron yield for this discharge is 1.6 times the average value. The soft X-ray production is 65% the average value. The anisotropy is 1.59. The diameter of the pinched column is 0.9 mm ($\pm 10\%$), 50% bigger than the average value of the measured radius in this category. The length of the column is 3.16 ($\pm 10\%$), 60% of the average value in this category. The $m=0$ instability is seen to be developed (bottom frame) and breaking the plasma column into two halves (middle and top).

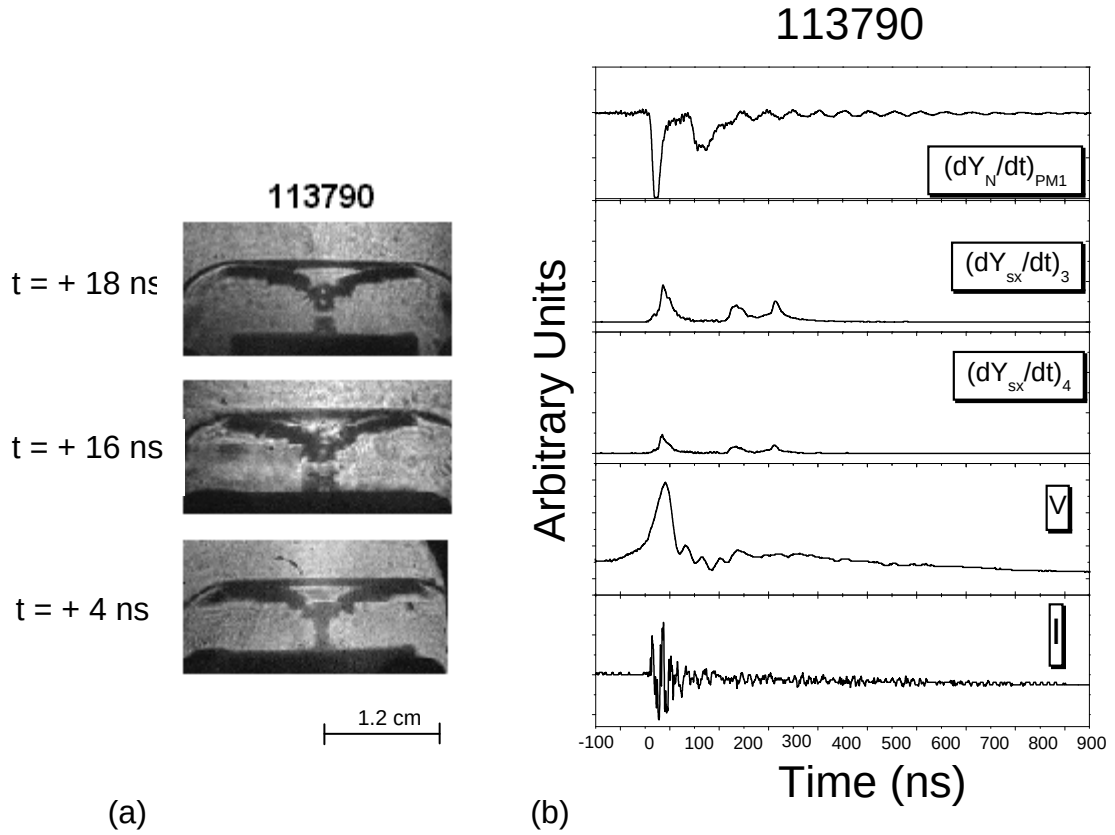


Figure 6.61: A three-frame sequence of a discharge during the focus phase; b) Corresponding signal profiles of dY_N/dt , dY_{HX}/dt , dY_{SX}/dt , V and I . $Y_n = 1.55 \times 10^8$.

For the discharges with a very high neutron yield (much higher than the average value), the column is usually broken at more than one place as shown in Figure 6.62. The neutron yield for this discharge is 1.7 times the average value, and the soft X-ray production is 65% of the average value. The anisotropy is 1.60. The plasma column is broken in three places due to the development of the $m=0$ instability. The wavelength of the instability (bottom frame) is 1.97 mm ($\pm 10\%$).

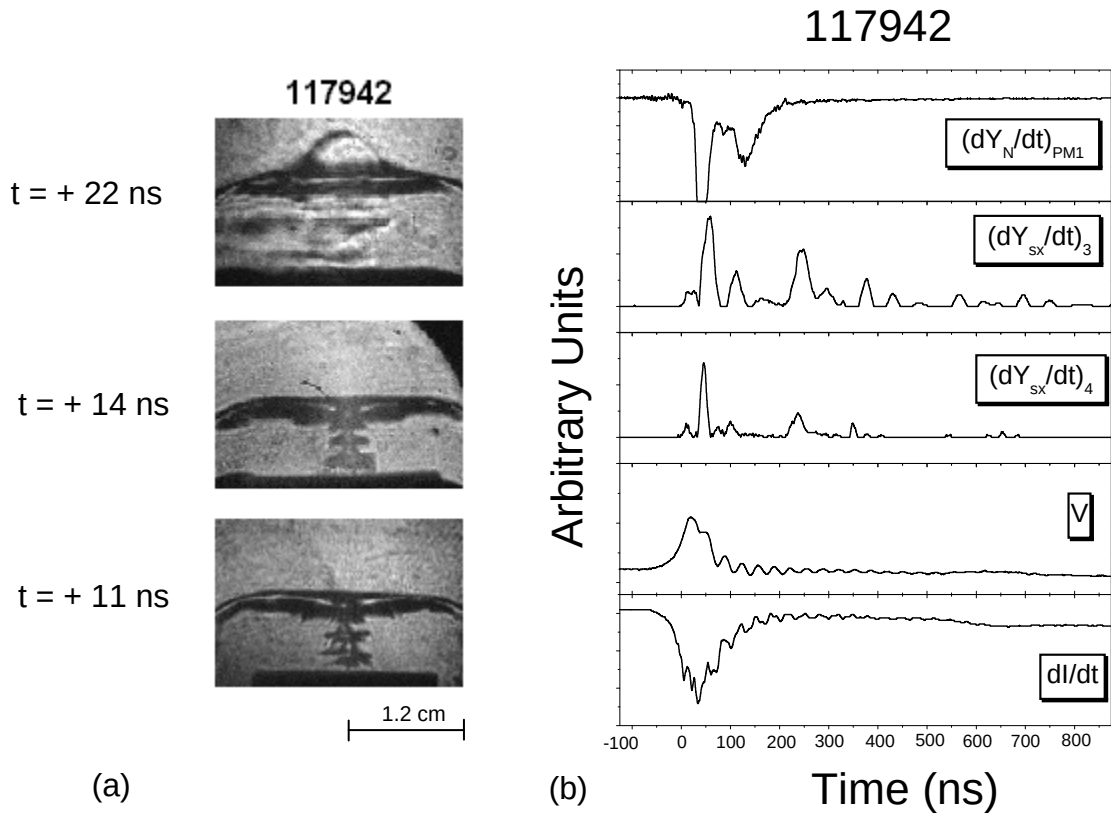


Figure 6.62: A three-frame sequence of a discharge during pre-focus phase; b) Corresponding signal profiles of dY_n/dt , dY_{HX}/dt , dY_{SX}/dt , V and dI/dt $Y_n = 1.65 \times 10^8$.

For the discharges with exceptionally high neutron yield, the dynamics after the compression are shown in Figure 6.63. For this discharge, the neutron yield is twice the average value, and the soft X-ray production is 40% of the average value. The anisotropy is 1.67. In the bottom frame, the left wing of the sheath seems to reach earlier than the right one. The column is broken in two places but the column seems to be compressed with an intense magnetic field.

The instant at which the $m=0$ instability starts to destroy the column, the neutron and X-ray emission begins. The breaking up of the column is characterised

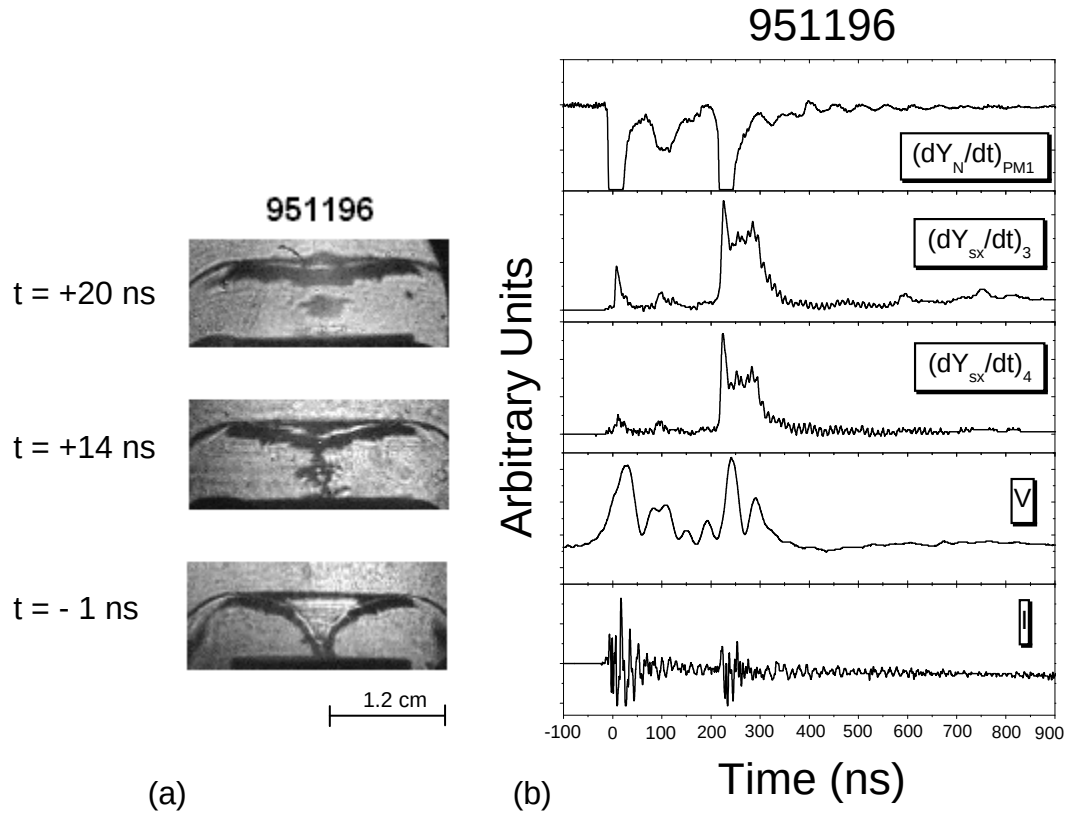


Figure 6.63: A three-frame sequence of a discharge during focus phase; b) Corresponding signal profiles of dY_N/dt , dY_{HX}/dt , dY_{SX}/dt , V and I . $Y_n = 2 \times 10^8$.

also by the development of microinstabilities. Some of them attain large intensities and decay either into the waves of possibly different frequencies, or by communicating their energy to the particles.

The breaking up may also correspond to the density fluctuation connected with the microturbulences. The density fluctuation and some other phenomena, like the non thermal emission of radiation, the heating and the acceleration of the particles have been evidenced. When the filament breaks up, the spectral analysis [200] of the scattered light indicates that the ion velocity distribution is no longer Maxwellian. Simultaneously, its intensity increases beyond the thermal level and this fact can be used to study the development of the microinstabilities in the plasma. These

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instabilities are attributed to the current which flows on the axis. The current creates a drift between the electrons and ions. The instabilities are related to the drift velocity as $V_D = 1/n_e e$. This means that with the increase of the drift velocity, the instability growth increases. In other words, the greater the drift velocity between the ion and electron, the greater the number of break-ups and higher the radiation output. Our data agrees with this assumption, because more than 65% of the recorded discharges with the high neutron yield show 2 and sometimes 3 break-ups.

Usually, the emitted radiation are attributed to the large internal density fluctuation together with the enhanced resistivity. Considering the waveform of dI/dt , the sharp dip corresponds the sharp increase in the plasma impedance. This is a necessary condition for a good operation of the machine.

The impedance of the plasma sheath consists of the inductance $L(t)$ and the resistance $R(t)$. Both vary during the propagation and the radial compression. The voltage between the two electrodes across the insulator is $V = d(LI)/dt + RI$. From the values given by magnetohydrodynamics computation for the electron temperature and the filament diameter, the resistance term is negligible compared to the inductance term. It is known, however, that the instabilities can increase the resistivity well above the Spitzer's expression for the plasma in equilibrium [200]. The dissipation process caused in the thermal plasma by collisions of the current driven electrons can be enhanced when instabilities occur. The anomalous resistance is caused by the collisions of the electrons with the waves when they reach high intensity level. A typical problem in the plasma focus studies is the role played by the anomalous resistivity in heating the plasma. This is important in heating the plasma in the connective phase of the discharge, following the break ups. This plasma appears to act

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as a source of deuterons, which are accelerated axially into the plasma formed further away from the anode, and produce neutrons.

Figure 6.64 presents a graph of the column radius as a function of the column length. It seems that the discharges with the longer column have larger radius.

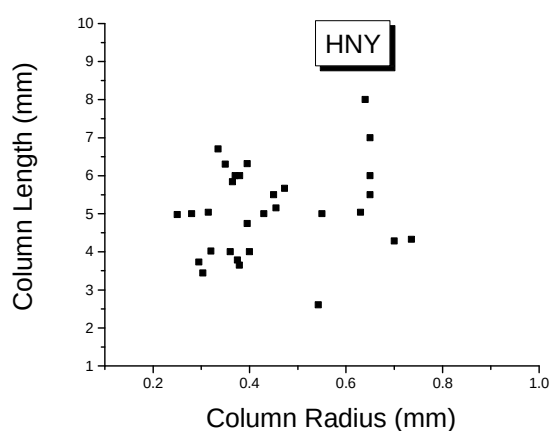


Figure 6.64: Column radius as a function of column length.

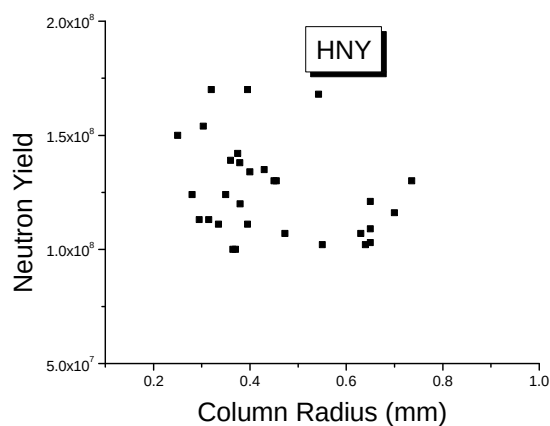


Figure 6.65: Neutron yield as a function of column radius.

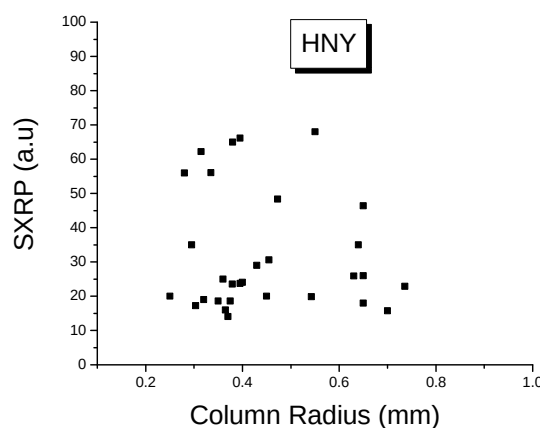


Figure 6.66: Soft X-ray production as a function of column radius

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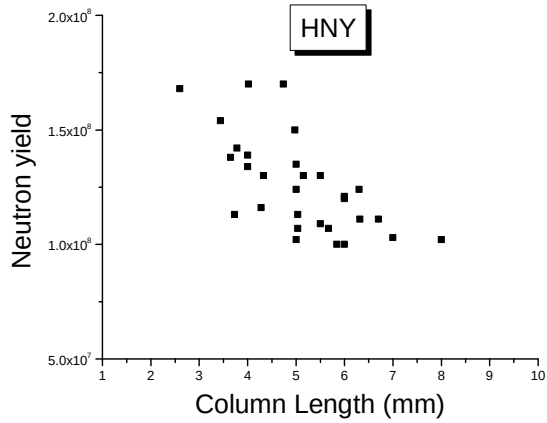


Figure 6.67: Neutron yield as a function of column length.

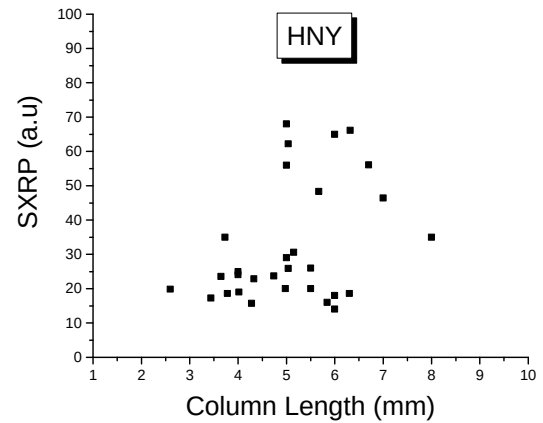


Figure 6.68: Soft X-ray production as a function of column length.

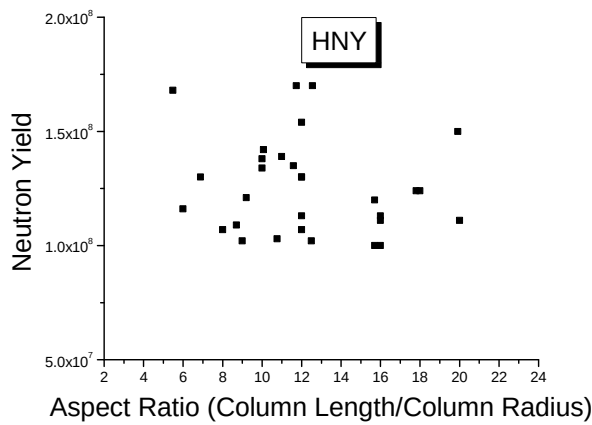


Figure 6.69: Neutron yield as a function of aspect ratio.

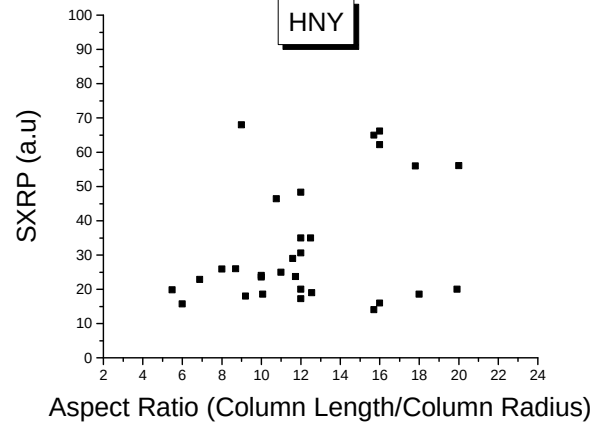


Figure 6.70: Soft X-ray production as a function of aspect ratio.

Figures 6.65 and 6.66 show the variation of the neutron yield and the soft X-ray production, respectively, with the column radius. It seems that the larger the radius, the lower the neutron yield and the higher the soft X-ray production.

The graphs in Figures 6.67 and 6.68 present the dependence of the neutron yield and the soft X-ray production, respectively, on the column length. The neutron

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yield is lower for the discharges with the longer column length and the soft X-ray production is higher.

Figures 6.69 and 6.70, respectively, show the graphs of the dependence of the neutron yield and the soft X-ray production on the aspect ratio. These graphs show that both the neutron yield and the soft X-ray production are higher for the discharges with larger aspect ratio.

From the above-discussion, it is clear that the pinch dimensions play an important role in the emission of the radiation (neutrons and X-rays) for each category. Table 6.4 gives the measured values of the column (R) radius, pinch length (L) and the aspect ratio (A) for the categories LNY and HNY.

Table 6.4: Pinch dimensions.

Category	Average Radius	Average Length	Average Aspect Ratio
	R (mm)	L (mm)	(L/R)
LNY	1.02 ($\pm$ 0.53)	6.75 ($\pm$ 0.23)	9.13
HNY	0.42 ($\pm$ 0.087)	5.08 ($\pm$ 0.29)	12.28

6.5.2.3 POST-FOCUS (DECAY) PHASE

After the maximum compression, the plasma column decays. The decay time depends on the development of the instabilities during the pinch phase.

A) LOW NEUTRON YIELD (LNY)

Figure 6.71 shows a three-frame sequence during the decay phase for category LNY. The neutron yield is 45% of the average value, and the soft X-ray production is 2 times the average value. The value of the anisotropy is 1.30.

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Usually, the decay time for the discharges in this category is longer. This is because of very slow growth rate of instabilities during the focus phase. The pinch lifetime is 29 ns ($\pm 10\%$), same as the average value for this category.

Figure 6.72 shows a three-frame sequence of a shadowgraph taken during the decay phase. The neutron yield for the shot is 90% of the average value and the soft X-ray production is same as the average value. The anisotropy is 1.48. The pinch lifetime is 25 ns ($\pm 10\%$), 87% of the average value for this category, and 30% longer than the average value for the HNY category. The reason, again, is attributed to the moderate growth rate of the instabilities.

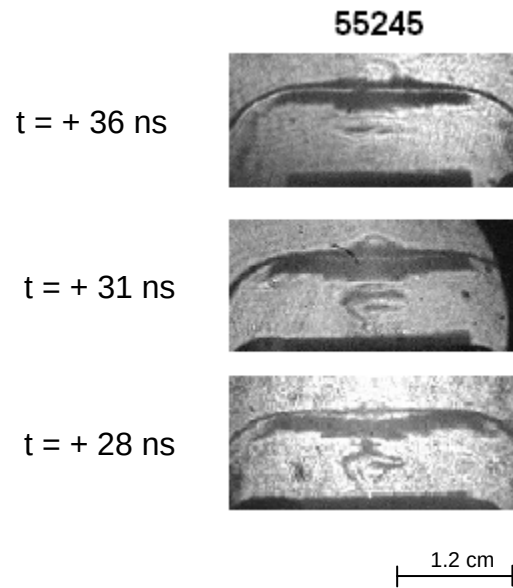


Figure 6.71: A three-frame sequence of a discharge during the post-focus phase. $Y_n = 0.45 \times 10^8$.

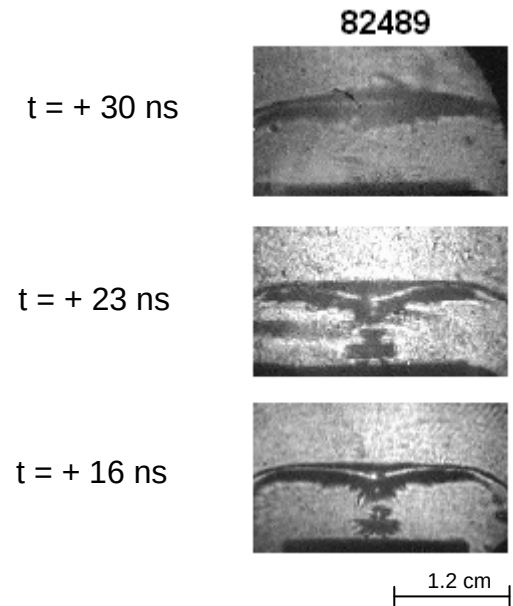


Figure 6.72: A three-frame sequence of a discharge during the post-focus phase. $Y_n = 0.9 \times 10^8$.

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Figure 6.73 shows a sequence taken some 500 ns after the pinch. In this discharge, a second focused column, along with the emission of the copper vapours is seen. Both the neutron yield and the soft X-ray production have the average values. The anisotropy for this discharge is 1.48. The electrical signal profiles corresponding to this discharge are also shown. The voltage probe signal shows a hump at the instant at which the shadowgraph is taken. This clearly indicates the existence of the second

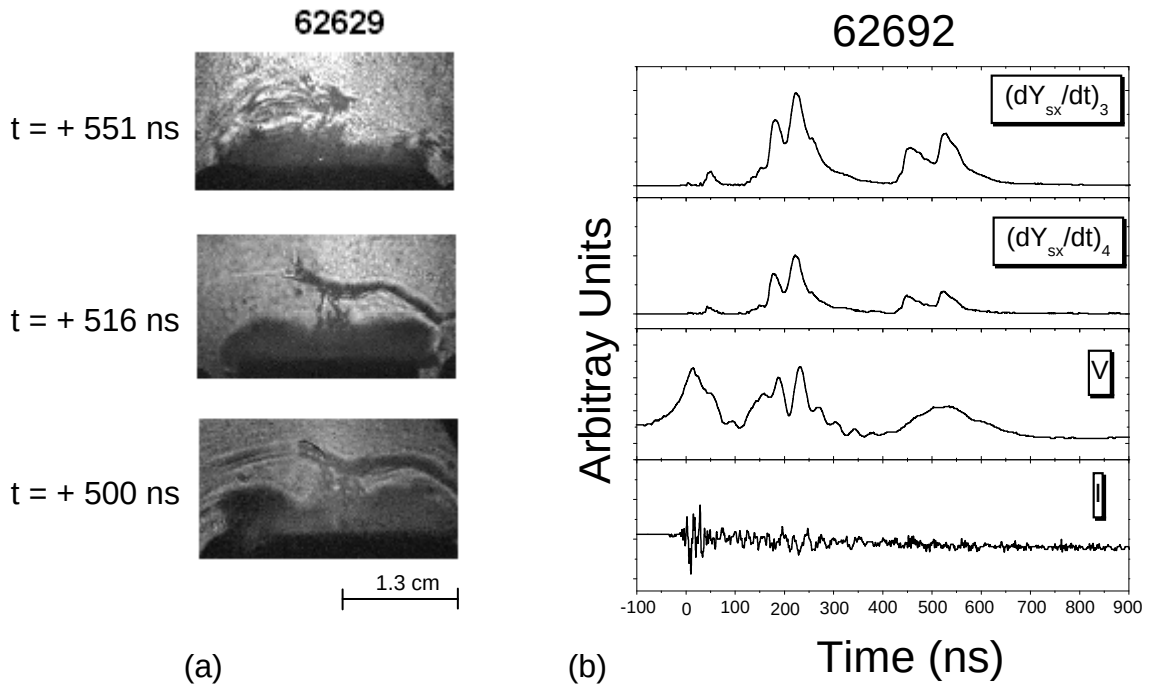


Figure 6.73: A three-frame sequence of a discharge during post-focus phase; b) Corresponding signal profiles of dY_{SX}/dt , V and I . $Y_n = 1.02 \times 10^8$.

pinch at this instant. The column is diffuse and the current sheath seems to be asymmetrical. The focus lasts for about 50 ns. There is also a strong soft X-ray emission during this time. This emission is partly due to the copper vapours from the anode material.

B) HIGH NEUTRON YIELD (HNY)

Figure 6.74 shows a sequence of a picture taken during the post-focus phase. The neutron yield for this discharge is 90% higher than the average value, and the soft X-ray production is 2 times lower than the average value. The plasma column in this

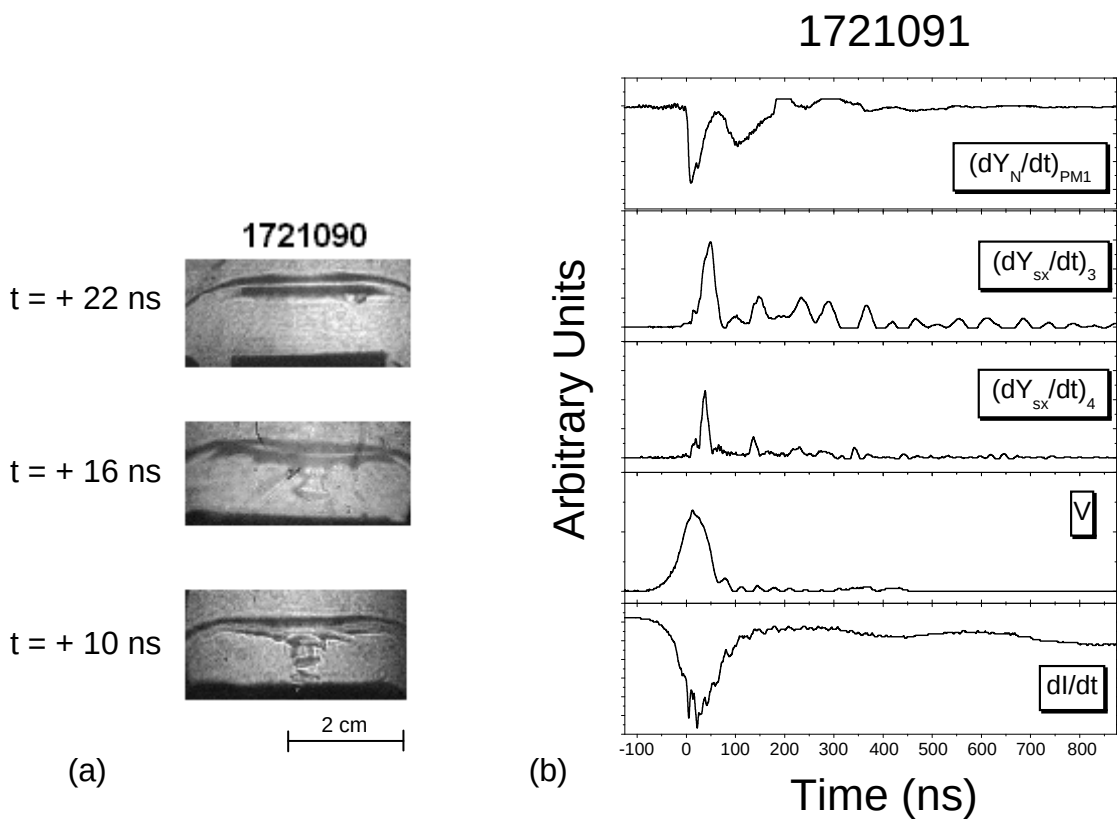


Figure 6.74: A three-frame sequence of a discharge during the post-focus phase; b) Corresponding signal profiles of dY_{SX}/dt , V and dI/dt . $Y_n = 1.9 \times 10^8$.

figure seems to be broken only on one side (bottom frame). This might be the combination of the $m=0$ and $m=1$ instabilities occurring simultaneously. The magnetic field might be acting in such a manner that the column is compressed only on one side. The pinch lifetime is 18 ns, 14% shorter than the average value in this

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category, and 40% shorter than the average value for the category LNY. The reason behind is the fast growth rate of the instabilities.

The shadowgraphic observations revealed that instabilities play a key role in the radiation emission from the plasma focus. Therefore, it is important to discuss the role of their growth rate in radiation emission and pinch lifetime.

6.5.2.4 INSTABILITIES GROWTH RATE AND PINCH LIFETIME

In this section, the growth rate of the instabilities and its correlation with the pinch lifetime and radiation emission is discussed. The growth rate $\gamma_m = (g/\lambda)^{1/2}$, (where g is the acceleration due to the characteristic force of the system and λ is the wavelength of the instability) [215, 216] for the $m=0$ instability was calculated. The results are again divided into two categories namely: LNY and HNY.

A) LOW NEUTRON YIELD (LNY)

Figure 6.75 shows a graph presenting a variation of the growth rate with the wavelength of the instabilities. The growth rate decrease with the increasing value of the wavelength.

Figure 6.76 presents the variation of the pinch lifetime with the growth rate of the $m=0$ instabilities. It is clear that the pinch lifetime is shorter for the higher values of the growth rate. It means that, for the discharges with very high growth rate, the column is ruptured very fast due to the strong $m=0$ instabilities. These instabilities grow very fast due to the magnetic field, which, in turn, accelerates the particles producing radiation.

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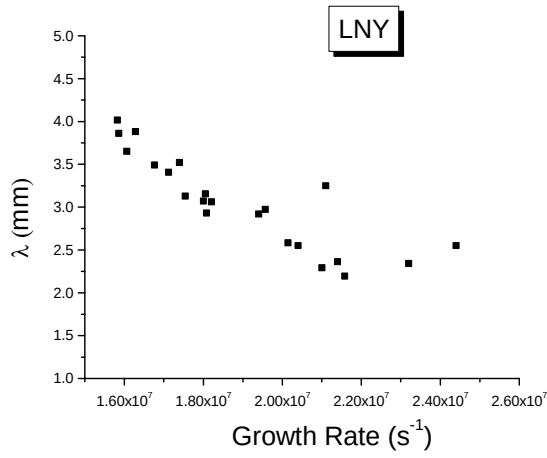


Figure 6.75: Growth rate as a function of wavelength.

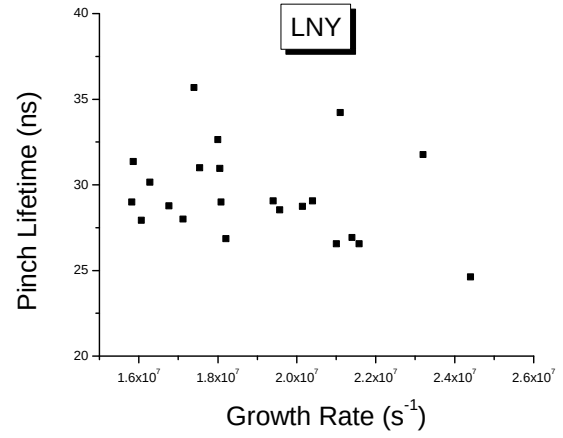


Figure 6.76: Pinch lifetime as a function of growth rate.

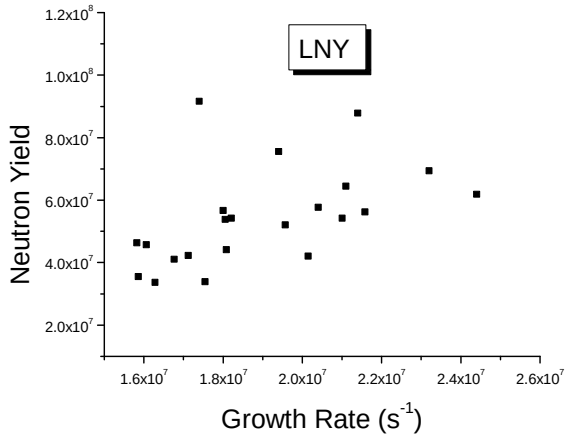


Figure 6.77: Neutron yield as a function of growth rate.

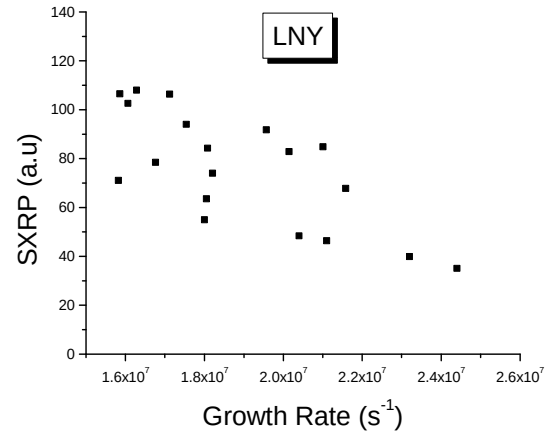


Figure 6.78: Soft X-ray production as a function of growth rate.

Figure 6.77, shows the graph of the variation of the neutron yield with the growth rate of the $m=0$ instabilities. The neutron yield is higher for the discharge with

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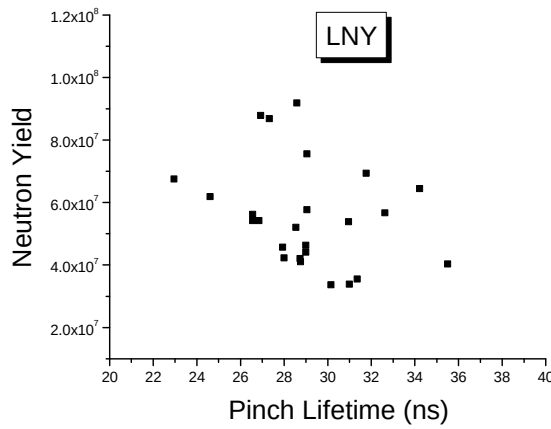


Figure 6.79: Neutron yield as a function of pinch lifetime.

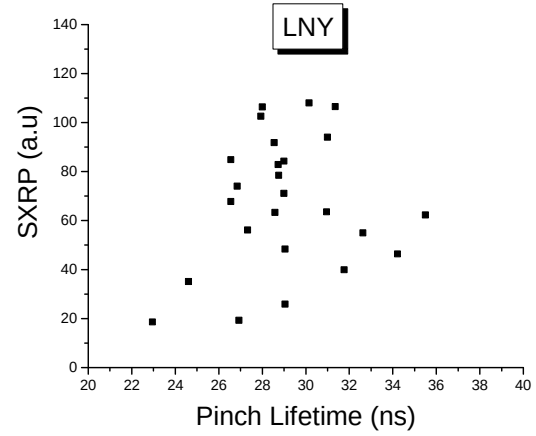


Figure 6.80: Soft X-ray production as a function of pinch lifetime.

the higher growth rate. Whereas, the soft X-ray production is lower for the discharges with the higher growth rate (Figure 6.78). It means that the higher growth rate sets the favourable conditions for the neutron yield and the lower growth rate favours the soft X-ray production. Regarding the growth rate of the instabilities, the conditions for the plasma focus to operate in the neutron-optimised mode are different from that for the soft X-ray-optimised mode. Figure 6.79 presents a graph of variation of the neutron yield with the pinch lifetime. It is clear that the neutron yield is higher for the shorter pinch lifetime. The reason behind this is the fast growth rate of the instabilities which enhances the neutron yield. As far as the dependence of the soft X-ray production on the pinch lifetime is concerned, the longer pinch lifetime favours the soft X-ray production (Figure 6.80). The reason could be attributed to the slow growth rate of the instabilities.

B) HIGH NEUTRON YIELD (HNY)

Figure 6.81 gives the graph of the variation of the growth rate with the wavelength of the $m=0$ instabilities. It is clear that for the discharges with the larger wavelength, the growth rate is slower, or vices versa.

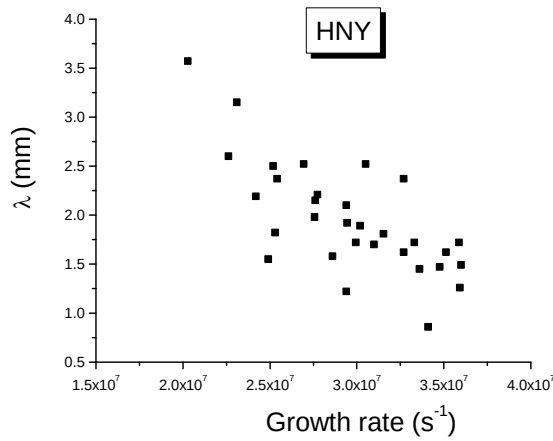


Figure 6.81: Growth rate as a function of wavelength.

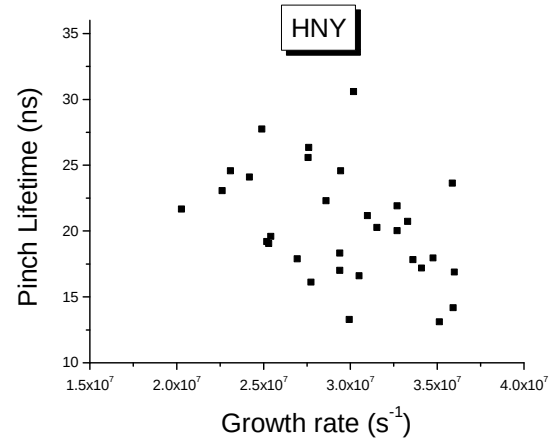


Figure 6.82: Pinch lifetime as a function of growth rate.

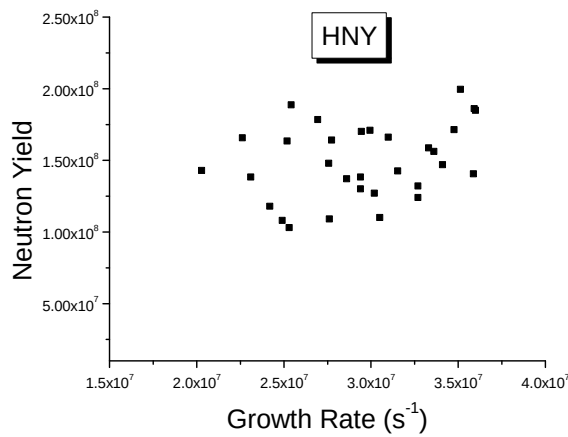


Figure 6.83: Neutron yield as a function of growth rate.

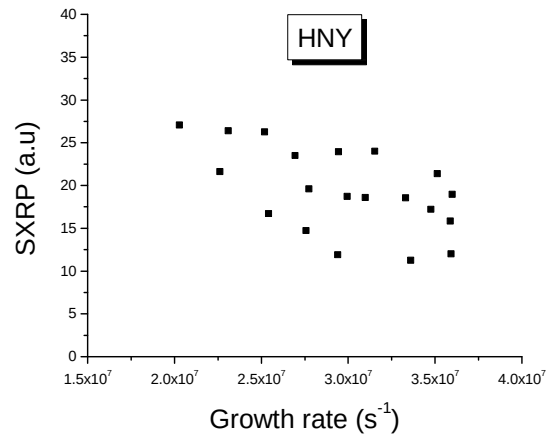


Figure 6.84: Soft X-ray production as a function of growth rate.

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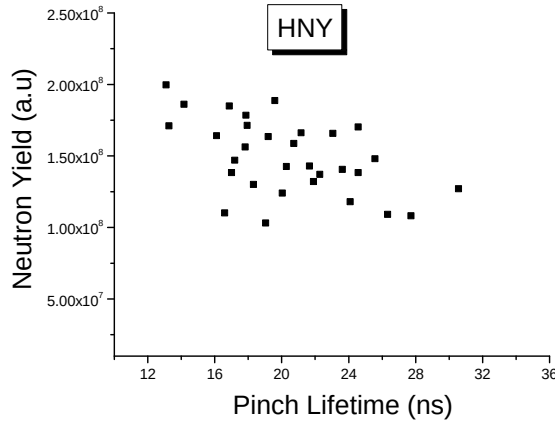


Figure 6.85: Neutron yield as function of pinch life time.

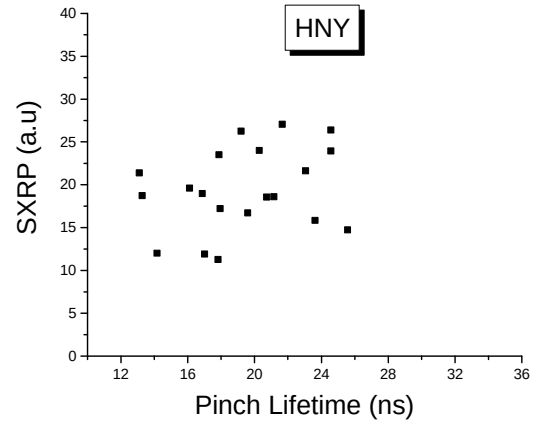


Figure 6.86: Soft x-ray production as a function of pinch lifetime.

Figure 6.82 presents a graph of the dependence of the pinch lifetime on the growth rate. The discharges with the higher growth rate have shorter pinch lifetime. This means that the column decays faster if the instabilities growth rate is higher. Figures 6.83 and 6.84 represent the graphs of the dependence of the neutron yield and the soft X-ray production, respectively, on the growth rate. The neutron yield is higher for the discharge with the higher growth rate, whereas the soft X-ray production is lower for the higher values of the growth rate.

Figures 6.85 and 6.86 show the graphs of the dependence of the neutron yield and the soft X-ray production, respectively, on the pinch lifetime. The discharges with the longer pinch lifetime exhibit low neutron yield while, for the soft X-ray production, longer pinch lifetime is favourable.

Table 6.5 presents the calculated values of the average growth rate and the pinch lifetime for the two categories.

Table 6.5: Growth rate and pinch lifetime for each category.

Category	Average Growth Rate	Average Pinch Lifetime
	$\gamma_m = (g/\lambda)^{1/2} \text{ (s}^{-1}\text{)}$	PLT (ns)
LNy	2.2E7 ($\pm 1.9\text{E6}$)	29 (± 2)
HNy	3.3E7 ($\pm 1\text{E6}$)	20 (± 3)

6.6 DYNAMICS OF THE RADIAL INWARDS SHOCK AND AXIAL ELONGATION OF THE CURRENT SHEATH

Our shadowgraphic system enabled us to perform some investigations on the focus dynamics. With the implementation of the system, the radial inwards shock motion and the axial elongation of the current sheath were determined.

Figure 6.87 shows the trajectories of the radial inward shock front for the two

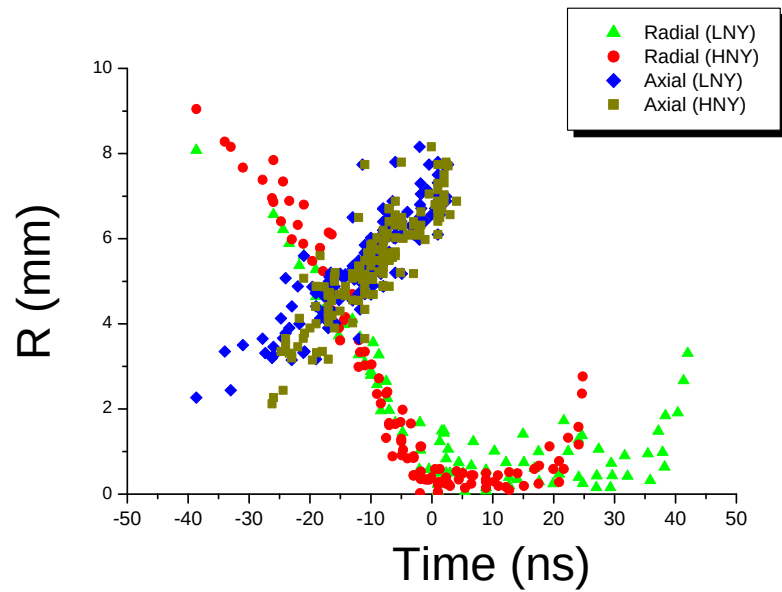


Figure 6.87: Experimentally measured trajectories of the radial inward shock front and the axial elongation of the sheath.

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categories (LNY and HNY). During the decay phase, a significant difference between the trajectories is seen. The discharges with a high neutron yield decay much faster than the low neutron yield discharges. The reason is attributable to the faster disruption of the pinched column due to the fast growth rate of the instabilities. The figure also shows the experimentally determined axial elongation of the current sheath during the radial phase for the two categories. The graphs in Figures 6.88 and 6.89 show the plots of the measured and calculated radial inwards shock velocities for the categories LNY and HNY, respectively. The velocities are calculated by taking the first derivative of the radial position of the inwards shock motion. Velocities up to 32 cm/ μ s for LNY were measured. For HNY, the velocities up to 42 cm/ μ s were determined. The higher value of the radial shock speed for the HNY category can be explained by the dependence of the shock velocity on the plasma temperature. The higher the temperature, the higher the shock speed (see Figure 4.4). The hot plasma is the source of radiation.

Table 6.6 presents the values of the radial inward shock speeds (both the experimentally measured and the calculated values) for the two categories.

Table 6.6: Experimental and calculated radial inward shock speeds for the two categories.

Category	Experimental Radial Inward Shock Speed	Calculated Radial Inward Shock Speed
	(cm/ μ s)	(cm/ μ s)
	from -40 ns to 0 ns	from -40 ns to 0 ns
LNY	13 - 32	12 - 37
HNY	14 - 42	14 - 45

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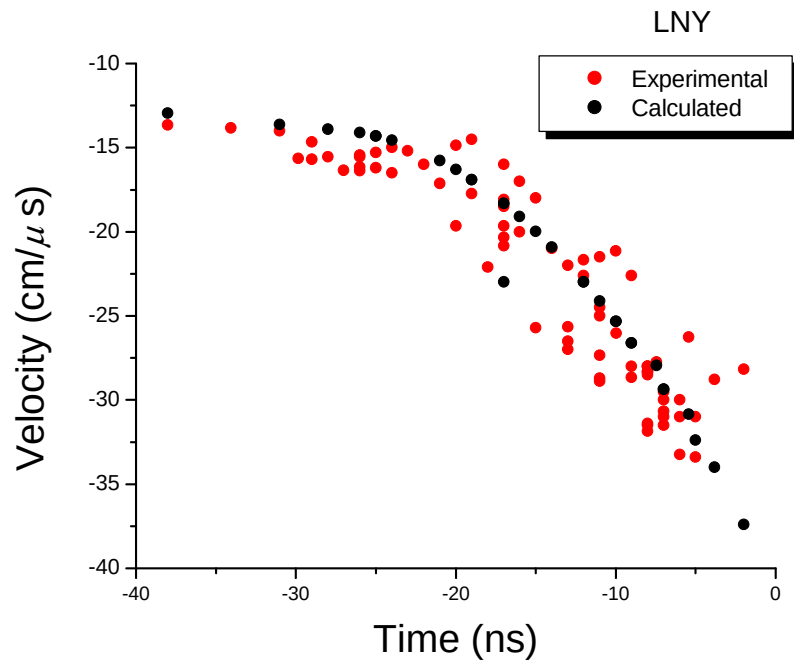


Figure 6.88: Radial inward shock speed (calculated and measured).

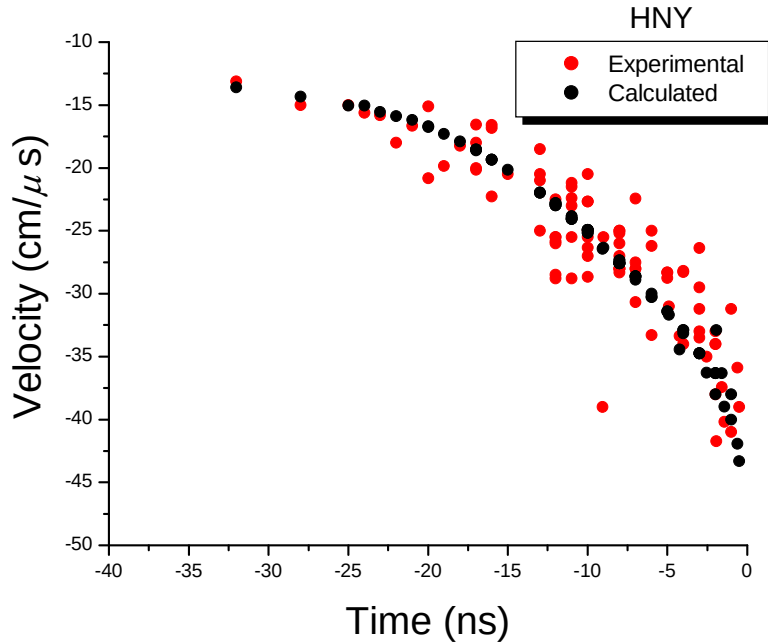


Figure 6.89: Radial inward shock speed (calculated and measured).

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The value of the shock speed for the category HNY are 10-30 % higher than that for the category LNY. This means that for the discharges with the high neutron yield, the shock moves faster than that for the discharges with low or average neutron yield.

6.6.1 COMPARISON BETWEEN EXPERIMENTAL AND THEORETICAL RESULTS

A theoretical model of plasma focus was developed along with the experiments. In this section, a comparison between the experimental and simulated results is presented.

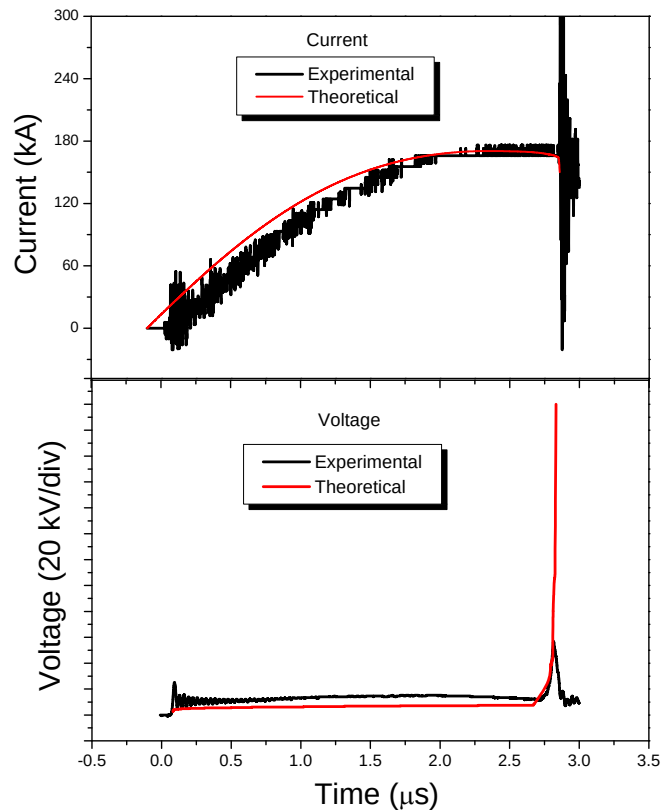


Figure 6.90: Comparison between the experimental (black) and simulated (red) for current and voltage signal profiles. Plasma Focus operated at 14.5 kV and 4 mbar of deuterium.

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Figure 6.90 compares the experimental and simulated results of the electrical diagnostics (Rogowski coil and high voltage probe) employed. The plots show that the current signal profile is in fairly good agreement with the simulated result. The simulated voltage curve is sharper with a larger amplitude as compared to the experimental one. The difference is explained by the poor frequency response of the resistive voltage probe.

Figure 6.91 presents the trajectories (experimental is the average of the LNY and HNY) of the radial inward shock (both experimental and theoretical), reflected shock (theoretical), and magnetic piston (theoretical). Our data seems to fit well with the simulated results.

When a shock front hits the axis a reflected shock develops and moves radially outwards. The piston continues to compress inwards until it hits the out going reflected shock front. At a certain point the piston meets the reflected shock. This point is called as the point of maximum gross compression, obtaining $r_{min} \sim 1$ mm. Our experimental observations using shadowgraphs show that after this point, the plasma radius remain at about the same value of r_{min} for a short period (about 20-35 ns depending on the growth rate of the instabilities) before the pinch disassembles rather violently.

Figure 6.92 presents the comparison of the experimental (average) and simulated axial elongation of the sheath. Our data fits well with the simulated results.

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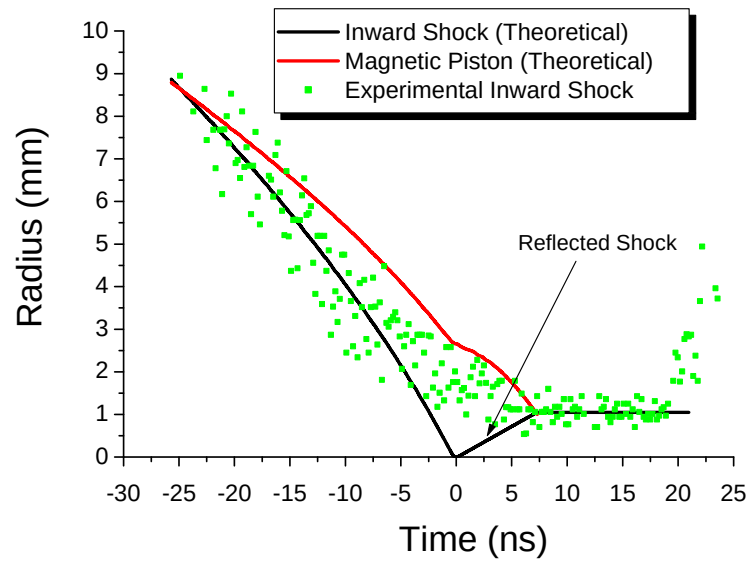


Figure 6.91: Experimental (green dots) and simulated (solid black) trajectories of the radial inward shock front, reflected shock (solid black), and magnetic piston (solid red). The experimental values represent the average of all the discharges.

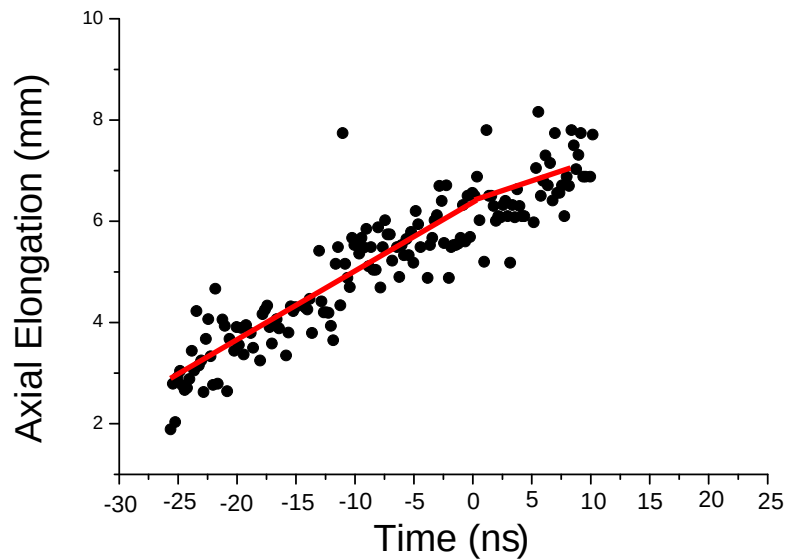


Figure 6.92: Experimental (black dots) and simulated (solid red) axial elongation of the current sheath. The experimental values represent the average of all the discharges.

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Experimentally determined radial inward shock speed and simulated results are presented in Figure 6.93. The comparison shows that both the results are in fairly good agreement.

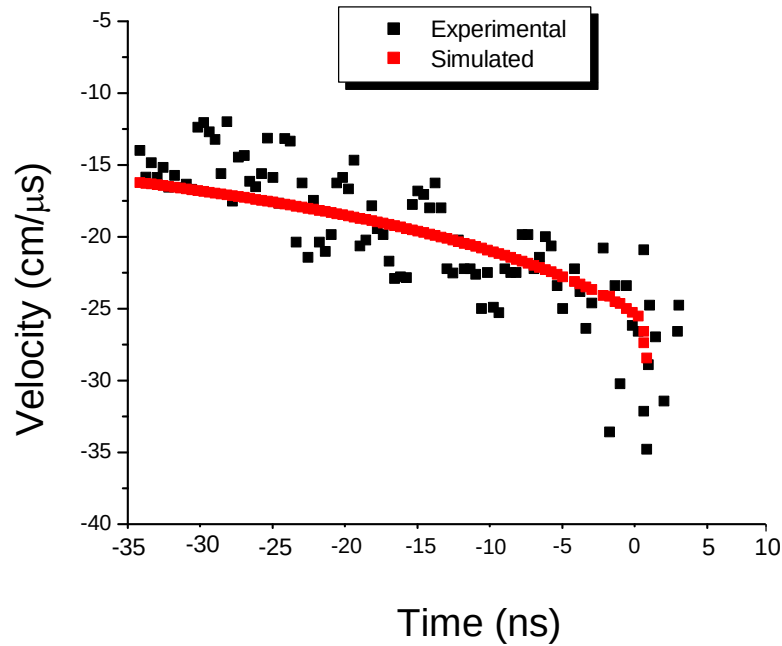


Figure 6.93: Experimental (black dots) and theoretical (red line) inward shock speeds.

The computational model gives 30 cm/μs as the final radial velocity of the plasma sheath. The length of the pinched column, about 7 mm, is in good agreement with the experimental results.

Despite some disagreements in absolute values of the discharge and the plasma parameters, the computational model has proven to be a powerful and useful instrument for simulating the evolution and gross dynamics of the discharge.

Chapter 7

CONCLUSIONS AND SUGGESTIONS FOR FURTHER WORK

The compression dynamics, discharge characteristics and the emission of radiation (neutrons, deuterons and X-rays) from a 3 kJ plasma focus were studied.

One of the objectives of the project was to obtain the energy spectra of the axially emitted deuteron beams on a shot-to-shot basis and to correlate the deuteron flux with the total neutron yield.

To attain this goal, a magnetic spectrometer was designed, constructed and implemented, successfully on the forward axis of our plasma focus device. The *CR-39* solid state nuclear track material was used in the plane of the spectrometer to detect the ions. More importantly, an automated scanning and measurement system was also successfully developed and employed to overcome the traditional time-consuming and laborious counting and measurement of ion tracks. This automated image analysis system was controlled by a PC running program written in *ALI* (analytical language for image processing) specifically developed for this application

The image processing system was designed to mimic the distinction between real and spurious tracks as judged by the human observer by making repeated comparisons of exposed and unexposed areas of detector surface and over as wide a selection of the detectors as endurance permitted. In this manner, the image

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processing algorithm and associated parameters were fine tuned. The selected algorithm comprises sequentially: a Sobel filter; binary threshold; binary fill, binary erode and dilate (opening operation).

This complex automated deuteron energy spectra analysis system represents the element of novelty of this work. The total neutron yield was monitored by an In foil activation detector placed axially on top of the focus chamber.

The simplest and most significant conclusion to be drawn from the results obtained to date is that there is a strong correlation between the neutron yield and the forwardly directed high energy deuteron beam emitted from the pinch region of the plasma focus. This correlation is present for all three pressures (3.5, 4.0 and 4.5 mbar). The deuteron energy spectrum was found to be $80 \text{ keV} < E_d < 250 \text{ keV}$ with $100 \text{ keV} < E_d < 140 \text{ keV}$ the most frequent energy range for each pressure.

The ion distribution is relatively well fitted with an exponential function of the form $\exp(E_d/B)$ with the values of B ranging from 40-100 keV.

The higher energy region of the spectra (above 125 keV) are well fitted by the empirical relation $dN/dE \propto E^{-x}$ with an x value broadly within the range of 3 to 4. For energies below 125 keV the spectra level off and then decrease towards lower energies. This is in contradiction with most theoretical calculations, which predict that the energy spectra should increase monotonically as deuteron energy decreases. Another difference occurs from the comparison of our findings with the results of previous workers who claim that deuterons in the approximate energy range of 50 to 75 keV are responsible for the bulk of neutron production via the beam-target interaction. This difference could be ascribed to the decreasing efficiency of the automated image processing system in recognising and counting tracks on the surface

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of the CR-39 detectors, as the range and specific energy loss of the particles decrease at lower energies.

A key objective of this project has been successfully attained in the development of an automated measurement system for etched ion tracks in CR-39. The development and implementation of this automated scanning system permits, we believe for the first time, a large number of individual energy spectra to be obtained.

Time-resolved investigations were performed on the emission of neutrons, soft X-rays and hard X-rays. These were correlated with the electrical discharge parameters and the characteristic time period of the focus, as well as with some of the emission parameters. The idea was to identify the focus operating regimes and the favorable condition for the optimum emission of radiation.

Optical measurements using the shadowgraphy method were carried out to study the plasma focus dynamics. A three-frame laser shadowgraphy system was successfully designed, developed and incorporated in our experiment to investigate the radial phase of the plasma focus.

It is worth mentioning the conceptualization and the implementation of the three-frame shadowgraphy system, which represents another significant effort put in the developmental work of this project. The discharge chamber was redesigned and modified; the lasers and optical benches were constructed; the triggering and synchronization circuits together with the computer- controlled image acquisition and storage systems were designed, built, accurately assembled and implemented. The two nitrogen lasers were designed and constructed. The design of the lasers was improved by using pseudo sparkgaps as switches. The lasers were more stable, reliable, and reproducible with an extremely low jitter of a few nanoseconds.

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The focus discharge and the diagnostics were triggered and controlled by a computer. Our data acquisition and analysis system enabled us to record, transfer and store data from all the diagnostics simultaneously. The time resolution achieved in our work was 1 nanosecond.

The correlation of all our diagnostic parameters helped us to identify two regimes of the plasma focus operated in deuterium. One is the single-compression regime corresponding to a single-compression of the pinch. The other one is the multiple-compression regime associated with two or more compressions of the focussed column. Our data showed that the neutron yield is greater in discharges exhibiting the single-compression regime.

An average neutron energy of $2.48 (\pm 0.04)$ MeV in the radial direction was determined using the peak-to-peak method of analysis of the data recorded in a time-of-flight arrangement. This result was combined with the time-resolved emission rate in the forward direction. An average neutron energy of $3.0 (\pm 0.09)$ MeV was estimated in support of the dominant effect of the beam-target mechanism responsible for the neutron production. These results correlate very well with our work on ion emission spectra. Deuterons of energy 100 keV are responsible for the production of 3.0 MeV neutrons in the forward direction.

The soft X-ray and the neutron emissions were correlated. While performing the neutron optimization of the device, it was observed that as the operating pressure increases from 3.0 mbar, the soft X-ray production decreases and the neutron emission is enhanced. The optimum neutron production pressure is 4.0 mbar. The further increase in the pressure causes a reduction in the neutron emission and an increase in the soft X-ray production.

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An average value of 1.48 for the neutron anisotropy was determined in our experiments. An anisotropy value of 1.53 was calculated for the discharges with high neutron yield (higher than the average value), and 1.43 for the discharges with low neutron production. For the neutron-optimized pressure, both the neutron yield and the anisotropy show their maximum values. At higher (higher than the optimum pressure) or lower (lower than the optimum pressure) pressures, there is a corresponding decrease both in the total neutron yield and the anisotropy.

For each focus discharge, the neutron anisotropy was correlated with the total neutron yield and the soft X-ray production. The correlation between the neutron yield and anisotropy reveals that, regardless of the operating pressure, the anisotropy is greater for discharges with higher neutron yield. The correlation between the anisotropy and the soft X-ray production shows that for the pressures other than 4 mbar (the neutron-optimized pressure), the variation of the anisotropy with the soft X-ray production is not significant. But, for the neutron-optimized pressure, the soft X-ray production decreases with the increasing value of the anisotropy. Thus, the increase in the neutron yield is associated with an increase in the anisotropy and decrease in the soft X-ray production.

The correlation of shadowgraphic images of the focus with the electrical parameters of the discharge, together with the X-ray and neutron emission characteristics, enabled us to identify the important periods of the focus (i.e. the radial compression, the instant when minimum pinch radius is attained, the pinch lifetime, and the post focus phenomena), and to correlate the development of instabilities with the radiation output. The implementation of this method allowed us *i)* to study the plasma dynamics during the compression phase, *ii)* to examine the temporal evolution

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of the plasma instabilities during the radial and pinch phases, *iii*) to monitor the radial implosion parameters (i.e. radial and axial shock positions, radial and axial shock speeds, etc.), *iv*) to determine the pinch lifetime and its geometrical characteristics, *v*) to correlate all this information with the radiation emission characteristics of the focus, and *vi*) to compare the experimental data with the results predicted by the theoretical model.

During the initial stage of the radial phase from -30 ns to -12 ns, velocities in the range of 10-23 cm/ μ s are determined. During the later stages of the radial phase (starting from -5 ns up till the compression), velocities of 32-42 cm/ μ s are observed in our experiment. At the instant at which the maximum compression occurs, i.e., at $t = 0$, a minimum pinch radius $r_{min} \approx 0.4 - 1.5$ mm, and a maximum column length of 6-10 mm, are observed.

The correlation of the radiation emission with the pinch characteristics shows that the neutron yield is higher for smaller pinch radius, and on the other hand, the soft X-ray production is lower. The compression of the column is one of the important parameter to set the favorable conditions for the plasma focus to work in the neutron-optimized mode. The smaller the radius, the greater the possibility for the development of the instabilities which create a strong electric field which, in turn, are responsible for the emission of the particle beams (deuterons and electrons).

The plasma column length is another parameter, which could be related to the emission of radiation. The neutron yield is lower for the discharges with longer pinch length, which appears to be more favorable for the soft X-ray production. However, the length of the column seems more closely related to the neutron production, rather than the soft X-ray emission. Our data also indicate that the column radius has more

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effect on the radiation emission (both soft X-rays and neutrons) compared to the pinch length.

The radiation output (both the neutron yield and the soft X-ray) appears higher for the discharges with larger aspect ratio. The scattered aspect of the plotted data show that there is no clear and conclusive correlation between the two geometrical parameters of the pinch and the radiation output. This, in turn, shows that there are many different ways in which the final pinch is formed and the radiation mechanism/s is/are not simple. Other parameters, like the compression dynamics, implosion velocity, sheath symmetry, development and evolution of instabilities, pinch lifetime (confinement time), number of particles in the pinch volume, plasma density and temperature are of greater importance in the neutron production mechanism/s and radiation emission.

The average column radius for the discharges with the low neutron yield is 1.02 mm. The average radius for the discharges exhibiting high neutron yield is 0.4 mm. The average determined pinch length for the discharges with the low neutron yield is 6.75 mm. The average length for the discharges exhibiting high neutron yield is 5.08 mm.

The growth rate of instabilities were also correlated with the radiation output.

The growth rate decrease with the increasing value of the instability's wavelength. The pinch lifetime is shorter for higher values of the instabilities growth rate. The neutron yield is higher for the discharge with higher growth rate. Whereas, the soft X-ray production is lower for the discharges with higher growth rate. It means that the higher growth rate sets the favorable conditions for the neutron production and the lower growth rate favors the soft X-ray emission. Regarding the growth rate

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of the instabilities, the conditions for the plasma focus to operate in the neutron-optimized mode are different from that for the soft X-ray optimized mode.

The neutron yield is higher for the shorter pinch lifetime, whereas longer pinch lifetime favors the soft X-ray production.

While investigating the radial trajectories, it is observed that the plasma radius remain at about the same value of $r_{\min}$ for a short period (about 20-35 ns before the pinch disassembles rather violently).

The measured values of the average growth rate for the discharges with the low neutron yield (LNY) is $2.2 \times 10^7 \text{s}^{-1}$, and for the discharges with high neutron yield (HNY) is $3.3 \times 10^7 \text{s}^{-1}$. The pinch lifetime for the two categories are 29 and 20 ns, respectively. Velocities from 13 up to 32 cm/ μs for LNY were measured. For HNY, radial implosion velocities from 14 up to 42 cm/ μs were determined.

Another scope of this project was to perform a computational modeling of the plasma focus by including the effect of radiation. The aim was to correlate the data with the experimental results.

The experimental current and voltage signals are in fairly good agreement with the simulated results. The experimental axial elongation of the sheath during the radial phase fits very well with the simulated results.

Experimentally determined radial inward shock speed and simulated results are also in fairly good agreement.

The computational model gives 30 cm/ μs for the final radial velocity of the plasma sheath and 7 mm for the length of the pinched column. These are also in good agreement with the experimental results.

SUGGESTION FOR FURTHER WORK

The study of the correlation between the energy spectra of individual plasma focus discharge and the associated neutron yield in order to better understand the role of the beam-target mechanism in plasma focus fusion can be extended. The efficiency of the detection system as function of deuteron energy is one aspect of the present measurement requiring careful study in future work. It is likely that the detection efficiency is dependent on many factors, some of which are difficult to quantify. It will clearly depend on the diameter and depth of the track, which, in turn, is related the bulk and track etch rates of *CR-39*, in addition to deuteron energy. Further work is required to ensure the consistency of etching conditions throughout a long sequence of experiments. Clearly the refinement of this image processing algorithm (and the performance of the hardware autofocus) over areas of varying track density, size, and surface quality is an essential area of further work.

Perhaps the most important limitation of the measurements made so far, is that each exposed position on the detector represents the cumulative spectrum of three sequential shots, which was necessary to obtain reasonable statistics in the derived spectrum. Clearly, in any attempt to correlate the characteristics of the deuteron spectrum with neutron yield, it is much preferable to obtain a deuteron spectrum for individual shots with sufficient statistical accuracy and energy resolution to give a meaningful spectrum. It is another aspect for further work in this area.

Further work is required in order to correctly deduce the energy spectrum of the deuterons emitted from the pinch region (of approximately 1mm diameter). In particular the deviation of deuteron paths from a straight line path (energy dependant) as they pass through the deuterium gas in the plasma focus chamber must be correctly

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dealt with in determining the probability for a deuteron passing through the pinhole to also pass through the slit. A Monte-Carlo simulation program will be required in order to calculate the associated correction factor as a function of deuteron energy.

Once correct deuteron energy spectra can be deduced for individual shots, it then becomes possible to calculate the expected contribution to neutron yield from the beam-target interaction. This requires assumptions about the width of the deuteron emission cone, the constancy of the deuteron energy spectrum throughout that cone, and the extrapolation of the spectrum to lower energies (most likely employing the empirical $dN/dE \propto E^{-x}$ formula). It will then be possible to compare this contribution to the neutron yield derived from the deuteron energy spectrum with the observed yield on a shot to shot basis. Such measurements should go a long way to resolving the question of whether the neutron production in the deuterium operated plasma focus discharge is predominantly of thermonuclear or beam-target origin.

Further study can be performed to establish a correlation between the deuteron spectra and the relativistic electron beams and their role in the emission of radiation (neutrons and X-rays).

The three-frame shadowgraphy system can be improved by taking the shadowgraphs in the same direction. This will give a clearer picture of the plasma dynamics. The three-frame shadowgraphic system together with the magnetic spectrometer and image analysis system can be simultaneously implemented to correlate the deuteron energy spectrum with the pinch lifetime, the geometrical characteristics of the pinch and the development of instabilities.

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CONCLUSIONS

- **Design, construction and implementation of a magnetic spectrometer**
- **Design, development and implementation of an automated deuteron energy spectra analysis system.**
- **Design, development and implementation of three-frame laser shadowgraphy system,.**

➤ The deuteron energy spectra were found to be $80 \text{ keV} < E_d < 250 \text{ keV}$ with $100 \text{ keV} < E_d < 140 \text{ keV}$ the most frequent energy range for each pressure.

➤ A strong correlation between the neutron yield and the forwardly directed high energy deuteron beam emitted from the pinch region of the plasma focus. This correlation is present for all three pressures (3.5, 4.0 and 4.5 mbar).

- The ion distribution is relatively well fitted with an exponential function of the form $\exp (E_d/B)$ with the values of B ranging from 40-100 keV.
- The higher energy region of the spectra (above 125 keV) are well fitted by the empirical relation $dN/dE \propto E^{-x}$ with an x value broadly within the range of 3 to 4.
- Two regimes of operation: One is the single-compression regime corresponding to a single-compression of the pinch. The other one is the multiple-compression

regime associated with two or more compressions of the focussed column. Our data showed that the neutron yield is greater in discharges exhibiting the single-compression regime.

➤ An average neutron energy of 2.48 ($\pm$ 0.04) MeV and 3.0 ($\pm$ 0.09) MeV in the radial direction and axial direction, respectively are observed

➤ Deuterons of energy 100 keV are responsible for the production of 3.0 MeV neutrons in the forward direction.

➤ **As the operating pressure increases from 3.0 mbar, the soft X-ray production decreases and the neutron emission is enhanced. The optimum neutron production pressure is 4.0 mbar. The further increase in the pressure causes a reduction in the neutron emission and an increase in the soft X-ray production.**

➤ **An average value of 1.48 for the neutron anisotropy. An anisotropy value of 1.53 was calculated for the discharges with high neutron yield (higher than the**

average value), and 1.43 for the discharges with low neutron production.

➤ The anisotropy is greater for discharges with higher neutron yield (regardless of the operating pressure)

➤ At the pressures other than 4 mbar (the neutron-optimized pressure), the variation of the anisotropy with the soft X-ray production is not significant. But, for the neutron-optimized pressure, the

soft X-ray production decreases with the increasing value of the anisotropy.

➤ During the initial stage of the radial phase from -30 ns to -12 ns, velocities of the sheath are in the range of 10-23 cm/μs

➤ . During the later stages of the radial phase (starting from -5 ns up till the compression), the radial inward velocities range is 32-42 cm/μs.

➤ At $t = 0$, a minimum pinch radius $r_{min} \approx 0.4 - 1.5$ mm, and a maximum column length is 6-10 mm.

➤ Neutron yield is higher for smaller pinch radius

➤ Soft X-ray production is lower for smaller pinch radius.

➤ The neutron yield is lower for the discharges with longer pinch length,

- **Longer pinch length is favorable for the soft X-ray production.**

- **The column radius has more effect on the radiation emission (both soft X-rays and neutrons) compared to the pinch length.**

- **The radiation output (both the neutron yield and the soft X-ray) appears higher for the discharges with larger aspect ratio.**

➤ The average column radius for the discharges with the low neutron yield is 1.02 mm. The average radius for the discharges exhibiting high neutron yield is 0.4 mm.

➤ The average determined pinch length for the discharges with the low neutron yield is 6.75 mm. The average length for the discharges exhibiting high neutron yield is 5.08 mm.

- **The growth rate decrease with the increasing value of the instability's wavelength.**
- **The pinch lifetime is shorter for higher values of the instabilities growth rate.**
- **The neutron yield is higher for the discharge with higher growth rate.**
- **The soft X-ray production is lower for the discharges with higher growth rate**

- **The neutron yield is higher for the shorter pinch lifetime, whereas longer pinch lifetime favors the soft X-ray production.**
- **The measured values of the average growth rate for the discharges with the low neutron yield (LNY) is $2.2 \times 10^7 \text{s}^{-1}$, and for the discharges with high neutron yield (HNY) is $3.3 \times 10^7 \text{s}^{-1}$. The pinch lifetime for the two categories are 29 and 20 ns, respectively.**

- **For HNY, radial implosion velocities from 14 up to 42 cm/ μ s were determined.**
- **The experimental current and voltage signals are in fairly good agreement with the simulated results.**
- **The experimental axial elongation of the sheath during the radial phase fits very well with the simulated results.**

➤ **Experimentally determined radial inward shock speed and simulated results are also in fairly good agreement.**

SUMMARY

This thesis reports the results of the investigations of the radiation (i.e. neutrons, deuterons, soft X-rays and hard X-rays) emitted from a 3 kJ Mather-type plasma focus device, National Institute of Education School of Science Plasma Focus Facility (*NIE-SSC-PFF*).

In order to study the correlation between the total neutron yield and the time-integrated energy spectra of the axially emitted deuteron beam, on a shot-to-shot basis, a magnetic spectrometer was successfully designed, constructed and implemented.

More importantly, an automated scanning and measurement system was also successfully developed and employed to overcome the traditional time-consuming and laborious counting and measurement of ion tracks.

Our analysis shows a clear correlation between the intensity of the axially emitted deuteron beam and the corresponding total neutron yield. The deuteron energy spectra range from 80 keV to 250 keV, with $100 \text{ keV} < E_d < 140 \text{ keV}$ as the most frequent energy range for each operating pressure. The experimental ion energy distribution is relatively well fitted by an exponential function of the form $\exp(-E_d/B)$. It is also noted that the spectra at higher energies are in good agreement with the empirical relation $dN/dE \propto E^{-x}$.

The time-resolved soft X-ray measurements employed two filtered *PIN* diodes. For some discharges, the electron temperature of the pinch was determined by employing the filter ratio method. The time-resolved hard X-ray and neutron

measurements were performed using three scintillator-photomultiplier systems in a time-of-flight arrangement.

An average neutron energy of $2.48 (\pm 0.04)$ MeV and $3.00 (\pm 0.09)$ MeV were determined in the radial and the axial directions, respectively. Deuterons of energy 100 keV were found to be responsible for the production of 3.00 MeV neutrons in the forward direction.

An average value of 1.48 for the neutron anisotropy was determined. The correlation of the neutron anisotropy with the total neutron yield revealed that, regardless of the operating pressure, the anisotropy is greater for discharges with higher neutron yield. The correlation between the anisotropy and the soft X-ray production shows that the anisotropy is lower for discharges with high soft X-ray production (regardless of the operating pressure).

The soft X-ray and the total neutron yield were also correlated. The discharges with high neutron yield exhibit low soft X-ray production. The higher operating pressures favour the soft X-ray production.

Two regimes of the plasma focus were identified namely: the single-compression regime and the multiple-compression. The neutron yield is greater in discharges exhibiting the single-compression regime.

A three-frame computer-controlled laser shadowgraphic system was designed, developed and implemented successfully.

The growth rate of the instabilities was correlated with the radiation output: higher growth rates favour neutron emission. A correlation of the instability growth rate with the pinch lifetime shows that the pinch lifetime is shorter for the discharges with higher growth rates. The radiation output is higher for the discharges with shorter

pinch lifetime. The pinch dimensions (column radius, column length and aspect ratio) are also correlated with the radiation output. The correlation shows that the neutron yield is higher for smaller radius, and the soft X-ray production is lower. The neutron yield is lower for the discharges with the longer pinch length, which appears to be more favorable for the soft X-ray production. The radiation output (both the neutron yield and the soft X-ray) appears higher for the discharges with larger aspect ratio.

The trajectories of the inward radial shock front and its axial elongation were determined from the shadowgraphic images. Radial and axial elongation velocities were also calculated. The discharges with higher radial implosion speeds exhibit higher radiation output.

A computational modeling of the plasma focus was also performed. The experimentally measured radial and axial elongation trajectories, along with the radial implosion speed were compared with the simulated results. The simulated results are in fairly good agreement with the experimental data.

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PLASMA FOCUS

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